MIL-HDBK-264(SH)
30 September 1980

MILITARY HANDBOOK

PROPERTIES OF STEEL SHAPES AND PLATE–BEAM COMBINATION USED IN SHIPBUILDING

FSC 19GP

☆U.S. GOVERNMENT PRINTING OFFICE: 1980—703-023/4437

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DEPARTMENT OF THE NAVY

NAVAL SEA SYSTEMS COMMAND

WASHINGTON, D.C. 20362

MIL-HDBK-264(SH)
Properties of Steel Shapes and Plate -
Beam Combination Used in Shipbuilding

1. This standardization handbook was developed by the Department of Defense in accordance with established procedure.

2. This publication was approved on 30 September 1980 for printing and inclusion in the military standardization handbook series.

3. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FOREWORD

The purpose of this handbook is to provide convenient tabulations of properties of structural steel sections which are readily available from the principal producers of structural steel shapes. The handbook is intended for use in the design of ship structure.

This handbook is the result of the combined efforts of the Surface Ship Structures Branch of the Naval Sea Systems Command (NAVSEA) and the Ship Structures Division of the David W. Taylor Naval Ship Research and Development Center (DTNSRDC).

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1. SCOPE

1.1 Scope. This handbook contains tabulations of the properties of the following items (using the 1978 American Institute of Steel Construction standard series of wide flange shapes):

Combined beam and plate:	Columns:	Structural shape catalogs:
Tees and plates	Wide flange shapes	Tees
Angles and plates	Tubes (iron pipe sizes)	Angles

1.2 Application. The tables for properties of combined beam and plate are intended to be used in the design of ship structure where the plate is considered to be acting as one of the flanges of the attached member. For properties of combined beam and plate based on earlier standard steel shapes see NAVSHIPS 250-443-1, Manual of Properties of Combined Beam and Plate, Part I - Tees and Angles, Part II - Flanged Plate.

2. REFERENCED DOCUMENTS

2.1 Issues of documents. (Not applicable).

2.2 Other publications. The following document forms a part of this handbook to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION, INC.
Manual of Steel Construction.

(Application for copies should be addressed to the American Institute of Steel Construction, Inc., 101 Park Avenue, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. DEFINITIONS

3.1 Symbol nomenclature and definitions. Symbol nomenclature and definitions shall be as follows:

A	= Area of the shape, in inches squared.
ALLOWABLE LENGTH	= Allowable length of a column based on $\frac{L}{R} = 60$, in feet.
ASH	= Shear area of shape only (depth of beam times thickness of web), inches squared.
C	= Channel CUT to an angle.
CTOE	= Distance from the neutral axis of the shape only to the toe of the shape, in inches.
D	= Depth of the beam or column, in inches.
I	= Inertia of the combined beam and plate or column, in inches to the fourth.

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IO	= Inertia of the shape only, in inches to the fourth.
R	= Radius of gyration, in inches cubed.
S	= Section modulus with respect to the indicated axis, in inches cubed.
SECTION MODULUS FLANGE	= Section modulus of combined beam and plate to the flange, in inches cubed.
SECTION MODULUS PLATE	= Section modulus of combined beam and plate to the plate, in inches cubed.
T	= Thickness of plate.
TF	= Thickness of flange, in inches.
TW	= Thickness of web, in inches.
WF	= Width of flange, in inches.
WT/FT	= Weight of shape only (after it is cut) in pounds per foot.
YF	= Distance from the neutral axis of the combined beam and plate to the flange, in inches.
YP	= Distance from the neutral axis of the combined beam and plate to the plate, in inches.

See figures 1 and 2 for a pictorial representation of some of these symbols and definitions.

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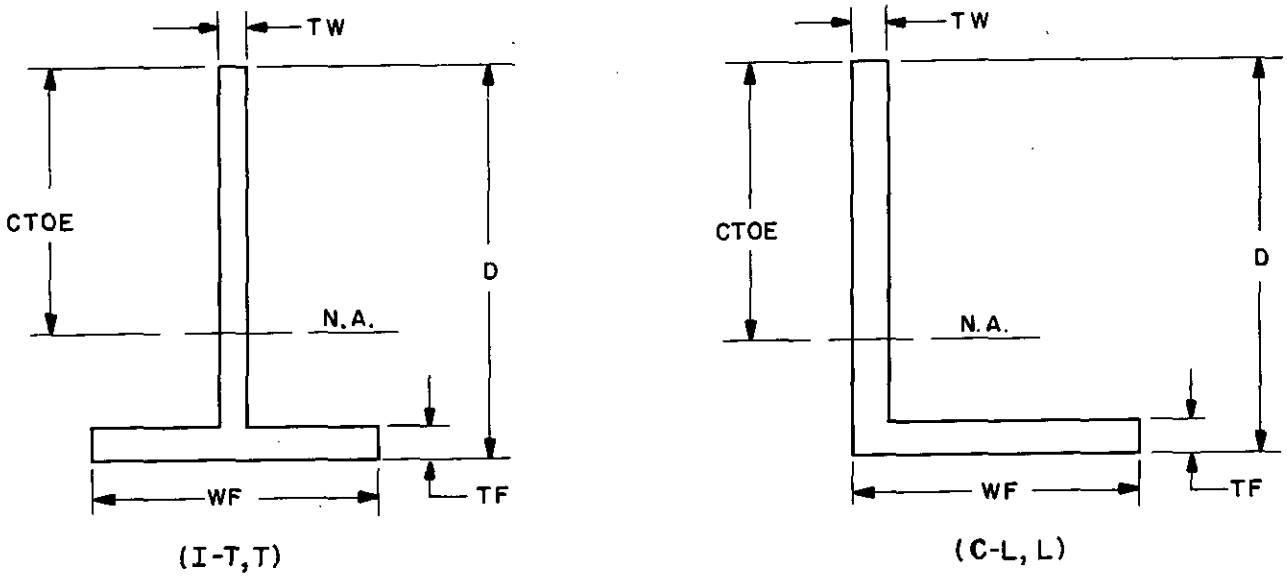


FIGURE 1. Typical structural shape (tee, angle).

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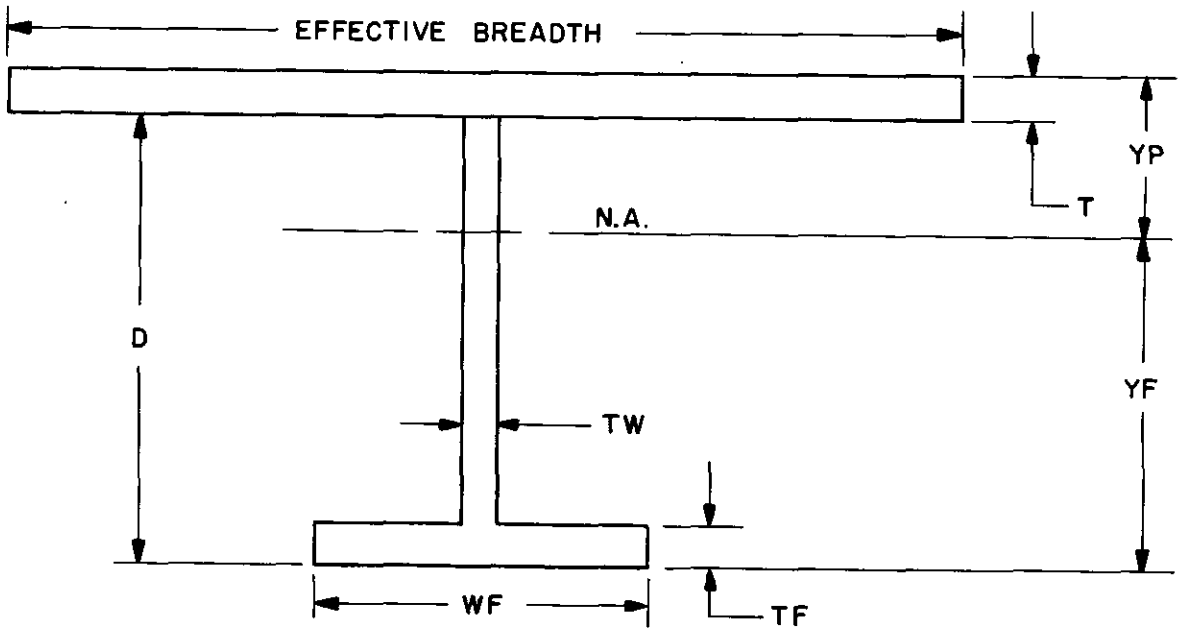


FIGURE 2. Typical tee-beam and plate combination.

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4. GENERAL REQUIREMENTS

4.1 Shapes. The steel shapes selected for inclusion in tables I thru XIV are those produced as or cut from U.S. commercial standards. The sizes were chosen based on experience of actual use.

4.1.1 Properties. The properties indicated in tables II thru XIV are for rolled shapes. These are produced by one of the following methods:

(a) By cutting off two sides of one flange of an I-beam even with the web, a tee-beam can be made having the same depth as the original beam (indicated as an I cut to a tee or I-T).
(b) By cutting through the middle of the web of an I-beam to make two tees having a depth of one-half that of the original beam (indicated as a T).
(c) By off-center cutting through the web of an I-beam (for nominal depths greater than 12 inches) a tee-beam can be made having a depth which may vary from a maximum equal to the original depth of the beam to a minimum equal to the width of the flange (indicated as a tee cut from a wide flange or CF W).
(d) By cutting off one flange of a channel even with the web, an angle can be made having the same depth as the original channel (indicated as a channel cut to an angle or C-L).

Weights and properties have been calculated from the theoretical dimensions which appear in the Manual of Steel Construction. Fillets or rounded edges have not been included in computing the weights and properties of all sections. Weights of all sections are computed on the bases that one cubic inch of steel weighs 0.2833 pounds.

4.2 Effective breadth of plating. The selection of an effective breadth of plating to be assumed acting in association with any shape (tee or angle) is influenced by the yield strength and the modulus of elasticity of the material. This is calculated by the formula $2\sqrt{E/Fy}\ t$ when E is the modulus of elasticity (lb/in^2a), Fy is the tensile yield strength (lb/in^2a) and t is the thickness of the plate. If the plating and beams have different yield strength, the lower value yield strength is used in the formula.

4.2.1 Values. The values of effective breadth of plating in tables VIII thru XI have been selected as 60t, 50t, 38t, and 35t for tee shapes and tables XII and XIII have been selected as 60t and 50t for angle shapes. These correspond to the actual values for MS (60t), HTS (50t), HY80 (38t), and HY100 (35t).

4.2.1.1 Tables VIII thru XIII apply only if the calculated stiffener spacing is equal to or greater than the effective breadth of plating. Where stiffener spacing is less than the nominal effective breadth, equivalent plating having the required area can be used by selecting an appropriate thickness. This method will result in only negligible error for most cases.

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4.3 Limitations for shape attachment to plate. In order to avoid unreasonable combinations of beams and plates, the following limitations have been imposed on selecting shapes for attachment to plates whose thickness is greater than 0.1875 inches:

(a) The neutral axis of the combined section must lie outside the plate.
(b) The thickness of the web must not exceed the thickness of the plate to which it is attached.
(c) For plates 0.1875 inches thick or less use shape sizes for plate of 0.2188 inches thick.

Preparing activity:
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TABLE I. Effective breadths and equivalent areas (60t, 50t, 38t, 35t).

EFFECTIVE BREADTHS AND EQUIVALENT AREAS										
PLATE			60T		50T		38T		35T	
LBS/FT2	DECIMAL	FRACTION	BREADTH	AREA	BREADTH	AREA	BREADTH	AREA	BREADTH	AREA
	IN	IN	IN	IN2	IN	IN2	IN	IN2	IN	IN2
5.100	.1250	1/8	7.50	.94	6.25	.78	4.75	.59	4.38	.55
6.375	.1563	5/32	9.38	1.46	7.81	1.22	5.94	.93	5.47	.85
7.650	.1875	3/16	11.25	2.11	9.38	1.76	7.13	1.34	6.56	1.23
8.925	.2188	7/32	13.13	2.87	10.94	2.39	8.31	1.82	7.66	1.67
10.200	.2500	1/4	15.00	3.75	12.50	3.13	9.50	2.38	8.75	2.19
11.475	.2813	9/32	16.88	4.75	14.06	3.96	10.69	3.01	9.84	2.77
12.750	.3125	5/16	18.75	5.86	15.63	4.88	11.88	3.71	10.94	3.42
14.025	.3438	11/32	20.63	7.09	17.19	5.91	13.06	4.49	12.03	4.14
15.300	.3750	3/8	22.50	8.44	18.75	7.03	14.25	5.34	13.13	4.92
17.850	.4375	7/16	26.25	11.48	21.88	9.57	16.63	7.27	15.31	6.70
20.400	.5000	1/2	30.00	15.00	25.00	12.50	19.00	9.50	17.50	8.75
22.950	.5625	9/16	33.75	18.98	28.13	15.82	21.38	12.02	19.69	11.07
25.500	.6250	5/8	37.50	23.44	31.25	19.53	23.75	14.84	21.88	13.67
28.050	.6875	11/16	41.25	28.36	34.38	23.63	26.13	17.95	24.06	16.54
30.600	.7500	3/4	45.00	33.75	37.50	28.13	28.50	21.38	26.25	19.69
35.700	.8750	7/8	52.50	45.94	43.75	38.28	33.25	29.09	30.63	26.80
40.800	1.0000	1.0	60.00	60.00	50.00	50.00	38.00	38.00	35.00	35.00
45.900	1.1250	1 1/8	67.50	75.94	56.25	63.28	42.75	48.09	39.38	44.30
51.000	1.2500	1 1/4	75.00	93.75	62.50	78.13	47.50	59.38	43.75	54.69
56.100	1.3750	1 3/8	82.50	113.44	68.75	94.53	52.25	71.84	48.13	66.17
61.200	1.5000	1 1/2	90.00	135.00	75.00	112.50	57.00	85.50	52.50	78.75
71.400	1.7500	1 3/4	105.00	183.75	87.50	153.13	66.50	116.38	61.25	107.19

EFFECTIVE BREADTHS AND EQUIVALENT AREAS

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TABLE I. Effective breadths and equivalent areas (60t, 50t, 38t, 35t). - Continued

EFFECTIVE BREADTHS AND EQUIVALENT AREAS										
PLATE			60T		50T		38T		35T	
LBS/FT2	DECIMAL	FRACTION	BREADTH	AREA	BREADTH	AREA	BREADTH	AREA	BREADTH	AREA
	IN	IN	IN	IN2	IN	IN2	IN	IN2	IN	IN2
81.600	2.0000	2.0	120.00	240.00	100.00	200.00	76.00	152.00	70.00	140.00
91.800	2.2500	2 1/4	135.00	303.75	112.50	253.13	85.50	192.38	78.75	177.19
102.000	2.5000	2 1/2	150.00	375.00	125.00	312.50	95.00	237.50	87.50	218.75
112.200	2.7500	2 3/4	165.00	453.75	137.50	378.13	104.50	287.38	96.25	264.69
122.400	3.0000	3.0	180.00	540.00	150.00	450.00	114.00	342.00	105.00	315.00
132.600	3.2500	3 1/4	195.00	633.75	162.50	528.13	123.50	401.38	113.75	369.69
142.800	3.5000	3 1/2	210.00	735.00	175.00	612.50	133.00	465.50	122.50	428.75
153.000	3.7500	3 3/4	225.00	843.75	187.50	703.13	142.50	534.38	131.25	492.19
163.200	4.0000	4.0	240.00	960.00	200.00	800.00	152.00	608.00	140.00	560.00

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TABLE II. Properties of wide flange columns.

PROPERTIES OF WIDE FLANGE COLUMNS

NOMINAL SIZE		WT/FT	A	ELASTIC PROPERTIES									
				AXIS X-X			AXIS Y-Y				FLANGE		
				I	S	R	I	S	R	D	WF	TF	TW	ALLOWABLE LENGTH
IN X IN X LBS/FT		LBS	IN2	IN4	IN3	IN	IN4	IN3	IN	IN	IN	IN	IN	FT
4 X 4	X 13 I	13.0	3.8	11.3	5.5	1.72	3.9	1.9	1.00	4.16	4.060	.345	.280	5.0
6 X 6	X 15 I	15.1	4.4	29.1	9.7	2.56	9.3	3.1	1.45	5.99	5.990	.260	.230	7.3
6 X 6	X 20 I	20.0	5.9	41.4	13.4	2.66	13.3	4.4	1.50	6.20	6.020	.365	.260	7.5
6 X 6	X 25 I	25.0	7.3	53.4	16.7	2.70	17.1	5.6	1.52	6.38	6.080	.455	.320	7.6
8 X 8	X 31 I	31.0	9.1	110.0	27.5	3.47	37.1	9.3	2.02	8.00	7.995	.435	.285	10.1
8 X 8	X 35 I	35.0	10.3	127.0	31.2	3.51	42.6	10.6	2.03	8.12	8.020	.495	.310	10.2
8 X 8	X 40 I	39.8	11.7	146.0	35.5	3.53	49.1	12.2	2.04	8.25	8.070	.560	.360	10.2
8 X 8	X 48 I	47.9	14.1	184.0	43.3	3.61	60.9	15.0	2.08	8.50	8.110	.685	.400	10.4
8 X 8	X 58 I	58.1	17.1	228.0	52.0	3.65	75.1	18.3	2.10	8.75	8.220	.810	.510	10.5
8 X 8	X 67 I	67.0	19.7	272.0	60.4	3.72	88.6	21.4	2.12	9.00	8.280	.935	.570	10.6
10 X 10	X 49 I	49.0	14.4	272.0	54.6	4.35	93.4	19.7	2.54	9.98	10.000	.560	.340	12.7
10 X 10	X 54 I	53.7	15.8	303.0	60.0	4.37	103.0	20.6	2.56	10.09	10.030	.615	.370	12.8
10 X 10	X 60 I	59.8	17.6	341.0	66.7	4.39	116.0	23.0	2.57	10.22	10.080	.680	.420	12.9
10 X 10	X 68 I	68.0	20.0	394.0	75.7	4.44	134.0	26.4	2.59	10.40	10.130	.770	.470	13.0
10 X 10	X 77 I	76.8	22.6	455.0	85.9	4.49	154.0	30.1	2.60	10.60	10.190	.870	.530	13.0
10 X 10	X 88 I	88.1	25.9	534.0	98.5	4.54	179.0	34.8	2.63	10.84	10.265	.990	.605	13.2
10 X 10	X 100 I	100.0	29.4	623.0	112.0	4.60	207.0	40.0	2.65	11.10	10.340	1.120	.680	13.3
10 X 10	X 112 I	111.9	32.9	716.0	126.0	4.66	236.0	45.3	2.68	11.36	10.415	1.250	.755	13.4
12 X 12	X 65 I	64.9	19.1	533.0	87.9	5.28	174.0	29.1	3.02	12.12	12.000	.605	.390	15.1
12 X 12	X 72 I	71.7	21.1	597.0	97.4	5.31	195.0	32.4	3.04	12.25	12.040	.670	.430	15.2
12 X 12	X 79 I	79.9	23.2	662.0	107.0	5.34	216.0	35.8	3.05	12.38	12.080	.735	.470	15.3
12 X 12	X 87 I	87.0	25.6	740.0	118.0	5.38	241.0	39.7	3.07	12.53	12.125	.810	.515	15.4
12 X 12	X 96 I	95.9	28.2	833.0	131.0	5.44	270.0	44.4	3.09	12.71	12.160	.900	.550	15.5

PROPERTIES OF WIDE FLANGE COLUMNS

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TABLE II. Properties of wide flange columns. - Continued

PROPERTIES OF WIDE FLANGE COLUMNS														
			ELASTIC PROPERTIES											
			AXIS X-X			AXIS Y-Y				FLANGE				
NOMINAL SIZE		WT/FT	A	I	S	R	I	S	R	D	WF	TF	TW	ALLOWABLE LENGTH
IN X IN X LBS/FT		LBS	IN2	IN4	IN3	IN	IN4	IN3	IN	IN	IN	IN	IN	FT
12 X 12	X 106 I	106.1	31.2	933.0	145.0	5.47	301.0	49.3	3.11	12.89	12.220	.990	.610	15.6
12 X 12	X 120 I	120.0	35.3	1070.0	163.0	5.51	345.0	56.0	3.13	13.12	12.320	1.105	.710	15.7
12 X 12	X 136 I	135.7	39.9	1240.0	186.0	5.58	398.0	64.2	3.16	13.41	12.400	1.250	.790	15.8
12 X 12	X 152 I	152.0	44.7	1430.0	209.0	5.66	454.0	72.9	3.19	13.71	12.480	1.400	.870	16.0
12 X 12	X 170 I	170.0	50.0	1650.0	235.0	5.74	517.0	82.3	3.22	14.03	12.570	1.560	.960	16.1
12 X 12	X 190 I	189.7	55.8	1890.0	263.0	5.82	589.0	93.0	3.25	14.38	12.670	1.735	1.060	16.3
12 X 12	X 210 I	210.1	61.8	2140.0	292.0	5.89	664.0	104.0	3.28	14.71	12.790	1.900	1.180	16.4
14 X 14 1/2	X 90 I	90.1	26.5	999.0	143.0	6.14	362.0	49.9	3.70	14.02	14.520	.710	.440	18.5
14 X 14 1/2	X 99 I	98.9	29.1	1110.0	157.0	6.17	402.0	55.2	3.71	14.16	14.565	.780	.485	18.6
14 X 14 1/2	X 109 I	108.8	32.0	1240.0	173.0	6.22	447.0	61.2	3.73	14.32	14.605	.860	.525	18.7
14 X 14 1/2	X 120 I	120.0	35.3	1380.0	190.0	6.24	495.0	67.5	3.74	14.48	14.670	.940	.590	18.7
14 X 14 1/2	X 132 I	131.9	38.8	1530.0	209.0	6.28	548.0	74.5	3.76	14.66	14.725	1.030	.645	18.8
14 X 16	X 145 I	145.2	42.7	1710.0	232.0	6.33	677.0	87.3	3.98	14.78	15.500	1.090	.680	19.9
14 X 16	X 159 I	158.8	46.7	1900.0	254.0	6.38	748.0	96.2	4.00	14.98	15.565	1.190	.745	20.0
14 X 16	X 176 I	176.1	51.8	2140.0	281.0	6.43	838.0	107.0	4.02	15.22	15.650	1.310	.830	20.1
14 X 16	X 193 I	193.1	56.8	2400.0	310.0	6.50	931.0	119.0	4.05	15.48	15.710	1.440	.890	20.3
14 X 16	X 211 I	210.8	62.0	2660.0	338.0	6.55	1030.0	130.0	4.07	15.72	15.800	1.560	.980	20.4
14 X 16	X 233 I	232.9	68.5	3010.0	375.0	6.63	1150.0	145.0	4.10	16.04	15.890	1.720	1.070	20.5
14 X 16	X 257 I	257.0	75.6	3400.0	415.0	6.71	1290.0	161.0	4.13	16.38	15.995	1.890	1.175	20.7
14 X 16	X 283 I	283.2	83.3	3840.0	459.0	6.79	1440.0	179.0	4.17	16.74	16.110	2.070	1.290	20.9
14 X 16	X 311 I	310.8	91.4	4330.0	506.0	6.88	1610.0	199.0	4.20	17.12	16.230	2.260	1.410	21.0
14 X 16	X 342 I	343.4	101.0	4900.0	559.0	6.98	1810.0	221.0	4.24	17.54	16.360	2.470	1.540	21.2
14 X 16	X 370 I	370.6	109.0	5440.0	607.0	7.07	1990.0	241.0	4.27	17.92	16.475	2.660	1.655	21.4
14 X 16	X 398 I	397.8	117.0	6000.0	656.0	7.16	2170.0	262.0	4.31	18.29	16.590	2.845	1.770	21.6
14 X 16	X 426 I	425.0	125.0	6600.0	707.0	7.26	2360.0	283.0	4.34	18.67	16.695	3.035	1.875	21.7

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TABLE III. Properties of structural steel tubes (standard iron pipe sizes).

PROPERTIES OF STRUCTURAL STEEL TUBES									
PIPE SIZE OUTSIDE DIA X WALL THICKNESS	WT/FT	A	I	S	R	DIAMETER OUTSIDE	DIAMETER INSIDE	WALL THICKNESS	ALLOWABLE LENGTH
IN X IN	LBS	IN2	IN4	IN3	IN	IN	IN	IN	FT
3.500 X 0.216	7.58	2.23	3.02	1.72	1.16	3.500	3.068	.216	5.80
4.000 X 0.226	9.11	2.68	4.79	2.39	1.34	4.000	3.548	.226	6.70
4.500 X 0.237	10.79	3.17	7.23	3.21	1.51	4.500	4.026	.237	7.55
5.563 X 0.258	14.62	4.30	15.20	5.45	1.88	5.563	5.047	.258	9.40
6.625 X 0.280	18.97	5.58	28.10	8.50	2.25	6.625	6.065	.280	11.25
8.625 X 0.322	28.55	8.40	72.50	16.80	2.94	8.625	7.981	.322	14.70
10.750 X 0.365	40.48	11.90	161.00	29.90	3.67	10.750	10.020	.365	18.35
12.750 X 0.375	49.56	14.60	279.00	43.80	4.38	12.750	12.000	.375	21.90

NOTE- THESE ARE STRUCTURAL STEEL TUBES IN STANDARD IRON PIPE SIZES

PROPERTIES OF STRUCTURAL STEEL TUBES

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TABLE IV. Properties of structural steel tubes (extra strong iron pipe sizes).

PROPERTIES OF STRUCTURAL STEEL TUBES									
PIPE SIZE OUTSIDE DIA x WALL THICKNESS	WT/FT	A	I	S	R	DIAMETER OUTSIDE	DIAMETER INSIDE	WALL THICKNESS	ALLOWABLE LENGTH
IN X IN	LBS	IN2	IN4	IN3	IN	IN	IN	IN	FT
3.500 X 0.300	10.25	3.02	3.89	2.23	1.14	3.500	2.900	.300	5.70
4.000 X 0.318	12.50	3.68	6.28	3.14	1.31	4.000	3.364	.318	6.55
4.500 X 0.337	14.98	4.41	9.61	4.27	1.48	4.500	3.826	.337	7.40
5.563 X 0.375	20.78	6.11	20.70	7.43	1.84	5.563	4.813	.375	9.20
6.625 X 0.432	28.57	8.40	40.50	12.20	2.19	6.625	5.761	.432	10.95
8.625 X 0.500	43.39	12.80	106.00	24.50	2.88	8.625	7.625	.500	14.40
10.750 X 0.500	54.74	16.10	212.00	39.40	3.63	10.750	9.750	.500	18.15
12.750 X 0.500	65.42	19.20	362.00	56.70	4.33	12.750	11.750	.500	21.65

NOTE- THESE ARE STRUCTURAL STEEL TUBES IN EXTRA-STRONG IRON PIPE SIZES

PROPERTIES OF STRUCTURAL STEEL TUBES

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TABLE V. Properties of structural steel tubes (double-extra strong iron pipe sizes).

PROPERTIES OF STRUCTURAL STEEL TUBES									
PIPE SIZE OUTSIDE DIA X WALL THICKNESS	WT/FT	A	I	S	R	DIAMETER OUTSIDE	DIAMETER INSIDE	WALL THICKNESS	ALLOWABLE LENGTH
IN X IN	LBS	IN2	IN4	IN3	IN	IN	IN	IN	FT
3.500 X 0.600	18.58	5.47	5.99	3.42	1.05	3.500	2.300	.600	5.25
4.500 X 0.674	27.54	8.10	15.30	6.79	1.37	4.500	3.152	.674	6.85
5.563 X 0.750	38.55	11.30	33.60	12.10	1.72	5.563	4.063	.750	8.60
6.625 X 0.864	53.16	15.60	66.30	20.00	2.06	6.625	4.897	.864	10.30
8.625 X 0.875	72.42	21.30	162.00	37.60	2.76	8.625	6.875	.875	13.80

NOTE- THESE ARE STRUCTURAL STEEL TUBES IN DOUBLE-EXTRA STRONG IRON PIPE SIZES

PROPERTIES OF STRUCTURAL STEEL TUBES

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TABLE VI. Steel structural shape catalog (I-T and T).

STEEL

STRUCTURAL SHAPE CATALOG (I-T AND T)

NOMINAL SIZE				A	CTOE	IO	D	TW	WF	TF
IN	X	IN	X LBS/FT	IN2	IN	IN4	IN	IN	IN	IN
4	X	4	X 5.0 T	1.44	2.98	2.1	3.95	.170	3.94	.205
4	X	4	X 6.5 T	1.88	2.96	2.9	4.00	.230	4.00	.255
4	X	4	X 7.5 T	2.18	3.05	3.3	4.06	.245	4.02	.315
4	X	5 1/4	X 9.0 T	2.59	3.23	3.4	4.07	.230	5.25	.330
4	X	4	X 13.0 I-T	2.47	3.09	3.9	4.16	.280	4.06	.345
5	X	4	X 6.0 T	1.73	3.55	4.3	4.94	.190	3.96	.210
5	X	4	X 7.5 T	2.17	3.61	5.4	5.00	.230	4.00	.270
5	X	4	X 9.5 T	2.46	3.73	6.0	5.06	.240	4.01	.330
5	X	4	X 9.5 T	2.77	3.83	6.7	5.17	.250	4.02	.395
5	X	5	X 16.0 I-T	2.92	3.87	6.4	5.01	.240	5.00	.360
5	X	5	X 19.0 I-T	3.44	3.98	7.7	5.15	.270	5.03	.430
6	X	4	X 7.0 T	2.04	4.17	7.6	5.96	.200	3.97	.225
6	X	4	X 9.0 T	2.32	4.24	8.6	6.00	.220	3.99	.265
6	X	4	X 9.0 I-T	1.81	4.22	6.5	5.90	.170	3.94	.215
6	X	4	X 9.5 T	2.75	4.42	10.0	6.08	.235	4.01	.350
6	X	4	X 11.0 T	3.20	4.51	11.7	6.16	.260	4.03	.425
6	X	4	X 12.0 I-T	2.44	4.26	9.2	6.03	.230	4.00	.280
6	X	6	X 15.0 I-T	2.88	4.49	10.0	5.99	.230	5.99	.260
6	X	4	X 16.0 I-T	3.16	4.56	12.2	6.28	.260	4.03	.405
6	X	6	X 20.0 I-T	3.71	4.75	13.0	6.20	.260	6.02	.365
7	X	5	X 11.0 T	3.18	5.08	14.7	6.87	.230	5.00	.335
7	X	5	X 13.0 T	3.78	5.21	17.3	6.96	.255	5.03	.420
7	X	6 3/4	X 15.0 T	4.36	5.33	18.9	6.92	.270	6.73	.385
7	X	6 3/4	X 17.0 T	4.93	5.44	20.8	6.99	.285	6.75	.455
7	X	6 3/4	X 19.0 T	5.51	5.50	23.2	7.05	.310	6.77	.515
7	X	8	X 21.5 T	6.16	5.50	21.9	6.83	.305	8.00	.530
7	X	8	X 24.0 T	6.92	5.53	24.9	6.90	.340	8.03	.595

STRUCTURAL SHAPE CATALOG (I-T AND T)

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TABLE VI. Steel structural shape catalog (I-T and T). - Continued

S T E E L

STRUCTURAL SHAPE CATALOG (I-T AND T)

NOMINAL SIZE						A	CTOE	IO	D	TW	WF	TF
IN	X	IN	X	LBS/FT		IN2	IN	IN4	IN	IN	IN	IN
8	X	4	X	10.0	I-T	2.11	5.35	14.2	7.89	.170	3.94	.205
8	X	4	X	13.0	I-T	2.80	5.32	19.2	7.99	.230	4.00	.255
8	X	5 1/2	X	13.0	T	3.77	5.73	23.4	7.85	.250	5.50	.345
8	X	4	X	15.0	I-T	3.17	5.51	22.2	8.11	.245	4.02	.315
8	X	5 1/2	X	15.5	T	4.49	5.90	27.3	7.94	.275	5.53	.440
8	X	5 1/4	X	18.0	I-T	3.53	5.90	23.8	8.14	.230	5.25	.330
8	X	7	X	18.0	T	5.22	6.03	30.4	7.93	.295	6.99	.430
8	X	7	X	20.0	T	5.82	6.18	33.1	8.01	.305	7.00	.505
8	X	5 1/4	X	21.0	I-T	4.09	6.08	27.7	8.28	.250	5.27	.400
8	X	7	X	22.5	T	6.56	6.20	37.8	8.07	.345	7.04	.565
8	X	6 1/2	X	24.0	I-T	4.44	6.08	25.7	7.93	.245	6.50	.400
8	X	7 1/8	X	25.0	T	7.30	6.23	42.2	8.13	.380	7.07	.630
8	X	6 1/2	X	28.0	I-T	5.23	6.15	31.0	8.06	.285	6.54	.465
8	X	7 1/8	X	28.5	T	8.32	6.27	48.7	8.22	.430	7.12	.715
8	X	8	X	31.0	I-T	5.63	6.25	31.6	8.00	.285	8.00	.435
9	X	6	X	17.5	T	5.08	6.43	39.8	8.85	.300	6.00	.425
9	X	6	X	20.0	T	5.81	6.64	44.6	8.95	.315	6.02	.525
10	X	4	X	12.0	I-T	2.67	6.37	28.2	9.87	.190	3.96	.210
10	X	4	X	15.0	I-T	3.32	6.49	35.8	9.99	.230	4.00	.270
10	X	4	X	17.0	I-T	3.67	6.71	40.3	10.11	.240	4.01	.330
10	X	4	X	19.0	I-T	4.05	6.93	45.2	10.24	.250	4.02	.395
10	X	5 3/4	X	22.0	I-T	4.42	7.28	47.4	10.17	.240	5.75	.360
10	X	5 3/4	X	26.0	I-T	5.11	7.51	55.1	10.33	.260	5.77	.440
10	X	5 3/4	X	30.0	I-T	5.95	7.59	65.5	10.47	.300	5.81	.510
10	X	8	X	33.0	I-T	6.16	7.38	55.3	9.73	.290	7.96	.435
10	X	8	X	39.0	I-T	7.19	7.61	64.7	9.92	.315	7.99	.530
10	X	8	X	45.0	I-T	8.29	7.77	75.8	10.10	.350	8.02	.620

STRUCTURAL SHAPE CATALOG (I-T AND T)

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TABLE VI. Steel structural shape catalog (I-T and T). - Continued

S T E E L

STRUCTURAL SHAPE CATALOG (I-T AND T)

NOMINAL SIZE						A	GTOE	IO	D	TW	WF	TF
IN	X	IN	X	LBS/FT		IN2	IN	IN4	IN	IN	IN	IN
12	X	4		X	14.0 I-T	3.23	7.49	49.5	11.91	.200	3.97	.225
12	X	4		X	16.0 I-T	3.64	7.61	56.5	11.99	.220	3.99	.265
12	X	4		X	19.0 I-T	4.18	7.95	66.7	12.16	.235	4.01	.350
12	X	4		X	22.0 I-T	4.80	8.14	78.1	12.31	.260	4.03	.425
12	X	6	1/2	X	26.0 I-T	5.19	8.82	80.2	12.22	.230	6.49	.380
12	X	6	1/2	X	30.0 I-T	5.96	8.92	93.2	12.34	.260	6.52	.440
12	X	6	1/2	X	35.0 I-T	7.01	9.03	111.4	12.50	.300	6.56	.520
12	X	8		X	40.0 I-T	7.49	9.00	102.8	11.94	.295	8.01	.515
12	X	8		X	45.0 I-T	8.47	9.03	118.8	12.06	.335	8.05	.575
12	X	8	1/8	X	50.0 I-T	9.44	9.11	134.6	12.19	.370	8.08	.640
12	X	10		X	53.0 I-T	9.71	9.31	129.0	12.06	.345	10.00	.575
12	X	10		X	58.0 I-T	10.56	9.47	140.1	12.19	.360	10.01	.640
14	X	5		X	22.0 I-T	4.76	9.12	97.4	13.74	.230	5.00	.335
14	X	5		X	26.0 I-T	5.55	9.39	115.5	13.91	.255	5.03	.420
14	X	6	3/4	X	30.0 I-T	6.22	9.61	127.3	13.84	.270	6.73	.385
14	X	6	3/4	X	34.0 I-T	6.92	9.86	142.3	13.98	.285	6.75	.455
14	X	6	3/4	X	38.0 I-T	7.70	9.99	159.6	14.10	.310	6.77	.515
14	X	8		X	43.0 I-T	8.24	10.08	153.7	13.66	.305	8.00	.530
14	X	8		X	48.0 I-T	9.26	10.15	175.2	13.79	.340	8.03	.595
16	X	5	1/2	X	26.0 I-T	5.73	10.27	153.4	15.69	.250	5.50	.345
16	X	5	1/2	X	31.0 I-T	6.68	10.61	181.9	15.88	.275	5.53	.440
16	X	7		X	36.0 I-T	7.56	10.87	204.1	15.86	.295	6.99	.430
16	X	7		X	40.0 I-T	8.26	11.18	224.4	16.01	.305	7.00	.505
16	X	7		X	45.0 I-T	9.34	11.21	257.1	16.13	.345	7.04	.565
16	X	7	1/8	X	50.0 I-T	10.39	11.30	289.3	16.26	.380	7.07	.630
16	X	7	1/8	X	57.0 I-T	11.85	11.39	335.2	16.43	.430	7.12	.715
16	X	10	1/4	X	67.0 I-T	12.99	12.11	342.9	16.33	.395	10.24	.665
16	X	10	1/4	X	77.0 I-T	15.00	12.19	404.1	16.52	.455	10.30	.760

STRUCTURAL SHAPE CATALOG (I-T AND T)

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TABLE VI. Steel structural shape catalog (I-T and T). - Continued

S T E E L

STRUCTURAL SHAPE CATALOG (I-T AND T)

NOMINAL SIZE				A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT				IN2	IN	IN4	IN	IN	IN	IN
16 X 10 3/8 X 89.0 I-T				17.40	12.30	480.2	16.75	.525	10.37	.875
18 X 6 X 35.0 I-T				7.73	11.56	262.8	17.70	.300	6.00	.425
18 X 6 X 40.0 I-T				8.63	11.96	298.2	17.90	.315	6.02	.525
18 X 7 1/2 X 50.0 I-T				10.46	12.39	360.9	17.99	.355	7.50	.570
18 X 7 1/2 X 60.0 I-T				12.53	12.59	440.7	18.24	.415	7.56	.695
18 X 7 5/8 X 71.0 I-T				14.93	12.66	536.4	18.47	.495	7.64	.810
18 X 11 1/8 X 86.0 I-T				17.00	13.43	578.5	18.39	.480	11.09	.770
18 X 11 1/8 X 97.0 I-T				19.18	13.56	662.8	18.59	.535	11.19	.870
18 X 11 1/4 X 106.0 I-T				21.02	13.58	738.6	18.73	.590	11.20	.940
18 X 11 1/4 X 119.0 I-T				23.67	13.74	847.1	18.97	.655	11.27	1.060
21 X 8 1/4 X 62.0 I-T				13.22	14.21	626.3	20.99	.400	8.24	.615
21 X 8 1/4 X 68.0 I-T				14.46	14.36	691.0	21.13	.430	8.27	.685
21 X 8 1/4 X 73.0 I-T				15.47	14.47	744.5	21.24	.455	8.30	.740
21 X 8 3/8 X 83.0 I-T				17.58	14.55	858.5	21.43	.515	8.36	.835
21 X 8 3/8 X 93.0 I-T				19.83	14.61	982.4	21.62	.580	8.42	.930
21 X 12 1/4 X 101.0 I-T				20.11	15.50	935.9	21.36	.500	12.29	.800
21 X 12 3/8 X 111.0 I-T				22.15	15.56	1043.4	21.51	.550	12.34	.875

STRUCTURAL SHAPE CATALOG (I-T AND T)

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TABLE VII. Steel structural shape catalog (C-L and L).

S T E E L

STRUCTURAL SHAPE CATALOG (C-L AND L)

NOMINAL SIZE IN X IN X IN X LBS/FT	A IN2	CTOE IN	IO IN4	D IN	TW IN	WF IN	TF IN
2 X 1 1/2 X 3/16 X 2.12 L	.62	1.36	.2	2.00	.187	1.50	.187
2 X 2 X 3/16 X 2.44 L	.71	1.43	.3	2.00	.187	2.00	.187
2 X 1 1/2 X 1/4 X 2.77 L	.81	1.34	.3	2.00	.250	1.50	.250
2 X 2 X 1/4 X 3.19 L	.94	1.41	.3	2.00	.250	2.00	.250
2 X 2 X 3/8 X 4.7 L	1.36	1.36	.5	2.00	.375	2.00	.375
2 1/2 X 2 X 3/16 X 2.75 L	.81	1.74	.5	2.50	.187	2.00	.187
2 1/2 X 2 X 1/4 X 3.62 L	1.06	1.71	.7	2.50	.250	2.00	.250
2 1/2 X 2 X 5/16 X 4.5 L	1.31	1.69	.8	2.50	.313	2.00	.313
2 1/2 X 2 X 3/8 X 5.3 L	1.55	1.67	.9	2.50	.375	2.00	.375
3 X 2 X 3/16 X 3.07 L	.90	2.03	.8	3.00	.187	2.00	.187
3 X 3 X 3/16 X 3.71 L	1.09	2.18	1.0	3.00	.187	3.00	.187
3 X 2 X 1/4 X 4.1 L	1.19	2.01	1.1	3.00	.250	2.00	.250
3 X 2 1/2 X 1/4 X 4.5 L	1.31	2.09	1.2	3.00	.250	2.50	.250
3 X 3 X 1/4 X 4.9 L	1.44	2.16	1.2	3.00	.250	3.00	.250
3 X 2 X 5/16 X 5.0 L	1.47	1.98	1.3	3.00	.313	2.00	.313
3 X 2 1/2 X 5/16 X 5.6 L	1.62	2.07	1.4	3.00	.313	2.50	.313
3 X 3 X 5/16 X 6.1 L	1.78	2.13	1.5	3.00	.313	3.00	.313
3 X 2 1/2 X 3/8 X 6.6 L	1.92	2.04	1.7	3.00	.375	2.50	.375
3 X 3 X 3/8 X 7.2 L	2.11	2.11	1.8	3.00	.375	3.00	.375
3 X 2 1/2 X 7/16 X 7.6 L	2.22	2.02	1.9	3.00	.438	2.50	.438
3 X 3 X 7/16 X 8.3 L	2.44	2.09	2.0	3.00	.438	3.00	.438
3 X 3 X 1/2 X 9.4 L	2.75	2.07	2.2	3.00	.500	3.00	.500
3 1/2 X 2 1/2 X 1/4 X 4.9 L	1.44	2.39	1.8	3.50	.250	2.50	.250
3 1/2 X 3 X 1/4 X 5.4 L	1.56	2.47	1.9	3.50	.250	3.00	.250
3 1/2 X 2 1/2 X 5/16 X 6.1 L	1.78	2.36	2.2	3.50	.313	2.50	.313
3 1/2 X 3 X 5/16 X 6.6 L	1.94	2.44	2.3	3.50	.313	3.00	.313
3 1/2 X 2 1/2 X 3/8 X 7.2 L	2.11	2.34	2.6	3.50	.375	2.50	.375

STRUCTURAL SHAPE CATALOG (C-L AND L)

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TABLE VII. Steel structural shape catalog (C-L and L). - Continued

S T E E L

STRUCTURAL SHAPE CATALOG (C-L AND L)

NOMINAL SIZE	A	CTOE	IO	D	TW	WF	TF
IN X IN X IN X LBS/FT	IN2	IN	IN4	IN	IN	IN	IN
3 1/2 X 3 X 3/8 X 7.9 L	2.30	2.42	2.7	3.50	.375	3.00	.375
4 X 3 X 1/4 X 5.8 L	1.69	2.76	2.8	4.00	.250	3.00	.250
4 X 3 1/2 X 1/4 X 6.2 L	1.81	2.84	2.9	4.00	.250	3.50	.250
4 X 4 X 1/4 X 6.6 L	1.94	2.91	3.0	4.00	.250	4.00	.250
4 X 3 X 5/16 X 7.2 L	2.09	2.74	3.4	4.00	.313	3.00	.313
4 X 3 1/2 X 5/16 X 7.7 L	2.25	2.82	3.6	4.00	.313	3.50	.313
4 X 4 X 5/16 X 8.2 L	2.41	2.88	3.7	4.00	.313	4.00	.313
4 X 3 X 3/8 X 8.5 L	2.48	2.72	4.0	4.00	.375	3.00	.375
4 X 3 1/2 X 3/8 X 9.1 L	2.67	2.79	4.2	4.00	.375	3.50	.375
4 X 4 X 3/8 X 9.8 L	2.86	2.86	4.4	4.00	.375	4.00	.375
4 X 3 X 7/16 X 9.8 L	2.87	2.70	4.5	4.00	.438	3.00	.438
4 X 3 X 1/2 X 11.1 L	3.25	2.67	5.0	4.00	.500	3.00	.500
4 X 4 X 7/16 X 11.3 L	3.31	2.84	5.0	4.00	.438	4.00	.438
4 X 3 1/2 X 1/2 X 11.9 L	3.50	2.75	5.3	4.00	.500	3.50	.500
4 X 4 X 1/2 X 12.8 L	3.75	2.82	5.6	4.00	.500	4.00	.500
4 X 4 X 5/8 X 15.7 L	4.61	2.77	6.7	4.00	.625	4.00	.625
4 X 4 X 3/4 X 18.5 L	5.44	2.73	7.7	4.00	.750	4.00	.750
5 X 3 X 1/4 X 6.6 L	1.94	3.34	5.1	5.00	.250	3.00	.250
5 X 3 1/2 X 1/4 X 7.0 L	2.06	3.44	5.4	5.00	.250	3.50	.250
5 X 3 X 5/16 X 8.2 L	2.41	3.32	6.3	5.00	.313	3.00	.313
5 X 3 1/2 X 5/16 X 8.7 L	2.56	3.41	6.6	5.00	.313	3.50	.313
5 X 3 X 3/8 X 9.8 L	2.86	3.30	7.4	5.00	.375	3.00	.375
5 X 3 1/2 X 3/8 X 10.4 L	3.05	3.39	7.8	5.00	.375	3.50	.375
5 X 3 X 7/16 X 11.3 L	3.31	3.27	8.4	5.00	.438	3.00	.438
5 X 3 1/2 X 7/16 X 12.0 L	3.53	3.37	8.9	5.00	.438	3.50	.438
5 X 5 X 3/8 X 12.3 L	3.61	3.61	8.7	5.00	.375	5.00	.375
5 X 3 X 1/2 X 12.8 L	3.75	3.25	9.5	5.00	.500	3.00	.500

STRUCTURAL SHAPE CATALOG (C-L AND L)

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TABLE VII. Steel structural shape catalog (C-L and L). - Continued

S T E E L

STRUCTURAL SHAPE CATALOG (C-L AND L)

NOMINAL SIZE IN X IN X IN X LBS/FT		A IN2	CTOE IN	ID IN4	O IN	TW IN	WF IN	TF IN
5 X 3 1/2 X 1/2 X 13.6	L	4.00	3.34	10.0	5.00	.500	3.50	.500
5 X 3 1/2 X 5/8 X 16.8	L	4.92	3.30	12.0	5.00	.625	3.50	.625
5 X 3 1/2 X 3/4 X 19.8	L	5.81	3.25	13.9	5.00	.750	3.50	.750
6 X 3 1/2 X 5/16 X 9.8	L	2.88	3.99	10.9	6.00	.313	3.50	.313
6 X 4 X 5/16 X 10.3	L	3.03	4.08	11.4	6.00	.313	4.00	.313
6 X 3 1/2 X X 15.3	C	3.26	4.05	12.1	6.00	.340	3.50	.385
6 X 3 1/2 X 3/8 X 11.7	L	3.42	3.96	12.9	6.00	.375	3.50	.375
6 X 4 X 3/8 X 12.3	L	3.61	4.06	13.5	6.00	.375	4.00	.375
6 X 3 1/2 X X 18.0	C	3.76	4.09	13.7	6.00	.379	3.50	.475
6 X 4 X 7/16 X 14.3	L	4.19	4.04	15.5	6.00	.438	4.00	.438
6 X 4 X 1/2 X 16.2	L	4.75	4.01	17.4	6.00	.500	4.00	.500
6 X 4 X 9/16 X 18.1	L	5.31	3.99	19.3	6.00	.563	4.00	.563
6 X 4 X 5/8 X 20.0	L	5.86	3.97	21.1	6.00	.625	4.00	.625
6 X 4 X 3/4 X 23.6	L	6.94	3.92	24.5	6.00	.750	4.00	.750
7 X 4 X 3/8 X 13.6	L	3.98	4.63	20.6	7.00	.375	4.00	.375
7 X 4 X 1/2 X 17.9	L	5.25	4.58	26.7	7.00	.500	4.00	.500
7 X 4 X 5/8 X 22.1	L	6.48	4.54	32.4	7.00	.625	4.00	.625
7 X 4 X 3/4 X 26.2	L	7.69	4.49	37.8	7.00	.750	4.00	.750
8 X 4 X 1/2 X 19.6	L	5.75	5.14	38.5	8.00	.500	4.00	.500
8 X 4 X 9/16 X 21.9	L	6.44	5.12	42.8	8.00	.563	4.00	.563
8 X 6 X 1/2 X 23.0	L	6.75	5.53	44.3	8.00	.500	6.00	.500
8 X 4 X 5/8 X 24.2	L	7.11	5.09	46.9	8.00	.625	4.00	.625
8 X 6 X 9/16 X 25.7	L	7.57	5.50	49.3	8.00	.563	6.00	.563
8 X 4 X 3/4 X 28.7	L	8.44	5.05	54.9	8.00	.750	4.00	.750
8 X 4 X 7/8 X 33.1	L	9.73	5.00	62.5	8.00	.875	4.00	.875
8 X 6 X 3/4 X 33.9	L	9.94	5.44	63.4	8.00	.750	6.00	.750
8 X 4 X 1 X 37.4	L	11.00	4.95	69.6	8.00	1.000	4.00	1.000

STRUCTURAL SHAPE CATALOG (C-L AND L)

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TABLE VII. Steel structural shape catalog (C-L and L). - Continued

S T E E L

STRUCTURAL SHAPE CATALOG (C-L AND L)

NOMINAL SIZE IN X IN X IN X LBS/FT					A IN2	CTOE IN	IO IN4	D IN	TW IN	WF IN	TF IN
8	X 6	X 7/8	X 39.1	L	11.48	5.39	72.3	8.00	.875	6.00	.875
8	X 6	X 1	X 44.2	L	13.00	5.35	80.8	8.00	1.000	6.00	1.000
9	X 4	X 1/2	X 21.3	L	6.25	5.69	53.2	9.00	.500	4.00	.500
9	X 4	X 9/16	X 23.8	L	7.00	5.67	59.2	9.00	.563	4.00	.563
9	X 4	X 5/8	X 26.3	L	7.73	5.64	64.9	9.00	.625	4.00	.625
10	X 2 5/8	X	X 15.3	C	3.43	6.43	36.5	10.00	.240	2.60	.436
10	X 3 1/2	X	X 21.9	C	4.81	6.54	50.9	10.00	.325	3.45	.503
10	X 2 5/8	X	X 20.0	C	4.82	6.02	50.1	10.00	.379	2.74	.436
10	X 3 1/2	X	X 24.9	C	5.51	6.49	57.9	10.00	.377	3.40	.575
10	X 3 1/2	X	X 25.3	C	5.81	6.28	61.2	10.00	.425	3.55	.500
10	X 4 1/2	X	X 28.5	C	6.28	6.52	65.9	10.00	.425	3.95	.575
10	X 3 1/2	X	X 28.3	C	6.51	6.26	68.1	10.00	.477	3.50	.575
10	X 4	X	X 33.6	C	7.78	6.23	81.3	10.00	.575	4.10	.575
10	X 4	X	X 41.1	C	9.99	5.96	102.3	10.00	.796	4.32	.575
12	X 3	X	X 20.7	C	4.72	7.62	72.2	12.00	.282	2.94	.501
12	X 3	X	X 25.0	C	5.98	7.28	90.0	12.00	.387	3.05	.501
12	X 3 1/2	X	X 30.9	C	7.20	7.43	108.7	12.00	.450	3.45	.600
12	X 3	X	X 30.0	C	7.45	7.03	109.6	12.00	.510	3.17	.501
12	X 3 1/2	X	X 32.9	C	7.80	7.32	117.0	12.00	.500	3.50	.600
12	X 4	X	X 35.0	C	7.92	7.65	119.6	12.00	.467	3.77	.700
12	X 3 1/2	X	X 37.0	C	9.00	7.14	133.2	12.00	.600	3.60	.600
12	X 4	X	X 40.0	C	9.39	7.39	140.7	12.00	.590	3.89	.700
12	X 4	X	X 45.0	C	10.85	7.20	160.6	12.00	.712	4.01	.700
12	X 4	X	X 50.0	C	12.33	7.06	180.3	12.00	.835	4.14	.700
13	X 4	X	X 31.8	C	7.09	8.43	127.1	13.00	.375	4.00	.610
13	X 4	X	X 35.0	C	8.02	8.21	143.4	13.00	.447	4.07	.610

STRUCTURAL SHAPE CATALOG (C-L AND L)

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TABLE VII. Steel structural shape catalog (C-L and L). - Continued

STEEL

STRUCTURAL SHAPE CATALOG (C-L AND L)

NOMINAL SIZE					A	CTOE	IO	D	TW	WF	TF
IN	X	IN	X	IN X LBS/FT	IN2	IN	IN4	IN	IN	IN	IN
13	X	4		X 40.0 C	9.49	7.94	167.8	13.00	.560	4.19	.610
13	X	4		X 50.0 C	12.44	7.60	213.9	13.00	.787	4.41	.610
15	X	3 3/8		X 33.9 C	7.95	9.26	188.3	15.00	.400	3.40	.650
15	X	3 3/8		X 40.0 C	9.75	8.94	226.6	15.00	.520	3.52	.650
15	X	4		X 50.0 C	12.06	9.08	281.3	15.00	.625	4.00	.797
15	X	3 3/8		X 50.0 C	12.69	8.60	286.5	15.00	.716	3.72	.650
15	X	4		X 53.2 C	13.01	8.97	301.2	15.00	.688	4.06	.797
15	X	4		X 56.4 C	13.96	8.87	321.0	15.00	.751	4.13	.797
15	X	4		X 59.6 C	14.89	8.78	340.0	15.00	.813	4.19	.797
15	X	4		X 62.8 C	15.83	8.71	359.1	15.00	.876	4.25	.797
15	X	4		X 65.9 C	16.74	8.64	377.5	15.00	.937	4.31	.797
15	X	4		X 69.1 C	17.69	8.58	396.3	15.00	1.000	4.37	.797
15	X	4		X 75.0 C	19.43	8.48	430.9	15.00	1.116	4.49	.797
18	X	4		X 42.7 C	10.29	10.85	348.8	18.00	.450	3.95	.625
18	X	4		X 45.8 C	11.19	10.70	375.9	18.00	.500	4.00	.625
18	X	4		X 51.9 C	12.99	10.46	429.0	18.00	.600	4.10	.625
18	X	4		X 58.0 C	14.79	10.29	480.9	18.00	.700	4.20	.625

STRUCTURAL SHAPE CATALOG (C-L AND L)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t).

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 7.500 IN.) PLATE WEIGHT = 5.100 LBS. (.1250 IN.) EFFECTIVE PLATE AREA = .938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X TN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.4	3.9	7.4	1.76	2.2	1.9	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.6	5.1	12.3	2.14	2.7	2.4	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.1	6.5	19.1	2.53	3.1	3.0	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	5.7	6.2	17.9	2.55	3.1	2.9	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	7.9	8.7	33.2	3.30	4.2	3.8	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.0	11.8	56.9	3.97	5.2	4.8	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	14.9	15.4	91.0	4.67	6.1	5.9	3.23	11.91	3.97	.225	.200	2.38

(60T) PLATE WEIGHT = 5.100 LBS. (.1250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 9.375 IN.) PLATE WEIGHT = 6.375 LBS. (.1563 IN.) EFFECTIVE PLATE AREA = 1.465 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.6	5.6	8.9	1.75	2.5	1.6	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.8	7.2	14.8	2.15	3.1	2.0	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.5	9.0	23.0	2.56	3.6	2.6	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	6.0	8.8	21.5	2.56	3.6	2.5	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.3	12.1	39.7	3.33	4.8	3.3	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.7	15.9	67.5	4.04	5.8	4.2	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	15.8	20.3	107.2	4.78	6.8	5.3	3.23	11.91	3.97	.225	.200	2.38

(60T) PLATE WEIGHT = 6.375 LBS. (.1563 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 11.250 IN.) PLATE WEIGHT = 7.650 LBS. (.1875 IN.) EFFECTIVE PLATE AREA = 2.109 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X LBS/FT			LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X 4	X 5.0	T		4.91	3.7	7.6	10.2	1.70	2.8	1.3	1.44	3.95	3.94	.205	.170	.67
5	X 4	X 6.0	T		5.88	5.0	9.8	17.0	2.10	3.4	1.7	1.73	4.94	3.96	.210	.190	.94
6	X 4	X 7.0	T		6.94	6.7	12.1	26.5	2.53	4.0	2.2	2.04	5.96	3.97	.225	.200	1.19
6	X 4	X 9.0	I-T		6.17	6.2	11.8	24.7	2.51	4.0	2.1	1.81	5.90	3.94	.215	.170	1.00
8	X 4	X 10.0	I-T		7.19	8.7	16.1	45.5	3.28	5.3	2.8	2.11	7.89	3.94	.205	.170	1.34
10	X 4	X 12.0	I-T		9.07	12.2	20.9	77.4	4.03	6.4	3.7	2.67	9.87	3.96	.210	.190	1.88
12	X 4	X 14.0	I-T		10.98	16.6	26.3	122.9	4.80	7.4	4.7	3.23	11.91	3.97	.225	.200	2.38

(60T) PLATE WEIGHT = 7.650 LBS. (.1875 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 13.125 IN.) PLATE WEIGHT = 8.925 LBS. (.2188 IN.) EFFECTIVE PLATE AREA = 2.871 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.7	9.9	11.3	1.62	3.0	1.1	1.44	3.95	3.94	.205	.170	.67
5	X	4	X	6.0	T	5.88	5.1	12.6	18.8	2.02	3.7	1.5	1.73	4.94	3.96	.210	.190	.94
6	X	4	X	7.0	T	6.94	6.9	15.6	29.5	2.45	4.3	1.9	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	9.0	I-T	6.17	6.3	15.3	27.4	2.42	4.3	1.8	1.81	5.90	3.94	.215	.170	1.00
8	X	4	X	10.0	I-T	7.19	8.9	20.8	50.5	3.18	5.7	2.4	2.11	7.89	3.94	.205	.170	1.34
10	X	4	X	12.0	I-T	9.07	12.6	26.7	86.2	3.95	6.9	3.2	2.67	9.87	3.96	.210	.190	1.88
12	X	4	X	14.0	I-T	10.98	17.2	33.2	137.3	4.74	8.0	4.1	3.23	11.91	3.97	.225	.200	2.38

(60T) PLATE WEIGHT = 8.925 LBS. (.2188 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 15.000 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 3.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.8	12.3	12.2	1.53	3.2	1.0	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.8	12.8	14.8	1.62	3.1	1.2	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.7	13.3	17.2	1.70	3.0	1.3	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.3	13.8	20.7	1.81	2.8	1.5	2.59	4.07	5.25	.330	.230	.94
5 X 4 X 6.0 T	5.88	5.2	15.8	20.3	1.93	3.9	1.3	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.6	16.5	24.6	2.04	3.8	1.5	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.7	17.0	28.1	2.13	3.7	1.7	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	8.9	17.5	31.6	2.20	3.6	1.8	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.6	17.4	32.6	2.21	3.4	1.9	2.92	5.01	5.00	.360	.240	1.20
6 X 4 X 7.0 T	6.94	7.0	19.5	32.0	2.35	4.6	1.6	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.1	20.0	35.9	2.43	4.5	1.8	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.4	19.2	29.6	2.31	4.6	1.5	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.0	20.9	42.8	2.57	4.3	2.0	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 12.0 I-T	8.30	8.5	20.3	37.6	2.46	4.4	1.9	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	10.9	21.0	44.7	2.60	4.1	2.1	2.88	5.99	5.99	.260	.230	1.38
7 X 5 X 11.0 T	10.81	13.3	24.4	61.3	2.97	4.6	2.5	3.18	6.87	5.00	.335	.230	1.58
8 X 4 X 10.0 I-T	7.19	9.1	26.1	54.7	3.06	6.0	2.1	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.5	27.2	66.8	3.19	5.8	2.5	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.4	28.7	87.8	3.42	5.0	3.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.6	28.4	76.9	3.33	5.7	2.7	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/4 X 18.0 I-T	12.00	16.8	29.5	89.8	3.51	5.3	3.0	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T	13.87	19.9	30.7	102.9	3.63	5.2	3.4	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.2	29.8	104.1	3.56	4.7	3.5	4.44	7.93	6.50	.400	.245	1.94

(60T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 15.000 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 3.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 X 12.0 I-T	9.07	12.9	33.3	94.0	3.83	7.3	2.8	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.1	34.9	112.7	3.99	7.0	3.2	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.5	36.2	127.1	4.14	6.9	3.5	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.2	37.5	142.1	4.27	6.7	3.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.3	38.4	158.8	4.41	6.3	4.1	4.42	10.17	5.75	.360	.240	2.44
12 X 4 X 14.0 I-T	10.98	17.6	41.2	150.1	4.64	8.5	3.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.1	42.4	166.9	4.75	8.3	3.9	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	24.3	44.6	195.4	4.97	8.0	4.4	4.18	12.16	4.01	.350	.235	2.86
12 X 6 1/2 X 26.0 I-T	17.64	35.6	47.8	254.5	5.34	7.2	5.3	5.19	12.22	6.49	.380	.230	2.81
14 X 5 X 22.0 I-T	16.18	31.8	52.3	276.7	5.70	8.7	5.3	4.76	13.74	5.00	.335	.230	3.16
16 X 5 1/2 X 26.0 I-T	19.49	41.8	62.2	398.4	6.48	9.5	6.4	5.73	15.69	5.50	.345	.250	3.92

(60T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 16.875 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 4.746 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.8	14.9	12.9	1.45	3.4	.9	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.9	15.5	15.8	1.55	3.3	1.0	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.8	16.2	18.5	1.63	3.2	1.1	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.4	16.9	22.5	1.75	3.0	1.3	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.5	16.8	20.9	1.70	3.2	1.2	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.3	19.2	21.6	1.83	4.1	1.1	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.7	20.0	26.4	1.95	4.0	1.3	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.8	20.7	30.3	2.05	3.9	1.5	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.0	21.4	34.3	2.14	3.8	1.6	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.8	21.3	35.5	2.15	3.6	1.7	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	11.7	22.2	41.6	2.25	3.6	1.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.1	23.8	34.2	2.24	4.8	1.4	2.04	5.96	3.37	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.2	24.4	38.5	2.33	4.7	1.6	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.5	23.4	31.5	2.19	4.8	1.3	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.2	25.5	46.2	2.48	4.5	1.8	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.0	26.4	53.1	2.59	4.4	2.0	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.6	24.7	40.4	2.37	4.7	1.6	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.0	25.7	48.4	2.52	4.4	1.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	11.9	26.8	54.1	2.62	4.5	2.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.0	27.5	62.8	2.73	4.2	2.3	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.5	29.8	66.6	2.90	4.9	2.2	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	16.4	30.9	77.6	3.02	4.7	2.5	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	19.5	31.5	86.8	3.09	4.4	2.8	4.36	6.92	6.73	.385	.270	1.87

(60T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 16.875 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 4.746 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 10.0 I-T	7.19	9.2	31.8	58.3	2.92	6.3	1.8	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.8	33.1	71.8	3.09	6.1	2.2	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.8	35.0	95.7	3.35	5.4	2.7	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.9	34.5	83.0	3.24	6.0	2.4	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.7	36.2	111.5	3.47	5.1	3.1	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.1	36.0	97.7	3.44	5.7	2.7	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T	13.87	20.3	37.3	112.6	3.57	5.5	3.0	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.7	36.4	114.6	3.53	5.1	3.2	4.44	7.93	6.50	.400	.245	1.94
10 X 4 X 12.0 I-T	9.07	13.1	40.5	100.6	3.68	7.7	2.5	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.4	42.4	121.5	3.88	7.4	2.9	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.9	44.0	137.6	4.04	7.3	3.1	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.7	45.5	154.5	4.19	7.1	3.4	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.8	46.6	173.6	4.35	6.7	3.7	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	30.6	48.5	199.2	4.50	6.5	4.1	5.11	10.33	5.77	.440	.260	2.69
12 X 4 X 14.0 I-T	10.98	18.0	50.0	161.4	4.50	9.0	3.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.5	51.4	180.1	4.63	8.8	3.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	24.9	54.0	212.0	4.87	8.5	3.9	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.2	56.2	241.8	5.03	8.3	4.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	36.4	57.9	279.4	5.30	7.7	4.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	41.7	59.8	310.1	5.38	7.4	5.2	5.96	12.34	6.52	.440	.260	3.21
14 X 5 X 22.0 I-T	16.18	32.6	63.1	301.3	5.63	9.2	4.8	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	39.0	65.9	347.9	5.81	8.9	5.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	45.4	67.6	383.2	5.91	8.4	5.7	6.22	13.84	6.73	.385	.270	3.74

(60T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 16.875 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 4.746 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	42.9	74.5	434.8	6.44	10.1	5.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	51.6	78.2	502.6	6.63	9.7	6.4	6.68	15.88	5.53	.440	.275	4.37

(60T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 18.750 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 5.859 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.9	17.5	13.6	1.36	3.5	.8	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.9	18.3	16.7	1.47	3.4	.9	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.9	19.2	19.7	1.56	3.3	1.0	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.5	20.1	24.1	1.69	3.2	1.2	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.6	19.9	22.3	1.63	3.4	1.1	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.3	22.7	22.7	1.73	4.3	1.0	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.7	23.8	27.9	1.87	4.1	1.2	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.9	24.7	32.2	1.97	4.1	1.3	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.2	25.5	36.6	2.06	4.0	1.4	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.9	25.4	38.0	2.08	3.8	1.5	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	11.9	26.6	44.8	2.20	3.8	1.7	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.2	28.3	36.0	2.13	5.0	1.3	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.3	29.1	40.7	2.23	4.9	1.4	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.6	27.8	33.1	2.08	5.0	1.2	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.3	30.4	49.2	2.39	4.8	1.6	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.2	31.5	56.9	2.51	4.7	1.8	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.8	29.4	42.8	2.27	4.9	1.5	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.2	30.7	51.7	2.43	4.6	1.7	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.1	32.0	57.9	2.53	4.8	1.8	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.2	32.9	67.8	2.66	4.5	2.1	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.7	35.7	71.2	2.81	5.2	2.0	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	16.7	36.9	83.5	2.94	5.0	2.3	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	19.8	37.7	94.0	3.03	4.7	2.5	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	22.8	38.6	104.8	3.12	4.6	2.7	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	25.5	39.4	114.0	3.17	4.5	2.9	5.51	7.05	6.77	.515	.310	2.19

(60T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 18.750 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 5.859 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	28.9	38.6	118.0	3.13	4.1	3.1	6.16	6.83	8.00	.530	.305	2.08
8 X 4 X 10.0 I-T	7.19	9.3	38.0	61.3	2.77	6.6	1.6	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.9	39.5	76.1	2.97	6.4	1.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.0	41.8	102.8	3.27	5.7	2.5	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.1	41.2	88.4	3.13	6.3	2.1	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	22.0	43.3	120.6	3.41	5.5	2.8	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.4	43.0	104.7	3.34	6.0	2.4	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	26.3	44.4	136.2	3.51	5.2	3.1	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	30.1	45.4	150.5	3.59	5.0	3.3	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.6	44.6	121.2	3.49	5.9	2.7	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	23.0	43.6	124.1	3.47	5.4	2.8	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	26.8	45.0	140.7	3.57	5.2	3.1	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	29.8	45.4	149.6	3.61	5.0	3.3	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	26.6	49.1	158.1	3.80	5.9	3.2	5.08	8.85	6.00	.425	.300	2.66
10 X 4 X 12.0 I-T	9.07	13.3	48.4	106.3	3.53	8.0	2.2	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.7	50.6	129.3	3.75	7.7	2.6	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.3	52.4	146.9	3.93	7.6	2.8	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	22.1	54.2	165.5	4.09	7.5	3.1	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.2	55.7	187.0	4.26	7.1	3.4	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	31.2	57.8	215.6	4.43	6.9	3.7	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	36.1	59.8	242.6	4.53	6.7	4.1	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	37.5	56.2	226.0	4.34	6.0	4.0	6.16	9.73	7.96	.435	.290	2.82

(60T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 18.750 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 5.859 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 14.0 I-T	10.98	18.3	59.6	171.3	4.34	9.3	2.9	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.9	61.3	191.7	4.49	9.2	3.1	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.4	64.3	226.8	4.75	8.9	3.5	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.8	66.7	259.7	4.94	8.7	3.9	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	37.0	69.1	302.1	5.23	8.2	4.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	42.5	71.1	336.6	5.34	7.9	4.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	49.8	73.8	380.9	5.44	7.7	5.2	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	54.4	71.5	378.4	5.32	7.0	5.3	7.49	11.94	8.01	.515	.295	3.52
14 X 5 X 22.0 I-T	16.18	33.2	75.0	323.5	5.52	9.7	4.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	39.8	78.2	375.2	5.73	9.4	4.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	46.3	80.0	415.1	5.86	9.0	5.2	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	52.9	82.6	460.8	6.00	8.7	5.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	59.1	84.9	501.9	6.08	8.5	5.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	65.4	83.5	512.3	6.03	7.8	6.1	8.24	13.66	8.00	.530	.305	4.17
16 X 5 1/2 X 26.0 I-T	19.49	43.8	88.2	468.4	6.36	10.7	5.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	52.8	92.3	543.7	6.59	10.3	5.9	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	61.7	95.1	605.2	6.72	9.8	6.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	69.7	97.9	664.6	6.86	9.5	6.8	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	64.3	105.6	720.1	7.28	11.2	6.8	7.73	17.70	6.00	.425	.300	5.31

(60T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 20.625 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 7.090 SQ. IN.

NOMINAL SIZE IN X TN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.9	20.0	14.1	1.29	3.6	.7	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.0	21.1	17.5	1.40	3.5	.8	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.9	22.2	20.7	1.49	3.5	.9	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.6	23.5	25.4	1.62	3.3	1.1	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.7	23.1	23.5	1.57	3.5	1.0	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.4	26.2	23.7	1.64	4.4	.9	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.8	27.7	29.2	1.78	4.3	1.1	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.0	28.8	33.9	1.88	4.2	1.2	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.3	29.8	38.6	1.98	4.2	1.3	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.0	29.8	40.2	2.00	4.0	1.4	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.0	31.2	47.7	2.13	4.0	1.5	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.3	32.9	37.6	2.03	5.2	1.1	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.4	33.9	42.7	2.13	5.1	1.3	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.7	32.3	34.5	1.97	5.2	1.1	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.4	35.6	51.8	2.29	5.0	1.5	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.4	36.9	60.2	2.42	4.9	1.6	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.9	34.3	44.9	2.17	5.1	1.3	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.3	35.9	54.5	2.34	4.8	1.5	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.3	37.5	61.2	2.44	5.0	1.6	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.4	38.7	72.1	2.58	4.7	1.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.9	41.9	75.3	2.71	5.4	1.8	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	16.9	43.4	88.8	2.86	5.3	2.0	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.1	44.4	100.5	2.96	5.0	2.3	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	23.2	45.5	112.6	3.06	4.9	2.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	25.9	46.4	122.9	3.12	4.7	2.7	5.51	7.05	6.77	.515	.310	2.19

(60T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 20.625 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 7.090 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	29.3	45.6	128.0	3.11	4.4	2.8	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	32.7	46.5	139.0	3.15	4.3	3.0	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.4	44.4	63.9	2.64	6.8	1.4	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.1	46.3	79.9	2.84	6.6	1.7	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.3	49.1	109.1	3.17	6.0	2.2	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.3	48.2	93.1	3.01	6.5	1.9	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	22.4	50.9	128.7	3.33	5.8	2.5	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.6	50.6	110.8	3.23	6.3	2.2	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	26.7	52.2	146.2	3.45	5.5	2.8	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	30.5	53.4	162.3	3.54	5.3	3.0	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.9	52.5	128.9	3.40	6.2	2.5	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	23.3	51.4	132.7	3.39	5.7	2.6	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	27.2	53.0	151.0	3.51	5.6	2.8	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	30.3	53.5	161.2	3.56	5.3	3.0	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	27.0	57.7	169.1	3.73	6.3	2.9	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	31.9	59.6	193.1	3.87	6.1	3.2	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.5	56.7	111.2	3.38	8.3	2.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.9	59.3	136.0	3.62	8.0	2.3	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.5	61.5	155.0	3.80	7.9	2.5	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	22.4	63.7	175.3	3.97	7.8	2.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.6	65.5	198.9	4.16	7.5	3.0	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	31.6	68.0	230.4	4.35	7.3	3.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	36.7	70.1	260.4	4.47	7.1	3.7	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	38.1	66.1	243.5	4.29	6.4	3.7	6.16	9.73	7.96	.435	.290	2.82

(60T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 20.625 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 7.090 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 39.0 I-T	24.45	45.6	68.7	281.2	4.44	6.2	4.1	7.19	9.92	7.99	.530	.315	3.12
12 X 4 X 14.0 I-T	10.98	18.6	70.0	179.8	4.17	9.7	2.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.2	71.9	202.0	4.34	9.5	2.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.7	75.4	240.0	4.61	9.3	3.2	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	30.2	78.2	275.9	4.82	9.1	3.5	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	37.6	81.2	322.7	5.13	8.6	4.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	43.2	83.5	360.9	5.26	8.4	4.3	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	50.6	86.4	410.1	5.39	8.1	4.7	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	55.3	83.8	409.2	5.30	7.4	4.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	61.8	86.0	446.0	5.35	7.2	5.2	8.47	12.06	8.05	.575	.335	4.04
14 X 5 X 22.0 I-T	16.18	33.7	88.0	343.4	5.38	10.2	3.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	40.5	91.6	400.2	5.63	9.9	4.4	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	47.1	93.7	444.4	5.78	9.4	4.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	53.8	96.5	495.0	5.94	9.2	5.1	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	60.2	99.0	540.5	6.05	9.0	5.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	66.6	97.5	554.0	6.01	8.3	5.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	74.4	100.2	603.5	6.07	8.1	6.0	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	44.6	103.1	499.1	6.24	11.2	4.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	53.8	107.7	581.7	6.50	10.8	5.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	62.9	110.8	650.0	6.66	10.3	5.9	7.56	15.36	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	71.0	114.0	715.7	6.83	10.1	6.3	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	65.6	122.7	771.6	7.21	11.8	6.3	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	76.4	127.5	871.3	7.44	11.4	6.8	8.63	17.90	6.02	.525	.315	5.64

(60T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T					4.91	4.0	22.4	14.6	1.21	3.7	.7	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T					6.40	5.0	23.9	18.2	1.33	3.6	.8	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T					7.42	6.0	25.3	21.5	1.42	3.6	.9	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T					8.82	7.7	26.9	26.7	1.55	3.5	1.0	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T					8.39	6.8	26.4	24.5	1.50	3.6	.9	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T					5.88	5.5	29.7	24.5	1.55	4.5	.8	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T					7.37	6.9	31.5	30.4	1.69	4.4	1.0	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T					8.36	8.1	33.0	35.3	1.80	4.4	1.1	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T					9.42	9.4	34.2	40.4	1.90	4.3	1.2	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T					9.91	10.1	34.3	42.2	1.93	4.2	1.2	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T					11.69	12.2	36.0	50.2	2.06	4.1	1.4	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T					6.94	7.3	37.6	38.9	1.93	5.3	1.0	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T					7.88	8.5	38.8	44.3	2.03	5.2	1.1	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T					6.17	6.7	36.8	35.6	1.86	5.3	1.0	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T					9.34	10.5	41.0	54.1	2.20	5.1	1.3	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T					10.89	12.5	42.6	63.1	2.33	5.1	1.5	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T					8.30	8.9	39.4	46.7	2.07	5.2	1.2	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T					9.78	11.4	41.4	57.0	2.24	5.0	1.4	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T					10.74	12.4	43.3	64.1	2.35	5.2	1.5	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T					12.63	15.6	44.8	76.0	2.50	4.9	1.7	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T					10.81	14.0	48.4	78.8	2.60	5.6	1.6	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T					12.85	17.1	50.3	93.5	2.77	5.5	1.9	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T					14.81	20.3	51.5	106.3	2.88	5.2	2.1	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T					16.77	23.4	52.8	119.6	2.99	5.1	2.3	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T					18.74	26.3	53.8	131.0	3.06	5.0	2.4	5.51	7.05	6.77	.515	.310	2.19

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	29.7	53.0	137.1	3.06	4.6	2.6	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	33.1	54.0	149.4	3.12	4.5	2.8	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.5	51.0	66.1	2.50	7.0	1.3	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.2	53.3	83.2	2.72	6.8	1.6	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.5	56.9	114.6	3.06	6.2	2.0	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.4	55.7	97.2	2.89	6.7	1.7	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	22.6	59.1	136.0	3.24	6.0	2.3	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.8	58.6	116.2	3.12	6.5	2.0	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	27.0	60.6	155.3	3.37	5.7	2.6	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	30.9	62.0	173.0	3.48	5.6	2.8	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	21.1	60.9	135.8	3.29	6.4	2.2	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	34.5	63.2	188.3	3.54	5.5	3.0	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	23.6	59.7	140.3	3.30	6.0	2.4	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	27.5	61.6	160.4	3.43	5.8	2.6	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	30.6	62.1	171.8	3.49	5.6	2.8	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	27.3	66.9	179.0	3.64	6.5	2.7	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	32.3	69.1	205.4	3.80	6.4	3.0	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.6	65.5	115.4	3.22	8.5	1.8	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.1	68.5	141.9	3.47	8.3	2.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.8	71.2	162.2	3.66	8.2	2.3	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	22.7	73.7	183.9	3.84	8.1	2.5	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.9	76.0	209.5	4.04	7.8	2.8	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	32.0	78.9	243.8	4.24	7.6	3.1	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	37.2	81.3	276.5	4.38	7.4	3.4	5.95	10.47	5.81	.510	.300	3.14

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 33.0 I-T	20.94	38.6	76.7	259.5	4.22	6.7	3.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	46.2	79.7	301.1	4.39	6.5	3.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	53.7	82.4	340.6	4.51	6.3	4.1	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	18.8	81.0	187.3	4.01	10.0	2.3	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.5	83.2	210.9	4.18	9.8	2.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.1	87.3	251.6	4.47	9.7	2.9	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	30.6	90.5	290.4	4.68	9.5	3.2	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	38.0	94.3	341.2	5.00	9.0	3.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	43.7	96.8	383.0	5.16	8.8	4.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	51.4	100.0	437.0	5.32	8.5	4.4	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	56.1	97.1	437.7	5.24	7.8	4.5	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	62.7	99.5	478.4	5.32	7.6	4.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	69.7	102.0	520.1	5.39	7.5	5.1	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	74.9	101.8	536.5	5.44	7.2	5.3	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	82.5	104.0	577.8	5.51	7.0	5.6	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	34.2	101.9	361.1	5.23	10.6	3.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	41.0	106.0	422.6	5.50	10.3	4.0	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	47.7	108.4	471.1	5.67	9.9	4.3	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	54.6	111.6	526.4	5.85	9.6	4.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	61.1	114.3	576.3	5.98	9.4	5.0	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	67.6	112.8	593.0	5.96	8.8	5.3	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	75.6	115.6	647.5	6.05	8.6	5.6	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	45.2	119.2	526.8	6.10	11.6	4.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	54.6	124.4	616.6	6.39	11.3	5.0	6.68	15.88	5.53	.440	.275	4.37

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 36.0 I-T	25.69	63.9	127.8	691.4	6.58	10.8	5.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	72.2	131.4	763.4	6.76	10.6	5.8	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	80.7	134.9	833.5	6.85	10.3	6.2	9.34	16.13	7.04	.565	.345	5.56
18 X 6 X 35.0 I-T	26.29	66.8	141.2	819.3	7.12	12.3	5.8	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	77.7	146.6	928.1	7.37	11.9	6.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	98.0	153.8	1099.1	7.63	11.2	7.1	10.46	17.99	7.50	.570	.355	6.39

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	4.0	26.8	15.4	1.09	3.8	.6	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.1	29.1	19.4	1.20	3.8	.7	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.1	31.1	23.0	1.30	3.8	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.9	33.7	28.7	1.43	3.7	.9	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.9	32.8	26.3	1.37	3.8	.8	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.5	36.3	25.9	1.40	4.7	.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	7.0	39.1	32.3	1.54	4.6	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.2	41.3	37.7	1.65	4.6	.9	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.5	43.2	43.4	1.75	4.6	1.0	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.3	43.4	45.4	1.78	4.4	1.0	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.4	46.0	54.5	1.91	4.4	1.2	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.5	46.7	41.2	1.75	5.5	.9	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.6	48.7	47.1	1.85	5.5	1.0	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.8	45.6	37.6	1.68	5.5	.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.7	52.0	57.9	2.02	5.4	1.1	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.7	54.3	68.0	2.15	5.3	1.3	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.1	49.5	49.7	1.89	5.5	1.0	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.6	52.6	61.1	2.06	5.3	1.2	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.6	55.2	69.0	2.17	5.5	1.2	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.8	57.5	82.5	2.33	5.2	1.4	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.3	62.0	84.7	2.40	5.9	1.4	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.4	64.8	101.3	2.58	5.8	1.6	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.7	66.6	116.1	2.71	5.6	1.7	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	23.9	68.6	131.6	2.83	5.5	1.9	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	26.8	70.0	145.1	2.92	5.4	2.1	5.51	7.05	6.77	.515	.310	2.19

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	30.3	69.1	153.1	2.95	5.1	2.2	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	33.9	70.5	168.0	3.02	5.0	2.4	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.6	64.3	69.7	2.26	7.2	1.1	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.4	67.8	88.5	2.49	7.1	1.3	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.8	73.4	123.9	2.85	6.6	1.7	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.7	71.3	104.1	2.66	7.1	1.5	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.0	76.5	148.3	3.05	6.4	1.9	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.1	75.5	125.1	2.89	6.9	1.7	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	27.6	78.7	170.8	3.20	6.2	2.2	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	31.5	80.8	191.6	3.33	6.1	2.4	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	21.5	78.8	147.3	3.08	6.8	1.9	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	35.2	82.2	209.8	3.41	6.0	2.6	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.0	77.5	153.1	3.10	6.4	2.0	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	39.0	83.7	228.0	3.48	5.8	2.7	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.0	80.0	176.5	3.25	6.3	2.2	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	44.1	85.6	251.9	3.57	5.7	2.9	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	31.2	80.9	190.1	3.33	6.1	2.3	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	27.9	86.7	195.9	3.44	7.0	2.3	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.0	89.7	226.6	3.62	6.9	2.5	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.8	83.8	122.3	2.94	8.8	1.5	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.4	88.1	151.7	3.20	8.7	1.7	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.1	91.8	174.2	3.39	8.7	1.9	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.1	95.3	198.4	3.57	8.6	2.1	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	27.4	98.6	227.4	3.78	8.3	2.3	4.42	10.17	5.75	.360	.240	2.44

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	32.6	102.6	266.6	4.01	8.2	2.6	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	37.9	105.6	304.5	4.18	8.0	2.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.4	100.0	287.2	4.03	7.3	2.9	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	47.2	103.9	336.2	4.24	7.1	3.2	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	55.0	107.4	383.1	4.40	7.0	3.6	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.1	104.4	199.5	3.68	10.4	1.9	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.9	107.5	225.8	3.86	10.3	2.1	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.6	113.1	271.0	4.16	10.2	2.4	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.3	117.3	314.8	4.40	10.1	2.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	38.7	122.8	372.6	4.73	9.6	3.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	44.6	126.0	421.1	4.91	9.4	3.3	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	52.5	130.0	484.1	5.12	9.2	3.7	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	57.3	126.5	488.2	5.07	8.5	3.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	64.2	129.3	536.5	5.18	8.4	4.1	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	71.5	132.3	586.0	5.29	8.2	4.4	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	76.7	132.4	607.1	5.35	7.9	4.6	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	84.6	135.1	657.0	5.46	7.8	4.9	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	34.8	132.3	391.1	4.91	11.2	3.0	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	41.9	137.7	461.1	5.20	11.0	3.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	48.8	140.8	517.2	5.40	10.6	3.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	55.9	145.0	581.3	5.62	10.4	4.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	62.6	148.3	639.7	5.77	10.2	4.3	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	69.2	146.5	662.4	5.79	9.6	4.5	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	77.5	149.9	727.1	5.92	9.4	4.8	9.26	13.79	8.03	.595	.340	4.69

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	46.2	154.7	574.2	5.78	12.4	3.7	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	55.9	161.2	677.2	6.11	12.1	4.2	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	65.5	165.5	764.4	6.34	11.7	4.6	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	74.0	170.2	848.4	6.55	11.5	5.0	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	82.9	174.0	930.6	6.68	11.2	5.3	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	92.1	178.1	1013.3	6.81	11.0	5.7	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	104.2	183.4	1120.9	6.93	10.8	6.1	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	126.9	187.7	1269.6	7.20	10.0	6.8	12.99	16.33	10.24	.665	.395	6.45
18 X 6 X 35.0 I-T	26.29	68.6	182.3	903.7	6.86	13.2	5.0	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	79.9	189.1	1029.5	7.15	12.9	5.4	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	100.8	197.7	1230.6	7.49	12.2	6.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	121.0	206.3	1424.7	7.70	11.8	6.9	12.53	18.24	7.56	.695	.415	7.57
21 X 8 1/4 X 62.0 I-T	44.94	141.3	240.1	1906.0	8.78	13.5	7.9	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	155.2	245.9	2051.9	8.89	13.2	8.3	14.46	21.13	8.27	.685	.430	9.09

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	4.1	30.3	16.2	.99	3.9	.5	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	5.2	33.6	20.4	1.10	3.9	.6	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	6.3	36.4	24.3	1.19	3.9	.7	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	8.0	40.0	30.5	1.32	3.8	.8	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	7.1	38.6	27.9	1.26	3.9	.7	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.6	42.1	27.1	1.27	4.8	.6	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	7.1	46.1	34.0	1.41	4.8	.7	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	8.4	49.1	39.8	1.51	4.7	.8	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	9.7	51.8	45.9	1.61	4.7	.9	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	10.5	52.3	48.1	1.64	4.6	.9	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	12.6	55.9	58.1	1.77	4.6	1.0	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.6	55.2	43.0	1.59	5.7	.8	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.7	58.0	49.4	1.69	5.6	.9	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.9	53.5	39.2	1.53	5.7	.7	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.9	62.6	60.9	1.85	5.6	1.0	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	12.9	66.1	71.9	1.99	5.6	1.1	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	9.2	59.2	52.2	1.73	5.6	.9	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	11.8	63.7	64.5	1.90	5.5	1.0	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	12.8	67.1	72.9	2.00	5.7	1.1	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	16.1	70.6	87.7	2.17	5.5	1.2	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	14.5	75.7	89.5	2.22	6.2	1.2	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	17.6	79.8	107.7	2.39	6.1	1.3	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	21.0	82.5	124.1	2.53	5.9	1.5	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	24.3	85.3	141.4	2.66	5.8	1.7	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	27.2	87.3	156.7	2.76	5.8	1.8	5.51	7.05	6.77	.515	.310	2.19

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T		20.94	30.8	86.6	166.5	2.81	5.4	1.9	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T		23.53	34.5	88.4	183.6	2.89	5.3	2.1	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T		7.19	9.7	77.1	72.6	2.06	7.4	.9	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T		9.52	12.6	82.4	92.8	2.28	7.4	1.1	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T		12.83	19.0	90.5	131.4	2.65	6.9	1.5	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T		10.79	14.9	87.2	109.5	2.45	7.4	1.3	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T		15.28	23.4	94.9	158.3	2.85	6.8	1.7	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T		12.00	18.3	93.0	132.2	2.67	7.2	1.4	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T		17.73	28.0	98.1	183.5	3.01	6.6	1.9	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T		19.79	32.0	101.0	207.0	3.15	6.5	2.0	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T		13.87	21.8	97.6	156.5	2.86	7.2	1.6	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T		22.32	35.8	103.0	227.8	3.25	6.4	2.2	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T		15.11	24.3	96.3	163.5	2.90	6.7	1.7	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T		24.83	39.7	104.9	248.7	3.34	6.3	2.4	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T		17.69	28.5	99.9	189.6	3.06	6.7	1.9	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T		28.28	45.0	107.3	276.4	3.44	6.1	2.6	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T		19.16	31.7	101.3	205.1	3.15	6.5	2.0	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T		17.26	28.3	108.0	209.7	3.23	7.4	1.9	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T		19.76	33.5	112.2	244.0	3.42	7.3	2.2	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T		9.07	14.0	102.2	127.7	2.69	9.1	1.2	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T		11.27	17.7	108.4	159.3	2.95	9.0	1.5	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T		12.48	20.4	113.4	183.6	3.14	9.0	1.6	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T		13.77	23.4	118.2	209.9	3.32	9.0	1.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T		15.04	27.8	122.9	241.6	3.53	8.7	2.0	4.42	10.17	5.75	.360	.240	2.44

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	33.1	128.2	285.0	3.76	8.6	2.2	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.6	132.3	327.5	3.95	8.5	2.5	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.0	125.4	310.0	3.83	7.8	2.5	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.0	130.6	365.6	4.06	7.6	2.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	56.0	135.1	419.4	4.24	7.5	3.1	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.4	126.9	209.0	3.39	10.8	1.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.2	133.2	237.5	3.57	10.7	1.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.0	140.8	286.5	3.86	10.6	2.0	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.8	146.4	334.4	4.11	10.5	2.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.2	154.1	397.9	4.44	10.1	2.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.3	158.2	452.2	4.64	10.0	2.9	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	53.4	163.3	523.3	4.88	9.8	3.2	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	58.2	159.3	530.4	4.86	9.1	3.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	65.4	162.7	585.9	5.00	9.0	3.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	72.9	166.2	642.9	5.13	8.8	3.9	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	78.1	166.7	668.2	5.20	8.6	4.0	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	86.2	170.2	726.2	5.33	8.4	4.3	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.4	165.5	414.9	4.58	11.7	2.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.6	172.5	492.2	4.89	11.6	2.9	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.6	176.7	555.1	5.11	11.2	3.1	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	56.8	182.1	626.9	5.35	11.0	3.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	63.7	186.2	692.9	5.53	10.9	3.7	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	70.4	184.4	721.2	5.57	10.2	3.9	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	79.0	188.4	795.4	5.73	10.1	4.2	9.26	13.79	8.03	.595	.340	4.69

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	47.0	194.0	612.7	5.44	13.0	3.2	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	56.9	202.3	727.2	5.79	12.8	3.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	66.6	207.7	825.5	6.05	12.4	4.0	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	75.4	213.6	920.1	6.29	12.2	4.3	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	84.6	218.1	1014.0	6.45	12.0	4.7	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	94.1	222.7	1108.5	6.61	11.8	5.0	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	106.7	228.8	1232.0	6.77	11.5	5.4	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	129.7	235.0	1406.7	7.09	10.8	6.0	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	148.3	241.9	1564.8	7.22	10.6	6.5	15.00	16.52	10.30	.760	.455	7.52
18 X 6 X 35.0 I-T	26.29	69.9	228.4	974.3	6.55	13.9	4.3	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	81.5	236.8	1115.5	6.87	13.7	4.7	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	103.1	247.2	1344.9	7.27	13.1	5.4	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	124.0	257.1	1567.3	7.55	12.6	6.1	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	145.2	266.6	1783.0	7.72	12.3	6.7	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	182.0	275.4	2069.9	8.04	11.4	7.5	17.00	18.39	11.09	.770	.480	8.83
21 X 8 1/4 X 62.0 I-T	44.94	144.9	298.4	2096.0	8.62	14.5	7.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	159.3	305.0	2263.2	8.77	14.2	7.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	170.7	310.1	2393.6	8.86	14.0	7.7	15.47	21.24	8.30	.740	.455	9.66
21 X 12 1/4 X 101.0 I-T	68.38	243.7	330.9	3067.8	9.35	12.6	9.3	20.11	21.36	12.29	.800	.500	10.68

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 6.5 T		6.40	5.3	37.2	21.3	1.01	4.0	.6	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T		7.42	6.4	40.8	25.5	1.10	4.0	.6	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T		8.82	8.1	45.5	32.0	1.22	3.9	.7	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T		8.39	7.2	43.7	29.2	1.17	4.1	.7	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T		5.88	5.7	46.7	28.1	1.17	4.9	.6	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T		7.37	7.2	52.0	35.4	1.29	4.9	.7	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T		8.36	8.5	56.0	41.5	1.39	4.9	.7	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T		9.42	9.8	59.7	48.0	1.49	4.9	.8	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T		9.91	10.6	60.5	50.4	1.52	4.7	.8	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T		11.69	12.8	65.3	61.1	1.65	4.8	.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T		6.94	7.7	62.6	44.6	1.46	5.8	.7	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T		7.88	8.9	66.4	51.3	1.55	5.8	.8	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T		6.17	7.0	60.2	40.6	1.40	5.8	.7	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T		9.34	11.0	72.6	63.5	1.71	5.8	.9	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T		10.89	13.1	77.2	75.2	1.84	5.7	1.0	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T		8.30	9.4	67.9	54.2	1.59	5.8	.8	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T		9.78	11.9	74.1	67.3	1.75	5.6	.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T		10.74	13.0	78.4	76.2	1.85	5.9	1.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T		12.63	16.3	83.4	92.1	2.01	5.7	1.1	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T		10.81	14.6	89.0	93.4	2.05	6.4	1.0	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T		12.85	17.8	94.6	112.9	2.23	6.3	1.2	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T		14.81	21.2	98.5	130.8	2.37	6.2	1.3	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T		16.77	24.6	102.4	149.6	2.50	6.1	1.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T		18.74	27.6	105.2	166.3	2.61	6.0	1.6	5.51	7.05	6.77	.515	.310	2.19

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	31.2	104.7	177.8	2.66	5.7	1.7	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	35.0	107.3	196.9	2.76	5.6	1.8	6.92	6.30	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.9	88.7	75.0	1.89	7.6	.8	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.8	96.2	96.3	2.10	7.6	1.0	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.3	107.6	137.5	2.46	7.1	1.3	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.1	102.6	114.0	2.27	7.6	1.1	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.7	113.8	166.5	2.66	7.0	1.5	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.5	110.4	138.1	2.48	7.5	1.3	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.3	118.2	194.1	2.83	6.9	1.6	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	32.4	122.2	219.8	2.98	6.8	1.8	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.1	116.6	164.0	2.67	7.4	1.4	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	36.3	124.9	242.9	3.08	6.7	1.9	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.6	115.6	172.1	2.71	7.0	1.5	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	40.3	127.4	266.3	3.18	6.6	2.1	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.8	120.4	200.5	2.88	7.0	1.7	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	45.7	130.6	297.3	3.30	6.5	2.3	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.1	122.5	217.6	2.97	6.8	1.8	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.7	130.1	221.0	3.03	7.7	1.7	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.0	135.8	258.5	3.23	7.6	1.9	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.2	120.1	132.1	2.47	9.3	1.1	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.9	128.6	165.6	2.72	9.3	1.3	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.7	135.2	191.3	2.91	9.3	1.4	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.7	141.6	219.3	3.09	9.3	1.5	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.1	148.0	253.3	3.29	9.0	1.7	4.42	10.17	5.75	.360	.240	2.44

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	33.5	155.2	300.1	3.53	9.0	1.9	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	39.0	160.5	346.5	3.73	8.9	2.2	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.4	152.4	329.0	3.62	8.1	2.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.6	159.3	390.3	3.86	8.0	2.5	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	56.7	165.0	450.2	4.06	7.9	2.7	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.6	153.6	216.7	3.12	11.1	1.4	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.4	159.3	246.9	3.30	11.0	1.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.3	169.4	298.9	3.59	11.0	1.8	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.2	176.8	350.3	3.84	10.9	2.0	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.7	187.2	418.5	4.16	10.5	2.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.8	192.6	477.8	4.38	10.4	2.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.1	199.1	555.9	4.62	10.3	2.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.0	194.7	565.9	4.62	9.6	2.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	66.3	198.9	627.7	4.78	9.5	3.2	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	74.8	203.3	691.6	4.93	9.4	3.4	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	79.2	204.3	720.7	5.01	9.1	3.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	87.5	208.6	786.1	5.16	9.0	3.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.8	200.5	434.2	4.28	12.1	2.2	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.1	209.7	517.6	4.59	12.0	2.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.2	215.3	586.2	4.82	11.7	2.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	57.5	222.2	664.7	5.07	11.6	3.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	64.6	227.4	737.5	5.26	11.4	3.2	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	71.3	225.6	770.7	5.32	10.8	3.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	80.2	230.5	853.6	5.50	10.6	3.7	9.26	13.79	8.03	.595	.340	4.69

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	47.6	236.1	644.1	5.10	13.5	2.7	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	57.7	246.6	768.4	5.47	13.3	3.1	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	67.6	253.7	876.4	5.75	13.0	3.5	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	76.5	261.1	980.4	6.00	12.8	3.8	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	86.0	266.4	1084.9	6.19	12.6	4.1	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	95.7	271.9	1190.5	6.37	12.4	4.4	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	108.7	278.9	1329.0	6.57	12.2	4.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	132.0	287.4	1527.7	6.91	11.6	5.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	151.1	295.2	1707.6	7.09	11.3	5.8	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	174.2	304.5	1918.4	7.26	11.0	6.3	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	71.0	278.7	1033.2	6.22	14.6	3.7	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	82.8	289.2	1188.1	6.56	14.4	4.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	104.8	301.9	1443.2	7.00	13.8	4.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	126.3	313.4	1692.6	7.33	13.4	5.4	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	148.2	323.9	1935.6	7.56	13.1	6.0	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	185.7	335.1	2264.8	7.93	12.2	6.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	209.0	344.2	2490.9	8.08	11.9	7.2	19.18	18.59	11.15	.870	.535	9.95
21 X 8 1/4 X 62.0 I-T	44.94	147.7	363.3	2263.4	8.38	15.3	6.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	162.6	370.8	2451.4	8.56	15.1	6.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	174.4	376.5	2598.3	8.68	14.9	6.9	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	196.6	386.7	2866.6	8.85	14.6	7.4	17.58	21.43	8.36	.835	.515	11.04
21 X 12 1/4 X 101.0 I-T	68.38	249.1	401.1	3368.9	9.28	13.5	8.4	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	272.2	409.6	3609.4	9.37	13.3	8.8	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 5 1/4 X 9.0 T	8.82	8.3	50.3	33.5	1.13	4.0	.7	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	7.4	47.9	30.5	1.09	4.1	.6	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 7.5 T	7.37	7.4	57.0	36.7	1.20	5.0	.6	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.6	61.9	43.1	1.29	5.0	.7	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	10.0	66.5	49.9	1.38	5.0	.8	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.8	67.7	52.5	1.41	4.9	.8	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	13.0	74.0	63.7	1.54	4.9	.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.8	68.7	46.1	1.35	5.9	.7	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	9.0	73.5	53.1	1.44	5.9	.7	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	7.1	65.6	41.9	1.29	5.9	.6	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	11.2	81.4	65.8	1.59	5.9	.8	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.3	87.5	78.1	1.71	5.9	.9	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.5	75.5	56.1	1.47	5.9	.7	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	12.1	83.4	69.8	1.63	5.8	.8	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.1	88.7	79.0	1.72	6.0	.9	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.5	95.4	95.9	1.88	5.8	1.0	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.8	101.2	96.8	1.91	6.5	1.0	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.0	108.7	117.4	2.08	6.5	1.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.5	114.0	136.4	2.22	6.3	1.2	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	24.8	119.2	156.5	2.35	6.3	1.3	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	27.9	123.1	174.6	2.46	6.3	1.4	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	31.6	123.1	187.4	2.52	5.9	1.5	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	35.4	126.6	208.3	2.62	5.9	1.6	6.92	6.90	8.03	.595	.340	2.35

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 10.0 I-T	7.19	10.0	98.8	77.1	1.74	7.7	.8	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.9	108.8	99.4	1.95	7.7	.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.5	124.0	142.6	2.29	7.3	1.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.2	117.0	117.8	2.10	7.7	1.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.9	132.3	173.5	2.49	7.3	1.3	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.7	127.0	143.0	2.30	7.6	1.1	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.6	138.3	203.0	2.66	7.1	1.5	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	32.8	143.7	230.6	2.81	7.0	1.6	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.3	135.2	170.4	2.49	7.6	1.3	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	36.8	147.3	255.8	2.92	7.0	1.7	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.8	134.6	179.3	2.54	7.2	1.3	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	40.8	150.7	281.3	3.02	6.9	1.9	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.1	141.0	209.6	2.71	7.2	1.5	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	46.3	154.9	315.3	3.15	6.8	2.0	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.4	144.0	228.1	2.80	7.0	1.6	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.0	152.3	230.6	2.84	8.0	1.5	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.4	159.8	270.8	3.04	7.9	1.7	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.3	136.5	135.9	2.28	9.5	1.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.1	147.9	170.8	2.53	9.5	1.2	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.9	156.5	197.7	2.70	9.5	1.3	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.9	164.6	227.1	2.87	9.5	1.4	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.3	173.1	262.9	3.07	9.3	1.5	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	33.8	182.5	312.6	3.31	9.2	1.7	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	39.5	189.6	362.4	3.51	9.2	1.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.9	180.2	344.9	3.41	8.4	1.9	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	49.1	189.2	411.2	3.66	8.4	2.2	7.19	9.92	7.99	.530	.315	3.12

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 45.0 I-T	28.19	57.4	196.6	476.5	3.88	8.3	2.4	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.8	177.4	223.1	2.89	11.3	1.3	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.7	185.1	254.7	3.07	11.2	1.4	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.6	198.0	309.2	3.35	11.2	1.6	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.5	207.8	363.5	3.59	11.2	1.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.0	221.2	435.6	3.90	10.9	2.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.3	228.4	499.0	4.12	10.8	2.2	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.7	236.8	583.3	4.38	10.7	2.5	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.6	232.0	595.7	4.39	10.0	2.6	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	67.1	237.4	663.3	4.56	9.9	2.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	74.9	242.9	733.3	4.72	9.8	3.0	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	80.1	244.5	765.7	4.81	9.6	3.1	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	88.5	250.0	837.9	4.96	9.5	3.4	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.1	236.4	450.1	4.00	12.5	1.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.6	248.2	538.7	4.31	12.4	2.2	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.7	255.6	612.1	4.54	12.1	2.4	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	58.2	264.5	696.2	4.79	12.0	2.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	65.3	271.1	774.9	4.99	11.9	2.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	72.1	269.5	812.6	5.06	11.3	3.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	81.1	275.6	903.3	5.26	11.1	3.3	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.1	280.1	670.0	4.79	13.9	2.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	58.3	293.5	802.7	5.16	13.8	2.7	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	68.3	302.5	919.1	5.45	13.4	3.0	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	77.4	311.9	1031.3	5.70	13.3	3.3	8.26	16.01	7.00	.505	.305	4.88

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 45.0 I-T	31.77	87.0	318.3	1145.3	5.91	13.2	3.6	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	97.0	325.0	1260.9	6.10	13.0	3.9	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	110.3	333.2	1413.2	6.33	12.8	4.2	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	133.8	344.4	1633.5	6.70	12.2	4.7	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	153.4	353.4	1834.2	6.91	12.0	5.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	177.1	364.0	2070.2	7.12	11.7	5.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	71.8	332.4	1082.6	5.89	15.1	3.3	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	83.8	345.5	1249.3	6.24	14.9	3.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	106.2	361.1	1527.5	6.71	14.4	4.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	128.2	374.6	1801.7	7.08	14.1	4.8	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	150.8	386.5	2070.9	7.35	13.7	5.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	188.8	400.7	2439.8	7.77	12.9	6.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	212.7	410.9	2693.3	7.95	12.7	6.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	230.9	418.3	2879.7	8.05	12.5	6.9	21.02	18.73	11.20	.940	.590	11.05
21 X 8 1/4 X 62.0 I-T	44.94	150.0	434.2	2409.8	8.11	16.1	5.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	165.2	442.8	2617.3	8.31	15.8	5.9	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	177.3	449.3	2779.9	8.45	15.7	6.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	200.2	460.6	3078.1	8.66	15.4	6.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	223.7	472.0	3376.3	8.83	15.1	7.2	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	253.5	478.4	3643.3	9.15	14.4	7.6	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	277.3	487.7	3913.3	9.27	14.1	8.0	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 4 X 9.5 T	9.42	10.2	72.3	51.7	1.29	5.1	.7	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	11.0	73.9	54.4	1.32	5.0	.7	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	13.2	81.6	66.2	1.44	5.0	.8	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 8.0 T	7.88	9.1	79.3	54.7	1.34	6.0	.7	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.5 T	9.34	11.3	88.9	67.9	1.48	6.0	.8	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.4	96.5	80.7	1.60	6.0	.8	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.6	81.7	57.9	1.37	6.0	.7	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	12.2	91.4	72.1	1.52	5.9	.8	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.3	97.8	81.7	1.61	6.1	.8	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.7	106.4	99.3	1.76	6.0	.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	15.0	112.2	99.8	1.78	6.7	.9	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.2	121.7	121.4	1.94	6.7	1.0	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.7	128.7	141.4	2.08	6.5	1.1	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.1	135.4	162.6	2.21	6.5	1.2	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	28.2	140.4	181.8	2.32	6.4	1.3	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	31.9	141.2	195.8	2.38	6.1	1.4	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	35.8	145.8	218.2	2.49	6.1	1.5	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	10.1	107.1	79.1	1.61	7.8	.7	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	13.1	119.8	102.2	1.81	7.8	.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.7	139.3	147.2	2.14	7.5	1.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.4	129.9	121.2	1.96	7.9	.9	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.2	149.9	179.5	2.34	7.4	1.2	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.9	142.4	147.3	2.15	7.8	1.0	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.9	157.9	210.7	2.51	7.3	1.3	5.22	7.93	6.99	.430	.295	2.34

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 7 X 20.0 T	19.79	33.1	164.9	239.9	2.65	7.2	1.5	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.5	152.8	175.9	2.33	7.8	1.2	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.1	169.7	266.9	2.76	7.2	1.6	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.1	152.8	185.5	2.38	7.4	1.2	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.3	174.1	294.3	2.87	7.1	1.7	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.4	161.1	217.6	2.55	7.4	1.4	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	46.9	179.6	331.0	3.00	7.1	1.8	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.7	165.1	237.2	2.64	7.3	1.4	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.3	173.9	238.8	2.67	8.2	1.4	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.7	183.6	281.3	2.87	8.1	1.5	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.4	151.1	139.2	2.12	9.6	.9	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.2	165.7	175.4	2.35	9.6	1.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.1	176.4	203.3	2.52	9.6	1.2	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.2	186.7	233.8	2.69	9.7	1.3	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.6	197.5	271.2	2.88	9.5	1.4	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.1	209.6	323.3	3.11	9.5	1.5	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	39.8	218.7	376.0	3.31	9.4	1.7	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.2	208.2	358.5	3.22	8.7	1.7	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	49.6	219.7	429.0	3.47	8.7	2.0	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	58.0	229.1	499.1	3.69	8.6	2.2	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.0	199.7	228.6	2.69	11.5	1.1	3.23	11.91	3.97	.225	.230	2.38
12 X 4 X 16.0 I-T	12.37	22.9	209.5	261.3	2.86	11.4	1.2	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.8	225.8	318.0	3.13	11.4	1.4	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.8	238.3	374.7	3.36	11.4	1.6	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.4	255.4	449.9	3.66	11.1	1.8	5.19	12.22	6.49	.380	.230	2.81

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 6 1/2 X 30.0 I-T	20.27	46.7	264.7	517.0	3.88	11.1	2.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	55.2	275.5	606.5	4.14	11.0	2.2	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	60.1	270.5	621.1	4.16	10.3	2.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	67.7	277.3	693.7	4.34	10.2	2.5	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	75.6	284.3	769.2	4.51	10.2	2.7	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	80.9	286.6	804.4	4.60	9.9	2.8	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	89.4	293.5	882.7	4.76	9.9	3.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.4	272.1	463.5	3.74	12.7	1.7	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.9	287.2	556.4	4.05	12.7	1.9	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	51.1	296.9	633.9	4.28	12.4	2.1	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	58.7	308.1	722.9	4.53	12.3	2.3	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	65.9	316.5	806.7	4.73	12.2	2.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	72.8	315.3	848.2	4.81	11.7	2.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	81.9	323.0	945.8	5.01	11.5	2.9	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.5	325.0	691.7	4.50	14.2	2.1	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	58.8	342.0	831.5	4.87	14.1	2.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	69.0	353.5	955.1	5.16	13.8	2.7	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	78.1	365.1	1074.4	5.42	13.8	2.9	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	87.9	373.1	1196.9	5.63	13.6	3.2	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	98.0	381.2	1321.4	5.84	13.5	3.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	111.6	391.1	1486.4	6.08	13.3	3.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	135.3	405.5	1725.9	6.46	12.8	4.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	155.3	415.9	1946.1	6.70	12.5	4.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	179.6	428.1	2206.0	6.94	12.3	5.2	17.40	16.75	10.37	.875	.525	8.79

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 6 X 35.0 I-T	26.29	72.6	388.6	1124.3	5.58	15.5	2.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	84.7	404.8	1301.3	5.93	15.4	3.2	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	107.3	424.1	1599.8	6.42	14.9	3.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	129.7	440.2	1896.6	6.81	14.6	4.3	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	152.8	453.8	2190.2	7.11	14.3	4.8	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	191.2	471.5	2595.7	7.56	13.6	5.5	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	215.8	483.1	2875.5	7.78	13.3	6.0	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	234.5	491.3	3081.9	7.90	13.1	6.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	264.0	504.7	3407.2	8.09	12.9	6.8	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	151.9	510.4	2537.3	7.81	16.7	5.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	167.4	520.4	2763.0	8.03	16.5	5.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	179.8	527.9	2940.3	8.19	16.4	5.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	203.2	540.6	3266.8	8.43	16.1	6.0	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	227.4	553.1	3594.4	8.64	15.8	6.5	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	257.2	562.5	3891.2	8.96	15.1	6.9	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	281.6	572.6	4190.3	9.11	14.9	7.3	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 5 X 19.0 I-T	11.69	13.4	88.1	68.5	1.36	5.1	.8	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 11.0 T	10.89	13.6	104.1	83.2	1.50	6.1	.8	3.20	6.16	4.03	.425	.260	1.60
6 X 6 X 15.0 I-T	9.78	12.4	98.1	74.2	1.42	6.0	.8	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.5	105.5	84.1	1.51	6.2	.8	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.9	116.0	102.5	1.65	6.1	.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	15.2	121.6	102.7	1.67	6.8	.8	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.5	133.3	125.0	1.82	6.8	.9	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	22.0	142.1	145.8	1.96	6.6	1.0	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.4	150.5	168.0	2.08	6.6	1.1	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	28.5	156.9	188.2	2.19	6.6	1.2	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	32.3	158.6	203.2	2.26	6.3	1.3	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	36.2	164.4	227.0	2.36	6.3	1.4	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 13.0 I-T	9.52	13.2	129.1	104.7	1.69	7.9	.8	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.9	153.0	151.3	2.01	7.6	1.0	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.6	141.1	124.4	1.84	8.0	.9	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.4	166.3	184.9	2.20	7.6	1.1	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.1	156.1	151.3	2.01	7.9	1.0	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	29.2	176.5	217.6	2.36	7.4	1.2	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.4	185.3	248.2	2.50	7.4	1.3	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.7	168.9	180.9	2.19	8.0	1.1	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.5	191.5	276.6	2.62	7.4	1.4	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.3	169.6	191.1	2.24	7.6	1.1	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.7	197.2	305.7	2.73	7.3	1.5	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.7	180.1	224.6	2.40	7.6	1.2	5.20	8.06	6.54	.465	.285	2.30

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 7 1/8 X 28.5 T	28.28	47.4	204.2	344.8	2.86	7.3	1.7	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.0	185.4	245.2	2.50	7.4	1.3	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.5	194.5	246.1	2.52	8.3	1.3	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.0	206.6	290.5	2.71	8.3	1.4	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.6	163.7	142.2	1.98	9.8	.9	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.4	181.5	179.5	2.20	9.8	1.0	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.3	194.6	208.2	2.36	9.8	1.1	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.4	207.1	239.7	2.52	9.8	1.2	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.8	220.5	278.4	2.70	9.7	1.3	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.4	235.6	332.7	2.93	9.7	1.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	40.2	247.3	387.8	3.13	9.7	1.6	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.6	235.6	370.4	3.05	8.9	1.6	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	50.0	250.0	444.6	3.30	8.9	1.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	58.5	261.9	518.8	3.51	8.9	2.0	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.1	219.8	233.4	2.51	11.6	1.1	3.23	11.91	3.97	.225	.200	2.36
12 X 4 X 16.0 I-T	12.37	23.1	232.1	267.2	2.67	11.6	1.2	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.0	252.1	325.6	2.93	11.6	1.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.1	267.8	384.4	3.16	11.6	1.4	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.7	288.8	462.3	3.45	11.4	1.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.0	300.8	532.5	3.66	11.3	1.8	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	55.7	314.5	626.5	3.92	11.3	2.0	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	60.6	309.5	643.0	3.95	10.6	2.1	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	68.3	318.1	720.0	4.13	10.5	2.3	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	76.3	326.8	800.4	4.30	10.5	2.4	9.44	12.19	8.08	.640	.370	4.51

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 10 X 53.0 I-T	33.01	81.6	330.1	838.1	4.39	10.3	2.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	90.2	338.6	921.7	4.56	10.2	2.7	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.7	306.8	475.0	3.51	12.9	1.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	44.3	325.8	571.5	3.81	12.9	1.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	51.5	338.2	652.6	4.04	12.7	1.9	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	59.1	352.2	745.8	4.28	12.6	2.1	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	66.5	362.8	834.1	4.49	12.6	2.3	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	73.3	362.2	878.8	4.57	12.0	2.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	82.6	371.8	982.6	4.76	11.9	2.6	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.9	369.8	710.2	4.24	14.5	1.9	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	59.3	391.0	856.2	4.60	14.4	2.2	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	69.5	405.5	986.0	4.89	14.2	2.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	78.7	420.0	1111.4	5.14	14.1	2.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	88.7	429.9	1241.4	5.37	14.0	2.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	98.9	439.8	1373.8	5.58	13.9	3.1	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	112.7	451.7	1550.1	5.83	13.7	3.4	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	136.5	469.8	1806.7	6.22	13.2	3.8	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	156.9	482.2	2044.8	6.48	13.0	4.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	181.6	496.4	2327.1	6.74	12.8	4.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	73.2	446.3	1159.9	5.29	15.9	2.6	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	85.4	466.1	1345.9	5.64	15.6	2.9	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	108.3	489.9	1662.3	6.13	15.3	3.4	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	131.0	509.3	1979.3	6.54	15.1	3.9	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	154.6	525.2	2295.5	6.87	14.8	4.4	14.93	18.47	7.64	.810	.495	9.14

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/8 X 86.0 I-T	57.79	193.4	547.0	2734.3	7.34	14.1	5.0	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	218.4	560.3	3038.9	7.58	13.9	5.4	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	237.5	569.4	3264.6	7.72	13.7	5.7	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	267.7	584.5	3620.5	7.94	13.5	6.2	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	153.5	591.2	2648.6	7.51	17.3	4.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	169.2	602.9	2890.8	7.74	17.1	4.8	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	181.8	611.6	3081.7	7.91	17.0	5.0	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	205.8	626.0	3434.8	8.18	16.7	5.5	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	230.4	640.0	3790.2	8.41	16.4	5.9	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	260.3	652.7	4113.8	8.74	15.8	6.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	285.2	663.9	4440.9	8.91	15.6	6.7	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 52.500 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 45.938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 6 3/4 X 15.0 T	14.81	22.4	164.4	154.0	1.75	6.9	.9	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.9	176.4	177.7	1.87	6.9	1.0	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	29.1	185.8	199.5	1.97	6.9	1.1	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	32.9	189.8	216.2	2.04	6.6	1.1	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	37.0	198.7	242.3	2.14	6.6	1.2	6.92	6.90	8.03	.595	.340	2.35
8 X 5 1/2 X 13.0 T	12.83	20.3	175.4	158.8	1.79	7.8	.9	3.77	7.85	5.50	.345	.250	1.96
8 X 5 1/2 X 15.5 T	15.28	24.9	194.1	194.5	1.96	7.8	1.0	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.5	178.1	158.4	1.79	8.1	.9	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	29.8	209.2	229.5	2.12	7.7	1.1	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	34.1	222.0	262.5	2.25	7.7	1.2	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	23.2	195.8	189.7	1.95	8.2	1.0	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	38.2	231.6	293.4	2.36	7.7	1.3	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.8	198.4	200.9	2.00	7.8	1.0	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	42.5	240.5	325.2	2.47	7.7	1.4	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	30.3	213.8	236.8	2.15	7.8	1.1	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	48.3	251.3	368.3	2.61	7.6	1.5	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.6	221.8	259.1	2.24	7.7	1.2	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	30.1	230.7	258.7	2.25	8.6	1.1	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.6	248.5	306.3	2.43	8.6	1.2	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 15.0 I-T	11.27	18.8	206.9	187.0	1.95	10.0	.9	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.7	224.6	217.0	2.09	10.0	1.0	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.8	241.8	250.1	2.24	10.1	1.0	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	29.3	260.7	290.9	2.40	9.9	1.1	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	35.0	282.6	348.5	2.61	10.0	1.2	5.11	10.33	5.77	.440	.260	2.69

(60T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 52.500 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 45.938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 30.0 I-T	20.23	40.8	300.3	407.7	2.80	10.0	1.4	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	42.2	286.6	390.4	2.74	9.2	1.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	50.8	308.2	470.7	2.98	9.3	1.5	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	59.4	326.0	551.6	3.19	9.3	1.7	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.5	252.6	242.1	2.22	11.8	1.0	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.4	270.0	277.4	2.37	11.8	1.0	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.5	298.1	338.7	2.60	11.9	1.1	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.6	320.9	400.8	2.81	11.9	1.2	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	41.2	350.6	483.0	3.07	11.7	1.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.7	369.0	558.1	3.28	11.7	1.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	56.4	390.1	659.6	3.53	11.7	1.7	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	61.4	385.8	679.2	3.57	11.1	1.8	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	69.3	399.2	763.6	3.75	11.0	1.9	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	77.5	412.4	852.0	3.92	11.0	2.1	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	82.8	417.9	893.8	4.01	10.8	2.1	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	91.5	430.7	986.3	4.18	10.8	2.3	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.2	370.3	494.3	3.12	13.3	1.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	44.9	398.6	596.6	3.40	13.3	1.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	52.2	417.7	683.4	3.62	13.1	1.6	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	59.9	438.5	783.3	3.85	13.1	1.8	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	67.4	454.6	878.9	4.05	13.0	1.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	74.3	456.1	929.1	4.14	12.5	2.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	83.8	470.9	1042.9	4.35	12.5	2.2	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	49.6	455.6	740.6	3.79	14.9	1.6	5.73	15.69	5.50	.345	.250	3.92

(60T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 52.500 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 45.938 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 31.0 I-T	22.70	60.1	487.3	896.4	4.13	14.9	1.8	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	70.5	509.4	1036.3	4.40	14.7	2.0	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	79.8	530.7	1171.6	4.65	14.7	2.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	90.0	545.9	1314.0	4.88	14.6	2.4	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	100.5	560.8	1459.8	5.09	14.5	2.6	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	114.6	578.3	1655.1	5.35	14.4	2.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	138.6	605.6	1940.3	5.74	14.0	3.2	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	159.5	623.3	2209.6	6.02	13.8	3.5	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	185.0	642.9	2531.6	6.32	13.7	3.9	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	74.2	562.3	1217.7	4.76	16.4	2.2	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	86.6	591.2	1418.2	5.10	16.4	2.4	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	109.9	626.8	1764.3	5.59	16.0	2.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	133.2	654.9	2115.6	6.02	15.9	3.2	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	157.4	677.2	2470.8	6.37	15.7	3.6	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	196.7	709.4	2967.1	6.87	15.1	4.2	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	222.5	727.3	3316.4	7.14	14.9	4.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	242.3	739.1	3577.4	7.31	14.8	4.8	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	273.5	758.7	3989.8	7.57	14.6	5.3	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	156.0	763.2	2831.8	6.92	18.2	3.7	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	172.1	779.5	3102.5	7.17	18.0	4.0	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	185.1	791.5	3317.0	7.35	17.9	4.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	209.8	810.6	3717.3	7.65	17.7	4.6	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	235.3	828.6	4123.1	7.92	17.5	5.0	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	265.1	849.1	4492.3	8.25	16.9	5.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	290.9	863.4	4871.0	8.46	16.7	5.6	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 60.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 60.000 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 6 3/4 X 19.0	T			18.74	29.8	208.8	209.8	1.79	7.0	1.0	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5	T			20.94	33.7	215.3	227.9	1.86	6.8	1.1	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0	T			23.53	37.8	227.6	255.8	1.96	6.8	1.1	6.92	6.90	8.03	.595	.340	2.35
8 X 7 X 18.0	T			17.73	30.4	235.0	240.3	1.92	7.9	1.0	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0	T			19.79	34.7	252.1	275.1	2.04	7.9	1.1	5.82	8.01	7.00	.505	.305	2.44
8 X 7 X 22.5	T			22.32	38.9	265.5	308.1	2.15	7.9	1.2	6.56	8.07	7.04	.565	.345	2.78
8 X 7 1/8 X 25.0	T			24.83	43.3	278.0	342.0	2.25	7.9	1.2	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0	I-T			17.69	30.9	240.4	247.8	1.95	8.0	1.0	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5	T			28.28	49.2	293.3	388.3	2.38	7.9	1.3	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0	I-T			19.16	34.3	251.4	271.4	2.03	7.9	1.1	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5	T			17.26	30.6	259.3	270.0	2.04	8.8	1.0	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0	T			19.76	36.3	283.0	320.1	2.21	8.8	1.1	5.81	8.95	6.02	.525	.315	2.82
10 X 5 3/4 X 22.0	I-T			15.04	29.8	292.0	302.1	2.17	10.1	1.0	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0	I-T			17.37	35.5	321.0	362.3	2.36	10.2	1.1	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0	I-T			20.23	41.5	345.3	424.6	2.54	10.2	1.2	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0	I-T			20.94	42.9	330.2	407.4	2.48	9.5	1.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0	I-T			24.45	51.6	359.9	492.4	2.71	9.6	1.4	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0	I-T			28.19	60.3	384.9	578.8	2.91	9.6	1.5	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 19.0	I-T			14.20	28.9	333.7	350.2	2.34	12.1	1.0	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0	I-T			16.33	34.1	363.9	414.9	2.53	12.2	1.1	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0	I-T			17.64	41.8	402.8	500.4	2.77	12.0	1.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0	I-T			20.27	48.3	428.7	579.4	2.96	12.0	1.4	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0	I-T			23.82	57.2	458.7	686.5	3.20	12.0	1.5	7.01	12.50	6.56	.520	.300	3.75

(60T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 60.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 60.000 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 8 X 40.0 I-T	25.48	62.2	455.9	708.6	3.24	11.4	1.6	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	70.2	475.5	798.8	3.42	11.4	1.7	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	78.5	494.4	893.6	3.59	11.4	1.8	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	83.8	502.8	938.5	3.67	11.2	1.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	92.7	521.0	1038.2	3.84	11.2	2.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.7	423.0	510.5	2.81	13.5	1.2	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.5	461.6	617.4	3.07	13.6	1.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	52.9	488.6	708.4	3.27	13.4	1.5	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	60.7	517.6	813.6	3.49	13.4	1.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	68.2	540.5	914.8	3.68	13.4	1.7	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	75.2	545.3	969.3	3.77	12.9	1.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	84.8	566.8	1091.1	3.97	12.9	1.9	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.2	531.7	765.3	3.41	15.3	1.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	60.8	575.8	928.6	3.73	15.3	1.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	71.3	607.6	1076.3	3.99	15.1	1.8	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	80.8	637.3	1219.2	4.23	15.1	1.9	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	91.1	659.8	1371.4	4.45	15.1	2.1	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	101.7	681.3	1527.6	4.66	15.0	2.2	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	116.1	706.6	1738.4	4.92	15.0	2.5	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	140.3	745.5	2046.1	5.29	14.6	2.7	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	161.6	770.7	2341.0	5.59	14.5	3.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	187.6	798.1	2696.2	5.90	14.4	3.4	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	75.1	673.3	1263.4	4.32	16.8	1.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	87.6	713.5	1475.0	4.64	16.8	2.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	111.3	764.6	1844.3	5.12	16.6	2.4	10.46	17.99	7.50	.570	.355	6.39

(60T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 60.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 60.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 7 1/2 X 60.0 I-T	42.61	134.9	804.8	2223.0	5.54	16.5	2.8	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	159.7	836.4	2610.3	5.90	16.3	3.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	199.4	882.1	3153.4	6.40	15.8	3.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	225.7	906.6	3540.5	6.69	15.7	3.9	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	246.0	922.3	3832.0	6.88	15.6	4.2	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	278.1	948.2	4294.0	7.16	15.4	4.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	158.0	942.9	2975.4	6.37	18.8	3.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	174.4	965.6	3269.3	6.63	18.7	3.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	187.6	982.2	3503.2	6.81	18.7	3.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	212.9	1008.3	3943.0	7.13	18.5	3.9	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	239.1	1032.3	4391.9	7.42	18.4	4.3	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	268.9	1062.1	4797.5	7.74	17.8	4.5	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	295.3	1081.0	5221.1	7.97	17.7	4.8	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 67.500 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 75.938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 7 1/8 X 25.0 T	24.83	44.2	308.7	357.6	2.07	8.1	1.2	7.30	8.13	7.07	.630	.380	3.09
8 X 7 1/8 X 28.5 T	28.28	50.1	328.7	406.5	2.20	8.1	1.2	8.32	8.22	7.12	.715	.430	3.53
10 X 5 3/4 X 30.0 I-T	20.23	42.1	381.1	440.0	2.32	10.4	1.2	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	43.6	365.1	423.0	2.27	9.7	1.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	52.4	403.1	511.8	2.48	9.8	1.3	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	61.2	435.8	602.6	2.67	9.8	1.4	8.29	10.10	8.02	.620	.350	3.54
12 X 6 1/2 X 26.0 I-T	17.64	42.4	443.8	516.1	2.52	12.2	1.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	49.0	477.5	598.2	2.70	12.2	1.3	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	58.0	517.1	710.0	2.93	12.3	1.4	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	63.0	516.6	734.1	2.97	11.6	1.4	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	71.1	543.2	828.9	3.13	11.7	1.5	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	79.5	568.9	928.8	3.30	11.7	1.6	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	84.9	580.5	976.4	3.38	11.5	1.7	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	93.8	605.1	1081.8	3.54	11.5	1.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	38.3	463.4	525.3	2.55	13.7	1.1	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	46.1	512.6	635.8	2.79	13.8	1.2	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	53.6	547.9	730.3	2.98	13.6	1.3	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	61.4	585.8	839.7	3.18	13.7	1.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	69.0	616.5	945.3	3.36	13.7	1.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	76.1	625.4	1003.2	3.45	13.2	1.6	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	85.8	654.9	1131.4	3.64	13.2	1.7	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.8	594.8	786.9	3.10	15.5	1.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	61.5	652.4	956.1	3.40	15.5	1.5	6.68	15.88	5.53	.440	.275	4.37

(60T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 67.500 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 75.938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 36.0 I-T	25.69	72.1	695.1	1189.9	3.65	15.4	1.6	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	81.6	734.4	1259.0	3.87	15.4	1.7	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	92.1	765.8	1418.9	4.08	15.4	1.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	102.9	795.6	1583.6	4.28	15.4	2.0	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	117.5	830.6	1806.8	4.54	15.4	2.2	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	141.8	883.4	2132.5	4.90	15.0	2.4	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	163.5	918.6	2448.5	5.19	15.0	2.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	189.9	956.2	2831.4	5.51	14.9	3.0	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	75.9	773.5	1301.4	3.94	17.1	1.7	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	88.6	826.8	1521.9	4.24	17.2	1.8	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	112.4	896.8	1909.7	4.70	17.0	2.1	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	136.4	952.4	2310.6	5.11	16.9	2.4	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	161.6	996.4	2724.2	5.48	16.9	2.7	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	201.6	1058.9	3305.5	5.96	16.4	3.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	228.4	1092.2	3724.3	6.26	16.3	3.4	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	249.1	1113.5	4042.0	6.46	16.2	3.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	281.8	1147.7	4546.4	6.76	16.1	4.0	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	159.7	1123.1	3091.5	5.89	19.4	2.8	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	176.3	1154.2	3404.2	6.14	19.3	2.9	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	189.7	1176.8	3654.1	6.32	19.3	3.1	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	215.5	1212.5	4126.7	6.64	19.2	3.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	242.2	1244.8	4611.9	6.94	19.0	3.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	271.9	1285.5	5046.9	7.25	18.6	3.9	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	298.9	1310.7	5508.9	7.49	18.4	4.2	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 75.000 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 93.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 45.0 I-T	28.19	62.2	477.9	624.6	2.47	10.0	1.3	8.29	10.10	8.02	.620	.350	3.54
12 X 6 1/2 X 35.0 I-T	23.82	58.8	564.3	731.7	2.69	12.5	1.3	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	63.9	566.5	757.4	2.74	11.9	1.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	72.0	600.5	856.1	2.89	11.9	1.4	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	80.5	633.4	960.3	3.05	11.9	1.5	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	85.9	648.4	1009.9	3.12	11.8	1.6	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	95.0	679.9	1120.1	3.28	11.8	1.6	10.56	12.19	10.01	.640	.360	4.39
14 X 6 3/4 X 30.0 I-T	21.16	54.3	594.8	750.7	2.74	13.8	1.3	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	62.2	641.4	863.4	2.93	13.9	1.3	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	69.9	680.2	972.8	3.10	13.9	1.4	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	77.0	693.7	1033.5	3.18	13.4	1.5	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	86.8	731.9	1166.9	3.37	13.4	1.6	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	51.4	644.0	806.9	2.85	15.7	1.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	62.3	715.0	980.9	3.13	15.8	1.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	72.9	769.1	1139.8	3.35	15.6	1.5	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	82.5	818.5	1293.8	3.56	15.7	1.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	93.1	859.9	1460.1	3.76	15.7	1.7	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	104.0	899.1	1631.8	3.96	15.7	1.8	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	118.8	945.5	1865.2	4.20	15.7	2.0	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	143.2	1014.0	2205.7	4.55	15.4	2.2	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	165.1	1061.5	2539.3	4.83	15.4	2.4	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	191.9	1111.9	2945.3	5.15	15.4	2.6	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	76.7	859.5	1334.9	3.63	17.4	1.6	7.73	17.70	6.00	.425	.300	5.31

(60T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 75.000 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 93.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 6 X 40.0 I-T	29.35	89.5	926.7	1562.5	3.91	17.5	1.7	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	113.5	1018.1	1965.4	4.34	17.3	1.9	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	137.8	1092.0	2384.5	4.74	17.3	2.2	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	163.3	1151.4	2819.9	5.09	17.3	2.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	203.6	1233.9	3432.8	5.57	16.9	2.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	230.8	1278.4	3878.3	5.86	16.8	3.0	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	251.8	1306.8	4218.2	6.06	16.8	3.2	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	285.0	1351.7	4759.1	6.37	16.7	3.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	161.2	1297.0	3188.4	5.46	19.8	2.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	178.0	1338.5	3516.6	5.70	19.8	2.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	191.6	1368.5	3779.7	5.88	19.7	2.8	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	217.7	1416.5	4279.7	6.20	19.7	3.0	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	244.9	1459.6	4795.5	6.50	19.6	3.3	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	274.6	1512.7	5254.4	6.79	19.1	3.5	20.11	21.36	12.29	.800	.500	10.66
21 X 12 3/8 X 111.0 I-T	75.30	301.9	1546.3	5749.1	7.04	19.0	3.7	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 82.500 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 113.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 8 1/8 X 50.0 I-T	32.11	81.6	687.0	989.8	2.84	12.1	1.4	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	87.1	705.4	1041.1	2.91	12.0	1.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	96.2	743.9	1155.3	3.05	12.0	1.6	10.56	12.19	10.01	.640	.360	4.39
14 X 8 X 43.0 I-T	28.02	78.0	749.5	1061.8	2.95	13.6	1.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	87.8	796.6	1199.7	3.13	13.7	1.5	9.26	13.79	8.03	.595	.340	4.69
16 X 7 X 36.0 I-T	25.69	73.8	828.8	1167.8	3.11	15.8	1.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	83.4	888.2	1325.9	3.30	15.9	1.5	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	94.1	940.0	1497.6	3.49	15.9	1.6	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	105.1	989.1	1675.2	3.68	15.9	1.7	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	120.0	1048.0	1917.2	3.91	16.0	1.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	144.6	1133.4	2269.9	4.24	15.7	2.0	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	166.7	1195.0	2618.2	4.52	15.7	2.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	193.8	1260.3	3043.9	4.82	15.7	2.4	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	77.6	929.9	1365.8	3.36	17.6	1.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	90.4	1011.1	1599.5	3.62	17.7	1.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	114.7	1125.1	2014.9	4.03	17.6	1.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	139.1	1219.3	2449.2	4.41	17.6	2.0	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	164.9	1296.5	2903.0	4.76	17.6	2.2	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	205.5	1401.7	3542.3	5.21	17.2	2.5	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	232.9	1459.5	4010.4	5.50	17.2	2.7	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	254.2	1496.8	4369.3	5.70	17.2	2.9	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	287.8	1554.7	4941.4	6.00	17.2	3.2	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	162.6	1459.2	3271.9	5.08	20.1	2.2	13.22	20.99	8.24	.615	.400	8.40

(60T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 82.500 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 113.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 1/4 X 68.0 I-T	49.15	179.6	1512.6	3613.2	5.32	20.1	2.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	193.3	1551.5	3887.2	5.49	20.1	2.5	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	219.7	1614.2	4410.4	5.80	20.1	2.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	247.2	1670.6	4952.2	6.10	20.0	3.0	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	276.9	1737.6	5430.7	6.38	19.6	3.1	20.11	21.36	12.29	.800	.500	10.66
21 X 12 3/8 X 111.0 I-T	75.30	304.6	1781.6	5953.3	6.63	19.5	3.3	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 90.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 135.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 45.0 I-T	31.77	95.2	1005.7	1533.2	3.26	16.1	1.5	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	106.2	1064.8	1715.7	3.44	16.1	1.6	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	121.3	1136.4	1965.1	3.66	16.2	1.7	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	146.0	1239.0	2328.2	3.97	16.0	1.9	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	168.3	1315.9	2689.2	4.23	16.0	2.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	195.6	1397.8	3131.7	4.53	16.0	2.2	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 40.0 I-T	29.35	91.4	1079.6	1634.4	3.37	17.9	1.5	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	115.8	1216.2	2060.6	3.76	17.8	1.7	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	140.4	1331.5	2507.9	4.12	17.9	1.9	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	166.5	1428.2	2977.3	4.46	17.9	2.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	207.3	1558.1	3639.1	4.89	17.6	2.3	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	235.0	1631.1	4126.5	5.17	17.6	2.5	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	256.5	1678.8	4501.8	5.37	17.5	2.7	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	290.5	1751.8	5100.8	5.67	17.6	2.9	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	164.0	1605.6	3346.4	4.75	20.4	2.1	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	181.1	1672.2	3698.6	4.97	20.4	2.2	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	194.9	1720.9	3982.1	5.14	20.4	2.3	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	221.6	1800.6	4524.9	5.45	20.4	2.5	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	249.4	1872.5	5089.0	5.73	20.4	2.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	279.2	1954.4	5584.0	6.00	20.0	2.9	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	307.1	2010.8	6130.5	6.25	20.0	3.0	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 105.000 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 183.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 1/4 X 77.0 I-T	50.98	171.7	1514.1	2817.4	3.77	16.4	1.9	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	199.4	1631.5	3287.5	4.04	16.5	2.0	17.40	16.75	10.37	.875	.525	8.79
18 X 7 5/8 X 71.0 I-T	50.75	169.7	1644.6	3110.9	3.96	18.3	1.9	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	211.0	1825.8	3808.8	4.36	18.1	2.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	239.1	1932.8	4327.8	4.62	18.1	2.2	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	261.0	2004.6	4729.9	4.81	18.1	2.4	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	295.6	2112.9	5372.9	5.09	18.2	2.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	166.9	1843.6	3479.6	4.20	20.9	1.9	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	184.2	1938.0	3849.6	4.41	20.9	2.0	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	198.2	2007.9	4148.2	4.56	20.9	2.1	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	225.4	2125.5	4723.0	4.84	21.0	2.2	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	253.7	2233.1	5323.2	5.11	21.0	2.4	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	283.4	2346.5	5844.2	5.35	20.6	2.5	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	311.9	2432.8	6429.6	5.59	20.6	2.6	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 120.000 IN.) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.) EFFECTIVE PLATE AREA = 240.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/8 X 97.0 I-T	65.20	243.5	2169.8	4507.2	4.17	18.5	2.1	19.18	19.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	265.7	2267.2	4930.4	4.35	18.6	2.2	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	300.8	2413.9	5608.2	4.61	18.6	2.3	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 3/8 X 83.0 I-T	59.78	229.3	2376.7	4899.3	4.36	21.4	2.1	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	258.0	2522.2	5527.8	4.61	21.4	2.2	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	287.8	2666.5	6068.6	4.83	21.1	2.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	316.6	2786.2	6684.4	5.05	21.1	2.4	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.)

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TABLE VIII. Properties of combined beam and plate, I-T and T (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(60T = 135.000 IN.) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.) EFFECTIVE PLATE AREA = 303.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 12 3/8 X 111.0 I-T	75.30	321.8	3062.8	6918.7	4.61	21.5	2.3	22.15	21.51	12.34	.875	.550	11.83

(60T) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t).

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 6.250 IN.) PLATE WEIGHT = 5.100 LBS. (.1250 IN.) EFFECTIVE PLATE AREA = .781 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.3	3.4	6.8	1.75	2.0	2.0	1.44	3.95	3.94	.205	.170	.67
5	X	4	X	6.0	T	5.88	4.5	4.4	11.3	2.13	2.5	2.6	1.73	4.94	3.96	.210	.190	.94
6	X	4	X	7.0	T	6.94	6.0	5.7	17.7	2.51	3.0	3.1	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	9.0	I-T	6.17	5.6	5.4	16.6	2.53	3.0	3.1	1.81	5.90	3.94	.215	.170	1.00
8	X	4	X	10.0	I-T	7.19	7.7	7.7	30.9	3.27	4.0	4.0	2.11	7.89	3.94	.205	.170	1.34
10	X	4	X	12.0	I-T	9.07	10.7	10.6	53.2	3.93	5.0	5.0	2.67	9.87	3.96	.210	.190	1.88
12	X	4	X	14.0	I-T	10.98	14.5	13.9	85.4	4.61	5.9	6.1	3.23	11.91	3.97	.225	.200	2.38

(50T) PLATE WEIGHT = 5.100 LBS. (.1250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 7.813 IN.) PLATE WEIGHT = 6.375 LBS. (.1563 IN.) EFFECTIVE PLATE AREA = 1.221 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.5	4.8	8.3	1.77	2.4	1.7	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.8	6.2	13.7	2.16	2.9	2.2	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.3	7.8	21.4	2.56	3.4	2.7	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	5.9	7.6	20.0	2.57	3.4	2.6	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.2	10.6	37.0	3.33	4.5	3.5	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.4	14.0	63.0	4.03	5.5	4.5	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	15.4	18.0	100.2	4.75	6.5	5.6	3.23	11.91	3.97	.225	.200	2.38

(50T) PLATE WEIGHT = 6.375 LBS. (.1563 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 9.375 IN.) PLATE WEIGHT = 7.650 LBS. (.1875 IN.) EFFECTIVE PLATE AREA = 1.758 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.6	6.5	9.6	1.73	2.7	1.5	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.9	8.4	15.9	2.14	3.2	1.9	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.6	10.4	24.8	2.56	3.8	2.4	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	6.1	10.1	23.2	2.55	3.8	2.3	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.5	13.9	42.6	3.32	5.0	3.1	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.9	18.2	72.5	4.05	6.1	4.0	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	16.2	23.0	115.0	4.80	7.1	5.0	3.23	11.91	3.97	.225	.200	2.38

(50T) PLATE WEIGHT = 7.650 LBS. (.1875 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 10.938 IN.) PLATE WEIGHT = 8.925 LBS. (.2188 IN.) EFFECTIVE PLATE AREA = 2.393 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.7	8.4	10.7	1.67	2.9	1.3	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	5.1	10.8	17.8	2.08	3.5	1.6	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.8	13.4	27.8	2.50	4.1	2.1	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	6.2	13.1	25.9	2.48	4.1	2.0	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.8	17.6	47.7	3.25	5.4	2.7	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	12.4	23.0	81.2	4.00	6.6	3.5	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	16.8	28.8	128.9	4.79	7.7	4.5	3.23	11.91	3.97	.225	.200	2.38

(50T) PLATE WEIGHT = 8.925 LBS. (.2188 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 12.500 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 3.125 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT						WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
						LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.8	10.5	11.7	1.60	3.1	1.1	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	4.7	10.9	14.1	1.68	3.0	1.3	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	5.6	11.4	16.2	1.75	2.9	1.4	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.2	11.8	19.4	1.84	2.7	1.6	2.59	4.07	5.25	.330	.230	.94
5	X	4	X	6.0	T	5.88	5.2	13.5	19.4	2.00	3.8	1.4	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	6.5	14.1	23.3	2.10	3.6	1.7	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	7.6	14.5	26.5	2.18	3.5	1.8	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	8.8	14.9	29.6	2.24	3.4	2.0	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	9.5	14.8	30.5	2.25	3.2	2.1	2.92	5.01	5.00	.360	.240	1.20
6	X	4	X	7.0	T	6.94	6.9	16.7	30.4	2.43	4.4	1.8	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.0	17.1	34.0	2.50	4.3	2.0	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.4	16.4	28.2	2.39	4.4	1.7	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	9.9	17.9	40.2	2.62	4.1	2.2	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	12.0	I-T	8.30	8.4	17.3	35.5	2.53	4.2	2.0	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	10.7	17.9	41.9	2.64	3.9	2.3	2.88	5.99	5.99	.260	.230	1.38
7	X	5	X	11.0	T	10.81	13.1	20.9	57.4	3.02	4.4	2.7	3.18	6.87	5.00	.335	.230	1.58
8	X	4	X	10.0	I-T	7.19	9.0	22.3	52.0	3.15	5.8	2.3	2.11	7.89	3.94	.205	.170	1.34
8	X	4	X	13.0	I-T	9.52	11.4	23.4	63.1	3.26	5.5	2.7	2.80	7.99	4.00	.255	.230	1.84
8	X	5 1/2	X	13.0	T	12.83	17.2	24.6	81.9	3.45	4.8	3.3	3.77	7.85	5.50	.345	.250	1.96
8	X	4	X	15.0	I-T	10.79	13.4	24.4	72.3	3.39	5.4	3.0	3.17	8.11	4.02	.315	.245	1.99
8	X	5 1/4	X	18.0	I-T	12.00	16.6	25.3	84.0	3.55	5.1	3.3	3.53	8.14	5.25	.330	.230	1.87
8	X	5 1/4	X	21.0	I-T	13.87	19.6	26.3	95.8	3.65	4.9	3.6	4.08	8.28	5.27	.400	.250	2.07
8	X	6 1/2	X	24.0	I-T	15.11	21.9	25.6	96.4	3.57	4.4	3.8	4.44	7.93	6.50	.400	.245	1.94

(50T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 12.500 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 3.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 X 12.0 I-T	9.07	12.7	28.5	88.9	3.92	7.0	3.1	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	15.8	30.1	106.1	4.06	6.7	3.5	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.2	31.2	119.3	4.19	6.5	3.8	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	20.8	32.4	133.0	4.31	6.4	4.1	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	24.9	33.1	147.9	4.43	6.0	4.5	4.42	10.17	5.75	.360	.240	2.44
12 X 4 X 14.0 I-T	10.98	17.3	35.4	141.6	4.72	8.2	4.0	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	19.7	36.7	157.0	4.82	8.0	4.3	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	23.9	38.6	183.1	5.01	7.7	4.7	4.18	12.16	4.01	.350	.235	2.86
12 X 6 1/2 X 26.0 I-T	17.64	35.0	41.4	236.4	5.33	6.8	5.7	5.19	12.22	6.49	.380	.230	2.81
14 X 5 X 22.0 I-T	16.18	31.2	45.3	258.7	5.73	8.3	5.7	4.76	13.74	5.00	.335	.230	3.16
16 X 5 1/2 X 26.0 I-T	19.49	40.9	54.3	371.9	6.48	9.1	6.9	5.73	15.69	5.50	.345	.250	3.92

(50T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 14.063 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.955 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.8	12.8	12.4	1.52	3.3	1.0	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	4.8	13.3	15.1	1.61	3.1	1.1	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	5.7	13.8	17.6	1.69	3.1	1.3	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.4	14.4	21.2	1.80	2.9	1.5	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	6.5	14.3	19.8	1.76	3.1	1.4	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.2	16.4	20.8	1.91	4.0	1.3	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	6.6	17.1	25.2	2.03	3.8	1.5	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	7.7	17.7	28.7	2.12	3.7	1.6	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	8.9	18.2	32.4	2.19	3.6	1.8	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	9.7	18.1	33.4	2.20	3.4	1.8	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	11.5	18.9	39.0	2.30	3.4	2.1	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.1	20.3	32.7	2.33	4.6	1.6	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.1	20.9	36.7	2.42	4.5	1.8	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.5	20.0	30.2	2.29	4.7	1.5	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.0	21.8	43.7	2.55	4.4	2.0	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	11.9	22.5	50.0	2.64	4.2	2.2	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	8.5	21.1	38.4	2.45	4.5	1.8	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	10.9	21.9	45.7	2.59	4.2	2.1	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	11.8	22.9	51.0	2.68	4.3	2.2	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	14.8	23.4	58.8	2.77	4.0	2.5	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	13.4	25.4	62.7	2.97	4.7	2.5	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	16.2	26.4	72.7	3.07	4.5	2.8	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	19.3	26.9	80.8	3.12	4.2	3.0	4.36	6.92	6.73	.385	.270	1.87

(50T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 14.063 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.955 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 10.0 I-T	7.19	9.1	27.2	55.8	3.03	6.1	2.1	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.6	28.4	68.2	3.18	5.9	2.4	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.5	29.9	89.9	3.41	5.1	3.0	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.7	29.5	78.5	3.32	5.7	2.7	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.4	31.0	104.0	3.51	4.9	3.4	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	16.9	30.7	91.9	3.50	5.4	3.0	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T	13.87	20.0	32.0	105.4	3.62	5.3	3.3	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.4	31.1	106.8	3.57	4.8	3.4	4.44	7.93	6.50	.400	.245	1.94
10 X 4 X 12.0 I-T	9.07	13.0	34.7	95.7	3.80	7.4	2.8	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.2	36.4	115.0	3.98	7.1	3.2	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.7	37.7	129.8	4.13	7.0	3.4	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.4	39.1	145.3	4.26	6.8	3.7	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.4	40.0	162.5	4.40	6.4	4.1	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	30.1	41.7	185.6	4.53	6.2	4.5	5.11	10.33	5.77	.440	.260	2.69
12 X 4 X 14.0 I-T	10.98	17.8	42.9	153.0	4.62	8.6	3.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.2	44.2	170.2	4.74	8.4	3.9	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.28	24.5	46.5	199.6	4.95	8.1	4.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	28.7	48.5	226.8	5.09	7.9	4.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	35.8	49.8	260.5	5.34	7.3	5.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	41.0	51.6	288.4	5.39	7.0	5.6	5.96	12.34	6.52	.440	.260	3.21
14 X 5 X 22.0 I-T	16.18	32.0	54.4	282.7	5.70	8.8	5.2	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	38.3	57.0	325.2	5.85	8.5	5.7	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	44.5	58.5	357.1	5.92	8.0	6.1	6.22	13.84	6.73	.385	.270	3.74

(50T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 14.063 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.955 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	42.1	64.6	407.1	6.48	9.7	6.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	50.6	68.0	469.0	6.64	9.3	6.9	6.68	15.88	5.53	.440	.275	4.37

(50T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 15.625 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 4.883 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.9	15.1	13.1	1.44	3.4	.9	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.9	15.7	16.1	1.54	3.3	1.0	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.8	16.4	18.8	1.63	3.2	1.1	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.5	17.2	22.9	1.75	3.1	1.3	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.6	17.0	21.2	1.70	3.2	1.2	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.3	19.5	21.9	1.82	4.1	1.1	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.7	20.4	26.8	1.95	4.0	1.3	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.8	21.1	30.7	2.05	3.9	1.5	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.1	21.8	34.8	2.13	3.8	1.6	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.8	21.7	36.0	2.15	3.7	1.7	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	11.8	22.7	42.3	2.25	3.6	1.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.1	24.2	34.6	2.24	4.8	1.4	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.2	24.8	39.0	2.33	4.7	1.6	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.6	23.8	31.9	2.18	4.9	1.3	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.2	26.0	46.8	2.48	4.6	1.8	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.1	26.9	53.9	2.58	4.5	2.0	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.7	25.1	40.9	2.36	4.7	1.6	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.1	26.1	49.1	2.52	4.4	1.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.0	27.3	54.9	2.61	4.6	2.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.1	28.0	63.8	2.72	4.2	2.3	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.6	30.4	67.5	2.89	5.0	2.2	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	16.5	31.5	78.7	3.01	4.8	2.5	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	19.6	32.1	88.1	3.09	4.5	2.7	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	22.6	32.9	97.8	3.16	4.3	3.0	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	25.2	33.6	106.0	3.19	4.2	3.2	5.51	7.05	6.77	.515	.310	2.19

(50T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 15.625 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 4.883 SQ. IN.

NOMINAL SIZE							WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
								FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT			LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8	X	21.5	T		20.94	28.5	32.9	109.0	3.14	3.8	3.3	6.16	6.83	8.00	.530	.305	2.08
8	X	4	X	10.0	I-T		7.19	9.2	32.4	59.0	2.90	6.4	1.8	2.11	7.89	3.94	.205	.170	1.34
8	X	4	X	13.0	I-T		9.52	11.8	33.8	72.7	3.08	6.1	2.2	2.80	7.99	4.00	.255	.230	1.84
8	X	5 1/2	X	13.0	T		12.83	17.8	35.7	97.1	3.35	5.4	2.7	3.77	7.85	5.50	.345	.250	1.96
8	X	4	X	15.0	I-T		10.79	13.9	35.2	84.1	3.23	6.0	2.4	3.17	8.11	4.02	.315	.245	1.99
8	X	5 1/2	X	15.5	T		15.28	21.8	37.0	113.1	3.47	5.2	3.1	4.49	7.94	5.53	.440	.275	2.18
8	X	5 1/4	X	18.0	I-T		12.00	17.2	36.7	99.0	3.43	5.8	2.7	3.53	8.14	5.25	.330	.230	1.87
8	X	7	X	18.0	T		17.73	26.0	37.9	127.1	3.55	4.9	3.4	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T		19.79	29.6	38.8	139.8	3.61	4.7	3.6	5.82	8.01	7.00	.505	.305	2.44
8	X	5 1/4	X	21.0	I-T		13.87	20.4	38.1	114.1	3.57	5.6	3.0	4.08	8.28	5.27	.400	.250	2.07
8	X	6 1/2	X	24.0	I-T		15.11	22.7	37.2	116.3	3.53	5.1	3.1	4.44	7.93	6.50	.400	.245	1.94
8	X	6 1/2	X	28.0	I-T		17.69	26.4	38.5	131.2	3.61	5.0	3.4	5.20	8.06	6.54	.465	.285	2.30
8	X	8	X	31.0	I-T		19.16	29.4	38.8	139.1	3.64	4.7	3.6	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T		17.26	26.2	42.1	148.0	3.85	5.6	3.5	5.08	8.85	6.00	.425	.300	2.66
10	X	4	X	12.0	I-T		9.07	13.2	41.3	101.7	3.67	7.7	2.5	2.67	9.87	3.96	.210	.190	1.88
10	X	4	X	15.0	I-T		11.27	16.5	43.3	123.0	3.87	7.5	2.8	3.32	9.99	4.00	.270	.230	2.30
10	X	4	X	17.0	I-T		12.48	19.0	44.9	139.2	4.03	7.3	3.1	3.67	10.11	4.01	.330	.240	2.43
10	X	4	X	19.0	I-T		13.77	21.8	46.4	156.4	4.18	7.2	3.4	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T		15.04	25.9	47.6	175.9	4.35	6.8	3.7	4.42	10.17	5.75	.360	.240	2.44
10	X	5 3/4	X	26.0	I-T		17.37	30.8	49.5	201.9	4.50	6.6	4.1	5.11	10.33	5.77	.440	.260	2.69
10	X	5 3/4	X	30.0	I-T		20.23	35.5	51.3	226.4	4.57	6.4	4.4	5.95	10.47	5.81	.510	.300	3.14
10	X	8	X	33.0	I-T		20.94	37.0	48.2	210.2	4.36	5.7	4.4	6.16	9.73	7.96	.435	.290	2.82

(50T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 15.625 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 4.883 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 14.0 I-T	10.98	18.1	51.0	163.2	4.48	9.0	3.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.6	52.5	182.1	4.62	8.8	3.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.0	55.1	214.5	4.87	8.6	3.9	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.3	57.3	244.7	5.03	8.4	4.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	36.5	59.2	283.1	5.30	7.7	4.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	41.9	61.1	314.3	5.38	7.5	5.1	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	49.0	63.6	354.5	5.46	7.2	5.6	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	53.5	61.5	350.6	5.32	6.6	5.7	7.49	11.94	8.01	.515	.295	3.52
14 X 5 X 22.0 I-T	16.18	32.7	64.4	304.9	5.62	9.3	4.7	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	39.2	67.3	352.2	5.81	9.0	5.2	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	45.5	69.0	388.2	5.91	8.5	5.6	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	52.0	71.2	429.7	6.03	8.3	6.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	58.0	73.4	467.0	6.09	8.1	6.4	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	64.2	72.1	474.8	6.01	7.4	6.6	8.24	13.66	8.00	.530	.305	4.17
16 X 5 1/2 X 26.0 I-T	19.49	43.1	76.1	440.1	6.44	10.2	5.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	51.8	79.8	508.9	6.63	9.8	6.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	60.6	82.4	564.6	6.74	9.3	6.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	68.4	85.0	618.5	6.86	9.0	7.3	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	63.1	91.8	673.4	7.31	10.7	7.3	7.73	17.70	6.00	.425	.300	5.31

(50T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 17.188 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 5.908 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.9	17.3	13.7	1.37	3.5	.8	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	4.9	18.2	16.9	1.47	3.4	.9	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	5.9	19.1	19.9	1.57	3.4	1.0	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.6	20.1	24.3	1.69	3.2	1.2	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	6.7	19.9	22.5	1.64	3.4	1.1	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.4	22.6	22.9	1.73	4.3	1.0	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	6.8	23.7	28.2	1.87	4.2	1.2	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	7.9	24.7	32.5	1.97	4.1	1.3	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	9.2	25.5	36.9	2.06	4.0	1.4	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	10.0	25.4	38.3	2.08	3.8	1.5	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	11.9	26.6	45.2	2.20	3.8	1.7	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.2	28.2	36.3	2.14	5.0	1.3	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.3	29.0	41.0	2.23	4.9	1.4	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.6	27.7	33.4	2.08	5.0	1.2	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.3	30.4	49.6	2.39	4.8	1.6	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	12.2	31.5	57.3	2.51	4.7	1.8	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	8.8	29.4	43.1	2.27	4.9	1.5	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	11.2	30.7	52.1	2.43	4.6	1.7	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	12.1	32.0	58.3	2.54	4.8	1.8	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	15.3	33.0	68.3	2.66	4.5	2.1	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	13.8	35.7	71.7	2.81	5.2	2.0	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	16.7	37.0	84.2	2.95	5.0	2.3	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	19.9	37.8	94.7	3.04	4.8	2.5	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	22.9	38.7	105.6	3.12	4.6	2.7	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	25.6	39.5	114.9	3.17	4.5	2.9	5.51	7.05	6.77	.515	.310	2.19

(50T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 17.188 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 5.908 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	29.0	38.8	118.9	3.14	4.1	3.1	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	32.2	39.6	128.7	3.17	4.0	3.3	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.3	37.9	61.7	2.77	6.6	1.6	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.0	39.5	76.6	2.97	6.4	1.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.1	41.9	103.5	3.27	5.7	2.5	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.1	41.2	89.0	3.13	6.3	2.2	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	22.1	43.4	121.4	3.42	5.5	2.8	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.4	43.1	105.3	3.34	6.0	2.4	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	26.4	44.5	137.2	3.51	5.2	3.1	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	30.1	45.6	151.6	3.59	5.0	3.3	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.7	44.8	122.0	3.50	5.9	2.7	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	23.1	43.8	125.0	3.48	5.4	2.9	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	26.9	45.2	141.7	3.57	5.3	3.1	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	29.9	45.6	150.7	3.61	5.0	3.3	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	26.7	49.3	159.1	3.81	6.0	3.2	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	31.5	50.9	180.8	3.93	5.7	3.6	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.3	48.4	106.9	3.53	8.0	2.2	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.7	50.7	130.0	3.75	7.8	2.6	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.3	52.6	147.7	3.93	7.6	2.8	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	22.1	54.4	166.4	4.09	7.5	3.1	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.3	55.9	188.1	4.27	7.1	3.4	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	31.3	58.1	216.9	4.44	6.9	3.7	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	36.2	60.0	244.1	4.54	6.7	4.1	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	37.6	56.5	227.5	4.34	6.0	4.0	6.16	9.73	7.96	.435	.290	2.82

(50T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 17.188 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 5.908 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 39.0 I-T	24.45	44.9	58.8	261.3	4.47	5.8	4.4	7.19	9.92	7.99	.530	.315	3.12
12 X 4 X 14.0 I-T	10.98	18.4	59.8	172.1	4.34	9.4	2.9	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.0	61.5	192.7	4.49	9.2	3.1	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.4	64.5	228.0	4.75	9.0	3.5	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.8	67.0	261.1	4.94	8.8	3.9	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	37.1	69.4	303.8	5.23	8.2	4.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	42.6	71.4	338.5	5.34	7.9	4.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	49.9	74.2	383.1	5.45	7.7	5.2	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	54.5	71.8	380.6	5.33	7.0	5.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	60.8	74.0	413.9	5.36	6.8	5.6	8.47	12.06	8.05	.575	.335	4.04
14 X 5 X 22.0 I-T	16.18	33.3	75.3	325.1	5.52	9.8	4.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	39.9	78.5	377.2	5.74	9.5	4.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	46.4	80.4	417.2	5.86	9.0	5.2	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	53.0	82.9	463.2	6.01	8.7	5.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	59.2	85.3	504.6	6.09	8.5	5.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	65.5	83.9	515.2	6.03	7.9	6.1	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	73.1	86.5	559.9	6.07	7.7	6.5	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	43.9	88.6	470.7	6.36	10.7	5.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	52.9	92.7	546.4	6.59	10.3	5.9	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	61.8	95.5	608.3	6.72	9.8	6.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	69.8	98.4	668.0	6.87	9.6	6.8	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	64.5	106.1	723.5	7.28	11.2	6.8	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	75.0	110.5	814.6	7.49	10.9	7.4	8.63	17.90	6.02	.525	.315	5.64

(50T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASW
IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T				4.91	4.0	19.6	14.2	1.30	3.6	.7	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T				6.40	5.0	20.7	17.6	1.41	3.5	.9	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T				7.42	6.0	21.8	20.8	1.50	3.5	1.0	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T				8.82	7.7	23.1	25.6	1.63	3.3	1.1	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T				8.39	6.8	22.7	23.6	1.58	3.5	1.0	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T				5.88	5.4	25.7	23.8	1.65	4.4	.9	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T				7.37	6.9	27.2	29.4	1.79	4.3	1.1	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T				8.36	8.0	28.3	34.0	1.89	4.2	1.2	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T				9.42	9.3	29.3	38.8	1.99	4.2	1.3	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T				9.91	10.1	29.3	40.4	2.02	4.0	1.4	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T				11.69	12.1	30.8	47.9	2.14	4.0	1.6	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T				6.94	7.3	32.3	37.7	2.04	5.2	1.2	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T				7.88	8.4	33.4	42.8	2.14	5.1	1.3	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T				6.17	6.7	31.7	34.6	1.98	5.2	1.1	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T				9.34	10.5	35.1	52.0	2.31	5.0	1.5	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T				10.89	12.4	36.4	60.4	2.43	4.9	1.7	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T				8.30	8.9	33.8	45.1	2.18	5.1	1.3	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T				9.78	11.3	35.4	54.7	2.35	4.8	1.5	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T				10.74	12.3	37.0	61.4	2.45	5.0	1.7	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T				12.63	15.5	38.2	72.3	2.59	4.7	1.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T				10.81	13.9	41.3	75.5	2.72	5.4	1.8	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T				12.85	16.9	42.9	89.0	2.87	5.3	2.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T				14.81	20.1	43.9	100.7	2.97	5.0	2.3	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T				16.77	23.2	45.0	112.8	3.07	4.9	2.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T				18.74	26.0	45.9	123.1	3.13	4.7	2.7	5.51	7.05	6.77	.515	.310	2.19

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	29.4	45.1	128.1	3.12	4.4	2.8	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	32.7	46.0	139.1	3.16	4.2	3.0	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.4	43.7	64.1	2.65	6.8	1.5	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.1	45.6	80.1	2.85	6.6	1.8	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	16.3	48.5	109.3	3.18	6.0	2.3	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.3	47.6	93.3	3.02	6.5	2.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	22.4	50.3	128.9	3.34	5.8	2.6	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.6	49.9	111.0	3.24	6.3	2.2	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	26.8	51.6	146.4	3.46	5.5	2.8	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	30.6	52.9	162.4	3.55	5.3	3.1	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.9	51.9	129.1	3.41	6.2	2.5	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	34.0	53.9	176.2	3.60	5.2	3.3	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	23.4	50.8	132.9	3.40	5.7	2.6	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	27.2	52.4	151.2	3.52	5.6	2.9	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	30.3	52.9	161.4	3.57	5.3	3.1	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	27.0	57.1	169.2	3.74	6.3	3.0	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	32.0	58.9	193.2	3.88	6.0	3.3	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.5	56.0	111.4	3.39	8.3	2.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.9	58.6	136.2	3.63	8.0	2.3	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.6	60.8	155.2	3.81	7.9	2.6	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	22.4	62.9	175.5	3.98	7.8	2.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.6	64.8	199.1	4.17	7.5	3.1	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	31.7	67.3	230.6	4.36	7.3	3.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	36.7	69.4	260.4	4.48	7.1	3.8	5.95	10.47	5.81	.510	.300	3.14

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X TN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 33.0 I-T	20.94	38.2	65.4	243.6	4.30	6.4	3.7	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	45.6	68.0	281.1	4.45	6.2	4.1	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	52.9	70.5	316.7	4.55	6.0	4.5	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	18.6	69.1	180.0	4.19	9.7	2.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.2	71.1	202.2	4.35	9.5	2.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.8	74.6	240.1	4.63	9.3	3.2	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	30.3	77.4	276.0	4.83	9.1	3.6	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	37.6	80.4	322.7	5.14	8.6	4.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	43.2	82.6	360.9	5.27	8.3	4.4	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	50.7	85.6	409.9	5.40	8.1	4.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	55.3	83.0	408.9	5.31	7.4	4.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	61.8	85.3	445.7	5.36	7.2	5.2	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	68.6	87.6	483.3	5.42	7.0	5.5	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	73.8	87.2	497.1	5.45	6.7	5.7	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	81.2	89.2	534.0	5.51	6.6	6.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	33.8	87.1	343.4	5.40	10.2	3.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	40.5	90.7	400.1	5.64	9.9	4.4	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	47.1	92.8	444.1	5.79	9.4	4.8	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	53.9	95.6	494.6	5.95	9.2	5.2	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	60.2	98.1	540.0	6.06	9.0	5.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	66.6	96.7	553.5	6.02	8.3	5.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	74.4	99.4	602.8	6.08	8.1	6.1	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	44.6	102.1	498.8	6.25	11.2	4.9	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	53.8	106.7	581.3	6.51	10.8	5.4	6.68	15.88	5.53	.440	.275	4.37

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 36.0 I-T	25.69	62.9	109.8	649.3	6.67	10.3	5.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	71.1	113.0	714.9	6.84	10.1	6.3	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	79.4	116.3	778.6	6.90	9.8	6.7	9.34	16.13	7.04	.565	.345	5.56
18 X 6 X 35.0 I-T	26.29	65.7	121.6	770.7	7.23	11.7	6.3	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	76.4	126.4	870.2	7.45	11.4	6.9	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	96.2	133.1	1025.6	7.66	10.7	7.7	10.46	17.99	7.50	.570	.355	6.39

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	4.0	23.7	15.1	1.17	3.7	.6	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.1	25.5	18.9	1.28	3.7	.7	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.1	27.1	22.4	1.38	3.7	.8	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.8	29.2	27.8	1.51	3.6	1.0	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.9	28.5	25.5	1.46	3.7	.9	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.5	31.8	25.3	1.50	4.6	.8	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	7.0	34.0	31.5	1.64	4.5	.9	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.2	35.7	36.6	1.75	4.5	1.0	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.5	37.3	42.0	1.85	4.4	1.1	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.3	37.4	43.9	1.87	4.3	1.2	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.3	39.5	52.5	2.01	4.3	1.3	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.4	40.6	40.2	1.86	5.4	1.0	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.6	42.1	45.8	1.96	5.3	1.1	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.8	39.7	36.7	1.80	5.4	.9	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.6	44.7	56.1	2.13	5.3	1.3	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.6	46.7	65.6	2.27	5.2	1.4	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.0	42.8	48.3	2.01	5.3	1.1	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.5	45.3	59.1	2.18	5.1	1.3	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.5	47.4	66.6	2.29	5.3	1.4	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.7	49.2	79.2	2.44	5.0	1.6	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.2	53.1	81.8	2.53	5.8	1.5	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.3	55.4	97.4	2.70	5.6	1.8	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.5	56.9	111.0	2.82	5.4	2.0	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	23.7	58.4	125.3	2.94	5.3	2.1	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	26.6	59.6	137.6	3.02	5.2	2.3	5.51	7.05	6.77	.515	.310	2.19

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT							WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
							LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8	X	21.5	T		20.94	30.1	58.8	144.6	3.03	4.8	2.5	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T		23.53	33.6	60.8	157.9	3.09	4.7	2.6	6.92	6.90	8.03	.595	.340	2.35
8	X	4	X	10.0	I-T		7.19	9.6	55.5	68.0	2.41	7.1	1.2	2.11	7.89	3.94	.205	.170	1.34
8	X	4	X	13.0	I-T		9.52	12.4	58.3	85.9	2.64	7.0	1.5	2.80	7.99	4.00	.255	.230	1.84
8	X	5 1/2	X	13.0	T		12.83	18.7	62.7	119.2	2.99	6.4	1.9	3.77	7.85	5.50	.345	.250	1.96
8	X	4	X	15.0	I-T		10.79	14.6	61.1	100.7	2.81	6.9	1.6	3.17	8.11	4.02	.315	.245	1.99
8	X	5 1/2	X	15.5	T		15.28	22.9	65.3	141.8	3.18	6.2	2.2	4.49	7.94	5.53	.440	.275	2.18
8	X	5 1/4	X	18.0	I-T		12.00	18.0	64.5	120.5	3.03	6.7	1.9	3.53	8.14	5.25	.330	.230	1.87
8	X	7	X	18.0	T		17.73	27.4	67.1	162.6	3.32	5.9	2.4	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T		19.79	31.3	68.8	181.6	3.43	5.8	2.6	5.82	8.01	7.00	.505	.305	2.44
8	X	5 1/4	X	21.0	I-T		13.87	21.4	67.3	141.3	3.22	6.6	2.1	4.08	8.28	5.27	.400	.250	2.07
8	X	7	X	22.5	T		22.32	34.9	70.1	198.1	3.50	5.7	2.8	6.56	8.07	7.04	.565	.345	2.78
8	X	6 1/2	X	24.0	I-T		15.11	23.8	66.0	146.4	3.23	6.2	2.2	4.44	7.93	6.50	.400	.245	1.94
8	X	7 1/8	X	25.0	T		24.83	38.6	71.3	214.6	3.57	5.6	3.0	7.30	8.13	7.07	.630	.380	3.09
8	X	6 1/2	X	28.0	I-T		17.69	27.8	68.2	167.9	3.37	6.0	2.5	5.20	8.06	6.54	.465	.285	2.30
8	X	7 1/8	X	28.5	T		28.28	43.5	73.0	236.1	3.63	5.4	3.2	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T		19.16	31.0	68.9	180.3	3.44	5.8	2.6	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T		17.26	27.6	74.0	186.9	3.57	6.8	2.5	5.08	8.85	6.00	.425	.300	2.66
9	X	6	X	20.0	T		19.76	32.7	76.5	215.1	3.74	6.6	2.8	5.81	8.95	6.02	.525	.315	2.82
10	X	4	X	12.0	I-T		9.07	13.7	71.9	118.9	3.12	8.7	1.7	2.67	9.87	3.96	.210	.190	1.88
10	X	4	X	15.0	I-T		11.27	17.3	75.4	146.7	3.37	8.5	1.9	3.32	9.99	4.00	.270	.230	2.30
10	X	4	X	17.0	I-T		12.48	20.0	78.5	168.0	3.56	8.4	2.1	3.67	10.11	4.01	.330	.240	2.43
10	X	4	X	19.0	I-T		13.77	22.9	81.4	190.8	3.74	8.3	2.3	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T		15.04	27.2	84.1	217.9	3.95	8.0	2.6	4.42	10.17	5.75	.360	.240	2.44

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	32.4	87.4	254.3	4.16	7.9	2.9	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	37.6	90.1	289.2	4.32	7.7	3.2	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.0	85.1	272.0	4.16	7.0	3.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	46.7	88.5	316.7	4.35	6.8	3.6	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	54.4	91.5	359.3	4.49	6.6	3.9	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.0	89.3	193.2	3.88	10.2	2.2	3.23	11.31	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.7	91.9	218.0	4.06	10.1	2.4	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.3	96.6	260.7	4.35	9.9	2.7	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.0	100.2	301.6	4.58	9.7	3.0	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	38.4	104.6	355.4	4.91	9.3	3.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	44.2	107.4	400.1	5.08	9.1	3.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	52.0	110.9	457.8	5.26	8.8	4.1	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	56.7	107.8	459.9	5.19	8.1	4.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	63.5	110.4	503.8	5.28	7.9	4.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	70.6	113.0	548.6	5.37	7.8	4.9	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	75.8	113.0	566.9	5.42	7.5	5.0	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	83.5	115.4	611.7	5.51	7.3	5.3	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	34.5	112.9	374.8	5.11	10.9	3.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	41.5	117.5	439.9	5.39	10.6	3.7	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	48.3	120.2	491.6	5.58	10.2	4.1	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	55.2	123.7	550.6	5.78	10.0	4.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	61.8	126.7	604.0	5.91	9.8	4.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	68.4	125.1	623.2	5.91	9.1	5.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	76.5	126.1	681.8	6.02	8.9	5.3	9.26	13.79	8.03	.595	.340	4.69

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	45.7	132.1	547.9	5.98	12.0	4.1	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	55.2	137.8	643.3	6.29	11.6	4.7	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	64.6	141.5	723.2	6.50	11.2	5.1	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	73.1	145.5	800.2	6.70	10.9	5.5	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	81.8	149.2	875.1	6.80	10.7	5.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	90.7	152.9	950.4	6.90	10.5	6.2	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	102.5	157.9	1048.5	7.00	10.2	6.6	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	124.9	161.3	1160.6	7.23	9.4	7.3	12.99	16.33	10.24	.665	.395	6.45
18 X 6 X 35.0 I-T	26.29	67.6	156.2	855.9	7.03	12.7	5.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	78.7	162.1	971.7	7.31	12.3	6.0	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	99.3	169.8	1154.9	7.59	11.6	6.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	119.0	177.9	1331.7	7.76	11.2	7.5	12.53	18.24	7.56	.695	.415	7.57
21 X 8 1/4 X 62.0 I-T	44.94	138.8	207.5	1782.3	8.84	12.8	8.6	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	152.3	213.0	1915.4	8.93	12.6	9.0	14.46	21.13	8.27	.685	.430	9.09

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T					4.91	4.1	27.2	15.9	1.07	3.9	.6	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T					6.40	5.2	29.8	20.0	1.18	3.8	.7	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T					7.42	6.2	32.1	23.8	1.27	3.8	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T					8.82	8.0	35.0	29.7	1.40	3.7	.8	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T					8.39	7.0	33.9	27.2	1.35	3.9	.8	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T					5.88	5.6	37.3	26.5	1.37	4.7	.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T					7.37	7.1	40.5	33.2	1.50	4.7	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T					8.36	8.3	42.9	38.8	1.61	4.7	.9	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T					9.42	9.6	45.1	44.7	1.71	4.6	1.0	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T					9.91	10.4	45.4	46.8	1.74	4.5	1.0	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T					11.69	12.5	48.4	56.2	1.88	4.5	1.2	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T					6.94	7.5	48.4	42.2	1.70	5.6	.9	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T					7.88	8.7	50.7	48.3	1.80	5.5	1.0	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T					6.17	6.9	47.1	38.4	1.64	5.6	.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T					9.34	10.8	54.4	59.3	1.97	5.5	1.1	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T					10.89	12.8	57.1	69.8	2.11	5.4	1.2	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T					8.30	9.2	51.6	50.9	1.85	5.5	1.0	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T					9.78	11.7	55.2	62.7	2.02	5.4	1.1	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T					10.74	12.7	58.0	70.8	2.13	5.6	1.2	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T					12.63	16.0	60.8	84.8	2.29	5.3	1.4	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T					10.81	14.4	65.3	86.9	2.35	6.0	1.3	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T					12.85	17.5	68.6	104.1	2.53	5.9	1.5	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T					14.81	20.9	70.7	119.6	2.66	5.7	1.7	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T					16.77	24.1	72.9	135.7	2.79	5.6	1.9	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T					18.74	27.0	74.6	149.8	2.88	5.5	2.0	5.51	7.05	6.77	.515	.310	2.19

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 6 X 21.5 T	20.94	30.6	73.8	158.5	2.91	5.2	2.1	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	34.2	75.3	174.1	2.99	5.1	2.3	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.7	67.1	71.2	2.21	7.3	1.1	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.5	71.3	90.5	2.43	7.2	1.3	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.9	77.7	127.1	2.80	6.7	1.6	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.8	75.2	106.5	2.61	7.2	1.4	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.2	81.3	152.5	3.00	6.6	1.9	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.2	79.9	128.2	2.83	7.0	1.6	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	27.8	83.8	176.0	3.15	6.3	2.1	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	31.8	86.2	197.7	3.28	6.2	2.3	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.67	21.7	83.6	151.1	3.02	7.0	1.8	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	35.5	87.8	216.9	3.37	6.1	2.5	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.2	82.4	157.5	3.05	6.5	1.9	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	39.4	89.4	236.0	3.45	6.0	2.6	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.3	85.3	181.8	3.20	6.4	2.1	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	44.5	91.5	261.2	3.54	5.9	2.9	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	31.5	86.4	196.1	3.29	6.2	2.3	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.1	92.4	201.5	3.39	7.2	2.2	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.3	95.8	233.5	3.57	7.0	2.4	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.9	88.2	124.8	2.87	9.0	1.4	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.6	93.2	155.0	3.13	8.8	1.7	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.3	97.3	176.1	3.32	8.8	1.8	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.3	101.2	203.1	3.50	8.7	2.0	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	27.6	105.0	233.1	3.71	8.5	2.2	4.42	10.17	5.75	.360	.240	2.44

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	32.9	109.4	273.8	3.94	8.3	2.5	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.3	112.8	313.4	4.12	8.2	2.8	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.7	106.9	296.0	3.98	7.5	2.8	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	47.6	111.2	347.2	4.20	7.3	3.1	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	55.4	115.0	396.5	4.37	7.2	3.4	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.3	110.7	203.5	3.60	10.6	1.8	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.0	114.1	230.6	3.78	10.5	2.0	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.8	120.4	277.2	4.08	10.4	2.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.5	125.1	322.5	4.32	10.2	2.6	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.0	131.3	382.3	4.65	9.8	2.9	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.0	134.8	432.8	4.84	9.6	3.2	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	53.0	139.1	498.6	5.06	9.4	3.6	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	57.7	135.6	503.7	5.02	8.7	3.7	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	64.8	138.6	554.4	5.14	8.6	4.0	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	72.1	141.7	606.4	5.26	8.4	4.3	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	77.4	142.0	628.9	5.32	8.1	4.4	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	85.3	144.9	681.4	5.44	8.0	4.7	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.1	141.3	400.3	4.82	11.4	2.8	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.2	147.1	472.9	5.12	11.2	3.2	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.1	150.7	531.3	5.33	10.8	3.5	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	56.3	155.2	598.1	5.55	10.6	3.9	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	63.1	158.8	659.0	5.71	10.4	4.2	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	69.7	157.0	683.6	5.74	9.8	4.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	78.2	160.6	751.4	5.88	9.6	4.7	9.26	13.79	8.03	.595	.340	4.69

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	46.6	165.4	588.6	5.68	12.6	3.6	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	56.3	172.5	695.5	6.02	12.3	4.0	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	66.0	177.2	786.4	6.26	11.9	4.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	74.6	182.2	873.9	6.49	11.7	4.8	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	83.6	186.3	960.0	6.63	11.5	5.2	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	92.9	190.5	1046.5	6.76	11.3	5.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	105.2	196.1	1159.2	6.90	11.0	5.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	128.0	201.0	1316.3	7.19	10.3	6.5	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	146.1	207.5	1459.3	7.29	10.0	7.0	15.00	16.52	10.30	.760	.455	7.52
18 X 6 X 35.0 I-T	26.29	69.1	195.1	928.9	6.78	13.4	4.8	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	80.5	202.3	1059.9	7.08	13.2	5.2	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	101.7	211.5	1270.2	7.44	12.5	6.0	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	122.2	220.6	1473.3	7.67	12.1	6.7	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	142.8	229.6	1669.9	7.80	11.7	7.3	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	179.1	236.9	1926.6	8.08	10.8	8.1	17.00	18.39	11.09	.770	.480	8.83
21 X 8 1/4 X 62.0 I-T	44.94	142.7	256.4	1970.1	8.75	13.8	7.7	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	156.7	262.5	2122.6	8.87	13.5	8.1	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	167.9	267.2	2241.6	8.95	13.4	8.4	15.47	21.24	8.30	.740	.455	9.66
21 X 12 1/4 X 101.0 I-T	68.38	239.5	285.9	2848.7	9.35	11.9	10.0	20.11	21.36	12.29	.800	.500	10.68

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 6.5 T	6.40	5.3	33.5	20.9	1.09	3.9	.6	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.3	36.4	25.0	1.18	3.9	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	8.1	40.3	31.3	1.30	3.9	.8	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	7.2	38.8	28.6	1.25	4.0	.7	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.7	41.9	27.7	1.26	4.8	.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	7.2	46.3	34.7	1.39	4.8	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.5	49.5	40.6	1.49	4.8	.8	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.8	52.5	46.9	1.59	4.8	.9	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.6	53.1	49.2	1.62	4.6	.9	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.7	57.0	59.4	1.76	4.7	1.0	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.7	55.5	43.9	1.57	5.7	.8	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.8	58.6	50.3	1.67	5.7	.9	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	7.0	53.6	39.9	1.50	5.7	.7	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	11.0	63.6	62.1	1.83	5.7	1.0	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.0	67.4	73.4	1.96	5.6	1.1	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.3	59.9	53.2	1.71	5.7	.9	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.9	64.8	65.8	1.88	5.5	1.0	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.9	68.4	74.3	1.98	5.8	1.1	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.2	72.3	89.6	2.14	5.5	1.2	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.6	77.4	91.1	2.19	6.3	1.2	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.8	81.9	109.8	2.37	6.2	1.3	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.1	84.9	126.7	2.51	6.0	1.5	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	24.4	88.0	144.4	2.64	5.9	1.6	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	27.4	90.2	160.1	2.74	5.8	1.8	5.51	7.05	6.77	.515	.310	2.19

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	31.0	89.6	170.4	2.78	5.5	1.9	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	34.8	91.7	188.1	2.88	5.4	2.1	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.8	78.0	73.7	2.03	7.5	.9	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.7	84.0	94.3	2.25	7.4	1.1	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.2	93.0	133.7	2.61	7.0	1.4	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.0	89.1	111.4	2.42	7.4	1.2	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.5	97.9	161.3	2.82	6.9	1.6	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.4	95.5	134.5	2.64	7.3	1.4	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.2	101.4	187.3	2.98	6.6	1.8	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	32.3	104.6	211.3	3.12	6.6	2.0	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	21.9	100.5	159.3	2.83	7.3	1.6	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	36.1	106.8	232.8	3.23	6.5	2.2	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.5	99.4	166.7	2.87	6.8	1.7	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	40.0	108.8	254.4	3.32	6.4	2.3	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.7	103.2	193.4	3.03	6.7	1.9	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	45.3	111.5	283.0	3.42	6.2	2.5	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	31.9	104.8	209.4	3.12	6.6	2.0	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.5	111.7	213.6	3.20	7.5	1.9	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.8	116.2	248.9	3.39	7.4	2.1	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.1	104.4	129.6	2.65	9.2	1.2	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.8	111.3	161.8	2.91	9.1	1.5	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.6	116.7	186.5	3.09	9.1	1.6	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.6	121.8	213.3	3.28	9.1	1.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	27.9	127.0	245.7	3.48	8.8	1.9	4.42	10.17	5.75	.360	.240	2.44

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	33.3	132.8	290.0	3.72	8.7	2.2	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.8	137.2	333.6	3.91	8.6	2.4	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.2	130.2	316.1	3.79	7.9	2.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.3	135.8	373.3	4.03	7.7	2.7	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	56.3	140.6	428.7	4.22	7.6	3.0	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.5	132.5	211.9	3.34	10.9	1.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.3	137.2	240.9	3.52	10.8	1.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.1	145.4	290.8	3.81	10.7	2.0	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.0	151.5	339.7	4.06	10.6	2.2	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.5	159.9	404.5	4.39	10.3	2.5	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.5	164.4	460.2	4.60	10.1	2.8	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	53.7	169.8	533.1	4.83	9.9	3.1	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	58.6	165.8	541.0	4.82	9.2	3.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	65.8	169.4	598.1	4.96	9.1	3.5	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	73.3	173.2	656.9	5.10	9.0	3.8	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	78.6	173.8	683.1	5.17	8.7	3.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	86.7	177.5	743.0	5.31	8.6	4.2	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.6	171.6	421.2	4.52	11.8	2.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.8	179.1	500.2	4.84	11.7	2.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.8	183.7	564.5	5.06	11.3	3.1	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	57.1	189.4	638.1	5.30	11.2	3.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	64.1	193.8	705.9	5.48	11.0	3.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	70.8	192.1	735.4	5.53	10.4	3.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	79.5	196.3	811.8	5.69	10.2	4.1	9.26	13.79	8.03	.595	.340	4.69

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	47.3	201.5	622.3	5.37	13.2	3.1	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	57.2	210.4	739.3	5.73	12.9	3.5	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	67.0	216.3	840.1	6.00	12.5	3.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	75.8	222.5	937.2	6.24	12.4	4.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	85.1	227.2	1033.6	6.41	12.1	4.5	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	94.6	232.1	1130.9	6.57	11.9	4.9	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	107.4	238.4	1258.0	6.74	11.7	5.3	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	130.5	245.1	1438.6	7.07	11.0	5.9	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	149.2	252.3	1601.8	7.21	10.7	6.3	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	171.7	260.9	1792.8	7.35	10.4	6.9	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	70.3	237.8	991.0	6.49	14.1	4.2	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	81.9	246.7	1135.7	6.82	13.9	4.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	103.6	257.7	1371.4	7.22	13.2	5.3	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	124.7	268.0	1600.3	7.51	12.8	6.0	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	146.1	277.7	1822.4	7.70	12.5	6.6	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	183.2	287.1	2119.2	8.04	11.6	7.4	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	205.9	295.5	2323.9	8.15	11.3	7.9	19.18	18.59	11.15	.870	.535	9.95
21 X 8 1/4 X 62.0 I-T	44.94	145.8	311.0	2139.2	8.58	14.7	6.9	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	160.3	317.8	2311.2	8.74	14.4	7.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	171.8	323.0	2445.5	8.84	14.2	7.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	193.5	332.7	2690.4	8.97	13.9	8.1	17.58	21.43	8.36	.835	.515	11.04
21 X 12 1/4 X 101.0 I-T	68.38	245.3	344.7	3141.9	9.35	12.8	9.1	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	267.7	352.8	3359.9	9.41	12.6	9.5	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 7.5 T				7.42	6.5	40.1	26.1	1.10	4.0	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T				8.82	8.3	45.0	32.8	1.22	4.0	.7	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T				8.39	7.3	43.1	29.9	1.17	4.1	.7	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T				5.88	5.8	45.8	28.7	1.16	4.9	.6	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T				7.37	7.3	51.2	36.1	1.29	4.9	.7	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T				8.36	8.6	55.4	42.3	1.39	4.9	.8	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T				9.42	9.9	59.2	48.9	1.48	4.9	.8	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T				9.91	10.8	60.0	51.4	1.51	4.8	.9	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T				11.69	12.9	65.2	62.2	1.65	4.8	1.0	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T				6.94	7.8	61.6	45.4	1.45	5.8	.7	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T				7.88	8.9	65.6	52.2	1.54	5.8	.8	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T				6.17	7.1	59.2	41.3	1.39	5.8	.7	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T				9.34	11.1	72.1	64.5	1.70	5.8	.9	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T				10.89	13.2	77.0	76.4	1.83	5.8	1.0	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T				8.30	9.5	67.2	55.1	1.58	5.8	.8	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T				9.78	12.0	73.7	68.4	1.75	5.7	.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T				10.74	13.1	78.1	77.4	1.85	5.9	1.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T				12.63	16.4	83.5	93.6	2.01	5.7	1.1	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T				10.81	14.7	88.8	94.8	2.04	6.4	1.1	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T				12.85	18.0	94.8	114.6	2.22	6.4	1.2	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T				14.81	21.4	99.0	132.7	2.36	6.2	1.3	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T				16.77	24.7	103.1	151.9	2.49	6.1	1.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T				18.74	27.8	106.2	168.9	2.60	6.1	1.6	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T				20.94	31.4	105.9	180.7	2.65	5.7	1.7	6.16	6.83	8.00	.530	.305	2.08

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT							WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
							LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8		X	24.0	T	23.53	35.2	108.7	200.2	2.75	5.7	1.8	6.92	6.90	8.03	.595	.340	2.35
8	X	4		X	10.0	I-T	7.19	9.9	87.8	76.0	1.87	7.6	.9	2.11	7.89	3.94	.205	.170	1.34
8	X	4		X	13.0	I-T	9.52	12.9	95.8	97.6	2.09	7.6	1.0	2.80	7.99	4.00	.255	.230	1.84
8	X	5	1/2	X	13.0	T	12.83	19.4	108.0	139.3	2.45	7.2	1.3	3.77	7.85	5.50	.345	.250	1.96
8	X	4		X	15.0	I-T	10.79	15.2	102.5	115.5	2.26	7.6	1.1	3.17	8.11	4.02	.315	.245	1.99
8	X	5	1/2	X	15.5	T	15.28	23.8	114.5	168.8	2.65	7.1	1.5	4.49	7.94	5.53	.440	.275	2.18
8	X	5	1/4	X	18.0	I-T	12.00	18.6	110.7	139.9	2.46	7.5	1.3	3.53	8.14	5.25	.330	.230	1.87
8	X	7		X	18.0	T	17.73	28.5	119.3	196.8	2.82	6.9	1.6	5.22	7.93	6.99	.430	.295	2.34
8	X	7		X	20.0	T	19.79	32.6	123.6	222.9	2.97	6.8	1.8	5.82	8.01	7.00	.505	.305	2.44
8	X	5	1/4	X	21.0	I-T	13.87	22.2	117.3	166.2	2.65	7.5	1.4	4.08	8.28	5.27	.400	.250	2.07
8	X	7		X	22.5	T	22.32	36.5	126.4	246.5	3.07	6.7	1.9	6.56	8.07	7.04	.565	.345	2.78
8	X	6	1/2	X	24.0	I-T	15.11	24.7	116.5	174.4	2.70	7.1	1.5	4.44	7.93	6.50	.400	.245	1.94
8	X	7	1/8	X	25.8	T	24.83	40.6	129.2	270.3	3.17	6.7	2.1	7.30	8.13	7.07	.630	.380	3.09
8	X	6	1/2	X	28.0	I-T	17.69	29.0	121.6	203.3	2.87	7.0	1.7	5.20	8.06	6.54	.465	.285	2.30
8	X	7	1/8	X	28.5	T	28.28	46.0	132.6	302.0	3.29	6.6	2.3	8.32	8.22	7.12	.715	.430	3.53
8	X	8		X	31.0	I-T	19.16	32.2	123.8	220.7	2.96	6.8	1.8	5.63	8.00	8.00	.435	.285	2.28
9	X	6		X	17.5	T	17.26	28.8	131.4	223.9	3.02	7.8	1.7	5.08	8.85	6.00	.425	.300	2.66
9	X	6		X	20.0	T	19.76	34.2	137.4	262.0	3.22	7.7	1.9	5.81	8.95	6.02	.525	.315	2.82
10	X	4		X	12.0	I-T	9.07	14.2	119.8	133.6	2.45	9.4	1.1	2.67	9.87	3.96	.210	.190	1.88
10	X	4		X	15.0	I-T	11.27	18.0	128.9	167.5	2.71	9.3	1.3	3.32	9.99	4.00	.270	.230	2.30
10	X	4		X	17.0	I-T	12.48	20.8	135.9	193.5	2.89	9.3	1.4	3.67	10.11	4.01	.330	.240	2.43
10	X	4		X	19.0	I-T	13.77	23.8	142.5	221.8	3.07	9.3	1.6	4.05	10.24	4.02	.395	.250	2.56
10	X	5	3/4	X	22.0	I-T	15.04	28.2	149.3	256.2	3.27	9.1	1.7	4.42	10.17	5.75	.360	.240	2.44
10	X	5	3/4	X	26.0	I-T	17.37	33.7	156.9	303.6	3.51	9.0	1.9	5.11	10.33	5.77	.440	.260	2.69

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 30.0 I-T	20.23	39.2	162.6	350.8	3.71	8.9	2.2	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.6	154.5	333.2	3.60	8.2	2.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.8	161.7	395.5	3.85	8.1	2.4	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	57.0	167.8	456.5	4.05	8.0	2.7	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.7	154.2	218.9	3.10	11.1	1.4	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.5	160.3	249.4	3.28	11.1	1.6	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.4	170.9	302.0	3.57	11.0	1.8	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.3	178.8	354.0	3.81	11.0	2.0	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.9	189.7	423.0	4.14	10.6	2.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.0	195.5	483.1	4.35	10.5	2.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.4	202.3	562.4	4.60	10.3	2.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.2	198.0	572.8	4.60	9.7	2.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	66.6	202.4	635.7	4.76	9.5	3.1	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	74.3	207.0	700.7	4.92	9.4	3.4	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	79.6	208.2	730.3	5.00	9.2	3.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	87.9	212.7	797.0	5.15	9.1	3.7	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.9	203.0	438.5	4.25	12.2	2.2	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.3	212.6	523.0	4.57	12.1	2.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.4	218.6	592.4	4.80	11.8	2.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	57.8	225.9	672.0	5.04	11.6	3.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	64.9	231.4	745.9	5.23	11.5	3.2	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	71.6	229.7	779.9	5.30	10.9	3.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	80.5	234.8	864.2	5.48	10.7	3.7	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	47.8	239.6	650.3	5.07	13.6	2.7	5.73	15.69	5.50	.345	.250	3.92

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 31.0 I-T	22.70	57.9	250.8	776.2	5.44	13.4	3.1	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	67.9	258.2	885.7	5.72	13.1	3.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	76.8	265.9	991.2	5.97	12.9	3.7	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	86.3	271.5	1097.3	6.16	12.7	4.0	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	96.1	277.2	1204.6	6.34	12.5	4.3	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	109.2	284.4	1345.3	6.55	12.3	4.7	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	132.5	293.4	1547.5	6.90	11.7	5.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	151.8	301.4	1730.6	7.08	11.4	5.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	175.0	310.9	1945.3	7.26	11.1	6.3	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	71.3	283.7	1043.7	6.19	14.6	3.7	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	83.1	294.7	1200.7	6.53	14.5	4.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	105.2	307.9	1459.6	6.98	13.9	4.7	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	126.8	319.8	1712.9	7.31	13.5	5.4	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	148.9	330.5	1960.0	7.54	13.2	5.9	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	186.5	342.2	2295.4	7.93	12.3	6.7	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	209.9	351.5	2525.5	8.08	12.0	7.2	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	227.6	358.5	2694.7	8.15	11.8	7.5	21.02	18.73	11.20	.940	.590	11.05
21 X 8 1/4 X 62.0 I-T	44.94	148.3	370.8	2289.7	8.36	15.4	6.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	163.2	378.5	2480.7	8.54	15.2	6.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	175.1	384.3	2629.9	8.67	15.0	6.8	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	197.4	394.8	2902.7	8.84	14.7	7.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	220.3	405.4	3175.2	8.98	14.4	7.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	250.1	409.6	3414.4	9.28	13.6	8.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	273.3	418.3	3659.1	9.37	13.4	8.7	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT						WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
						LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	5 1/4	X	9.0	T	8.82	8.4	49.0	34.2	1.14	4.1	.7	2.59	4.07	5.25	.330	.230	.94
5	X	4	X	8.5	T	8.36	8.7	60.3	43.9	1.30	5.0	.7	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	10.1	65.0	50.8	1.39	5.0	.8	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	10.9	66.2	53.4	1.42	4.9	.8	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	13.1	72.5	64.8	1.55	4.9	.9	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.9	66.7	46.8	1.35	5.9	.7	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	9.1	71.5	53.9	1.44	5.9	.8	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.5	T	9.34	11.3	79.5	66.7	1.59	5.9	.8	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	13.4	85.7	79.2	1.72	5.9	.9	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	9.6	73.5	57.0	1.48	5.9	.8	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	12.2	81.6	70.8	1.63	5.8	.9	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	13.3	86.9	80.1	1.73	6.0	.9	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	16.6	93.9	97.2	1.89	5.9	1.0	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	14.9	99.3	98.0	1.91	6.6	1.0	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	18.2	107.0	118.8	2.08	6.5	1.1	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	21.6	112.6	138.0	2.22	6.4	1.2	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	25.0	118.0	158.4	2.35	6.3	1.3	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	28.1	121.9	176.6	2.46	6.3	1.4	5.51	7.05	6.77	.515	.310	2.19
7	X	8	X	21.5	T	20.94	31.8	122.2	189.6	2.52	6.0	1.6	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T	23.53	35.6	125.8	210.7	2.63	5.9	1.7	6.92	6.90	8.03	.595	.340	2.35
8	X	4	X	10.0	I-T	7.19	10.0	96.2	78.0	1.74	7.8	.8	2.11	7.89	3.94	.205	.170	1.34
8	X	4	X	13.0	I-T	9.52	13.0	106.5	100.5	1.95	7.7	.9	2.80	7.99	4.00	.255	.230	1.84
8	X	5 1/2	X	13.0	T	12.83	19.6	122.2	144.2	2.29	7.4	1.2	3.77	7.85	5.50	.345	.250	1.96

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 15.0 I-T	10.79	15.3	114.8	119.1	2.11	7.8	1.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.1	130.7	175.3	2.50	7.3	1.3	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 16.0 I-T	12.00	18.8	125.1	144.5	2.31	7.7	1.2	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.8	137.1	205.1	2.67	7.1	1.5	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.0	142.6	233.8	2.81	7.1	1.6	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.4	133.5	172.1	2.49	7.7	1.3	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.0	146.4	258.4	2.93	7.0	1.8	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.0	133.1	181.1	2.54	7.3	1.4	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.1	149.9	284.2	3.03	6.9	1.9	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.3	139.7	211.8	2.71	7.2	1.5	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	46.6	154.3	318.6	3.16	6.8	2.1	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.6	142.8	230.5	2.81	7.1	1.6	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.1	150.9	232.8	2.85	8.0	1.5	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.5	158.6	273.3	3.05	7.9	1.7	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.4	133.9	137.1	2.28	9.5	1.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.2	145.6	172.4	2.53	9.5	1.2	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.0	154.3	199.5	2.70	9.5	1.3	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.1	162.7	229.0	2.88	9.5	1.4	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.5	171.4	265.2	3.07	9.3	1.5	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.0	181.1	315.2	3.31	9.3	1.7	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	39.6	188.5	365.4	3.51	9.2	1.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.0	179.3	347.9	3.42	8.5	1.9	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	49.3	188.5	414.7	3.67	8.4	2.2	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	57.6	196.1	480.6	3.88	8.3	2.5	8.29	10.10	8.02	.620	.350	3.54

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 14.0 I-T	10.98	19.9	174.9	224.8	2.69	11.3	1.3	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.8	182.8	256.6	3.07	11.3	1.4	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.7	196.1	311.5	3.35	11.3	1.6	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.6	206.2	366.2	3.59	11.2	1.8	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.2	220.0	438.7	3.90	10.9	2.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.5	227.5	502.6	4.12	10.8	2.2	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.9	236.2	587.5	4.38	10.7	2.5	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.8	231.5	600.2	4.39	10.0	2.6	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	67.3	237.1	668.3	4.56	9.9	2.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	75.1	242.8	738.9	4.73	9.8	3.0	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	80.4	244.5	771.5	4.81	9.6	3.2	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	88.9	250.1	844.3	4.97	9.5	3.4	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.3	234.8	453.1	4.00	12.5	1.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.7	247.0	542.2	4.31	12.4	2.2	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.9	254.8	616.1	4.54	12.1	2.4	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	58.3	263.9	700.9	4.79	12.0	2.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	65.5	270.7	780.1	4.99	11.9	2.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	72.3	269.3	818.1	5.07	11.3	3.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	81.4	275.6	909.5	5.26	11.2	3.3	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.3	279.0	674.0	4.79	14.0	2.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	58.5	292.9	807.5	5.16	13.8	2.8	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	68.6	302.2	924.7	5.45	13.5	3.1	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	77.6	311.8	1037.6	5.70	13.4	3.3	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	87.3	318.5	1152.4	5.91	13.2	3.6	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	97.2	325.3	1268.8	6.11	13.0	3.9	10.39	16.26	7.07	.630	.380	6.18

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 1/8 X 57.0 I-T	40.28	110.6	333.8	1422.2	6.33	12.9	4.3	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	134.1	345.3	1644.0	6.70	12.3	4.8	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	153.9	354.4	1846.2	6.91	12.8	5.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	177.6	365.2	2083.9	7.13	11.7	5.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	72.1	332.2	1088.7	5.89	15.1	3.3	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	84.0	345.6	1256.5	6.24	15.0	3.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	106.5	361.7	1536.4	6.71	14.4	4.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	128.5	375.5	1812.4	7.08	14.1	4.8	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	151.2	387.5	2083.4	7.35	13.8	5.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	189.3	402.1	2454.9	7.77	13.0	6.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	213.3	412.4	2710.2	7.96	12.7	6.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	231.5	419.9	2898.0	8.06	12.5	6.9	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	260.4	432.1	3193.6	8.22	12.3	7.4	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	150.4	435.4	2423.0	8.11	16.1	5.6	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	165.6	444.2	2631.8	8.31	15.9	5.9	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	177.8	450.8	2795.5	8.46	15.7	6.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	200.7	462.2	3095.5	8.67	15.4	6.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	224.3	473.7	3395.6	8.84	15.1	7.2	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	254.2	480.4	3664.6	9.15	14.4	7.6	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	278.0	489.7	3936.4	9.27	14.2	8.0	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 4 X 9.5 T	9.42	10.3	69.9	52.6	1.38	5.1	.8	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	11.1	71.5	55.3	1.33	5.0	.8	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	13.3	79.0	67.1	1.46	5.1	.8	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 9.5 T	9.34	11.4	85.9	68.8	1.49	6.0	.8	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.5	93.4	81.7	1.62	6.0	.9	3.20	6.16	4.03	.425	.260	1.60
6 X 6 X 15.0 I-T	9.78	12.3	88.4	73.0	1.53	5.9	.8	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.4	94.7	82.7	1.63	6.2	.9	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.8	103.3	100.5	1.78	6.0	1.0	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	15.1	108.7	100.9	1.80	6.7	.9	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.4	118.3	122.6	1.96	6.7	1.0	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.9	125.3	142.8	2.10	6.5	1.1	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.3	132.1	164.2	2.23	6.5	1.2	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	28.4	137.2	183.4	2.34	6.5	1.3	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	32.1	138.1	197.6	2.40	6.1	1.4	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	36.0	142.7	220.1	2.51	6.1	1.5	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	10.2	103.1	80.0	1.63	7.9	.8	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	13.1	115.9	103.2	1.83	7.8	.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.8	135.4	148.5	2.16	7.5	1.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.5	125.9	122.4	1.98	7.9	1.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.3	146.1	181.1	2.36	7.5	1.2	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.0	138.3	148.7	2.17	7.8	1.1	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	29.1	154.2	212.4	2.52	7.3	1.4	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.3	161.3	241.8	2.67	7.3	1.5	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.6	148.8	177.4	2.35	7.8	1.2	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.3	166.1	268.9	2.78	7.2	1.6	6.56	8.07	7.04	.565	.345	2.78

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 6 1/2 X 24.0 I-T	15.11	25.2	148.9	187.1	2.40	7.4	1.3	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.5	170.7	296.4	2.89	7.1	1.7	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.6	157.3	219.3	2.57	7.4	1.4	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	47.1	176.2	333.3	3.02	7.1	1.9	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.9	161.4	239.1	2.66	7.3	1.5	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.4	169.9	240.6	2.69	8.2	1.4	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.9	179.6	283.3	2.89	8.1	1.6	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.5	146.3	140.3	2.13	9.7	1.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.3	160.9	176.8	2.37	9.6	1.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.2	171.6	204.8	2.54	9.7	1.2	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.3	181.9	235.4	2.71	9.7	1.3	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.7	192.8	273.0	2.90	9.5	1.4	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.3	204.9	325.3	3.13	9.5	1.6	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	40.0	214.2	378.2	3.33	9.5	1.8	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.4	204.0	360.7	3.24	8.7	1.8	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	49.8	215.6	431.5	3.50	8.7	2.0	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	58.2	225.1	501.7	3.71	8.6	2.2	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.0	194.1	230.0	2.71	11.5	1.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.0	204.0	262.9	2.88	11.5	1.3	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.9	220.4	319.8	3.15	11.5	1.5	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.9	233.0	376.7	3.38	11.4	1.6	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.5	250.1	452.2	3.68	11.2	1.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.8	259.6	519.5	3.90	11.1	2.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	55.4	270.6	609.2	4.16	11.0	2.3	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	60.3	265.9	623.9	4.19	10.3	2.3	7.49	11.94	8.01	.515	.295	3.52

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 8 X 45.0 I-T	28.61	67.9	272.8	696.6	4.36	10.3	2.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	75.9	279.8	772.3	4.53	10.2	2.8	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	81.2	282.3	807.6	4.62	9.9	2.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	89.7	289.2	885.9	4.79	9.9	3.1	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.6	266.3	465.7	3.76	12.7	1.7	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	44.1	281.6	558.8	4.07	12.7	2.0	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	51.3	291.4	636.5	4.30	12.4	2.2	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	58.8	302.8	725.7	4.55	12.3	2.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	66.1	311.3	809.7	4.75	12.2	2.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	73.0	310.3	851.2	4.84	11.7	2.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	82.1	318.1	949.0	5.04	11.6	3.0	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.7	318.9	694.3	4.53	14.3	2.2	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	59.0	336.1	834.4	4.90	14.1	2.5	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	69.2	347.7	958.2	5.18	13.9	2.8	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	78.3	359.5	1077.6	5.44	13.8	3.0	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	88.2	367.6	1200.3	5.66	13.6	3.3	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	98.2	375.8	1324.9	5.86	13.5	3.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	111.9	385.8	1489.9	6.11	13.3	3.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	135.5	400.3	1729.4	6.49	12.8	4.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	155.6	410.9	1949.5	6.72	12.5	4.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	179.9	423.2	2209.3	6.97	12.3	5.2	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	72.7	382.5	1127.5	5.61	15.5	2.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	84.8	398.7	1304.7	5.96	15.4	3.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	107.6	418.3	1603.3	6.45	14.9	3.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	130.0	434.6	1900.1	6.84	14.6	4.4	12.53	18.24	7.56	.695	.415	7.57

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 7 5/8 X 71.0 I-T	50.75	153.1	448.3	2193.6	7.14	14.3	4.9	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	191.6	466.1	2598.7	7.59	13.6	5.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	216.2	477.8	2878.2	7.80	13.3	6.0	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	234.8	486.0	3084.3	7.92	13.1	6.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	264.4	499.5	3409.1	8.11	12.9	6.8	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	152.1	504.3	2540.7	7.84	16.7	5.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	167.6	514.3	2766.2	8.06	16.5	5.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	180.0	521.9	2943.3	8.22	16.3	5.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	203.5	534.6	3269.5	8.46	16.1	6.1	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	227.7	547.2	3596.5	8.66	15.8	6.6	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	257.5	556.6	3892.8	8.98	15.1	7.0	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	282.0	566.7	4191.3	9.13	14.9	7.4	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
6 X 6 X 20.0 I-T	12.63	17.2	118.9	106.6	1.59	6.2	.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 13.0 T	12.85	18.8	137.0	129.5	1.75	6.9	.9	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	22.3	147.3	151.2	1.88	6.8	1.0	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.8	157.3	174.3	2.01	6.8	1.1	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	29.0	164.9	195.4	2.11	6.7	1.2	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	32.8	167.7	211.3	2.18	6.4	1.3	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	36.8	174.8	236.3	2.29	6.4	1.4	6.92	6.90	8.03	.595	.340	2.35
8 X 5 1/2 X 13.0 T	12.83	20.2	157.8	156.3	1.93	7.7	1.0	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.9	143.8	128.4	1.76	8.1	.9	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.8	173.3	191.1	2.11	7.7	1.1	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.4	160.5	156.1	1.93	8.0	1.0	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	29.7	185.5	225.1	2.27	7.6	1.2	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.9	196.0	257.0	2.41	7.6	1.3	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	23.1	175.3	186.7	2.10	8.1	1.1	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	38.1	203.6	286.8	2.53	7.5	1.4	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.7	177.0	197.4	2.15	7.7	1.1	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	42.3	210.7	317.3	2.64	7.5	1.5	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	30.1	189.5	232.3	2.31	7.7	1.2	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	48.0	219.3	358.4	2.77	7.5	1.6	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.5	195.9	253.8	2.40	7.6	1.3	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.9	204.5	254.0	2.42	8.5	1.2	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.5	218.9	300.1	2.61	8.5	1.4	5.81	8.95	6.02	.525	.315	2.82

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 X 12.0 I-T	9.07	14.8	165.9	146.2	1.89	9.9	.9	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.7	186.5	184.5	2.11	9.9	1.0	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.6	201.3	214.0	2.26	9.9	1.1	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.7	215.7	246.4	2.41	10.0	1.1	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	29.2	231.4	286.3	2.59	9.8	1.2	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.6	249.2	342.4	2.81	9.8	1.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	40.7	263.4	399.6	3.01	9.8	1.5	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	42.1	251.3	382.2	2.93	9.1	1.5	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	50.6	268.6	459.6	3.18	9.1	1.7	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	59.2	282.9	537.1	3.40	9.1	1.9	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.4	226.8	239.1	2.40	11.7	1.1	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.3	241.2	273.8	2.56	11.7	1.1	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.4	264.5	333.8	2.80	11.8	1.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.4	283.0	394.4	3.03	11.8	1.4	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	41.1	307.5	474.6	3.30	11.6	1.5	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.5	322.2	547.3	3.52	11.5	1.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	56.2	339.0	645.0	3.77	11.5	1.9	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	61.2	334.6	663.1	3.81	10.8	2.0	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	69.0	345.2	743.7	3.99	10.8	2.2	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	77.1	355.7	827.9	4.17	10.7	2.3	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	82.4	360.0	867.6	4.25	10.5	2.4	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	91.1	370.2	955.4	4.42	10.5	2.6	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.1	325.6	486.5	3.36	13.1	1.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	44.7	348.4	586.0	3.66	13.1	1.7	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	52.0	363.6	670.0	3.88	12.9	1.8	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	59.7	380.5	766.6	4.12	12.8	2.0	6.92	13.98	6.75	.455	.285	3.98

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
14 X 6 3/4 X 38.0 I-T	26.17	67.1	393.3	858.4	4.32	12.8	2.2	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	74.0	393.8	905.8	4.41	12.2	2.3	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	83.4	405.6	1014.3	4.62	12.2	2.5	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	49.4	397.1	727.5	4.07	14.7	1.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	59.8	422.6	878.3	4.42	14.7	2.1	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	70.2	440.3	1013.0	4.70	14.4	2.3	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	79.5	457.5	1143.2	4.96	14.4	2.5	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	89.6	469.6	1279.0	5.18	14.3	2.7	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	99.9	481.6	1417.7	5.40	14.2	2.9	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	113.9	495.9	1602.8	5.65	14.1	3.2	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	137.8	517.7	1872.5	6.04	13.6	3.6	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	158.5	532.3	2124.6	6.31	13.4	4.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	183.6	548.6	2424.4	6.60	13.2	4.4	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	73.8	485.4	1190.6	5.09	16.1	2.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	86.2	508.9	1383.5	5.43	16.1	2.7	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	109.3	537.5	1713.7	5.93	15.7	3.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	132.3	560.5	2046.4	6.35	15.5	3.7	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	156.2	579.0	2380.1	6.69	15.2	4.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	195.3	605.0	2844.4	7.17	14.6	4.7	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	220.7	620.2	3168.5	7.43	14.4	5.1	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	240.2	630.4	3409.3	7.58	14.2	5.4	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	270.9	647.3	3789.4	7.82	14.0	5.9	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	154.9	652.1	2737.1	7.29	17.7	4.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	170.9	665.7	2992.0	7.53	17.5	4.5	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	183.6	675.7	3193.3	7.71	17.4	4.7	15.47	21.24	8.30	.740	.455	9.66

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 3/8 X 83.0 I-T	59.78	208.0	692.0	3567.0	7.99	17.2	5.2	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	233.1	707.6	3944.1	8.24	16.9	5.6	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	262.9	723.4	4287.8	8.57	16.3	5.9	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	288.3	735.9	4636.9	8.76	16.1	6.3	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 50.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 50.000 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	6 3/4	X	17.0	T	16.77	26.4	177.6	183.5	1.83	7.0	1.0	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	29.6	188.0	205.9	1.93	7.0	1.1	5.51	7.05	6.77	.515	.310	2.19
7	X	8	X	21.5	T	20.94	33.5	192.9	223.4	1.99	6.7	1.2	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.8	T	23.53	37.6	203.0	250.4	2.10	6.7	1.2	6.92	6.90	8.03	.595	.340	2.35
8	X	5 1/2	X	15.5	T	15.28	25.3	194.9	200.2	1.92	7.9	1.0	4.49	7.94	5.53	.440	.275	2.18
8	X	7	X	18.0	T	17.73	30.2	211.4	236.2	2.07	7.8	1.1	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T	19.79	34.6	225.7	270.1	2.20	7.8	1.2	5.82	8.01	7.00	.505	.305	2.44
8	X	7	X	22.5	T	22.32	38.8	236.6	302.1	2.31	7.8	1.3	6.56	8.07	7.04	.565	.345	2.78
8	X	6 1/2	X	24.0	I-T	15.11	26.2	199.3	206.7	1.95	7.9	1.0	4.44	7.93	6.50	.400	.245	1.94
8	X	7 1/8	X	25.0	T	24.83	43.1	246.7	335.0	2.42	7.8	1.4	7.30	8.13	7.07	.630	.380	3.09
8	X	6 1/2	X	28.0	I-T	17.69	30.7	216.2	243.6	2.10	7.9	1.1	5.20	8.06	6.54	.465	.285	2.30
8	X	7 1/8	X	28.5	T	28.28	48.9	259.0	379.6	2.55	7.8	1.5	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T	19.16	34.1	225.2	266.6	2.19	7.8	1.2	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T	17.26	30.5	233.2	265.7	2.20	8.7	1.1	5.08	8.85	6.00	.425	.300	2.66
9	X	6	X	20.0	T	19.76	36.1	252.9	314.5	2.37	8.7	1.2	5.81	8.95	6.02	.525	.315	2.82
10	X	4	X	19.0	I-T	13.77	25.2	242.4	256.2	2.18	10.2	1.1	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T	15.04	29.7	262.9	297.8	2.34	10.0	1.1	4.42	10.17	5.75	.360	.240	2.44
10	X	5 3/4	X	26.0	I-T	17.37	35.4	287.1	356.8	2.54	10.1	1.2	5.11	10.33	5.77	.440	.260	2.69
10	X	5 3/4	X	30.0	I-T	20.23	41.3	306.9	417.5	2.73	10.1	1.4	5.95	10.47	5.81	.510	.300	3.14
10	X	8	X	33.0	I-T	20.94	42.7	293.3	400.2	2.67	9.4	1.4	6.16	9.73	7.96	.435	.290	2.82
10	X	8	X	39.8	I-T	24.45	51.4	317.6	482.7	2.91	9.4	1.5	7.19	9.92	7.99	.530	.315	3.12
10	X	8	X	45.0	I-T	28.19	60.1	337.8	566.2	3.12	9.4	1.7	8.29	10.10	8.02	.620	.350	3.54
12	X	4	X	16.0	I-T	12.37	23.7	270.0	283.4	2.30	11.9	1.0	3.64	11.99	3.99	.265	.220	2.64

(50T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 50.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 50.000 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 19.0 I-T	14.20	28.8	300.4	345.8	2.53	12.0	1.2	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	34.0	325.6	409.2	2.73	12.1	1.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	41.6	358.1	493.0	2.99	11.8	1.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	48.2	379.1	570.0	3.19	11.8	1.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	57.0	403.2	674.0	3.44	11.8	1.7	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	62.0	399.8	694.8	3.48	11.2	1.7	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	69.9	415.4	781.6	3.66	11.2	1.9	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	78.2	430.5	872.8	3.83	11.2	2.0	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	83.5	437.1	915.9	3.92	11.0	2.1	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	92.4	451.7	1011.4	4.09	11.0	2.2	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.6	377.0	503.7	3.03	13.4	1.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.3	408.7	608.2	3.31	13.4	1.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	52.7	430.5	697.0	3.52	13.2	1.6	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	60.5	454.1	799.3	3.75	13.2	1.8	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	68.0	472.5	897.3	3.94	13.2	1.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	75.0	475.4	949.3	4.04	12.7	2.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	84.5	492.5	1066.5	4.24	12.6	2.2	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.0	469.0	754.1	3.68	15.1	1.6	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	60.6	504.8	913.2	4.01	15.1	1.8	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	71.1	530.3	1056.5	4.28	14.9	2.0	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	80.4	554.4	1195.0	4.53	14.9	2.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	90.7	572.2	1341.4	4.75	14.8	2.3	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	101.3	589.4	1491.4	4.97	14.7	2.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	115.5	609.6	1692.9	5.23	14.7	2.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	139.6	640.7	1986.9	5.62	14.2	3.1	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	160.8	661.0	2265.9	5.90	14.1	3.4	15.00	16.52	10.30	.760	.455	7.52

(50T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 50.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 50.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 3/8 X 89.0 I-T	59.17	186.5	683.3	2600.2	6.21	13.9	3.8	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	74.8	586.5	1240.3	4.64	16.6	2.1	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	87.3	619.2	1445.4	4.97	16.6	2.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	110.7	660.0	1800.9	5.46	16.3	2.7	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	134.2	692.3	2162.8	5.88	16.1	3.1	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	158.7	717.9	2530.2	6.24	15.9	3.5	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	198.2	754.6	3043.9	6.74	15.4	4.0	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	224.2	774.7	3406.9	7.02	15.2	4.4	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	244.3	787.9	3678.8	7.20	15.1	4.7	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	275.9	809.6	4108.8	7.47	14.9	5.1	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	157.1	809.0	2892.9	6.76	18.4	3.6	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	173.4	827.5	3172.3	7.02	18.3	3.8	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	186.4	841.1	3394.0	7.20	18.2	4.0	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	211.4	862.6	3808.6	7.51	18.0	4.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	237.3	882.7	4229.9	7.78	17.8	4.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	267.1	906.1	4612.3	8.11	17.3	5.1	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	293.1	922.0	5006.8	8.33	17.1	5.4	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 56.250 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 63.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 24.0 T	23.53	38.4	226.4	263.4	1.94	6.9	1.2	6.92	6.90	8.03	.595	.340	2.35
8 X 7 X 20.0 T	19.79	35.3	249.7	282.3	2.02	8.0	1.1	5.82	8.01	7.00	.505	.305	2.44
8 X 7 X 22.5 T	22.32	39.5	263.9	316.1	2.13	8.0	1.2	6.56	8.07	7.04	.565	.345	2.78
8 X 7 1/8 X 25.0 T	24.83	43.9	277.4	350.9	2.23	8.0	1.3	7.30	8.13	7.07	.630	.380	3.09
8 X 7 1/8 X 28.5 T	28.28	49.9	293.8	398.4	2.36	8.0	1.4	8.32	8.22	7.12	.715	.430	3.53
9 X 6 X 20.0 T	19.76	36.8	280.5	327.8	2.18	8.9	1.2	5.81	8.95	6.02	.525	.315	2.82
10 X 5 3/4 X 26.0 I-T	17.37	36.0	317.4	370.0	2.33	10.3	1.2	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	42.0	343.2	433.4	2.50	10.3	1.3	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	43.4	328.5	416.3	2.45	9.6	1.3	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	52.1	360.1	503.0	2.67	9.6	1.4	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	61.0	387.0	591.2	2.87	9.7	1.5	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 22.0 I-T	16.33	34.5	359.4	422.7	2.49	12.3	1.2	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	42.2	399.9	509.4	2.73	12.1	1.3	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	48.8	427.7	589.7	2.92	12.1	1.4	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	57.7	460.1	698.9	3.15	12.1	1.5	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	62.8	458.4	721.8	3.19	11.5	1.6	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	70.8	479.8	813.7	3.37	11.5	1.7	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	79.2	500.6	910.5	3.54	11.5	1.8	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	84.6	509.8	956.4	3.62	11.3	1.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	93.5	529.6	1058.2	3.79	11.3	2.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	38.1	418.7	519.0	2.76	13.6	1.2	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.9	459.7	627.5	3.02	13.7	1.4	5.55	13.91	5.03	.420	.255	3.55

(50T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 56.250 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 63.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
14 X 6 3/4 X 30.0 I-T	21.16	53.4	488.8	720.1	3.22	13.5	1.5	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	61.2	520.0	827.0	3.43	13.5	1.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	68.8	544.9	929.9	3.62	13.5	1.7	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	75.8	551.2	985.7	3.71	13.0	1.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	85.5	574.8	1109.9	3.91	13.0	1.9	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.6	531.2	776.9	3.36	15.4	1.5	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	61.3	578.6	942.5	3.67	15.4	1.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	71.9	613.3	1092.6	3.93	15.2	1.8	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	81.3	645.4	1237.9	4.16	15.2	1.9	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	91.8	670.4	1392.8	4.38	15.2	2.1	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	102.4	694.2	1552.0	4.59	15.1	2.2	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	117.0	722.1	1767.0	4.85	15.1	2.4	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	141.2	764.6	2080.6	5.22	14.7	2.7	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	162.7	792.6	2382.2	5.52	14.6	3.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	188.9	822.8	2745.8	5.83	14.5	3.3	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	75.6	680.8	1281.4	4.25	16.9	1.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	88.2	724.3	1496.2	4.56	17.0	2.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	112.0	780.5	1871.9	5.04	16.7	2.4	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	135.8	824.9	2257.9	5.46	16.6	2.7	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	160.7	860.0	2653.5	5.82	16.5	3.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	200.6	910.1	3208.3	6.32	16.0	3.5	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	227.2	937.0	3604.6	6.61	15.9	3.8	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	247.6	954.3	3903.7	6.80	15.8	4.1	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	279.9	982.4	4377.6	7.10	15.6	4.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	158.9	969.3	3019.4	6.28	19.0	3.1	13.22	20.99	8.24	.615	.400	8.40

(50T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 56.250 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 63.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 1/4 X 68.0 I-T	49.15	175.5	994.3	3319.1	6.53	18.9	3.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	188.7	1012.5	3557.8	6.72	18.9	3.5	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	214.2	1041.3	4007.2	7.04	18.7	3.8	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	240.6	1067.6	4466.6	7.33	18.6	4.2	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	270.4	1100.1	4860.6	7.65	18.0	4.4	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	297.0	1120.7	5315.0	7.89	17.9	4.7	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 62.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 78.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 7 1/8 X 28.5 T	28.28	50.9	323.0	416.1	2.19	8.2	1.3	8.32	8.22	7.12	.715	.430	3.53
10 X 8 X 39.0 I-T	24.45	53.0	395.5	521.8	2.47	9.9	1.3	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	61.9	429.3	614.0	2.67	9.9	1.4	8.29	10.10	8.02	.620	.350	3.54
12 X 6 1/2 X 30.0 I-T	20.27	49.5	467.0	608.0	2.69	12.3	1.3	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	58.5	508.0	721.3	2.91	12.3	1.4	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	63.6	508.5	746.0	2.95	11.7	1.5	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	71.7	536.4	842.2	3.12	11.7	1.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	80.2	563.3	943.7	3.28	11.8	1.7	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	85.6	575.5	991.9	3.36	11.6	1.7	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	94.6	601.3	1098.9	3.52	11.6	1.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 26.0 I-T	18.87	46.5	500.6	645.4	2.78	13.9	1.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	54.1	537.0	741.1	2.96	13.7	1.4	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	61.9	576.1	851.8	3.16	13.8	1.5	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	69.6	608.1	958.8	3.34	13.8	1.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	76.7	618.2	1017.7	3.43	13.3	1.6	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	86.5	649.2	1147.6	3.62	13.3	1.8	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	51.2	582.2	797.5	3.08	15.6	1.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	62.0	641.6	968.6	3.38	15.6	1.5	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	72.7	686.2	1124.2	3.62	15.5	1.6	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	82.2	727.1	1274.9	3.84	15.5	1.8	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	92.8	760.4	1436.9	4.05	15.5	1.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	103.6	791.9	1603.7	4.26	15.5	2.0	10.39	16.26	7.07	.630	.380	6.18

(50T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 62.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 78.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 1/8 X 57.0 I-T	40.28	118.3	829.2	1829.9	4.51	15.5	2.2	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	142.7	884.8	2159.7	4.87	15.1	2.4	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	164.5	922.5	2480.2	5.16	15.1	2.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	191.0	962.7	2868.8	5.48	15.0	3.0	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	76.4	764.8	1316.9	3.92	17.2	1.7	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	89.1	820.2	1539.7	4.21	17.3	1.9	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	113.1	894.2	1932.0	4.67	17.1	2.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	137.2	953.4	2337.9	5.08	17.0	2.5	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	162.5	1000.6	2757.1	5.44	17.0	2.8	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	202.8	1066.9	3346.2	5.93	16.5	3.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	229.7	1102.5	3771.0	6.23	16.4	3.4	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	250.5	1125.2	4093.6	6.43	16.3	3.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	283.4	1161.5	4605.9	6.73	16.3	4.0	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	160.5	1127.3	3124.8	5.85	19.5	2.8	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	177.3	1160.6	3441.4	6.10	19.4	3.0	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	190.7	1184.6	3694.4	6.28	19.4	3.1	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	216.6	1222.9	4173.3	6.60	19.3	3.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	243.5	1257.5	4665.3	6.90	19.2	3.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	273.3	1300.2	5105.4	7.21	18.7	3.9	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	300.3	1327.2	5574.3	7.46	18.6	4.2	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 68.750 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 94.531 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 8 X 40.0 I-T	25.48	64.5	549.7	768.9	2.75	11.9	1.4	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	72.7	584.1	868.7	2.90	11.9	1.5	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	81.3	617.4	974.1	3.06	12.0	1.6	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	86.7	632.7	1024.3	3.13	11.8	1.6	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	95.8	664.6	1135.6	3.29	11.9	1.7	10.56	12.19	10.01	.640	.360	4.39
14 X 6 3/4 X 34.0 I-T	23.54	62.7	621.7	875.0	2.94	13.9	1.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	70.5	660.8	985.4	3.10	14.0	1.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	77.6	675.1	1046.9	3.19	13.5	1.6	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	87.5	714.0	1181.7	3.37	13.5	1.7	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 31.0 I-T	22.70	62.8	692.9	992.8	3.13	15.8	1.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	73.5	747.5	1153.2	3.36	15.7	1.5	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	83.1	797.4	1308.5	3.57	15.7	1.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	93.8	839.7	1476.3	3.77	15.7	1.8	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	104.7	879.9	1649.6	3.97	15.8	1.9	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	119.5	927.6	1885.2	4.21	15.8	2.0	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	144.0	997.6	2228.5	4.55	15.5	2.2	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	166.1	1046.8	2565.1	4.84	15.4	2.5	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	193.0	1099.0	2975.0	5.16	15.4	2.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	77.3	836.3	1349.2	3.63	17.5	1.6	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	90.1	904.2	1578.6	3.91	17.5	1.7	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	114.2	997.6	1984.8	4.35	17.4	2.0	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	138.6	1073.7	2407.4	4.74	17.4	2.2	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	164.2	1135.4	2846.6	5.10	17.3	2.5	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	204.7	1220.3	3464.5	5.57	16.9	2.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	231.9	1266.5	3913.8	5.87	16.9	3.1	19.18	18.59	11.15	.870	.535	9.95

(50T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 68.750 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 94.531 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/4 X 106.0 I-T	71.48	253.1	1296.2	4256.8	6.07	16.8	3.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	286.4	1342.7	4802.3	6.37	16.8	3.6	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	162.0	1278.3	3215.2	5.46	19.8	2.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	178.9	1321.3	3546.0	5.70	19.8	2.7	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	192.5	1352.4	3811.0	5.89	19.8	2.8	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	218.7	1402.3	4315.0	6.20	19.7	3.1	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	246.0	1447.3	4835.0	6.50	19.7	3.3	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	275.8	1501.6	5296.8	6.80	19.2	3.5	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	303.2	1536.6	5795.6	7.05	19.1	3.8	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 75.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 112.500 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 8 1/8 X 50.0 I-T	32.11	82.4	662.7	1003.2	2.87	12.2	1.5	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	87.8	680.9	1055.0	2.94	12.0	1.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	97.0	719.0	1170.1	3.08	12.1	1.6	10.56	12.19	10.01	.640	.360	4.39
14 X 8 X 48.0 I-T	31.50	88.5	768.5	1213.9	3.16	13.7	1.6	9.26	13.79	8.03	.595	.340	4.69
16 X 7 X 40.0 I-T	28.09	84.0	855.8	1340.0	3.33	15.9	1.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	94.8	907.3	1513.0	3.52	16.0	1.7	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	105.8	956.3	1691.7	3.71	16.0	1.8	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	120.8	1015.2	1935.4	3.95	16.0	1.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	145.4	1100.3	2290.2	4.27	15.7	2.1	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	167.7	1162.3	2640.7	4.55	15.7	2.3	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	194.8	1228.2	3068.9	4.86	15.8	2.5	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	78.1	895.0	1379.6	3.39	17.7	1.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	91.0	975.1	1614.6	3.65	17.7	1.7	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	115.3	1088.7	2032.6	4.07	17.6	1.9	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	139.9	1183.0	2469.5	4.44	17.7	2.1	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	165.8	1261.0	2926.0	4.79	17.6	2.3	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	206.5	1366.6	3568.5	5.25	17.3	2.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	234.1	1425.1	4038.8	5.54	17.3	2.8	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.8 I-T	71.48	255.4	1463.1	4399.5	5.74	17.2	3.0	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	289.2	1521.7	4974.1	6.04	17.2	3.3	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	163.4	1418.4	3294.9	5.12	20.2	2.3	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	180.5	1472.3	3637.8	5.35	20.2	2.5	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	194.2	1511.5	3913.1	5.53	20.2	2.6	15.47	21.24	8.30	.740	.455	9.66

(50T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 75.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 112.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 3/8 X 83.0 I-T	59.78	220.7	1575.2	4438.7	5.84	20.1	2.8	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	248.3	1632.5	4982.8	6.14	20.1	3.1	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	278.1	1699.4	5462.9	6.42	19.6	3.2	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	305.8	1744.2	5987.5	6.67	19.6	3.4	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 87.500 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 153.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 1/8 X 57.0 I-T	40.28	123.5	1155.1	2027.9	3.51	16.4	1.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	148.3	1270.1	2401.3	3.80	16.2	1.9	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	170.9	1359.8	2774.4	4.06	16.2	2.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	198.6	1456.3	3232.7	4.35	16.3	2.2	17.40	16.75	10.37	.875	.525	8.79
18 X 7 1/2 X 60.0 I-T	42.61	142.6	1362.9	2581.2	3.95	18.1	1.9	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	169.0	1476.1	3065.7	4.27	18.1	2.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	210.1	1626.6	3748.0	4.69	17.8	2.3	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	238.1	1713.8	4252.9	4.97	17.9	2.5	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	260.0	1771.6	4642.7	5.16	17.9	2.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	294.3	1859.3	5265.2	5.46	17.9	2.8	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	166.2	1656.2	3434.6	4.54	20.7	2.1	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	183.5	1734.2	3797.0	4.76	20.7	2.2	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	197.5	1791.7	4089.0	4.92	20.7	2.3	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	224.4	1887.3	4649.6	5.22	20.7	2.5	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	252.6	1974.2	5233.3	5.50	20.7	2.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	282.4	2068.4	5742.3	5.76	20.3	2.8	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	310.7	2137.5	6309.3	6.00	20.3	3.0	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 100.000 IN.) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.) EFFECTIVE PLATE AREA = 200.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 3/8 X 89.0 I-T	59.17	202.6	1636.8	3379.8	3.94	16.7	2.1	17.40	16.75	10.37	.875	.525	8.79
18 X 11 1/8 X 86.0 I-T	57.79	214.0	1834.0	3906.9	4.24	18.3	2.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	242.4	1952.2	4439.0	4.50	18.3	2.3	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	264.5	2032.4	4851.9	4.69	18.3	2.4	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	299.4	2153.3	5512.3	4.96	18.4	2.6	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 68.0 I-T	49.15	186.8	1935.3	3939.5	4.29	21.1	2.0	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	200.9	2011.6	4244.5	4.44	21.1	2.1	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	228.2	2141.6	4832.6	4.71	21.2	2.3	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	256.8	2261.7	5447.3	4.98	21.2	2.4	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	286.7	2384.0	5978.4	5.21	20.9	2.5	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	315.4	2481.5	6578.7	5.44	20.9	2.7	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.)

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TABLE IX. Properties of combined beam and plate, I-T and T (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(50T = 112.500 IN.) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.) EFFECTIVE PLATE AREA = 253.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/4 X 106.0 I-T	71.48	269.4	2239.4	5045.5	4.29	18.7	2.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	304.8	2394.2	5737.0	4.55	18.8	2.4	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 3/8 X 93.0 I-T	67.42	261.3	2488.2	5644.4	4.55	21.6	2.3	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	291.3	2636.6	6192.9	4.76	21.3	2.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	320.3	2764.1	6820.3	4.98	21.3	2.5	22.15	21.51	12.34	.875	.550	11.83

(50T) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t).

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 4.750 IN.) PLATE WEIGHT = 5.100 LBS. (.1250 IN.) EFFECTIVE PLATE AREA = .594 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	x	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	x	5.0	T	4.91	3.2	2.7	6.0	1.72	1.9	2.2	1.44	3.95	3.94	.205	.170	.67
5	X	4	x	6.0	T	5.88	4.4	3.7	10.1	2.08	2.3	2.8	1.73	4.94	3.96	.210	.190	.94
6	X	4	x	7.0	T	6.94	5.8	4.7	15.9	2.45	2.7	3.3	2.04	5.96	3.97	.225	.200	1.19
6	X	4	x	9.0	I-T	6.17	5.4	4.5	14.7	2.47	2.7	3.3	1.81	5.90	3.94	.215	.170	1.00
8	X	4	x	10.0	I-T	7.19	7.5	6.5	27.8	3.20	3.7	4.3	2.11	7.89	3.94	.205	.170	1.34
10	X	4	x	12.0	I-T	9.07	10.3	9.1	48.3	3.85	4.7	5.3	2.67	9.87	3.96	.210	.190	1.88
12	X	4	x	14.0	I-T	10.98	14.0	12.1	78.1	4.52	5.6	6.4	3.23	11.91	3.97	.225	.200	2.38

(38T) PLATE WEIGHT = 5.100 LBS. (.1250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 5.938 IN.) PLATE WEIGHT = 6.375 LBS. (.1563 IN.) EFFECTIVE PLATE AREA = .928 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0 T	4.91	3.4	3.8	7.4	1.77	2.2	1.9	1.44	3.95	3.94	.205	.170	.67
5	X	4	X	6.0 T	5.88	4.6	5.0	12.3	2.15	2.7	2.4	1.73	4.94	3.96	.210	.190	.94
6	X	4	X	7.0 T	6.94	6.1	6.4	19.1	2.54	3.1	3.0	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	9.0 I-T	6.17	5.7	6.1	17.9	2.55	3.1	2.9	1.81	5.90	3.94	.215	.170	1.00
8	X	4	X	10.0 I-T	7.19	7.9	8.6	33.2	3.30	4.2	3.9	2.11	7.89	3.94	.205	.170	1.34
10	X	4	X	12.0 I-T	9.07	11.0	11.7	56.8	3.98	5.2	4.9	2.67	9.87	3.96	.210	.190	1.88
12	X	4	X	14.0 I-T	10.98	14.9	15.2	90.8	4.67	6.1	6.0	3.23	11.91	3.97	.225	.200	2.38

(38T) PLATE WEIGHT = 6.375 LBS. (.1563 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 7.125 IN.) PLATE WEIGHT = 7.650 LBS. (.1875 IN.) EFFECTIVE PLATE AREA = 1.336 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.5	5.1	8.7	1.77	2.4	1.7	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.8	6.7	14.3	2.16	3.0	2.2	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.4	8.3	22.3	2.57	3.5	2.7	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	5.9	8.1	20.9	2.57	3.5	2.6	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.3	11.2	38.5	3.34	4.6	3.4	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.6	14.9	65.4	4.04	5.7	4.4	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	15.6	19.0	103.9	4.77	6.6	5.5	3.23	11.91	3.97	.225	.200	2.38

(38T) PLATE WEIGHT = 7.650 LBS. (.1875 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 8.313 IN.) PLATE WEIGHT = 8.925 LBS. (.2188 IN.) EFFECTIVE PLATE AREA = 1.818 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.6	6.6	9.8	1.73	2.7	1.5	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	5.0	8.6	16.2	2.14	3.3	1.9	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.6	10.6	25.2	2.56	3.8	2.4	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	6.1	10.4	23.6	2.55	3.8	2.3	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.6	14.2	43.3	3.32	5.1	3.0	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	12.0	18.6	73.6	4.05	6.1	4.0	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	16.3	23.5	116.7	4.81	7.2	5.0	3.23	11.91	3.97	.225	.200	2.38

(38T) PLATE WEIGHT = 8.925 LBS. (.2188 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 9.500 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 2.375 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T		4.91	3.7	8.3	10.8	1.68	2.9	1.3	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T		6.40	4.7	8.6	12.9	1.74	2.8	1.5	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T		7.42	5.5	9.0	14.7	1.80	2.7	1.6	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T		8.82	7.1	9.3	17.4	1.87	2.4	1.9	2.59	4.07	5.25	.330	.230	.94
5 X 4 X 6.0 T		5.88	5.1	10.7	17.9	2.09	3.5	1.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T		7.37	6.4	11.1	21.2	2.16	3.3	1.9	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T		8.36	7.4	11.5	24.0	2.23	3.2	2.1	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T		9.42	8.6	11.8	26.7	2.28	3.1	2.3	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T		9.91	9.3	11.7	27.3	2.27	2.9	2.3	2.92	5.01	5.00	.360	.240	1.20
6 X 4 X 7.0 T		6.94	6.8	13.2	27.9	2.51	4.1	2.1	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T		7.88	7.8	13.6	31.0	2.57	4.0	2.3	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T		6.17	6.3	12.9	26.0	2.49	4.1	2.0	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T		9.34	9.6	14.2	36.3	2.66	3.8	2.6	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 12.0 I-T		8.30	8.2	13.8	32.3	2.59	3.9	2.3	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T		9.78	10.5	14.2	37.7	2.68	3.6	2.7	2.88	5.99	5.99	.260	.230	1.38
7 X 5 X 11.0 T		10.81	12.8	16.6	51.5	3.05	4.0	3.1	3.18	6.87	5.00	.335	.230	1.58
8 X 4 X 10.0 I-T		7.19	8.8	17.7	47.7	3.26	5.4	2.7	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T		9.52	11.1	18.7	57.4	3.33	5.2	3.1	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T		12.83	16.7	19.7	73.3	3.45	4.4	3.7	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T		10.79	13.1	19.5	65.4	3.43	5.0	3.4	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/4 X 18.0 I-T		12.00	16.2	20.2	75.4	3.57	4.7	3.7	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T		13.87	19.1	21.1	85.5	3.64	4.5	4.0	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T		15.11	21.3	20.5	85.4	3.54	4.0	4.2	4.44	7.93	6.50	.400	.245	1.94

(38T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 9.500 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 2.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 X 12.0 I-T	9.07	12.4	22.8	81.2	4.01	6.6	3.6	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	15.4	24.2	96.3	4.11	6.3	4.0	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	17.7	25.2	107.8	4.22	6.1	4.3	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	20.2	26.2	119.7	4.32	5.9	4.6	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	24.2	26.7	132.2	4.41	5.5	4.9	4.42	10.17	5.75	.360	.240	2.44
12 X 4 X 14.0 I-T	10.98	16.9	28.6	128.9	4.79	7.6	4.5	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	19.1	29.7	142.4	4.87	7.4	4.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	23.2	31.4	165.3	5.02	7.1	5.3	4.18	12.16	4.01	.350	.235	2.86
12 X 6 1/2 X 26.0 I-T	17.64	33.9	33.6	210.6	5.28	6.2	6.3	5.19	12.22	6.49	.380	.230	2.81
14 X 5 X 22.0 I-T	16.16	30.3	37.0	232.9	5.71	7.7	6.3	4.76	13.74	5.00	.335	.230	3.16
16 X 5 1/2 X 26.0 I-T	19.49	39.6	44.8	334.9	6.43	8.5	7.5	5.73	15.69	5.50	.345	.250	3.92

(38T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 10.688 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.006 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T				4.91	3.8	10.1	11.6	1.62	3.1	1.2	1.44	3.35	3.94	.205	.170	.67
4 X 4 X 6.5 T				6.40	4.7	10.5	14.0	1.69	2.9	1.3	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T				7.42	5.7	10.9	16.2	1.76	2.9	1.5	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T				8.82	7.3	11.3	19.2	1.85	2.6	1.7	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T				8.39	6.4	11.3	18.1	1.82	2.8	1.6	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T				5.88	5.2	13.0	19.3	2.02	3.7	1.5	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T				7.37	6.5	13.5	23.2	2.12	3.6	1.7	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T				8.36	7.6	14.0	26.3	2.19	3.5	1.9	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T				9.42	8.8	14.4	29.4	2.26	3.4	2.0	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T				9.91	9.5	14.3	30.2	2.26	3.2	2.1	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T				11.69	11.3	14.9	35.0	2.33	3.1	2.3	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T				6.94	6.9	16.0	30.2	2.45	4.4	1.9	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T				7.88	8.0	16.5	33.7	2.52	4.2	2.0	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T				6.17	6.4	15.8	28.1	2.41	4.4	1.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T				9.34	9.9	17.2	39.9	2.63	4.0	2.3	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T				10.89	11.6	17.8	45.3	2.70	3.9	2.5	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T				8.38	8.4	16.7	35.2	2.54	4.2	2.1	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T				9.78	10.7	17.3	41.5	2.66	3.9	2.4	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T				10.74	11.5	18.1	46.2	2.74	4.0	2.5	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T				12.63	14.5	18.5	52.7	2.80	3.6	2.8	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T				10.81	13.1	20.1	56.8	3.03	4.3	2.8	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T				12.85	15.8	20.9	65.3	3.10	4.1	3.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T				14.81	18.8	21.3	72.0	3.13	3.8	3.4	4.36	6.92	6.73	.385	.270	1.87

(38T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 10.688 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.006 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 10.0 I-T	7.19	9.0	21.4	51.6	3.18	5.8	2.4	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.4	22.5	62.5	3.28	5.5	2.8	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.1	23.8	81.0	3.46	4.7	3.4	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.4	23.5	71.6	3.40	5.3	3.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	20.8	24.7	93.0	3.52	4.5	3.8	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	16.6	24.4	83.1	3.57	5.0	3.4	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T	13.87	19.6	25.4	94.7	3.66	4.8	3.7	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	21.8	24.7	95.2	3.57	4.4	3.9	4.44	7.93	6.50	.400	.245	1.94
10 X 4 X 12.0 I-T	9.07	12.7	27.5	88.1	3.94	7.0	3.2	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	15.8	29.0	105.0	4.08	6.7	3.6	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.2	30.2	118.0	4.20	6.5	3.9	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	20.8	31.3	131.5	4.32	6.3	4.2	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	24.8	32.0	146.1	4.43	5.9	4.6	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	29.3	33.5	165.9	4.52	5.7	5.0	5.11	10.33	5.77	.440	.260	2.69
12 X 4 X 14.0 I-T	10.98	17.3	34.2	140.2	4.74	8.1	4.1	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	19.7	35.4	155.3	4.83	7.9	4.4	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	23.8	37.4	181.0	5.02	7.6	4.8	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	27.8	39.2	204.9	5.12	7.4	5.2	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	34.9	40.1	233.1	5.33	6.7	5.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	39.8	41.7	257.3	5.36	6.5	6.2	5.96	12.34	6.52	.440	.260	3.21
14 X 5 X 22.0 I-T	16.18	31.1	43.9	255.4	5.74	8.2	5.8	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	37.2	46.3	292.6	5.85	7.9	6.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	43.2	47.6	319.9	5.89	7.4	6.7	6.22	13.84	6.73	.385	.270	3.74

(38T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 10.688 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.006 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	40.8	52.7	367.1	6.48	9.0	7.0	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	49.0	55.8	421.5	6.60	8.6	7.6	6.68	15.88	5.53	.440	.275	4.37

(38T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 11.875 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 3.711 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.8	12.0	12.4	1.55	3.2	1.0	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	4.8	12.5	15.0	1.64	3.1	1.2	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	5.8	13.0	17.4	1.72	3.0	1.3	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.4	13.5	20.9	1.82	2.8	1.5	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	6.5	13.5	19.6	1.78	3.0	1.5	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.3	15.4	20.6	1.94	3.9	1.3	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	6.6	16.1	24.9	2.06	3.8	1.5	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	7.7	16.7	28.4	2.14	3.7	1.7	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	8.9	17.1	31.9	2.22	3.6	1.9	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	9.7	17.0	32.9	2.23	3.4	1.9	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	11.5	17.8	38.3	2.31	3.3	2.1	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.1	19.1	32.3	2.37	4.6	1.7	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.1	19.6	36.2	2.45	4.5	1.8	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.5	18.8	29.9	2.33	4.6	1.6	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.0	20.5	43.1	2.58	4.3	2.1	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	11.9	21.2	49.2	2.67	4.2	2.3	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	8.5	19.9	37.9	2.48	4.4	1.9	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	10.9	20.6	45.0	2.61	4.1	2.2	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	11.8	21.6	50.2	2.70	4.3	2.3	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	14.8	22.1	57.7	2.79	3.9	2.6	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	13.4	24.0	61.6	2.99	4.6	2.6	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	16.2	24.9	71.3	3.09	4.4	2.9	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	19.2	25.4	79.1	3.13	4.1	3.1	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	22.1	26.0	87.2	3.18	4.0	3.4	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	24.6	26.6	94.1	3.19	3.8	3.5	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE X. *Properties of combined beam and plate, I-T and T (38t).* - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 11.875 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 3.711 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	27.8	26.0	96.0	3.12	3.5	3.7	6.16	6.83	8.00	.530	.305	2.08
8 X 4 X 10.0 I-T	7.19	9.1	25.6	55.1	3.07	6.0	2.2	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.6	26.7	67.2	3.21	5.8	2.5	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.5	28.2	88.1	3.43	5.0	3.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.7	27.9	77.2	3.35	5.7	2.8	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.3	29.3	101.8	3.52	4.8	3.5	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	16.9	29.0	90.2	3.53	5.3	3.1	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	25.4	30.1	113.5	3.57	4.5	3.8	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	28.9	30.8	124.2	3.61	4.3	4.0	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.0	30.2	103.3	3.64	5.2	3.4	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.3	29.4	104.5	3.58	4.7	3.6	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	25.8	30.5	117.2	3.63	4.5	3.8	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.76	28.8	30.7	123.5	3.64	4.3	4.0	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	25.6	33.6	133.0	3.89	5.2	4.0	5.08	8.85	6.00	.425	.300	2.66
10 X 4 X 12.0 I-T	9.07	12.9	32.7	94.3	3.85	7.3	2.9	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.1	34.4	113.1	4.01	7.0	3.3	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.6	35.7	127.4	4.16	6.9	3.6	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.3	37.0	142.5	4.28	6.7	3.9	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.3	37.9	159.1	4.42	6.3	4.2	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	30.0	39.5	181.5	4.54	6.0	4.6	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	34.6	41.1	202.6	4.58	5.9	4.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	36.1	38.5	187.0	4.35	5.2	4.9	6.16	9.73	7.96	.435	.290	2.82

(38T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 11.875 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 3.711 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 14.0 I-T	10.98	17.7	40.5	150.5	4.66	8.5	3.7	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.1	41.8	167.2	4.77	8.3	4.0	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	24.4	44.0	195.7	4.98	8.0	4.4	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	28.5	46.0	222.2	5.11	7.8	4.8	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	35.7	47.2	254.7	5.35	7.1	5.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	40.8	49.0	281.6	5.40	6.9	5.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	47.6	51.3	316.3	5.43	6.6	6.2	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	52.8	49.5	310.8	5.27	6.0	6.3	7.49	11.94	8.01	.515	.295	3.52
14 X 5 X 22.0 I-T	16.18	31.9	51.6	276.9	5.72	8.7	5.4	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	38.1	54.1	318.2	5.86	8.3	5.9	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	44.3	55.6	349.0	5.93	7.9	6.3	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	50.5	57.6	384.8	6.01	7.6	6.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	56.3	59.6	417.2	6.05	7.4	7.0	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	62.4	58.5	421.6	5.94	6.8	7.2	8.24	13.66	8.00	.530	.305	4.17
16 X 5 1/2 X 26.0 I-T	19.49	41.8	61.4	398.3	6.49	9.5	6.5	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	50.3	64.8	458.4	6.64	9.1	7.1	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	58.7	67.1	506.6	6.71	8.6	7.5	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	66.3	69.4	553.2	6.80	8.3	8.0	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	61.0	75.2	606.8	7.28	9.9	8.1	7.73	17.70	6.00	.425	.300	5.31

(38T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 13.063 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 4.490 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.9	13.9	13.0	1.48	3.4	.9	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	4.9	14.5	15.9	1.58	3.2	1.1	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	5.8	15.1	18.6	1.67	3.2	1.2	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.5	15.9	22.5	1.78	3.0	1.4	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	6.6	15.7	20.9	1.73	3.2	1.3	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.3	18.0	21.7	1.87	4.1	1.2	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	6.7	18.8	26.4	1.99	3.9	1.4	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	7.8	19.5	30.2	2.09	3.9	1.6	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	9.1	20.1	34.1	2.17	3.8	1.7	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	9.8	20.0	35.3	2.18	3.6	1.8	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	11.7	21.0	41.3	2.28	3.5	2.0	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.1	22.3	34.1	2.29	4.8	1.5	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.2	22.9	38.4	2.37	4.7	1.7	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.6	21.9	31.5	2.24	4.8	1.4	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.2	24.0	46.0	2.52	4.5	1.9	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	12.0	24.9	52.8	2.62	4.4	2.1	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	8.7	23.2	40.2	2.41	4.6	1.7	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	11.1	24.2	48.1	2.56	4.3	2.0	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	12.0	25.3	53.8	2.65	4.5	2.1	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	15.0	25.9	62.3	2.75	4.1	2.4	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	13.6	28.1	66.0	2.93	4.9	2.3	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	16.4	29.2	76.8	3.05	4.7	2.6	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	19.6	29.8	85.7	3.11	4.4	2.9	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	22.5	30.5	95.0	3.17	4.2	3.1	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	25.1	31.2	102.8	3.21	4.1	3.3	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 13.063 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 4.490 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X TN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 6 X 21.5 T	20.94	28.3	30.5	105.4	3.15	3.7	3.5	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	31.4	31.3	113.6	3.15	3.6	3.6	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.2	29.9	58.1	2.97	6.3	1.9	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.8	31.3	71.3	3.13	6.1	2.3	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.8	33.1	94.7	3.39	5.3	2.9	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.9	32.6	82.3	3.28	5.9	2.5	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.7	34.3	110.1	3.50	5.1	3.2	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.2	34.0	96.7	3.47	5.6	2.8	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	25.9	35.2	123.4	3.57	4.8	3.5	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	29.5	36.0	135.5	3.63	4.6	3.8	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.3	35.3	111.3	3.60	5.5	3.1	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.7	34.5	113.1	3.56	5.0	3.3	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	26.3	35.7	127.4	3.63	4.8	3.6	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	29.3	36.0	134.8	3.65	4.6	3.7	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	26.1	39.1	143.9	3.88	5.5	3.7	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	30.8	40.4	162.4	3.97	5.3	4.0	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.1	38.3	99.8	3.73	7.6	2.6	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.4	40.1	120.4	3.93	7.3	3.0	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.9	41.6	136.1	4.08	7.2	3.3	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.7	43.1	152.6	4.23	7.0	3.5	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.8	44.2	171.3	4.38	6.6	3.9	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	30.6	46.0	196.2	4.52	6.4	4.3	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	35.3	47.8	219.6	4.59	6.2	4.6	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	36.8	44.8	203.6	4.37	5.5	4.5	6.16	9.73	7.96	.435	.290	2.82

(38T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 13.063 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 4.490 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 39.0 I-T	24.45	43.8	46.8	232.3	4.46	5.3	5.0	7.19	9.92	7.99	.530	.315	3.12
12 X 4 X 14.0 I-T	10.98	18.0	47.3	159.8	4.55	8.9	3.4	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.5	48.8	178.1	4.68	8.7	3.7	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	24.9	51.3	209.3	4.91	8.4	4.1	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.1	53.4	238.4	5.07	8.2	4.5	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	36.3	55.1	275.0	5.33	7.6	5.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	41.6	56.9	304.9	5.40	7.3	5.4	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	48.6	59.4	343.3	5.47	7.1	5.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	53.1	57.4	338.9	5.32	6.4	5.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	59.2	59.4	367.6	5.33	6.2	6.2	8.47	12.06	8.05	.575	.335	4.04
14 X 5 X 22.0 I-T	16.18	32.5	60.0	297.0	5.67	9.1	5.0	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	38.9	62.7	342.4	5.84	8.8	5.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	45.2	64.4	376.8	5.93	8.3	5.9	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	51.6	66.6	416.5	6.04	8.1	6.3	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	57.6	68.7	452.3	6.09	7.9	6.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	63.8	67.4	459.0	6.00	7.2	6.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	71.0	69.8	497.7	6.02	7.0	7.1	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	42.8	71.0	428.0	6.47	10.0	6.0	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	51.4	74.6	494.0	6.65	9.6	6.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	60.1	77.1	547.4	6.74	9.1	7.1	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	67.8	79.6	599.0	6.85	8.8	7.5	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	62.5	86.1	653.5	7.31	10.5	7.6	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	72.7	89.9	733.1	7.47	10.1	8.2	8.63	17.90	6.02	.525	.315	5.64

(38T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 14.250 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 5.344 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	3.9	15.8	13.6	1.41	3.5	.9	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	5.0	16.6	16.7	1.52	3.4	1.0	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	5.9	17.4	19.6	1.61	3.3	1.1	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.6	18.3	23.8	1.73	3.1	1.3	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	6.7	18.1	22.1	1.68	3.3	1.2	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.4	20.6	22.7	1.79	4.2	1.1	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	6.8	21.6	27.7	1.92	4.1	1.3	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	7.9	22.5	31.9	2.02	4.0	1.4	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	9.2	23.2	36.2	2.11	3.9	1.6	2.77	5.12	4.02	.395	.250	1.26
5	X	5	X	16.0	I-T	9.91	10.0	23.1	37.5	2.13	3.8	1.6	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	11.9	24.3	44.1	2.24	3.7	1.8	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.2	25.7	35.7	2.20	4.9	1.4	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.3	26.4	40.3	2.29	4.8	1.5	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.6	25.2	32.9	2.14	5.0	1.3	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.3	27.7	48.6	2.45	4.7	1.8	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	12.2	28.7	56.0	2.56	4.6	1.9	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	8.8	26.8	42.3	2.33	4.8	1.6	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	11.2	27.9	50.9	2.49	4.5	1.8	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	12.1	29.2	57.0	2.59	4.7	2.0	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	15.2	30.0	66.5	2.71	4.4	2.2	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	13.7	32.6	70.0	2.87	5.1	2.2	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	16.7	33.8	81.9	3.00	4.9	2.4	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	19.8	34.5	91.9	3.08	4.6	2.7	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	22.8	35.4	102.2	3.15	4.5	2.9	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	25.5	36.1	111.0	3.20	4.4	3.1	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 14.250 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 5.344 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	28.8	35.4	114.5	3.15	4.0	3.2	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	32.0	36.2	123.6	3.17	3.9	3.4	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.3	34.5	60.7	2.85	6.5	1.8	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.9	36.1	75.1	3.04	6.3	2.1	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	18.0	38.2	100.8	3.32	5.6	2.6	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.1	37.6	87.0	3.20	6.2	2.3	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	22.0	39.7	117.7	3.46	5.3	3.0	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.4	39.3	102.7	3.40	5.9	2.6	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	26.3	40.7	132.7	3.54	5.0	3.3	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	30.0	41.7	146.2	3.62	4.9	3.5	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.6	40.9	118.6	3.55	5.8	2.9	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	33.3	42.6	157.9	3.64	4.7	3.7	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	23.0	39.9	121.2	3.52	5.3	3.0	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	26.8	41.3	137.0	3.60	5.1	3.3	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	29.8	41.6	145.4	3.64	4.9	3.5	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	26.5	45.1	154.1	3.85	5.8	3.4	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	31.3	46.6	174.6	3.96	5.6	3.7	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.3	44.2	104.7	3.62	7.9	2.4	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.7	46.3	127.0	3.83	7.6	2.7	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.2	48.0	144.0	4.00	7.5	3.0	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	22.0	49.7	162.0	4.15	7.4	3.3	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.2	51.1	182.6	4.32	7.0	3.6	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	31.1	53.1	210.0	4.48	6.8	4.0	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	35.9	55.0	235.8	4.57	6.6	4.3	5.95	10.47	5.81	.510	.300	3.14

(38T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 14.250 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 5.344 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 33.0 I-T	20.94	37.4	51.7	219.4	4.37	5.9	4.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	44.6	53.9	251.3	4.48	5.6	4.7	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	51.7	56.0	281.5	4.54	5.4	5.0	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	18.3	54.6	168.2	4.43	9.2	3.1	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.8	56.2	188.0	4.58	9.0	3.3	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.3	59.1	221.8	4.83	8.8	3.8	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.6	61.4	253.5	5.00	8.6	4.1	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	36.9	63.5	294.0	5.28	8.0	4.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	42.3	65.5	327.0	5.38	7.7	5.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	49.5	68.1	369.2	5.47	7.5	5.4	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	54.1	66.0	366.0	5.34	6.8	5.5	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	60.3	68.0	397.6	5.36	6.6	5.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	66.8	70.2	429.8	5.39	6.4	6.1	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	71.9	69.7	440.1	5.41	6.1	6.3	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	79.0	71.4	471.2	5.44	6.0	6.6	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	33.1	69.0	315.6	5.59	9.5	4.6	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	39.6	72.1	365.2	5.79	9.2	5.1	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	46.0	73.9	403.2	5.90	8.8	5.5	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	52.6	76.3	446.9	6.04	8.5	5.9	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	58.7	78.5	486.1	6.11	8.3	6.2	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	65.0	77.2	495.3	6.04	7.6	6.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	72.5	79.7	537.7	6.07	7.4	6.7	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	43.6	81.4	455.9	6.42	10.5	5.6	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	52.4	85.4	528.0	6.63	10.1	6.2	6.68	15.88	5.53	.440	.275	4.37

(38T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 14.250 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 5.344 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 36.0 I-T	25.69	61.3	88.1	586.7	6.74	9.6	6.7	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.8 I-T	26.09	69.2	90.8	643.4	6.88	9.3	7.1	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	77.1	93.9	699.0	6.90	9.1	7.4	9.34	16.13	7.04	.565	.345	5.56
18 X 6 X 35.0 I-T	26.29	63.8	98.0	698.6	7.31	10.9	7.1	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	74.2	102.1	785.4	7.50	10.6	7.7	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	93.3	108.1	920.0	7.63	9.9	8.5	10.46	17.99	7.50	.570	.355	6.39

(38T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 16.625 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 7.273 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	4.0	19.5	14.6	1.29	3.6	.7	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.1	20.7	18.1	1.40	3.6	.9	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.0	21.9	21.3	1.50	3.5	1.0	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.8	23.3	26.2	1.63	3.4	1.1	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.8	22.9	24.2	1.58	3.5	1.1	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.5	25.8	24.3	1.64	4.4	.9	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.9	27.3	30.0	1.78	4.3	1.1	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.1	28.6	34.8	1.89	4.3	1.2	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.4	29.7	39.7	1.99	4.2	1.3	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.2	29.7	41.3	2.01	4.1	1.4	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.2	31.3	49.0	2.14	4.0	1.6	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.4	32.6	38.4	2.03	5.2	1.2	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.5	33.7	43.6	2.13	5.1	1.3	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.7	31.9	35.3	1.97	5.2	1.1	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.5	35.6	53.0	2.30	5.0	1.5	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.5	37.0	61.6	2.42	4.9	1.7	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.0	34.2	45.9	2.17	5.1	1.3	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.4	35.9	55.8	2.34	4.9	1.6	2.88	5.99,	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.4	37.6	62.6	2.45	5.1	1.7	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.6	38.9	73.8	2.59	4.7	1.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.0	42.0	76.9	2.71	5.5	1.8	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.1	43.7	90.8	2.87	5.3	2.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.3	44.8	102.7	2.97	5.1	2.3	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	23.4	45.9	115.1	3.07	4.9	2.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	26.2	46.9	125.8	3.14	4.8	2.7	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 16.625 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 7.273 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T		20.94	29.6	46.1	131.0	3.12	4.4	2.8	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T		23.53	33.0	47.1	142.3	3.17	4.3	3.0	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T		7.19	9.5	44.2	65.1	2.63	6.9	1.5	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T		9.52	12.2	46.3	81.4	2.84	6.7	1.8	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T		12.83	18.4	49.5	111.3	3.17	6.0	2.2	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T		10.79	14.4	48.4	94.9	3.01	6.6	2.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T		15.28	22.6	51.4	131.3	3.34	5.8	2.6	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T		12.00	17.8	50.9	112.9	3.23	6.4	2.2	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T		17.73	27.0	52.8	149.3	3.46	5.5	2.8	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T		19.79	30.6	54.1	165.7	3.56	5.4	3.1	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T		13.87	21.1	53.0	131.5	3.40	6.2	2.5	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T		22.32	34.3	55.1	179.9	3.61	5.2	3.3	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T		15.11	23.5	51.9	135.4	3.40	5.8	2.6	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T		24.83	37.9	56.2	193.8	3.65	5.1	3.4	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T		17.69	27.4	53.6	154.2	3.52	5.6	2.9	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T		28.28	42.6	57.7	212.1	3.69	5.0	3.7	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T		19.16	30.5	54.1	164.7	3.57	5.4	3.0	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T		17.26	27.2	58.3	172.3	3.74	6.3	3.0	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T		19.76	32.2	60.3	196.9	3.88	6.1	3.3	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T		9.07	13.6	56.9	113.0	3.37	8.3	2.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T		11.27	17.1	59.7	138.3	3.61	8.1	2.3	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T		12.48	19.7	62.0	157.7	3.80	8.0	2.5	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T		13.77	22.6	64.2	178.3	3.97	7.9	2.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T		15.04	26.8	66.2	202.4	4.16	7.6	3.1	4.42	10.17	5.75	.360	.240	2.44

(38T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 16.625 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 7.273 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	31.9	68.8	234.5	4.35	7.4	3.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	36.9	71.0	265.1	4.48	7.2	3.7	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	38.4	67.0	248.2	4.30	6.5	3.7	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	45.9	69.7	286.6	4.45	6.2	4.1	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	53.3	72.2	323.1	4.56	6.1	4.5	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	18.7	70.5	182.5	4.17	9.8	2.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.4	72.5	205.1	4.34	9.6	2.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.9	76.2	243.7	4.61	9.4	3.2	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	30.4	79.1	280.2	4.82	9.2	3.5	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	37.8	82.3	327.9	5.13	8.7	4.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	43.5	84.6	366.9	5.26	8.4	4.3	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	51.0	87.6	417.0	5.40	8.2	4.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	55.7	85.1	416.4	5.31	7.5	4.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	62.2	87.3	454.0	5.37	7.3	5.2	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	69.0	89.7	492.5	5.43	7.1	5.5	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	74.2	89.4	506.8	5.46	6.8	5.7	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	81.7	91.4	544.7	5.53	6.7	6.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	33.9	89.1	348.4	5.38	10.3	3.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	40.7	92.8	406.2	5.63	10.0	4.4	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	47.4	95.0	451.3	5.78	9.5	4.8	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	54.2	97.9	502.8	5.95	9.3	5.1	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	60.5	100.5	549.2	6.06	9.1	5.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	67.0	99.0	563.3	6.03	8.4	5.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	74.9	101.8	613.7	6.09	8.2	6.0	9.26	13.79	8.03	.595	.340	4.69

(38T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 16.625 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 7.273 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	44.9	104.5	506.2	6.24	11.3	4.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	54.1	109.3	590.2	6.50	10.9	5.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	63.3	112.4	659.7	6.67	10.4	5.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	71.5	115.7	726.7	6.84	10.2	6.3	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	79.8	119.1	791.7	6.90	9.9	6.6	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	88.3	122.5	857.1	6.97	9.7	7.0	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	99.6	127.2	942.4	7.02	9.5	7.4	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	121.7	129.5	1051.7	7.20	8.6	8.1	12.99	16.33	10.24	.665	.395	6.45
18 X 6 X 35.0 I-T	26.29	66.0	124.5	782.5	7.22	11.9	6.3	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	76.8	129.4	883.9	7.46	11.5	6.8	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	96.7	136.2	1042.5	7.67	10.8	7.7	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	115.6	143.7	1196.4	7.77	10.4	8.3	12.53	18.24	7.56	.695	.415	7.57
21 X 8 1/4 X 62.0 I-T	44.94	134.7	168.3	1603.3	8.85	11.9	9.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	147.7	173.4	1719.9	8.90	11.6	9.9	14.46	21.13	8.27	.685	.430	9.09

(38T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 19.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 9.500 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	4.1	22.8	15.4	1.19	3.8	.7	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	5.2	24.6	19.2	1.30	3.7	.8	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	6.2	26.3	22.8	1.40	3.7	.9	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	7.9	28.4	28.3	1.53	3.6	1.0	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	7.0	27.7	26.0	1.47	3.7	.9	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.6	30.7	25.7	1.51	4.6	.8	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	7.0	33.0	31.9	1.65	4.5	1.0	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	8.3	34.8	37.1	1.76	4.5	1.1	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	9.6	36.3	42.6	1.86	4.4	1.2	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	10.4	36.5	44.4	1.89	4.3	1.2	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	12.4	38.6	53.1	2.03	4.3	1.4	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.5	39.4	40.6	1.88	5.4	1.0	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.6	41.0	46.3	1.98	5.4	1.1	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.8	38.5	37.2	1.81	5.4	1.0	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.7	43.7	56.6	2.15	5.3	1.3	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	12.7	45.6	66.2	2.28	5.2	1.5	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	9.1	41.7	48.8	2.02	5.4	1.2	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	11.6	44.2	59.7	2.20	5.1	1.4	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	12.6	46.4	67.2	2.30	5.3	1.5	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	15.8	48.3	79.9	2.46	5.0	1.7	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	14.3	52.0	82.5	2.55	5.8	1.6	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	17.4	54.4	98.2	2.72	5.7	1.8	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	20.7	55.9	111.9	2.84	5.4	2.0	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	23.8	57.5	126.2	2.96	5.3	2.2	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	26.7	58.7	138.6	3.04	5.2	2.4	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 19.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 9.500 SQ. IN.

NOMINAL SIZE							WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
								FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT			LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8	X	21.5	T		20.94	30.2	58.0	145.6	3.05	4.8	2.5	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T		23.53	33.7	59.2	159.0	3.11	4.7	2.7	6.92	6.90	8.03	.595	.340	2.35
8	X	4	X	10.0	I-T		7.19	9.6	54.1	68.6	2.43	7.1	1.3	2.11	7.89	3.94	.205	.170	1.34
8	X	4	X	13.0	I-T		9.52	12.4	57.0	86.6	2.65	7.0	1.5	2.80	7.99	4.00	.255	.230	1.84
8	X	5 1/2	X	13.0	T		12.83	18.7	61.6	120.0	3.01	6.4	1.9	3.77	7.85	5.50	.345	.250	1.96
8	X	4	X	15.0	I-T		10.79	14.7	59.9	101.4	2.83	6.9	1.7	3.17	8.11	4.02	.315	.245	1.99
8	X	5 1/2	X	15.5	T		15.28	23.0	64.2	142.8	3.19	6.2	2.2	4.49	7.94	5.53	.440	.275	2.18
8	X	5 1/4	X	18.0	I-T		12.00	18.1	63.3	121.4	3.05	6.7	1.9	3.53	8.14	5.25	.330	.230	1.87
8	X	7	X	18.0	T		17.73	27.5	66.0	163.6	3.33	6.0	2.5	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T		19.79	31.4	67.8	182.7	3.45	5.8	2.7	5.82	8.01	7.00	.505	.305	2.44
8	X	5 1/4	X	21.0	I-T		13.87	21.5	66.1	142.2	3.24	6.6	2.2	4.08	8.28	5.27	.400	.250	2.07
8	X	7	X	22.5	T		22.32	35.0	69.1	199.3	3.52	5.7	2.9	6.56	8.07	7.04	.565	.345	2.78
8	X	6 1/2	X	24.0	I-T		15.11	23.9	65.0	147.3	3.25	6.2	2.3	4.44	7.93	6.50	.400	.245	1.94
8	X	7 1/8	X	25.0	T		24.83	38.8	70.4	215.8	3.58	5.6	3.1	7.30	8.13	7.07	.630	.380	3.09
8	X	6 1/2	X	28.0	I-T		17.69	27.9	67.2	168.9	3.39	6.0	2.5	5.20	8.06	6.54	.465	.285	2.30
8	X	7 1/8	X	28.5	T		28.28	43.7	72.1	237.3	3.65	5.4	3.3	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T		19.16	31.1	67.9	181.3	3.46	5.8	2.7	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T		17.26	27.7	72.9	187.9	3.59	6.8	2.6	5.08	8.85	6.00	.425	.300	2.66
9	X	6	X	20.0	T		19.76	32.8	75.4	216.2	3.76	6.6	2.9	5.81	8.95	6.02	.525	.315	2.82
10	X	4	X	12.0	I-T		9.07	13.8	70.3	119.6	3.14	8.7	1.7	2.67	9.87	3.96	.210	.190	1.88
10	X	4	X	15.0	I-T		11.27	17.4	74.0	147.5	3.39	8.5	2.0	3.32	9.99	4.00	.270	.230	2.30
10	X	4	X	17.0	I-T		12.48	20.1	77.1	168.9	3.58	8.4	2.2	3.67	10.11	4.01	.330	.240	2.43
10	X	4	X	19.0	I-T		13.77	23.0	80.0	191.8	3.76	8.3	2.4	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T		15.04	27.3	82.8	218.9	3.97	8.0	2.6	4.42	10.17	5.75	.360	.240	2.44

(38T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 19.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 9.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	32.5	86.2	255.4	4.18	7.9	3.0	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	37.7	88.9	290.4	4.34	7.7	3.3	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.2	84.0	273.2	4.18	7.0	3.3	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	46.9	87.4	318.0	4.36	6.8	3.6	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	54.5	90.5	360.6	4.50	6.6	4.0	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.0	87.7	194.1	3.90	10.2	2.2	3.23	11.91	3.37	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.8	90.3	219.0	4.08	10.1	2.4	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.4	95.1	261.8	4.37	9.9	2.8	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.1	98.7	302.8	4.60	9.7	3.1	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	38.5	103.2	356.7	4.93	9.3	3.5	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	44.3	106.0	401.4	5.09	9.1	3.8	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	52.1	109.6	459.1	5.27	8.8	4.2	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	56.9	106.6	461.2	5.21	8.1	4.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	63.7	109.2	505.1	5.30	7.9	4.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	70.8	111.8	549.9	5.39	7.8	4.9	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	76.0	111.8	568.2	5.44	7.5	5.1	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	83.7	114.2	613.0	5.53	7.3	5.4	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	34.6	111.3	376.0	5.14	10.9	3.4	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	41.6	116.0	441.2	5.41	10.6	3.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	48.4	118.7	492.9	5.60	10.2	4.2	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	55.4	122.3	551.9	5.80	10.0	4.5	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	62.0	125.3	605.3	5.93	9.8	4.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	68.5	123.7	624.5	5.93	9.1	5.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	76.7	126.8	683.1	6.03	8.9	5.4	9.26	13.79	8.03	.595	.340	4.69

(38T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 19.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 9.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	45.8	130.5	549.2	6.00	12.0	4.2	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	55.3	136.2	644.6	6.31	11.6	4.7	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	64.8	140.0	724.5	6.52	11.2	5.2	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	73.2	144.0	801.4	6.72	10.9	5.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	81.9	147.7	876.3	6.82	10.7	5.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	90.8	151.4	951.5	6.92	10.5	6.3	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	102.7	156.4	1049.5	7.01	10.2	6.7	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	125.1	159.9	1181.3	7.25	9.4	7.4	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	142.5	165.8	1304.3	7.30	9.2	7.9	15.00	16.52	10.30	.760	.455	7.52
18 X 6 X 35.0 I-T	26.29	67.7	154.5	857.1	7.05	12.7	5.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	78.9	160.4	972.8	7.32	12.3	6.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	99.5	168.2	1155.8	7.61	11.6	6.9	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	119.1	176.3	1332.3	7.78	11.2	7.6	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	138.8	184.8	1503.6	7.85	10.8	8.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	174.3	190.5	1719.1	8.05	9.9	9.0	17.00	18.39	11.09	.770	.480	8.83
21 X 8 1/4 X 62.0 I-T	44.94	139.0	205.7	1782.4	8.86	12.8	8.7	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	152.5	211.2	1915.3	8.94	12.6	9.1	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	163.2	215.6	2019.0	8.99	12.4	9.4	15.47	21.24	8.30	.740	.455	9.66
21 X 12 1/4 X 101.0 I-T	68.38	232.5	231.7	2536.8	9.26	10.9	10.9	20.11	21.36	12.29	.800	.500	10.68

(38T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 21.375 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 12.023 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	4	X	5.0	T	4.91	4.2	25.6	16.1	1.09	3.9	.6	1.44	3.95	3.94	.205	.170	.67
4	X	4	X	6.5	T	6.40	5.3	28.2	20.3	1.21	3.8	.7	1.88	4.00	4.00	.255	.230	.92
4	X	4	X	7.5	T	7.42	6.3	30.4	24.1	1.30	3.8	.8	2.18	4.06	4.02	.315	.245	.99
4	X	5 1/4	X	9.0	T	8.82	8.0	33.2	30.0	1.43	3.7	.9	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	7.1	32.2	27.5	1.38	3.9	.9	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	6.0	T	5.88	5.7	35.2	26.9	1.40	4.7	.8	1.73	4.94	3.96	.210	.190	.94
5	X	4	X	7.5	T	7.37	7.2	38.3	33.6	1.54	4.7	.9	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	8.4	40.7	39.1	1.64	4.7	1.0	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	9.7	42.8	45.0	1.74	4.6	1.1	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	10.5	43.2	47.1	1.78	4.5	1.1	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	12.6	46.0	56.6	1.91	4.5	1.2	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.6	45.9	42.5	1.74	5.6	.9	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	8.8	48.1	48.6	1.84	5.6	1.0	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	6.9	44.5	38.8	1.67	5.6	.9	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	10.9	51.7	59.7	2.01	5.5	1.2	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	12.9	54.4	70.2	2.15	5.4	1.3	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	9.3	49.0	51.3	1.88	5.5	1.0	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	11.8	52.5	63.1	2.06	5.4	1.2	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	12.8	55.2	71.1	2.16	5.6	1.3	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	16.1	58.0	85.1	2.33	5.3	1.5	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	14.5	62.2	87.2	2.40	6.0	1.4	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	17.6	65.4	104.4	2.57	5.9	1.6	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	21.0	67.5	119.7	2.70	5.7	1.8	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	24.2	69.7	135.7	2.83	5.6	1.9	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	27.1	71.4	149.7	2.92	5.5	2.1	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 21.375 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 12.023 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	30.7	70.7	158.3	2.95	5.2	2.2	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	34.4	72.2	173.7	3.03	5.1	2.4	6.92	6.98	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.8	63.7	71.5	2.25	7.3	1.1	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.6	67.8	90.9	2.48	7.2	1.3	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.0	74.2	127.3	2.84	6.7	1.7	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.9	71.6	106.8	2.65	7.2	1.5	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.3	77.7	152.5	3.04	6.5	2.0	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.3	76.2	128.4	2.87	7.0	1.7	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	27.9	80.2	175.8	3.19	6.3	2.2	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	31.9	82.6	197.3	3.33	6.2	2.4	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	21.8	79.9	151.2	3.06	7.0	1.9	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	35.7	84.2	216.2	3.41	6.1	2.6	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.2	78.8	157.5	3.09	6.5	2.0	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	39.5	85.8	235.1	3.49	6.0	2.7	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.4	81.6	181.6	3.25	6.4	2.2	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	44.6	87.8	259.9	3.57	5.8	3.0	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	31.6	82.7	195.7	3.33	6.2	2.4	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.2	88.4	201.2	3.43	7.1	2.3	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.4	91.8	232.9	3.61	7.0	2.5	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.0	84.0	125.1	2.92	8.9	1.5	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.6	86.9	155.1	3.18	8.8	1.7	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.4	93.0	178.2	3.37	8.8	1.9	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.3	96.8	203.1	3.55	8.7	2.1	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	27.7	100.5	232.8	3.76	8.4	2.3	4.42	10.17	5.75	.360	.240	2.44

(38T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 21.375 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 12.023 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	33.0	104.8	273.1	3.99	8.3	2.6	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.3	108.2	312.3	4.17	8.1	2.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.8	102.5	294.9	4.03	7.4	2.9	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	47.7	106.8	345.5	4.24	7.2	3.2	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	55.5	110.5	394.1	4.40	7.1	3.6	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.3	105.7	203.6	3.65	10.5	1.9	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.1	109.1	230.5	3.84	10.4	2.1	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.8	115.2	276.8	4.13	10.3	2.4	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.6	119.8	321.7	4.37	10.2	2.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.0	125.9	381.0	4.70	9.8	3.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.0	129.4	430.9	4.89	9.6	3.3	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	53.0	133.6	495.8	5.10	9.4	3.7	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	57.8	130.2	500.6	5.06	8.7	3.8	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	64.8	133.2	550.5	5.18	8.5	4.1	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	72.2	136.3	601.7	5.29	8.3	4.4	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	77.4	136.6	623.6	5.36	8.1	4.6	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	85.4	139.4	675.3	5.47	7.9	4.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.16	35.1	135.4	399.1	4.88	11.4	2.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.3	141.2	470.9	5.18	11.1	3.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.2	144.7	528.7	5.38	10.7	3.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	56.3	149.1	594.6	5.60	10.6	4.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	63.1	152.6	654.7	5.76	10.4	4.3	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	69.8	151.0	678.6	5.79	9.7	4.5	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	78.2	154.5	745.3	5.92	9.5	4.8	9.26	13.79	8.03	.595	.340	4.69

(38T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 21.375 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 12.023 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	46.6	158.9	585.9	5.74	12.6	3.7	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	56.3	165.8	691.5	6.08	12.3	4.2	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	66.0	170.4	781.2	6.32	11.8	4.6	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	74.6	175.3	867.4	6.54	11.6	4.9	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	83.6	179.4	952.1	6.68	11.4	5.3	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	92.8	183.6	1037.2	6.80	11.2	5.7	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	105.1	189.1	1148.1	6.93	10.9	6.1	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	127.9	193.8	1301.9	7.21	10.2	6.7	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	146.0	200.2	1442.2	7.31	9.9	7.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	167.5	208.0	1606.5	7.39	9.6	7.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	69.1	187.7	922.5	6.83	13.3	4.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	80.5	194.8	1051.6	7.14	13.1	5.4	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	101.7	203.9	1258.5	7.48	12.4	6.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	122.0	212.8	1458.2	7.71	12.0	6.9	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	142.5	221.8	1651.4	7.83	11.6	7.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	178.8	228.9	1902.6	8.10	10.6	8.3	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	200.6	236.5	2078.9	8.16	10.4	8.8	19.18	18.59	11.15	.870	.535	9.95
21 X 8 1/4 X 62.0 I-T	44.94	142.4	247.6	1949.0	8.79	13.7	7.9	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	156.4	253.6	2098.9	8.90	13.4	8.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	167.5	258.3	2215.8	8.98	13.2	8.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	188.3	267.3	2429.2	9.06	12.9	9.1	17.58	21.43	8.36	.835	.515	11.04
21 X 12 1/4 X 101.0 I-T	68.38	238.9	276.7	2810.5	9.35	11.8	10.2	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	260.3	284.3	2999.6	9.37	11.5	10.5	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 23.750 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 14.844 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 6.5 T		6.40	5.4	31.2	21.2	1.13	3.9	.7	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T		7.42	6.4	34.0	25.3	1.22	3.9	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T		8.82	8.2	37.6	31.6	1.35	3.9	.8	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T		8.39	7.2	36.2	28.9	1.29	4.0	.8	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T		5.88	5.8	39.0	27.9	1.30	4.8	.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T		7.37	7.3	43.1	35.0	1.43	4.8	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T		8.36	8.5	46.2	40.9	1.54	4.8	.9	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T		9.42	9.9	49.0	47.2	1.64	4.8	1.0	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T		9.91	10.7	49.5	49.5	1.67	4.6	1.0	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T		11.69	12.8	53.3	59.6	1.81	4.7	1.1	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T		6.94	7.7	51.7	44.2	1.62	5.7	.9	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T		7.88	8.9	54.6	50.6	1.72	5.7	.9	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T		6.17	7.0	49.9	40.2	1.55	5.7	.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T		9.34	11.0	59.3	62.4	1.88	5.7	1.1	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T		10.89	13.1	62.9	73.5	2.02	5.6	1.2	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T		8.30	9.4	55.8	53.5	1.76	5.7	1.0	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T		9.78	11.9	60.5	66.0	1.93	5.5	1.1	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T		10.74	13.0	63.8	74.5	2.03	5.7	1.2	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T		12.63	16.3	67.6	89.6	2.20	5.5	1.3	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T		10.81	14.6	72.2	91.3	2.25	6.2	1.3	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T		12.85	17.8	76.5	109.7	2.43	6.2	1.4	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T		14.81	21.2	79.4	126.4	2.57	6.0	1.6	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T		16.77	24.5	82.4	143.9	2.70	5.9	1.7	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T		18.74	27.5	84.5	159.4	2.80	5.8	1.9	5.51	7.05	6.77	.515	.310	2.19

(38T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 23.750 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 14.844 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	31.1	84.0	169.4	2.84	5.4	2.0	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	34.9	86.0	186.7	2.93	5.4	2.2	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.9	72.7	74.0	2.09	7.5	1.0	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.8	78.3	94.5	2.31	7.4	1.2	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.2	86.9	133.6	2.68	6.9	1.5	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.1	83.2	111.4	2.49	7.4	1.3	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.6	91.6	160.8	2.88	6.8	1.8	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.5	89.2	134.4	2.70	7.3	1.5	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.3	95.0	186.4	3.05	6.6	2.0	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	32.3	98.0	210.0	3.19	6.5	2.1	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.0	94.0	158.9	2.90	7.2	1.7	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	36.2	100.1	231.1	3.29	6.4	2.3	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.5	93.0	166.1	2.93	6.8	1.8	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	40.1	102.1	252.2	3.37	6.3	2.5	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.7	96.7	192.4	3.10	6.7	2.0	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	45.4	104.7	280.0	3.48	6.2	2.7	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.0	98.2	208.1	3.19	6.5	2.1	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.6	104.6	212.6	3.27	7.4	2.0	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.8	108.9	247.2	3.46	7.3	2.3	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.1	97.4	129.6	2.72	9.2	1.3	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.8	104.0	161.6	2.98	9.1	1.6	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.6	109.1	186.1	3.17	9.0	1.7	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.6	114.0	212.6	3.35	9.0	1.9	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.0	118.9	244.6	3.56	8.7	2.1	4.42	10.17	5.75	.360	.240	2.44

(38T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 23.750 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 14.844 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	33.4	124.5	288.2	3.80	8.6	2.3	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.8	128.7	331.1	3.99	8.5	2.6	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.3	122.1	313.6	3.86	7.8	2.6	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.3	127.5	369.5	4.10	7.6	2.9	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	56.4	132.0	423.6	4.28	7.5	3.2	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.5	123.9	211.5	3.42	10.8	1.7	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.3	128.4	240.1	3.60	10.7	1.9	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.2	136.2	289.5	3.90	10.7	2.1	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.0	142.0	337.7	4.15	10.6	2.4	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.5	149.9	401.6	4.48	10.2	2.7	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.6	154.2	456.2	4.68	10.0	3.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	53.7	159.4	527.6	4.91	9.8	3.3	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	58.6	155.7	534.9	4.89	9.1	3.4	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	65.8	159.2	590.5	5.03	9.0	3.7	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	73.3	162.9	647.8	5.16	8.8	4.0	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	78.6	163.5	673.2	5.24	8.6	4.1	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	86.7	167.0	731.4	5.37	8.4	4.4	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.6	160.8	418.5	4.62	11.8	2.6	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.8	168.0	496.2	4.93	11.6	3.0	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.8	172.5	559.3	5.15	11.2	3.2	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	57.1	177.9	631.4	5.39	11.1	3.5	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	64.0	182.2	697.7	5.56	10.9	3.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	70.7	180.6	726.1	5.61	10.3	4.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	79.4	184.7	800.5	5.76	10.1	4.3	9.26	13.79	8.03	.595	.340	4.69

(38T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 23.750 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 14.844 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	47.3	189.2	617.0	5.48	13.1	3.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	57.2	197.7	731.8	5.83	12.8	3.7	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	67.0	203.4	830.5	6.09	12.4	4.1	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	75.8	209.3	925.3	6.33	12.2	4.4	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	85.0	213.9	1019.3	6.49	12.0	4.8	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	94.5	218.7	1114.0	6.64	11.8	5.1	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	107.2	224.8	1237.6	6.81	11.5	5.5	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	130.2	231.1	1412.5	7.12	10.8	6.1	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	148.9	238.1	1570.6	7.26	10.5	6.6	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	171.2	246.5	1755.5	7.38	10.3	7.1	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	70.2	223.7	979.4	6.59	13.9	4.4	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	81.8	232.3	1120.9	6.91	13.7	4.8	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	103.5	242.9	1350.5	7.31	13.1	5.6	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	124.4	252.9	1573.1	7.58	12.6	6.2	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	145.7	262.5	1788.7	7.75	12.3	6.8	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	182.6	271.4	2075.3	8.07	11.4	7.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	205.1	279.6	2273.5	8.17	11.1	8.1	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	222.0	286.0	2419.4	8.21	10.9	8.5	21.02	18.73	11.20	.940	.590	11.05
21 X 8 1/4 X 62.0 I-T	44.94	145.3	293.8	2101.6	8.65	14.5	7.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	159.7	300.4	2268.7	8.80	14.2	7.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	171.2	305.5	2399.0	8.90	14.0	7.9	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	192.7	315.0	2636.6	9.02	13.7	8.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	214.6	324.8	2874.2	9.10	13.4	8.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	244.3	326.4	3072.0	9.37	12.6	9.4	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	266.5	334.5	3283.2	9.42	12.3	9.8	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 26.125 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 17.961 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 7.5 T	7.42	6.5	37.1	26.4	1.14	4.0	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	8.3	41.6	33.0	1.27	4.0	.8	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	7.4	39.8	30.2	1.22	4.1	.8	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 7.5 T	7.37	7.4	47.3	36.4	1.34	4.9	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.7	51.1	42.6	1.44	4.9	.8	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	10.0	54.6	49.2	1.54	4.9	.9	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.8	55.4	51.6	1.57	4.8	.9	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	13.0	60.1	62.4	1.71	4.8	1.0	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.8	56.8	45.7	1.51	5.8	.8	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	9.0	60.4	52.4	1.61	5.8	.9	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	7.1	54.5	41.6	1.45	5.8	.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	11.2	66.4	64.7	1.77	5.8	1.0	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.3	70.9	76.5	1.90	5.8	1.1	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.5	61.9	55.4	1.65	5.8	.9	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	12.1	67.9	68.6	1.81	5.7	1.0	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.2	71.9	77.5	1.92	5.9	1.1	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.5	76.9	93.6	2.08	5.7	1.2	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.8	81.8	94.8	2.12	6.4	1.2	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.0	87.3	114.4	2.29	6.3	1.3	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.5	91.2	132.3	2.43	6.2	1.5	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	24.8	95.0	151.1	2.57	6.1	1.6	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	27.9	97.8	167.8	2.67	6.0	1.7	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	31.5	97.6	179.2	2.73	5.7	1.8	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	35.3	100.1	198.2	2.82	5.6	2.0	6.92	6.90	8.03	.595	.340	2.35

(38T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 26.125 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 17.961 SQ. IN.

NOMINAL SIZE							WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
								FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT			LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8	X	4		X	10.0	I-T	7.19	10.0	80.8	76.2	1.95	7.6	.9	2.11	7.89	3.94	.205	.170	1.34
8	X	4		X	13.0	I-T	9.52	12.9	88.2	97.7	2.17	7.6	1.1	2.80	7.99	4.00	.255	.230	1.84
8	X	5	1/2	X	13.0	T	12.83	19.5	99.4	139.0	2.53	7.1	1.4	3.77	7.85	5.50	.345	.250	1.96
8	X	4		X	15.0	I-T	10.79	15.2	94.4	115.4	2.34	7.6	1.2	3.17	8.11	4.02	.315	.245	1.99
8	X	5	1/2	X	15.5	T	15.28	23.9	105.5	168.0	2.74	7.0	1.6	4.49	7.94	5.53	.440	.275	2.18
8	X	5	1/4	X	18.0	I-T	12.00	18.7	101.9	139.6	2.55	7.5	1.4	3.53	8.14	5.25	.330	.230	1.87
8	X	7		X	18.0	T	17.73	28.6	109.9	195.5	2.90	6.8	1.8	5.22	7.93	6.99	.430	.295	2.34
8	X	7		X	20.0	T	19.79	32.7	113.9	221.1	3.05	6.8	1.9	5.82	8.01	7.00	.505	.305	2.44
8	X	5	1/4	X	21.0	I-T	13.87	22.3	108.0	165.5	2.74	7.4	1.5	4.08	8.28	5.27	.400	.250	2.07
8	X	7		X	22.5	T	22.32	36.6	116.6	244.1	3.15	6.7	2.1	6.56	8.07	7.04	.565	.345	2.78
8	X	6	1/2	X	24.0	I-T	15.11	24.8	107.2	173.6	2.78	7.0	1.6	4.44	7.93	6.50	.400	.245	1.94
8	X	7	1/8	X	25.0	T	24.83	40.7	119.1	267.2	3.25	6.6	2.2	7.30	8.13	7.07	.630	.380	3.09
8	X	6	1/2	X	28.0	I-T	17.69	29.1	112.0	201.9	2.95	6.9	1.8	5.20	8.06	6.54	.465	.285	2.30
8	X	7	1/8	X	28.5	T	28.28	46.0	122.3	296.0	3.37	6.5	2.4	8.32	8.22	7.12	.715	.430	3.53
8	X	8		X	31.0	I-T	19.16	32.3	114.1	218.9	3.05	6.8	1.9	5.63	8.00	8.00	.435	.285	2.28
9	X	6		X	17.5	T	17.26	28.9	121.0	222.4	3.11	7.7	1.8	5.08	8.85	6.00	.425	.300	2.66
9	X	6		X	20.0	T	19.76	34.2	126.8	259.8	3.31	7.6	2.1	5.81	8.95	6.02	.525	.315	2.82
10	X	4		X	12.0	I-T	9.07	14.3	110.2	133.6	2.54	9.3	1.2	2.67	9.87	3.96	.210	.190	1.88
10	X	4		X	15.0	I-T	11.27	18.0	118.6	167.1	2.80	9.3	1.4	3.32	9.99	4.00	.270	.230	2.30
10	X	4		X	17.0	I-T	12.48	20.8	125.1	192.8	2.99	9.3	1.5	3.67	10.11	4.01	.330	.240	2.43
10	X	4		X	19.0	I-T	13.77	23.9	131.2	220.7	3.17	9.2	1.7	4.05	10.24	4.02	.395	.250	2.56
10	X	5	3/4	X	22.0	I-T	15.04	28.3	137.5	254.6	3.37	9.0	1.9	4.42	10.17	5.75	.360	.240	2.44
10	X	5	3/4	X	26.0	I-T	17.37	33.7	144.6	301.2	3.61	8.9	2.1	5.11	10.33	5.77	.440	.260	2.69
10	X	5	3/4	X	30.0	I-T	20.23	39.3	149.9	347.4	3.81	8.8	2.3	5.95	10.47	5.81	.510	.300	3.14
10	X	8		X	33.0	I-T	20.94	40.7	142.4	329.8	3.70	8.1	2.3	6.16	9.73	7.96	.435	.290	2.82
10	X	8		X	39.0	I-T	24.45	48.9	149.1	390.6	3.94	8.0	2.6	7.19	9.92	7.99	.530	.315	3.12

(38T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 26.125 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 17.961 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 45.0 I-T	28.19	57.1	154.8	449.8	4.14	7.9	2.9	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.7	141.9	218.2	3.21	11.1	1.5	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.6	147.6	248.3	3.39	11.0	1.7	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.4	157.4	300.3	3.68	10.9	1.9	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.3	164.7	351.4	3.93	10.9	2.1	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.9	174.8	419.2	4.26	10.5	2.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.1	180.2	478.8	4.47	10.4	2.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.4	186.7	555.3	4.72	10.2	3.0	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.2	182.6	564.9	4.71	9.5	3.1	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	66.6	186.8	625.9	4.87	9.4	3.3	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	74.3	191.2	688.8	5.01	9.3	3.6	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	79.6	192.2	717.3	5.09	9.0	3.7	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	87.8	196.4	781.6	5.23	8.9	4.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.0	187.1	435.1	4.38	12.1	2.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.3	196.1	517.9	4.69	12.0	2.6	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.4	201.6	585.8	4.92	11.6	2.9	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	57.8	208.4	663.4	5.16	11.5	3.2	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	64.8	213.6	735.3	5.35	11.3	3.4	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	71.6	212.0	767.8	5.41	10.7	3.6	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	80.4	216.9	849.4	5.59	10.6	3.9	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	47.8	221.0	643.6	5.21	13.5	2.9	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	57.8	231.4	766.7	5.58	13.3	3.3	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	67.8	238.4	873.3	5.85	12.9	3.7	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	76.7	245.6	975.9	6.10	12.7	4.0	8.26	16.01	7.00	.505	.305	4.88

(38T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 26.125 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 17.961 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 45.0 I-T	31.77	86.2	250.9	1078.7	6.29	12.5	4.3	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	95.9	256.4	1182.5	6.46	12.3	4.6	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	108.9	263.3	1318.4	6.65	12.1	5.0	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	132.2	271.5	1512.8	6.99	11.4	5.6	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	151.3	279.3	1688.6	7.16	11.2	6.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	174.3	288.5	1894.4	7.32	10.9	6.6	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	71.1	262.1	1028.9	6.33	14.5	3.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	82.9	272.4	1181.7	6.67	14.2	4.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	104.9	284.9	1432.4	7.10	13.7	5.0	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	126.4	296.2	1676.9	7.42	13.3	5.7	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	148.3	306.7	1914.8	7.63	12.9	6.2	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	185.7	317.6	2235.9	8.00	12.0	7.0	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	209.0	326.5	2456.4	8.13	11.8	7.5	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	226.5	333.3	2618.4	8.20	11.6	7.9	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	254.1	344.1	2873.3	8.31	11.3	8.4	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	147.7	343.9	2240.0	8.48	15.2	6.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	162.5	351.2	2424.0	8.65	14.9	6.9	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	174.3	356.8	2567.6	8.76	14.7	7.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	196.4	367.0	2829.8	8.92	14.4	7.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	219.0	377.4	3091.6	9.04	14.1	8.2	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	248.9	380.8	3318.6	9.34	13.3	8.7	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	271.8	389.3	3552.9	9.41	13.1	9.1	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 28.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 21.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 5 1/4 X 9.0 T	8.82	8.5	45.0	34.4	1.20	4.1	.8	2.59	4.07	5.25	.330	.230	.94
5 X 4 X 8.5 T	8.36	8.8	55.3	44.1	1.36	5.0	.8	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	10.2	59.5	51.0	1.45	5.0	.9	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	11.0	60.6	53.6	1.49	4.9	.9	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	13.2	66.3	64.9	1.62	4.9	1.0	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.9	61.1	47.1	1.42	5.9	.8	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	9.1	65.5	54.1	1.51	5.9	.8	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.5 T	9.34	11.3	72.7	66.9	1.67	5.9	.9	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.4	78.3	79.3	1.80	5.9	1.0	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.6	67.3	57.2	1.55	5.9	.9	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	12.3	74.6	70.9	1.71	5.8	1.0	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.3	79.4	80.2	1.81	6.0	1.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.7	85.6	97.1	1.97	5.8	1.1	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	15.0	90.7	98.0	2.00	6.5	1.1	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.3	97.6	118.5	2.17	6.5	1.2	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.7	102.6	137.5	2.31	6.3	1.3	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.1	107.4	157.5	2.45	6.3	1.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	28.2	111.0	175.3	2.55	6.2	1.6	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	31.9	111.2	187.9	2.61	5.9	1.7	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	35.8	114.5	208.4	2.71	5.8	1.8	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	10.1	87.9	78.3	1.83	7.7	.9	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	13.1	97.2	100.6	2.04	7.7	1.0	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.7	111.4	143.8	2.39	7.3	1.3	3.77	7.85	5.50	.345	.250	1.96

(38T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 28.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 21.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 15.0 I-T	10.79	15.4	104.7	119.0	2.20	7.7	1.1	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.26	24.1	119.1	174.4	2.60	7.2	1.5	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.80	18.9	114.0	144.1	2.41	7.6	1.3	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.9	124.8	203.6	2.77	7.0	1.6	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.1	129.8	230.9	2.91	7.0	1.8	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.5	121.6	171.4	2.59	7.6	1.4	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.0	133.2	255.6	3.02	6.9	1.9	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.0	121.2	180.1	2.64	7.2	1.5	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.1	136.4	280.6	3.13	6.8	2.1	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.4	127.2	210.2	2.81	7.2	1.7	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	46.6	140.4	314.0	3.25	6.7	2.2	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.7	130.0	228.4	2.91	7.0	1.8	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.2	137.4	231.1	2.96	7.9	1.7	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.6	144.4	270.8	3.16	7.8	1.9	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.4	122.0	137.1	2.39	9.5	1.1	2.67	9.87	3.96	.210	.190	1.86
10 X 4 X 15.0 I-T	11.27	18.2	132.6	171.9	2.64	9.4	1.3	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.0	140.6	198.7	2.82	9.4	1.4	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.1	148.1	227.9	2.99	9.5	1.5	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.5	156.0	263.4	3.20	9.2	1.7	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.0	164.8	312.6	3.44	9.2	1.9	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	39.7	171.5	361.6	3.64	9.1	2.1	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.1	163.1	344.1	3.54	8.4	2.1	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	49.4	171.5	409.1	3.78	8.3	2.4	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	57.7	178.4	472.9	3.99	8.2	2.7	8.29	10.10	8.02	.620	.350	3.54

(38T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 28.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 21.375 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 14.0 I-T	10.98	19.9	159.2	224.1	3.02	11.3	1.4	3.23	11.91	3.97	.225	.200	2.36
12 X 4 X 16.0 I-T	12.37	22.8	166.4	255.5	3.20	11.2	1.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.7	178.4	309.6	3.48	11.2	1.7	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.7	187.6	363.3	3.73	11.1	1.9	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.2	200.0	434.5	4.04	10.8	2.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.5	206.9	496.9	4.26	10.7	2.4	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.9	214.8	579.5	4.52	10.6	2.7	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.8	210.6	591.2	4.53	9.9	2.8	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	67.3	215.7	657.0	4.69	9.8	3.0	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	75.1	220.9	725.2	4.85	9.7	3.3	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	80.4	222.4	756.5	4.93	9.4	3.4	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	88.8	227.6	826.5	5.09	9.3	3.6	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.3	213.6	449.3	4.15	12.4	2.1	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.7	224.7	536.6	4.46	12.3	2.4	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.9	231.8	608.7	4.70	12.0	2.6	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	58.3	240.1	691.2	4.94	11.9	2.9	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	65.5	246.3	768.2	5.14	11.7	3.1	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	72.3	245.0	804.4	5.21	11.1	3.3	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	81.3	250.9	892.7	5.40	11.0	3.6	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.3	253.8	666.6	4.96	13.8	2.6	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	58.4	266.5	796.9	5.33	13.6	3.0	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	68.5	275.1	910.7	5.61	13.3	3.3	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	77.5	283.8	1020.3	5.87	13.2	3.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	87.1	290.1	1131.2	6.07	13.0	3.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	97.0	296.4	1243.4	6.26	12.8	4.2	10.39	16.26	7.07	.630	.380	6.18

(38T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 28.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 21.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 1/8 X 57.0 I-T	40.28	110.3	304.4	1390.9	6.47	12.6	4.6	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	133.8	314.7	1603.4	6.83	12.0	5.1	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	153.3	323.4	1796.4	7.03	11.7	5.6	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	176.9	333.6	2022.9	7.22	11.4	6.1	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	71.9	302.5	1072.1	6.07	14.9	3.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	83.9	314.7	1235.0	6.42	14.7	3.9	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	106.2	329.6	1505.2	6.88	14.2	4.6	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	128.1	342.6	1770.4	7.23	13.8	5.2	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	150.5	354.1	2029.9	7.48	13.5	5.7	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	188.4	367.3	2383.8	7.88	12.7	6.5	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	212.2	377.1	2626.6	8.05	12.4	7.0	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	230.2	384.4	2805.0	8.13	12.2	7.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	258.7	396.0	3085.8	8.28	11.9	7.8	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	149.8	397.6	2364.9	8.27	15.8	5.9	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	164.9	405.8	2565.1	8.46	15.6	6.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	176.9	412.1	2721.6	8.60	15.4	6.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	199.6	423.0	3008.1	8.79	15.1	7.1	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	222.9	434.1	3294.4	8.94	14.8	7.6	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	252.8	439.7	3548.6	9.25	14.0	8.1	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	276.3	448.7	3806.7	9.35	13.8	8.5	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 33.250 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 29.094 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 5 X 19.0 I-T	11.69	13.6	76.9	69.6	1.46	5.1	.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 11.0 T	10.89	13.8	90.8	84.3	1.62	6.1	.9	3.20	6.16	4.03	.425	.260	1.60
6 X 6 X 15.0 I-T	9.78	12.6	85.6	75.3	1.54	6.0	.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.7	91.9	85.2	1.63	6.2	.9	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	17.1	101.0	103.5	1.78	6.1	1.0	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	15.3	105.8	103.7	1.79	6.8	1.0	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.7	115.8	125.9	1.96	6.7	1.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	22.2	123.4	146.6	2.09	6.6	1.2	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.6	130.6	168.5	2.23	6.6	1.3	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	28.8	136.2	188.3	2.33	6.5	1.4	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	32.6	137.6	202.9	2.40	6.2	1.5	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	36.5	142.6	226.1	2.51	6.2	1.6	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 13.0 I-T	9.52	13.4	112.2	105.8	1.82	7.9	.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	20.1	132.8	152.1	2.15	7.6	1.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.7	122.6	125.4	1.97	8.0	1.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.6	144.3	185.4	2.35	7.5	1.3	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.3	135.4	152.1	2.16	7.9	1.1	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	29.5	153.0	217.5	2.52	7.4	1.4	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.7	160.6	247.6	2.66	7.3	1.5	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.9	146.5	181.5	2.34	7.9	1.2	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.8	166.0	275.3	2.78	7.3	1.7	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.5	147.1	191.5	2.39	7.5	1.3	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	42.0	171.0	303.5	2.89	7.2	1.8	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.9	156.2	224.4	2.56	7.5	1.4	5.20	8.06	6.54	.465	.285	2.30

(38T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 33.250 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 29.094 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 7 1/8 X 28.5 T	28.28	47.6	177.1	341.4	3.02	7.2	1.9	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.3	160.7	244.7	2.65	7.4	1.5	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.7	168.6	245.9	2.68	8.3	1.5	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.3	179.0	289.4	2.88	8.2	1.6	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.7	142.0	143.2	2.12	9.7	1.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.6	157.4	180.3	2.36	9.7	1.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.4	168.6	208.8	2.52	9.7	1.2	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.5	179.4	240.0	2.69	9.8	1.3	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	29.0	191.0	278.2	2.88	9.6	1.5	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.6	204.0	331.6	3.11	9.6	1.6	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	40.4	214.2	385.5	3.32	9.5	1.8	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.8	204.0	368.0	3.23	8.8	1.8	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	50.3	216.6	440.3	3.48	8.8	2.0	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	58.7	226.9	512.1	3.70	8.7	2.3	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.3	190.3	234.1	2.69	11.6	1.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.2	200.9	267.5	2.86	11.5	1.3	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.2	218.3	325.2	3.13	11.5	1.5	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.2	231.8	383.1	3.36	11.5	1.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.8	249.9	459.7	3.66	11.3	1.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.2	260.4	528.3	3.88	11.2	2.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	55.8	272.4	619.7	4.14	11.1	2.3	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	60.8	268.0	635.1	4.17	10.4	2.4	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	68.5	275.6	709.4	4.35	10.4	2.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	76.5	283.2	786.7	4.52	10.3	2.8	9.44	12.19	8.08	.640	.370	4.51

(38T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 33.250 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 29.094 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 10 X 53.0 I-T	33.01	81.8	286.0	822.8	4.60	10.1	2.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	90.4	293.4	902.9	4.77	10.0	3.1	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.8	265.5	472.9	3.74	12.8	1.8	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	44.4	282.0	567.5	4.05	12.8	2.0	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	51.7	292.8	646.5	4.28	12.5	2.2	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	59.3	305.0	737.3	4.52	12.4	2.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	66.6	314.3	822.8	4.73	12.4	2.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	73.5	313.8	865.5	4.81	11.8	2.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	82.7	322.2	965.3	5.02	11.7	3.0	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	49.0	320.1	704.3	4.50	14.4	2.2	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	59.4	338.7	846.6	4.86	14.3	2.5	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	69.6	351.3	972.5	5.15	14.0	2.8	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	78.8	364.0	1094.0	5.41	13.9	3.0	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	88.7	372.8	1219.0	5.63	13.7	3.3	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	98.9	381.7	1346.0	5.84	13.6	3.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	112.6	392.4	1514.3	6.08	13.4	3.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	136.4	408.0	1758.7	6.46	12.9	4.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	156.7	419.2	1983.7	6.71	12.7	4.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	181.1	432.1	2249.4	6.96	12.4	5.2	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	73.2	386.8	1143.4	5.57	15.6	3.0	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	85.4	404.2	1323.4	5.92	15.5	3.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	108.2	425.2	1627.5	6.41	15.0	3.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	130.8	442.6	1930.0	6.81	14.8	4.4	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	154.1	457.1	2229.6	7.12	14.5	4.9	14.93	18.47	7.64	.810	.495	9.14

(38T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 33.250 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 29.094 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/8 X 86.0 I-T	57.79	192.8	476.2	2643.5	7.57	13.7	5.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	217.5	488.3	2929.2	7.79	13.5	6.0	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	236.4	496.9	3140.1	7.92	13.3	6.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	266.1	510.8	3472.3	8.11	13.0	6.8	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	153.0	514.3	2578.4	7.81	16.9	5.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	168.6	524.9	2808.3	8.03	16.7	5.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	181.1	532.8	2988.9	8.19	16.5	5.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	204.8	546.1	3321.7	8.44	16.2	6.1	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	229.1	559.1	3655.7	8.64	16.0	6.5	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	259.0	569.4	3958.6	8.97	15.3	7.0	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	283.6	579.9	4263.8	9.12	15.0	7.4	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 38.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 38.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 5 X 13.0 T	12.85	19.1	130.4	132.6	1.78	6.9	1.0	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	22.7	140.7	154.7	1.91	6.8	1.1	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	26.2	150.6	178.1	2.04	6.8	1.2	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	29.4	158.4	199.5	2.14	6.8	1.3	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	33.2	161.4	215.8	2.21	6.5	1.3	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	37.3	168.7	241.2	2.32	6.5	1.4	6.92	6.90	8.03	.595	.340	2.35
8 X 5 1/2 X 13.0 T	12.83	20.5	150.2	159.6	1.95	7.8	1.1	3.77	7.85	5.50	.345	.250	1.96
8 X 5 1/2 X 15.5 T	15.28	25.1	165.7	194.9	2.14	7.8	1.2	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.7	152.6	159.3	1.96	8.1	1.0	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	30.0	178.0	229.4	2.30	7.6	1.3	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	34.3	188.6	261.7	2.44	7.6	1.4	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	23.4	167.3	190.3	2.13	8.1	1.1	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	38.5	196.4	291.9	2.56	7.6	1.5	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	26.0	169.3	201.3	2.18	7.7	1.2	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	42.8	203.7	322.8	2.67	7.5	1.6	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	30.5	181.9	236.6	2.34	7.8	1.3	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	48.6	212.5	364.5	2.81	7.5	1.7	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.9	188.4	258.5	2.43	7.6	1.4	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	30.3	196.2	258.4	2.45	8.5	1.3	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.9	210.7	305.1	2.64	8.5	1.4	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 15.0 I-T	11.27	18.9	177.1	187.8	2.13	9.9	1.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.8	191.7	217.6	2.29	10.0	1.1	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	25.0	206.0	250.4	2.44	10.0	1.2	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	29.5	221.6	290.7	2.62	9.9	1.3	4.42	10.17	5.75	.360	.240	2.44

(38T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 38.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 38.000 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	35.2	239.6	347.3	2.84	9.9	1.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	41.0	254.0	405.2	3.04	9.9	1.6	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	42.5	242.5	387.8	2.96	9.1	1.6	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	51.0	260.1	465.9	3.21	9.1	1.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	59.7	274.7	544.3	3.43	9.1	2.0	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.6	215.6	242.7	2.43	11.8	1.1	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.6	229.9	277.7	2.58	11.8	1.2	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.6	253.1	338.3	2.83	11.8	1.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.7	271.9	399.4	3.05	11.8	1.5	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	41.4	296.4	480.2	3.33	11.6	1.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.9	311.5	553.6	3.55	11.6	1.8	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	56.6	328.7	652.2	3.81	11.5	2.0	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	61.7	324.8	670.5	3.84	10.9	2.1	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	69.5	335.9	751.8	4.02	10.8	2.2	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	77.6	346.7	836.7	4.20	10.8	2.4	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	83.0	351.1	876.7	4.29	10.6	2.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	91.7	361.6	965.1	4.46	10.5	2.7	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.4	313.2	492.0	3.39	13.2	1.6	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.0	336.5	592.3	3.69	13.1	1.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	52.4	352.1	676.9	3.91	12.9	1.9	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	60.1	369.2	774.2	4.15	12.9	2.1	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	67.5	382.4	866.6	4.35	12.8	2.3	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	74.5	383.4	914.5	4.45	12.3	2.4	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	83.9	395.6	1023.8	4.65	12.2	2.6	9.26	13.79	8.03	.595	.340	4.69

(38T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 38.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 38.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	49.7	384.1	734.3	4.10	14.8	1.9	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	60.2	410.1	886.1	4.45	14.7	2.2	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	70.6	428.3	1021.7	4.74	14.5	2.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	79.9	445.9	1152.6	4.99	14.4	2.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	90.0	458.5	1289.2	5.22	14.3	2.8	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	100.4	470.9	1428.7	5.43	14.2	3.0	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	114.5	485.6	1614.7	5.69	14.1	3.3	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	138.5	507.8	1885.6	6.08	13.6	3.7	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	159.3	522.8	2138.7	6.35	13.4	4.1	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	184.4	539.6	2439.7	6.64	13.2	4.5	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	74.2	472.7	1199.8	5.12	16.2	2.5	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	86.6	496.6	1393.7	5.47	16.1	2.8	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	109.8	526.0	1725.5	5.97	15.7	3.3	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	132.9	549.6	2059.6	6.38	15.5	3.7	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	156.9	568.7	2394.5	6.73	15.3	4.2	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	196.1	595.3	2860.4	7.21	14.6	4.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	221.6	610.7	3185.4	7.46	14.4	5.2	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	241.1	621.2	3426.8	7.62	14.2	5.5	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	271.9	638.3	3807.7	7.86	14.0	6.0	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	155.5	640.5	2751.8	7.33	17.7	4.3	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	171.5	654.4	3007.4	7.57	17.5	4.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	184.3	664.6	3209.3	7.75	17.4	4.8	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.76	208.7	681.3	3584.0	8.03	17.2	5.3	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	233.9	697.2	3962.0	8.28	16.9	5.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	263.8	713.2	4306.2	8.61	16.3	6.0	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	289.3	725.9	4655.9	8.80	16.1	6.4	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 42.750 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 48.094 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	6 3/4	X	19.0	T	18.74	30.0	177.0	209.9	1.98	7.0	1.2	5.51	7.05	6.77	.515	.310	2.19
7	X	8	X	21.5	T	20.94	33.9	182.0	227.6	2.05	6.7	1.3	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T	23.53	38.1	191.7	254.8	2.15	6.7	1.3	6.92	6.90	8.03	.595	.340	2.35
8	X	7	X	18.0	T	17.73	30.6	198.9	240.2	2.12	7.8	1.2	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T	19.79	35.0	212.6	274.4	2.26	7.8	1.3	5.82	8.01	7.00	.505	.305	2.44
8	X	7	X	22.5	T	22.32	39.2	223.2	306.7	2.37	7.8	1.4	6.56	8.07	7.04	.565	.345	2.78
8	X	7 1/8	X	25.0	T	24.83	43.6	233.0	339.8	2.48	7.8	1.5	7.30	8.13	7.07	.630	.380	3.09
8	X	6 1/2	X	28.0	I-T	17.69	31.1	203.4	247.7	2.16	8.0	1.2	5.20	8.06	6.54	.465	.285	2.30
8	X	7 1/8	X	28.5	T	28.28	49.5	245.1	384.7	2.61	7.8	1.6	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T	19.16	34.5	212.1	270.9	2.25	7.8	1.3	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T	17.26	30.9	219.2	269.8	2.25	8.7	1.2	5.08	8.85	6.00	.425	.300	2.66
9	X	6	X	20.0	T	19.76	36.5	238.2	319.0	2.43	8.7	1.3	5.81	8.95	6.02	.525	.315	2.82
10	X	4	X	19.0	I-T	13.77	25.4	227.2	260.0	2.23	10.2	1.1	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T	15.04	30.0	246.8	301.9	2.40	10.1	1.2	4.42	10.17	5.75	.360	.240	2.44
10	X	5 3/4	X	26.0	I-T	17.37	35.7	270.0	361.3	2.61	10.1	1.3	5.11	10.33	5.77	.440	.260	2.69
10	X	5 3/4	X	30.0	I-T	20.23	41.7	289.3	422.3	2.80	10.1	1.5	5.95	10.47	5.81	.510	.300	3.14
10	X	8	X	33.0	I-T	20.94	43.1	276.6	405.0	2.73	9.4	1.5	6.16	9.73	7.96	.435	.290	2.82
10	X	8	X	39.0	I-T	24.45	51.8	300.1	488.0	2.97	9.4	1.6	7.19	9.92	7.99	.530	.315	3.12
10	X	8	X	45.0	I-T	28.19	60.6	319.8	571.7	3.18	9.4	1.8	8.29	10.10	8.02	.620	.350	3.54
12	X	4	X	16.0	I-T	12.37	24.0	252.6	287.2	2.36	12.0	1.1	3.64	11.99	3.99	.265	.220	2.64
12	X	4	X	19.0	I-T	14.20	29.1	281.7	350.0	2.59	12.0	1.2	4.18	12.16	4.01	.350	.235	2.86
12	X	4	X	22.0	I-T	16.33	34.2	305.9	413.7	2.80	12.1	1.4	4.80	12.31	4.03	.425	.260	3.20
12	X	6 1/2	X	26.0	I-T	17.64	42.0	337.2	497.9	3.06	11.9	1.5	5.19	12.22	6.49	.380	.230	2.81

(38T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 42.750 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 48.094 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 6 1/2 X 30.0 I-T	20.27	48.5	357.6	575.2	3.26	11.9	1.6	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	57.4	381.2	679.5	3.51	11.8	1.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	62.5	378.4	700.4	3.55	11.2	1.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	70.4	393.7	787.4	3.73	11.2	2.0	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	78.7	408.5	878.6	3.91	11.2	2.2	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	84.1	415.0	921.7	3.99	11.0	2.2	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	92.9	429.3	1017.2	4.16	10.9	2.4	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.9	354.5	508.5	3.10	13.4	1.4	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.6	385.2	613.4	3.38	13.4	1.6	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	53.1	406.5	702.4	3.60	13.2	1.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	60.8	429.5	804.9	3.82	13.2	1.9	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	68.4	447.5	903.0	4.02	13.2	2.0	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	75.4	450.8	955.1	4.12	12.7	2.1	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	85.0	467.6	1072.3	4.32	12.6	2.3	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.3	442.5	759.5	3.76	15.1	1.7	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	60.9	477.4	918.9	4.10	15.1	1.9	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	71.4	502.4	1062.3	4.37	14.9	2.1	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.8 I-T	28.09	80.9	525.9	1200.8	4.62	14.9	2.3	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	91.2	543.6	1347.1	4.84	14.8	2.5	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	101.7	560.5	1496.8	5.06	14.7	2.7	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	116.0	580.5	1697.8	5.32	14.6	2.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	140.2	611.0	1990.6	5.71	14.2	3.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	161.4	631.2	2268.1	6.00	14.1	3.6	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	187.1	653.2	2600.2	6.30	13.9	4.0	17.40	16.75	10.37	.875	.525	8.79

(38T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 42.750 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 48.094 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT LBS	SECTION MODULUS FLANGE IN3	SECTION MODULUS PLATE IN3	I IN4	R IN	YF IN	YP IN	BEAM DIMENSIONS A IN2	D IN	WF IN	TF IN	TW IN	ASH IN2
18 X 6 X 35.0 I-T	26.29	75.1	556.1	1246.1	4.72	16.6	2.2	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	87.6	587.9	1451.1	5.06	16.6	2.5	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	111.2	628.2	1805.9	5.55	16.2	2.9	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	134.7	660.1	2166.5	5.98	16.1	3.3	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	159.2	685.5	2531.9	6.34	15.9	3.7	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	198.9	721.6	3042.3	6.84	15.3	4.2	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	224.9	741.5	3402.2	7.11	15.1	4.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	244.9	754.6	3671.5	7.29	15.0	4.9	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	276.5	776.0	4097.0	7.56	14.8	5.3	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	157.6	772.3	2894.3	6.87	18.4	3.7	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	173.9	790.7	3172.1	7.12	18.2	4.0	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	186.9	804.0	3392.3	7.31	18.1	4.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	211.9	825.4	3803.8	7.61	17.9	4.6	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	237.8	845.4	4221.3	7.88	17.8	5.0	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	267.7	868.1	4600.3	8.21	17.2	5.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	293.8	883.9	4990.6	8.43	17.0	5.6	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 47.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 59.375 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 24.0 T		23.53	38.9	211.1	267.7	2.01	6.9	1.3	6.92	6.90	8.03	.595	.340	2.35
8 X 7 X 22.5 T		22.32	40.0	245.8	320.5	2.20	8.0	1.3	6.56	8.07	7.04	.565	.345	2.78
8 X 7 1/8 X 25.0 T		24.83	44.4	258.4	355.5	2.31	8.0	1.4	7.30	8.13	7.07	.630	.380	3.09
8 X 7 1/8 X 28.5 T		28.28	50.4	273.8	403.1	2.44	8.0	1.5	8.32	8.22	7.12	.715	.430	3.53
9 X 6 X 20.0 T		19.76	37.2	260.9	332.1	2.26	8.9	1.3	5.81	8.95	6.02	.525	.315	2.82
10 X 5 3/4 X 26.0 I-T		17.37	36.3	294.8	374.3	2.41	10.3	1.3	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T		20.23	42.3	319.0	438.0	2.59	10.3	1.4	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T		20.94	43.8	305.5	420.9	2.53	9.6	1.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T		24.45	52.6	335.2	507.8	2.76	9.7	1.5	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T		28.19	61.5	360.5	596.0	2.97	9.7	1.7	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 22.0 I-T		16.33	34.8	333.4	427.0	2.58	12.3	1.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T		17.64	42.5	371.2	513.9	2.82	12.1	1.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T		20.27	49.2	397.4	594.5	3.02	12.1	1.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T		23.82	58.1	428.0	703.7	3.26	12.1	1.6	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T		25.48	63.3	426.6	726.6	3.30	11.5	1.7	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T		28.81	71.3	446.9	818.4	3.47	11.5	1.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T		32.11	79.7	466.5	914.9	3.65	11.5	2.0	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T		33.01	85.1	475.2	960.7	3.73	11.3	2.0	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T		35.92	94.1	494.0	1062.0	3.90	11.3	2.2	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T		16.18	38.4	388.4	523.6	2.86	13.6	1.3	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T		18.87	46.2	426.9	632.3	3.12	13.7	1.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T		21.16	53.7	454.2	724.9	3.32	13.5	1.6	6.22	13.84	6.73	.385	.270	3.74

(38T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 47.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 59.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
14 X 6 3/4 X 34.0 I-T	23.54	61.6	483.6	831.8	3.54	13.5	1.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	69.2	507.2	934.6	3.73	13.5	1.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	76.3	513.2	990.2	3.83	13.0	1.9	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	86.0	535.6	1114.0	4.03	13.0	2.1	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.9	493.4	781.7	3.46	15.4	1.6	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	61.6	538.0	947.3	3.79	15.4	1.8	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	72.2	570.7	1097.1	4.05	15.2	1.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	81.7	601.1	1242.0	4.29	15.2	2.1	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	92.2	624.8	1396.3	4.51	15.1	2.2	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	102.9	647.4	1554.7	4.72	15.1	2.4	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	117.4	674.0	1768.2	4.98	15.1	2.6	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	141.8	714.2	2079.4	5.36	14.7	2.9	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	163.3	741.0	2377.8	5.65	14.6	3.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	189.5	769.9	2737.8	5.97	14.4	3.6	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	76.0	633.8	1285.6	4.38	16.9	2.0	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	88.6	674.8	1499.8	4.70	16.9	2.2	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	112.4	728.1	1873.5	5.18	16.7	2.6	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	136.3	770.5	2256.6	5.60	16.6	2.9	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	161.2	804.2	2648.1	5.97	16.4	3.3	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	201.2	851.7	3196.4	6.47	15.9	3.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	227.7	877.5	3587.1	6.76	15.8	4.1	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	248.2	894.1	3881.3	6.95	15.6	4.3	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	280.4	921.1	4347.0	7.23	15.5	4.7	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	159.3	906.0	3013.7	6.44	18.9	3.3	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	175.9	929.9	3310.2	6.70	18.8	3.6	14.46	21.13	8.27	.685	.430	9.09

(38T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 47.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 59.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 1/4 X 73.0 I-T	52.58	189.2	947.3	3546.1	6.88	18.7	3.7	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	214.6	975.0	3989.7	7.20	18.6	4.1	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	241.0	1000.4	4442.1	7.49	18.4	4.4	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	270.9	1030.8	4850.4	7.81	17.9	4.7	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	297.5	1050.7	5277.1	8.05	17.7	5.0	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 52.250 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 71.844 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8	X	7 1/8	X	28.5	T	28.28	51.4	298.5	420.7	2.29	8.2	1.4	8.32	8.22	7.12	.715	.430	3.53
10	X	8	X	39.0	I-T	24.45	53.4	364.9	526.5	2.58	9.9	1.4	7.19	9.92	7.99	.530	.315	3.12
10	X	8	X	45.0	I-T	28.19	62.4	395.9	618.6	2.78	9.9	1.6	8.29	10.10	8.02	.620	.350	3.54
12	X	6 1/2	X	30.0	I-T	20.27	49.8	430.3	612.6	2.81	12.3	1.4	5.96	12.34	6.52	.440	.260	3.21
12	X	6 1/2	X	35.0	I-T	23.82	58.9	468.0	725.9	3.03	12.3	1.6	7.01	12.50	6.56	.520	.300	3.75
12	X	8	X	40.0	I-T	25.48	64.1	468.5	750.6	3.08	11.7	1.6	7.49	11.94	8.01	.515	.295	3.52
12	X	8	X	45.0	I-T	28.81	72.2	494.1	846.5	3.25	11.7	1.7	8.47	12.06	8.05	.575	.335	4.04
12	X	8 1/8	X	50.0	I-T	32.11	80.7	518.9	947.5	3.41	11.7	1.8	9.44	12.19	8.08	.640	.370	4.51
12	X	10	X	53.0	I-T	33.01	86.1	530.1	995.5	3.49	11.6	1.9	9.71	12.06	10.00	.575	.345	4.16
12	X	10	X	58.0	I-T	35.92	95.2	553.8	1101.9	3.66	11.6	2.0	10.56	12.19	10.01	.640	.360	4.39
14	X	5	X	26.0	I-T	18.87	46.8	460.9	650.0	2.90	13.9	1.4	5.55	13.91	5.03	.420	.255	3.55
14	X	6 3/4	X	30.0	I-T	21.16	54.4	494.4	745.7	3.09	13.7	1.5	6.22	13.84	6.73	.385	.270	3.74
14	X	6 3/4	X	34.0	I-T	23.54	62.3	530.3	856.3	3.30	13.7	1.6	6.92	13.98	6.75	.455	.285	3.98
14	X	6 3/4	X	38.0	I-T	26.17	70.0	559.8	963.0	3.48	13.8	1.7	7.70	14.10	6.77	.515	.310	4.37
14	X	8	X	43.0	I-T	28.02	77.2	569.1	1021.7	3.57	13.2	1.8	8.24	13.66	8.00	.530	.305	4.17
14	X	8	X	48.0	I-T	31.50	86.9	597.7	1151.0	3.77	13.2	1.9	9.26	13.79	8.03	.595	.340	4.69
16	X	5 1/2	X	26.0	I-T	19.49	51.5	535.7	802.1	3.22	15.6	1.5	5.73	15.69	5.50	.345	.250	3.92
16	X	5 1/2	X	31.0	I-T	22.70	62.3	590.3	973.0	3.52	15.6	1.6	6.68	15.88	5.53	.440	.275	4.37
16	X	7	X	36.0	I-T	25.69	73.0	631.4	1128.3	3.77	15.4	1.8	7.56	15.86	6.99	.430	.295	4.68
16	X	7	X	40.0	I-T	28.09	82.6	669.0	1278.4	3.99	15.5	1.9	8.26	16.01	7.00	.505	.305	4.88
16	X	7	X	45.0	I-T	31.77	93.2	699.7	1439.5	4.21	15.4	2.1	9.34	16.13	7.04	.565	.345	5.56
16	X	7 1/8	X	50.0	I-T	35.34	104.0	728.8	1605.2	4.42	15.4	2.2	10.39	16.26	7.07	.630	.380	6.18

(38T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 52.250 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 71.844 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 1/8 X 57.0 I-T	40.28	118.7	763.3	1829.5	4.68	15.4	2.4	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	143.2	814.4	2156.2	5.04	15.1	2.6	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	165.0	849.4	2472.6	5.34	15.0	2.9	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	191.6	886.6	2855.4	5.66	14.9	3.2	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	76.8	703.5	1320.6	4.07	17.2	1.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	89.5	754.5	1542.4	4.38	17.2	2.0	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	113.5	822.7	1932.1	4.85	17.0	2.3	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	137.7	877.5	2334.2	5.26	17.0	2.7	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	163.0	921.3	2748.3	5.63	16.9	3.0	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	203.3	982.5	3329.1	6.12	16.4	3.4	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	230.2	1015.6	3746.6	6.42	16.3	3.7	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	251.0	1036.8	4062.9	6.61	16.2	3.9	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	283.8	1070.7	4564.4	6.91	16.1	4.3	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	160.9	1037.6	3115.6	6.05	19.4	3.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	177.7	1068.5	3428.2	6.30	19.3	3.2	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	191.1	1090.8	3677.7	6.49	19.2	3.4	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	217.0	1126.5	4149.1	6.81	19.1	3.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	243.8	1158.8	4632.2	7.11	19.0	4.0	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	273.7	1198.0	5065.1	7.42	18.5	4.2	20.11	21.36	12.29	.800	.500	10.66
21 X 12 3/8 X 111.0 I-T	75.30	300.7	1223.2	5524.1	7.67	18.4	4.5	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 57.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 85.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 45.0 I-T	28.19	63.4	426.0	640.3	2.61	10.1	1.5	8.29	10.10	8.02	.620	.350	3.54
12 X 8 X 40.0 I-T	25.48	65.0	503.7	773.4	2.88	11.9	1.5	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	73.2	534.8	872.9	3.05	11.9	1.6	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	81.8	564.9	977.9	3.21	12.0	1.7	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	87.2	578.7	1027.7	3.29	11.8	1.8	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	96.3	607.5	1138.4	3.44	11.8	1.9	10.56	12.19	10.01	.640	.360	4.39
14 X 6 3/4 X 34.0 I-T	23.54	63.1	569.3	879.5	3.08	13.9	1.5	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	70.9	604.6	989.6	3.26	14.0	1.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	78.1	617.4	1050.8	3.35	13.5	1.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	87.9	652.5	1185.0	3.54	13.5	1.8	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 31.0 I-T	22.70	63.1	634.0	997.2	3.29	15.8	1.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	73.9	683.4	1157.1	3.53	15.7	1.7	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	83.5	728.5	1311.8	3.74	15.7	1.8	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	94.2	766.7	1478.7	3.95	15.7	1.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	105.1	802.9	1650.7	4.15	15.7	2.1	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	120.0	846.0	1884.2	4.40	15.7	2.2	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	144.6	909.1	2224.1	4.75	15.4	2.4	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	166.6	953.6	2556.3	5.04	15.3	2.7	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	193.5	1000.7	2959.6	5.36	15.3	3.0	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	77.6	764.0	1352.7	3.81	17.4	1.8	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	90.4	825.4	1581.1	4.10	17.5	1.9	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	114.6	909.7	1984.4	4.55	17.3	2.2	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	139.0	978.5	2402.9	4.95	17.3	2.5	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	164.6	1034.2	2836.4	5.31	17.2	2.7	14.93	18.47	7.64	.810	.495	9.14

(38T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 57.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 85.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/8 X 86.0 I-T	57.79	205.2	1110.9	3445.2	5.80	16.8	3.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	232.5	1152.7	3886.2	6.09	16.7	3.4	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	253.6	1179.6	4222.8	6.30	16.7	3.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	286.8	1221.9	4755.3	6.60	16.6	3.9	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	162.4	1164.0	3204.8	5.70	19.7	2.8	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	179.3	1202.9	3531.2	5.94	19.7	2.9	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	192.9	1231.0	3792.4	6.13	19.7	3.1	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	219.1	1275.3	4287.9	6.45	19.6	3.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	246.3	1317.2	4798.0	6.75	19.5	3.6	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	276.2	1366.0	5251.9	7.05	19.0	3.8	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	303.6	1397.9	5739.4	7.30	18.9	4.1	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 66.500 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 116.375 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 45.0 I-T	31.77	96.3	874.4	1550.7	3.51	16.1	1.8	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	107.4	925.2	1733.1	3.70	16.1	1.9	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	122.5	986.9	1981.8	3.93	16.2	2.0	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.16	147.4	1075.3	2343.2	4.26	15.9	2.2	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	169.8	1141.5	2701.2	4.53	15.9	2.4	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 83.0 I-T	59.17	197.3	1212.2	3138.6	4.84	15.9	2.6	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 40.0 I-T	29.35	92.3	937.9	1652.0	3.64	17.9	1.8	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	116.9	1055.6	2077.6	4.05	17.8	2.0	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	141.7	1154.9	2522.8	4.42	17.8	2.2	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	167.8	1238.3	2988.4	4.77	17.8	2.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	208.9	1350.2	3642.9	5.23	17.4	2.7	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	236.6	1413.4	4122.9	5.52	17.4	2.9	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	258.2	1454.7	4491.4	5.72	17.4	3.1	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	292.3	1518.0	5078.1	6.02	17.4	3.3	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	165.2	1391.0	3357.4	5.09	20.3	2.4	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	182.4	1448.5	3706.3	5.32	20.3	2.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	196.2	1490.6	3986.6	5.50	20.3	2.7	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	223.0	1559.6	4522.1	5.81	20.3	2.9	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	250.8	1621.9	5076.7	6.11	20.2	3.1	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	280.7	1692.3	5564.3	6.39	19.8	3.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	308.7	1741.3	6099.4	6.64	19.8	3.5	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 76.000 IN.) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.) EFFECTIVE PLATE AREA = 152.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 1/4 X 67.0 I-T	44.18	150.4	1205.8	2450.7	3.85	16.3	2.0	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	173.2	1295.2	2829.3	4.12	16.3	2.2	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	201.1	1391.9	3294.0	4.41	16.4	2.4	17.40	16.75	10.37	.875	.525	8.79
18 X 7 1/2 X 60.0 I-T	42.61	144.5	1292.5	2630.8	4.00	18.2	2.0	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	171.1	1405.5	3121.8	4.32	18.2	2.2	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	212.5	1555.2	3812.2	4.75	17.9	2.5	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	240.7	1643.1	4323.2	5.03	18.0	2.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	262.7	1701.9	4717.9	5.22	18.0	2.8	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	297.4	1790.8	5347.6	5.52	18.0	3.0	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	168.0	1574.5	3490.6	4.60	20.8	2.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	185.5	1652.5	3857.2	4.81	20.8	2.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	199.5	1710.1	4152.4	4.98	20.8	2.4	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	226.7	1806.7	4719.4	5.28	20.8	2.6	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	255.0	1894.9	5309.5	5.56	20.8	2.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	285.0	1988.5	5822.8	5.82	20.4	2.9	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	313.5	2059.1	6395.8	6.06	20.4	3.1	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 85.500 IN.) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.) EFFECTIVE PLATE AREA = 192.375 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/8 X 86.0 I-T	57.79	216.4	1720.3	3967.9	4.35	18.3	2.3	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	245.0	1833.9	4504.4	4.61	18.4	2.5	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	267.3	1911.5	4920.6	4.80	18.4	2.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	302.5	2028.4	5585.6	5.08	18.5	2.8	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 73.0 I-T	52.58	203.0	1883.9	4304.9	4.55	21.2	2.3	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	230.6	2009.1	4897.5	4.83	21.2	2.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	259.3	2125.2	5516.6	5.10	21.3	2.6	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	289.3	2242.8	6050.3	5.34	20.9	2.7	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	318.2	2336.7	6654.2	5.57	20.9	2.8	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.)

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TABLE X. Properties of combined beam and plate, I-T and T (38t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(38T = 95.000 IN.) PLATE WEIGHT = 102.000 LBS. (2.5000 IN.) EFFECTIVE PLATE AREA = 237.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/4 X 119.0 I-T	80.48	307.9	2226.3	5807.5	4.72	18.9	2.6	23.67	18.97	11.27	1.060	.655	12.43
21 X 12 1/4 X 101.0 I-T	68.38	294.0	2448.4	6262.4	4.93	21.3	2.6	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	323.2	2568.0	6892.2	5.15	21.3	2.7	22.15	21.51	12.34	.875	.550	11.83

(38T) PLATE WEIGHT = 102.000 LBS. (2.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t).

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 4.375 IN.) PLATE WEIGHT = 5.100 LBS. (.1250 IN.) EFFECTIVE PLATE AREA = .547 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T				4.91	3.2	2.6	5.8	1.71	1.8	2.3	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T				5.88	4.3	3.5	9.7	2.07	2.3	2.8	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T				6.94	5.7	4.5	15.3	2.44	2.7	3.4	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T				6.17	5.3	4.2	14.2	2.46	2.7	3.4	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T				7.19	7.4	6.2	26.9	3.18	3.7	4.4	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T				9.07	10.2	8.7	47.0	3.82	4.6	5.4	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T				10.98	13.8	11.7	76.2	4.49	5.5	6.5	3.23	11.91	3.97	.225	.200	2.38

(35T) PLATE WEIGHT = 5.100 LBS. (.1250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE T-T AND T

(35T = 5.469 IN.) PLATE WEIGHT = 6.375 LBS. (.1563 IN.) EFFECTIVE PLATE AREA = .854 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.4	3.6	7.1	1.76	2.1	2.0	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.8 T	5.88	4.6	4.7	11.9	2.14	2.6	2.5	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.1	6.0	18.5	2.53	3.0	3.1	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	5.7	5.8	17.3	2.54	3.1	3.0	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	7.8	8.1	32.1	3.29	4.1	3.9	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	10.9	11.1	55.1	3.96	5.1	5.0	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	14.7	14.6	88.2	4.65	6.0	6.1	3.23	11.91	3.97	.225	.200	2.38

(35T) PLATE WEIGHT = 6.375 LBS. (.1563 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 6.563 IN.) PLATE WEIGHT = 7.650 LBS. (.1875 IN.) EFFECTIVE PLATE AREA = 1.230 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.5	4.8	8.4	1.77	2.4	1.8	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.8	6.2	13.9	2.16	2.9	2.2	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.4	7.8	21.6	2.57	3.4	2.8	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	5.9	7.6	20.2	2.57	3.4	2.7	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.2	10.5	37.3	3.34	4.5	3.5	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.4	14.0	63.4	4.03	5.5	4.5	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	15.5	18.0	100.8	4.75	6.5	5.6	3.23	11.91	3.97	.225	.200	2.38

(35T) PLATE WEIGHT = 7.650 LBS. (.1875 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 7.656 IN.) PLATE WEIGHT = 8.925 LBS. (.2188 IN.) EFFECTIVE PLATE AREA = 1.675 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.6	6.2	9.5	1.75	2.6	1.5	1.44	3.95	3.94	.205	.170	.67
5 X 4 X 6.0 T	5.88	4.9	8.0	15.7	2.15	3.2	2.0	1.73	4.94	3.96	.210	.190	.94
6 X 4 X 7.0 T	6.94	6.6	9.9	24.5	2.57	3.7	2.5	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 9.0 I-T	6.17	6.1	9.7	22.9	2.56	3.8	2.4	1.81	5.90	3.94	.215	.170	1.00
8 X 4 X 10.0 I-T	7.19	8.5	13.3	42.1	3.33	5.0	3.2	2.11	7.89	3.94	.205	.170	1.34
10 X 4 X 12.0 I-T	9.07	11.9	17.5	71.4	4.06	6.0	4.1	2.67	9.87	3.96	.210	.190	1.88
12 X 4 X 14.0 I-T	10.98	16.1	22.1	113.2	4.80	7.0	5.1	3.23	11.91	3.97	.225	.200	2.38

(35T) PLATE WEIGHT = 8.925 LBS. (.2188 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 8.750 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 2.188 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T		4.91	3.7	7.7	10.5	1.70	2.8	1.4	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T		6.40	4.6	8.1	12.5	1.75	2.7	1.5	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T		7.42	5.5	8.4	14.3	1.81	2.6	1.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T		8.82	7.1	8.6	16.8	1.87	2.4	1.9	2.59	4.07	5.25	.330	.230	.94
5 X 4 X 6.0 T		5.88	5.1	9.9	17.4	2.11	3.4	1.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T		7.37	6.3	10.4	20.6	2.18	3.3	2.0	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T		8.36	7.4	10.7	23.2	2.24	3.1	2.2	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T		9.42	8.5	11.0	25.8	2.28	3.0	2.3	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T		9.91	9.2	10.9	26.3	2.27	2.9	2.4	2.92	5.01	5.00	.360	.240	1.20
6 X 4 X 7.0 T		6.94	6.8	12.3	27.1	2.53	4.0	2.2	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T		7.88	7.7	12.7	30.1	2.58	3.9	2.4	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T		6.17	6.2	12.1	25.3	2.51	4.1	2.1	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T		9.34	9.6	13.3	35.2	2.67	3.7	2.7	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 12.0 I-T		8.30	8.2	12.9	31.3	2.60	3.8	2.4	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T		9.78	10.4	13.3	36.5	2.68	3.5	2.7	2.88	5.99	5.99	.260	.230	1.38
7 X 5 X 11.0 T		10.81	12.7	15.5	49.8	3.05	3.9	3.2	3.18	6.87	5.00	.335	.230	1.58
8 X 4 X 10.0 I-T		7.19	8.7	16.5	46.4	3.29	5.3	2.8	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T		9.52	11.0	17.5	55.7	3.34	5.1	3.2	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T		12.83	16.6	18.5	70.8	3.45	4.3	3.8	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T		10.79	12.9	18.3	63.4	3.44	4.9	3.5	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/4 X 18.0 I-T		12.00	16.0	18.9	72.8	3.57	4.5	3.8	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T		13.87	18.9	19.8	82.5	3.63	4.4	4.2	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T		15.11	21.1	19.2	82.2	3.52	3.9	4.3	4.44	7.93	6.50	.400	.245	1.94

(35T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 8.750 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 2.188 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 X 12.0 I-T	9.07	12.3	21.4	78.9	4.03	6.4	3.7	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	15.2	22.7	93.4	4.12	6.1	4.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	17.5	23.7	104.4	4.22	6.0	4.4	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	20.0	24.6	115.9	4.31	5.8	4.7	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	23.9	25.1	127.8	4.40	5.3	5.1	4.42	10.17	5.75	.360	.240	2.44
12 X 4 X 14.0 I-T	10.98	16.7	26.8	125.1	4.81	7.5	4.7	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	19.0	27.9	138.1	4.87	7.3	5.0	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	22.9	29.6	160.2	5.02	7.0	5.4	4.18	12.16	4.01	.350	.235	2.86
12 X 6 1/2 X 26.0 I-T	17.64	33.6	31.7	203.4	5.25	6.0	6.4	5.19	12.22	6.49	.380	.230	2.81
14 X 5 X 22.0 I-T	16.18	29.9	34.9	225.5	5.70	7.5	6.5	4.76	13.74	5.00	.335	.230	3.16
16 X 5 1/2 X 26.0 I-T	19.49	39.1	42.4	324.5	6.40	8.3	7.6	5.73	15.69	5.50	.345	.250	3.92

(35T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 9.844 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 2.769 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.8	9.4	11.4	1.64	3.0	1.2	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.7	9.8	13.6	1.71	2.9	1.4	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.6	10.2	15.7	1.78	2.8	1.5	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.2	10.5	18.6	1.86	2.6	1.8	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.3	10.6	17.6	1.83	2.8	1.7	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.2	12.1	18.9	2.05	3.7	1.6	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.5	12.6	22.5	2.14	3.5	1.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.6	13.0	25.5	2.21	3.4	2.0	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	8.7	13.4	28.5	2.27	3.3	2.1	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.5	13.3	29.2	2.27	3.1	2.2	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	11.2	13.9	33.8	2.33	3.0	2.4	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	6.9	15.0	29.5	2.48	4.3	2.0	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	7.9	15.4	32.8	2.54	4.1	2.1	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.3	14.7	27.4	2.45	4.3	1.9	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	9.8	16.1	38.7	2.65	4.0	2.4	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	11.5	16.6	43.9	2.71	3.8	2.6	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.3	15.6	34.3	2.56	4.1	2.2	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	10.7	16.1	40.2	2.67	3.8	2.5	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	11.4	16.9	44.8	2.75	3.9	2.6	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	14.4	17.3	50.9	2.80	3.5	2.9	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.0	18.8	55.0	3.04	4.2	2.9	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	15.7	19.5	63.1	3.10	4.0	3.2	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	18.7	19.9	69.5	3.12	3.7	3.5	4.36	6.92	6.73	.385	.270	1.87

(35T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 9.844 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 2.769 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 10.0 I-T	7.19	8.9	20.0	50.4	3.21	5.7	2.5	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.3	21.1	60.8	3.30	5.4	2.9	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.0	22.2	78.3	3.46	4.6	3.5	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.3	22.0	69.5	3.42	5.2	3.2	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	20.7	23.2	89.8	3.52	4.3	3.9	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	16.4	22.8	80.4	3.57	4.9	3.5	3.53	8.14	5.25	.330	.230	1.87
8 X 5 1/4 X 21.0 I-T	13.87	19.4	23.8	91.5	3.66	4.7	3.8	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	21.7	23.1	91.8	3.57	4.2	4.0	4.44	7.93	6.50	.400	.245	1.94
10 X 4 X 12.0 I-T	9.07	12.6	25.7	85.8	3.97	6.8	3.3	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	15.7	27.2	102.1	4.10	6.5	3.8	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.0	28.3	114.5	4.22	6.3	4.0	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	20.6	29.4	127.4	4.32	6.2	4.3	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	24.6	30.0	141.3	4.43	5.7	4.7	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	29.1	31.4	160.2	4.51	5.5	5.1	5.11	10.33	5.77	.440	.260	2.69
12 X 4 X 14.0 I-T	10.98	17.2	32.1	136.3	4.77	7.9	4.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	19.5	33.2	150.8	4.85	7.7	4.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	23.6	35.1	175.6	5.03	7.4	5.0	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	27.6	36.8	198.5	5.12	7.2	5.4	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	34.6	37.6	225.3	5.32	6.5	6.0	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	39.5	39.3	248.4	5.33	6.3	6.3	5.96	12.34	6.52	.440	.260	3.21
14 X 5 X 22.0 I-T	16.18	30.8	41.3	247.6	5.73	8.0	6.0	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	36.8	43.6	283.3	5.84	7.7	6.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	42.8	44.9	309.4	5.87	7.2	6.9	6.22	13.84	6.73	.385	.270	3.74

(35T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 9.844 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 2.769 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	40.4	49.7	355.7	6.47	8.8	7.2	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	48.5	52.7	408.1	6.57	8.4	7.7	6.68	15.88	5.53	.440	.275	4.37

(35T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 10.938 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 3.418 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.8	11.2	12.1	1.58	3.2	1.1	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.8	11.6	14.7	1.66	3.1	1.3	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.7	12.1	17.0	1.74	3.0	1.4	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.4	12.6	20.3	1.84	2.8	1.6	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.4	12.6	19.0	1.80	3.0	1.5	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.2	14.4	20.1	1.98	3.8	1.4	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.6	15.0	24.3	2.08	3.7	1.6	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.7	15.5	27.6	2.17	3.6	1.8	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	8.9	16.0	31.0	2.24	3.5	1.9	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.6	15.9	31.9	2.24	3.3	2.0	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	11.5	16.6	37.1	2.33	3.2	2.2	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.0	17.8	31.6	2.41	4.5	1.8	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.1	18.3	35.3	2.48	4.4	1.9	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.4	17.5	29.3	2.36	4.5	1.7	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.0	19.1	41.9	2.61	4.2	2.2	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	11.8	19.8	47.8	2.69	4.1	2.4	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.5	18.5	36.9	2.51	4.3	2.0	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	10.9	19.2	43.7	2.64	4.0	2.3	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	11.7	20.1	48.7	2.72	4.2	2.4	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	14.7	20.6	55.9	2.80	3.8	2.7	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.3	22.4	59.9	3.01	4.5	2.7	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	16.1	23.2	69.0	3.10	4.3	3.0	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	19.1	23.7	76.5	3.14	4.0	3.2	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	21.9	24.3	84.1	3.17	3.8	3.5	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	24.4	24.9	90.7	3.19	3.7	3.6	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 10.938 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 3.418 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	27.5	24.3	92.2	3.10	3.3	3.8	6.16	6.83	8.00	.530	.305	2.08
8 X 4 X 10.0 I-T	7.19	9.1	23.8	53.8	3.12	5.9	2.3	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.5	25.0	65.5	3.24	5.7	2.6	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.4	26.3	85.4	3.45	4.9	3.2	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.6	26.0	75.1	3.38	5.5	2.9	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.1	27.4	98.5	3.53	4.7	3.6	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	16.8	27.1	87.5	3.55	5.2	3.2	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	25.2	28.1	109.6	3.56	4.3	3.9	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	28.7	28.8	119.7	3.60	4.2	4.2	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	19.8	28.2	100.0	3.65	5.0	3.5	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.1	27.4	101.0	3.58	4.6	3.7	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	25.6	28.5	113.1	3.62	4.4	4.0	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	28.6	28.7	119.0	3.63	4.2	4.1	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	25.4	31.4	128.6	3.89	5.1	4.1	5.08	8.85	6.00	.425	.300	2.66
10 X 4 X 12.0 I-T	9.07	12.8	30.5	92.0	3.89	7.2	3.0	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.0	32.1	110.1	4.04	6.9	3.4	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.5	33.4	123.9	4.18	6.7	3.7	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.1	34.6	138.3	4.30	6.6	4.0	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.2	35.4	154.2	4.43	6.1	4.4	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	29.8	36.9	175.5	4.54	5.9	4.8	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	34.3	38.6	195.7	4.57	5.7	5.1	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	35.8	36.0	180.3	4.34	5.0	5.0	6.16	9.73	7.96	.435	.290	2.82

(35T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 10.938 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 3.418 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 14.0 I-T	10.98	17.6	37.9	146.6	4.70	8.4	3.9	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.0	39.1	162.7	4.80	8.1	4.2	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	24.2	41.2	190.1	5.00	7.9	4.6	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	28.3	43.1	215.5	5.12	7.6	5.0	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	35.4	44.2	246.4	5.35	7.0	5.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	40.5	45.9	272.2	5.39	6.7	5.9	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	47.1	48.2	305.4	5.41	6.5	6.3	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	51.6	46.5	299.5	5.24	5.8	6.4	7.49	11.94	8.01	.515	.295	3.52
14 X 5 X 22.0 I-T	16.18	31.6	48.4	268.6	5.73	8.5	5.6	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	37.8	50.8	308.3	5.86	8.2	6.1	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	43.9	52.3	337.7	5.92	7.7	6.5	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	50.1	54.2	371.9	6.00	7.4	6.9	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	55.7	56.2	403.1	6.02	7.2	7.2	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	61.8	55.0	406.7	5.91	6.6	7.4	8.24	13.66	8.00	.530	.305	4.17
16 X 5 1/2 X 26.0 I-T	19.49	41.5	57.7	386.2	6.50	9.3	6.7	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	49.8	61.0	444.0	6.63	8.9	7.3	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	58.2	63.3	490.2	6.68	8.4	7.7	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	65.6	65.5	534.9	6.77	8.2	8.2	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	60.4	71.0	588.0	7.26	9.7	8.3	7.73	17.70	6.00	.425	.300	5.31

(35T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 12.031 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 4.136 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T		4.91	3.9	13.0	12.8	1.51	3.3	1.0	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T		6.40	4.9	13.5	15.6	1.61	3.2	1.2	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T		7.42	5.8	14.1	18.1	1.69	3.1	1.3	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T		8.82	7.5	14.8	21.9	1.80	2.9	1.5	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T		8.39	6.6	14.7	20.4	1.76	3.1	1.4	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T		5.88	5.3	16.7	21.3	1.90	4.0	1.3	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T		7.37	6.7	17.5	25.8	2.02	3.9	1.5	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T		8.36	7.8	18.2	29.5	2.12	3.8	1.6	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T		9.42	9.0	18.7	33.3	2.19	3.7	1.8	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T		9.91	9.8	18.6	34.4	2.21	3.5	1.8	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T		11.69	11.7	19.5	40.1	2.30	3.4	2.1	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T		6.94	7.1	20.8	33.4	2.33	4.7	1.6	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T		7.88	8.2	21.4	37.5	2.41	4.6	1.8	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T		6.17	6.5	20.5	30.9	2.28	4.7	1.5	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T		9.34	10.1	22.4	44.8	2.55	4.4	2.0	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T		10.89	12.0	23.2	51.4	2.65	4.3	2.2	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T		8.30	8.6	21.7	39.3	2.45	4.6	1.8	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T		9.78	11.0	22.5	46.9	2.59	4.3	2.1	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T		10.74	11.9	23.6	52.3	2.68	4.4	2.2	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T		12.63	14.9	24.2	60.4	2.77	4.0	2.5	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T		10.81	13.5	26.2	64.3	2.96	4.8	2.5	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T		12.85	16.4	27.2	74.6	3.07	4.6	2.7	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T		14.81	19.4	27.8	83.0	3.13	4.3	3.0	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T		16.77	22.3	28.5	91.8	3.18	4.1	3.2	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T		18.74	24.9	29.1	99.2	3.21	4.0	3.4	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 12.031 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 4.136 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	28.1	28.5	101.5	3.14	3.6	3.6	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	31.2	29.2	109.2	3.14	3.5	3.7	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.2	27.9	56.9	3.02	6.2	2.0	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.7	29.2	69.7	3.17	5.9	2.4	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.7	30.8	92.0	3.41	5.2	3.0	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	13.8	30.4	80.3	3.31	5.8	2.6	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.5	32.0	106.7	3.52	5.0	3.3	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.1	31.7	94.1	3.50	5.5	3.0	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	25.7	32.8	119.3	3.57	4.6	3.6	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	29.3	33.6	130.8	3.62	4.5	3.9	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.2	33.0	108.0	3.63	5.3	3.3	4.08	8.28	5.27	.400	.250	2.07
8 X 6 1/2 X 24.0 I-T	15.11	22.5	32.1	109.6	3.57	4.9	3.4	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	26.2	33.3	123.2	3.63	4.7	3.7	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	29.1	33.6	130.1	3.65	4.5	3.9	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	25.9	36.6	139.4	3.89	5.4	3.8	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	30.5	37.8	156.9	3.97	5.1	4.2	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.1	35.7	97.6	3.79	7.5	2.7	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.3	37.5	117.4	3.97	7.2	3.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	18.8	38.9	132.5	4.12	7.0	3.4	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.5	40.3	148.4	4.26	6.9	3.7	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	25.6	41.3	166.3	4.41	6.5	4.0	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	30.4	43.0	190.0	4.53	6.3	4.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	35.8	44.7	212.5	4.59	6.1	4.7	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	36.5	41.9	196.6	4.37	5.4	4.7	6.16	9.73	7.96	.435	.290	2.82

(35T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 12.031 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 4.136 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 39.0 I-T	24.45	43.5	43.8	223.9	4.45	5.1	5.1	7.19	9.92	7.99	.530	.315	3.12
12 X 4 X 14.0 I-T	10.98	17.9	44.2	156.0	4.60	8.7	3.5	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.4	45.6	173.6	4.73	8.5	3.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	24.7	47.9	203.7	4.95	8.3	4.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	28.9	50.0	231.6	5.09	8.0	4.6	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	36.1	51.5	266.4	5.35	7.4	5.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	41.3	53.3	295.1	5.41	7.1	5.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	48.2	55.7	331.8	5.46	6.9	6.0	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	52.7	53.8	326.9	5.30	6.2	6.1	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	58.6	55.7	354.4	5.30	6.0	6.4	8.47	12.06	8.05	.575	.335	4.04
14 X 5 X 22.0 I-T	16.18	32.3	56.1	288.5	5.70	8.9	5.1	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	38.6	58.8	332.2	5.86	8.6	5.7	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	44.9	60.3	365.0	5.94	8.1	6.0	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	51.2	62.4	403.0	6.04	7.9	6.5	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	57.1	64.5	437.3	6.08	7.7	6.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	63.2	63.3	442.9	5.98	7.0	7.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	70.4	65.7	480.1	5.99	6.8	7.3	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	42.4	66.6	415.4	6.49	9.8	6.2	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	51.0	70.1	478.8	6.65	9.4	6.8	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	59.6	72.5	529.9	6.73	8.9	7.3	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	67.2	74.9	579.3	6.84	8.6	7.7	8.26	16.01	7.00	.505	.305	4.88
18 X 6 X 35.0 I-T	26.29	61.9	81.1	633.4	7.31	10.2	7.8	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	71.9	84.8	709.9	7.46	9.9	8.4	8.63	17.90	6.02	.525	.315	5.64

(35T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 13.125 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 4.922 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	3.9	14.8	13.4	1.45	3.4	.9	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	4.9	15.5	16.4	1.55	3.3	1.1	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	5.9	16.2	19.2	1.64	3.3	1.2	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.6	17.0	23.3	1.76	3.1	1.4	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.6	16.9	21.6	1.71	3.3	1.3	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.4	19.2	22.3	1.83	4.2	1.2	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.8	20.2	27.2	1.96	4.0	1.3	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	7.9	20.9	31.2	2.06	3.9	1.5	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.1	21.6	35.3	2.14	3.9	1.6	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	9.9	21.5	36.6	2.16	3.7	1.7	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	11.8	22.6	42.9	2.27	3.6	1.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.2	23.9	35.1	2.24	4.9	1.5	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.3	24.6	39.5	2.34	4.8	1.6	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.6	23.5	32.3	2.19	4.9	1.4	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.3	25.8	47.5	2.49	4.6	1.8	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.2	26.8	54.6	2.59	4.5	2.0	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.7	25.0	41.5	2.37	4.7	1.7	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.2	26.0	49.7	2.53	4.5	1.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.1	27.2	55.6	2.62	4.6	2.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.2	28.0	64.6	2.74	4.3	2.3	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	13.7	30.3	68.3	2.90	5.0	2.3	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	16.6	31.5	79.7	3.03	4.8	2.5	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	19.7	32.1	89.2	3.10	4.5	2.8	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	22.7	32.9	99.0	3.17	4.4	3.0	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	25.3	33.6	107.3	3.21	4.2	3.2	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 13.125 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 4.922 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	28.6	33.0	110.4	3.16	3.9	3.3	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	31.8	33.7	119.1	3.17	3.7	3.5	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.3	32.2	59.6	2.91	6.4	1.9	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	11.9	33.6	73.5	3.08	6.2	2.2	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	17.9	35.6	98.1	3.36	5.5	2.8	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.0	35.1	85.0	3.24	6.1	2.4	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	21.9	37.0	114.3	3.48	5.2	3.1	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	17.3	36.6	100.1	3.44	5.8	2.7	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	26.1	37.9	128.5	3.56	4.9	3.4	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	29.8	38.8	141.4	3.63	4.7	3.6	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	20.5	38.1	115.3	3.58	5.6	3.0	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	33.1	39.7	152.5	3.64	4.6	3.8	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	22.9	37.2	117.6	3.54	5.1	3.2	4.44	7.93	6.50	.400	.245	1.94
8 X 6 1/2 X 28.0 I-T	17.69	26.6	38.5	132.7	3.62	5.0	3.4	5.20	8.06	6.54	.465	.285	2.30
8 X 8 X 31.0 I-T	19.16	29.6	38.8	140.6	3.65	4.8	3.6	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	26.3	42.1	149.5	3.87	5.7	3.6	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	31.1	43.5	169.1	3.97	5.4	3.9	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.2	41.2	102.6	3.68	7.8	2.5	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	16.6	43.2	124.1	3.88	7.5	2.9	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	19.1	44.8	140.5	4.04	7.3	3.1	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	21.9	46.4	157.8	4.19	7.2	3.4	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	26.0	47.7	177.5	4.36	6.8	3.7	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	30.9	49.6	203.7	4.51	6.6	4.1	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	35.7	51.4	228.4	4.58	6.4	4.4	5.95	10.47	5.81	.510	.300	3.14

(35T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 13.125 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 4.922 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 33.0 I-T	20.94	37.2	48.3	212.2	4.38	5.7	4.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	44.3	50.3	242.6	4.48	5.5	4.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	51.2	52.4	271.3	4.53	5.3	5.2	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	18.2	50.9	164.5	4.49	9.1	3.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	20.7	52.5	183.6	4.63	8.9	3.5	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.1	55.1	216.2	4.87	8.6	3.9	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	29.4	57.4	246.7	5.04	8.4	4.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	36.7	59.3	285.3	5.31	7.8	4.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	42.0	61.2	316.9	5.40	7.5	5.2	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	49.1	63.8	357.3	5.47	7.3	5.6	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	53.7	61.7	353.5	5.34	6.6	5.7	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	59.8	63.7	383.6	5.35	6.4	6.0	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	66.2	65.8	414.5	5.37	6.3	6.3	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	71.3	65.3	423.8	5.38	5.9	6.5	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	78.3	66.9	453.4	5.41	5.8	6.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	32.8	64.5	307.1	5.63	9.4	4.8	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	39.3	67.4	354.8	5.82	9.0	5.3	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	45.7	69.1	391.1	5.92	8.6	5.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	52.2	71.4	432.8	6.04	8.3	6.1	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	58.2	73.6	470.4	6.11	8.1	6.4	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	64.5	72.3	478.4	6.03	7.4	6.6	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	71.8	74.8	519.0	6.05	7.2	6.9	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	43.2	76.2	443.0	6.45	10.3	5.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	52.0	80.0	512.3	6.65	9.9	6.4	6.68	15.88	5.53	.440	.275	4.37

(35T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 13.125 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 4.922 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 36.0 I-T	25.69	60.8	82.6	568.4	6.75	9.4	6.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	68.6	85.2	622.7	6.87	9.1	7.3	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	76.4	88.3	676.2	6.88	8.9	7.7	9.34	16.13	7.04	.565	.345	5.56
18 X 6 X 35.8 I-T	26.29	63.3	92.0	677.6	7.32	10.7	7.4	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	73.5	96.0	760.9	7.49	10.4	7.9	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	92.4	101.9	890.0	7.61	9.6	8.7	10.46	17.99	7.50	.570	.355	6.39

(35T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 15.313 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 6.699 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	4.0	18.3	14.4	1.33	3.6	.8	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.1	19.5	17.8	1.44	3.5	.9	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.0	20.5	21.0	1.54	3.5	1.0	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.7	21.8	25.7	1.66	3.3	1.2	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	6.8	21.4	23.8	1.61	3.5	1.1	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.5	24.1	24.0	1.69	4.4	1.0	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	6.9	25.6	29.5	1.83	4.3	1.2	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.1	26.7	34.1	1.93	4.2	1.3	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.4	27.7	38.9	2.03	4.2	1.4	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.1	27.7	40.4	2.05	4.0	1.5	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.1	29.1	47.9	2.17	3.9	1.6	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.3	30.4	37.9	2.08	5.2	1.2	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.5	31.5	42.9	2.18	5.1	1.4	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.7	29.9	34.8	2.02	5.2	1.2	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.5	33.2	52.0	2.35	5.0	1.6	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.4	34.5	60.3	2.47	4.8	1.7	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	8.9	31.9	45.1	2.22	5.1	1.4	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.4	33.5	54.7	2.39	4.8	1.6	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.3	35.0	61.3	2.49	5.0	1.8	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.5	36.2	72.1	2.63	4.6	2.0	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.0	39.2	75.3	2.76	5.4	1.9	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.0	40.7	88.7	2.91	5.2	2.2	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.2	41.7	100.1	3.01	5.0	2.4	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	23.3	42.8	112.0	3.10	4.8	2.6	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	26.0	43.6	122.1	3.16	4.7	2.8	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 15.313 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 6.699 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8	X	21.5	T	20.94	29.4	42.9	126.9	3.14	4.3	3.0	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T	23.53	32.8	43.8	137.6	3.18	4.2	3.1	6.92	6.90	8.03	.595	.340	2.35
8	X	4	X	10.0	I-T	7.19	9.5	41.3	64.1	2.70	6.8	1.6	2.11	7.89	3.94	.205	.170	1.34
8	X	4	X	13.0	I-T	9.52	12.2	43.2	80.0	2.90	6.6	1.9	2.80	7.99	4.00	.255	.230	1.84
8	X	5 1/2	X	13.0	T	12.83	18.4	46.1	108.8	3.22	5.9	2.4	3.77	7.85	5.50	.345	.250	1.96
8	X	4	X	15.0	I-T	10.79	14.3	45.1	93.1	3.07	6.5	2.1	3.17	8.11	4.02	.315	.245	1.99
8	X	5 1/2	X	15.5	T	15.28	22.4	47.9	128.0	3.38	5.7	2.7	4.49	7.94	5.53	.440	.275	2.18
8	X	5 1/4	X	18.0	I-T	12.00	17.7	47.4	110.5	3.29	6.2	2.3	3.53	8.14	5.25	.330	.230	1.87
8	X	7	X	18.0	T	17.73	26.8	49.1	145.2	3.49	5.4	3.0	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T	19.79	30.6	50.3	160.8	3.58	5.3	3.2	5.82	8.01	7.00	.505	.305	2.44
8	X	5 1/4	X	21.0	I-T	13.87	21.0	49.3	128.4	3.45	6.1	2.6	4.08	8.28	5.27	.400	.250	2.07
8	X	7	X	22.5	T	22.32	34.1	51.4	174.3	3.63	5.1	3.4	6.56	8.07	7.04	.565	.345	2.78
8	X	6 1/2	X	24.0	I-T	15.11	23.4	48.3	131.9	3.44	5.6	2.7	4.44	7.93	6.50	.400	.245	1.94
8	X	7 1/8	X	25.0	T	24.83	37.6	52.4	187.6	3.66	5.0	3.6	7.30	8.13	7.07	.630	.380	3.09
8	X	6 1/2	X	28.0	I-T	17.69	27.3	49.9	149.9	3.55	5.5	3.0	5.20	8.06	6.54	.465	.285	2.30
8	X	7 1/8	X	28.5	T	28.28	42.3	53.8	205.0	3.69	4.8	3.8	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T	19.16	30.4	50.4	159.9	3.60	5.3	3.2	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T	17.26	27.1	54.4	167.8	3.77	6.2	3.1	5.08	8.85	6.00	.425	.300	2.66
9	X	6	X	20.0	T	19.76	32.0	56.2	191.3	3.91	6.0	3.4	5.81	8.95	6.02	.525	.315	2.82
10	X	4	X	12.0	I-T	9.07	13.5	53.0	111.1	3.44	8.2	2.1	2.67	9.87	3.96	.210	.190	1.88
10	X	4	X	15.0	I-T	11.27	17.0	55.6	135.6	3.68	8.0	2.4	3.32	9.99	4.00	.270	.230	2.30
10	X	4	X	17.0	I-T	12.48	19.6	57.8	154.4	3.86	7.9	2.7	3.67	10.11	4.01	.330	.240	2.43
10	X	4	X	19.0	I-T	13.77	22.4	59.9	174.3	4.03	7.8	2.9	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T	15.04	26.7	61.7	197.5	4.21	7.4	3.2	4.42	10.17	5.75	.360	.240	2.44

(35T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 15.313 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 6.699 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	31.7	64.1	228.4	4.40	7.2	3.6	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	36.7	66.2	257.6	4.51	7.0	3.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	38.2	62.4	240.9	4.33	6.3	3.9	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	45.6	64.9	277.6	4.47	6.1	4.3	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	52.9	67.3	312.3	4.56	5.9	4.6	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	18.6	65.7	179.1	4.25	9.6	2.7	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.2	67.6	200.9	4.41	9.5	3.0	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	25.8	71.0	238.3	4.68	9.2	3.4	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	30.3	73.8	273.6	4.88	9.0	3.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	37.6	76.7	319.4	5.18	8.5	4.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	43.2	78.9	356.7	5.31	8.3	4.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	50.7	81.8	404.7	5.43	8.0	4.9	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	55.3	79.3	403.4	5.33	7.3	5.1	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.61	61.8	81.5	439.2	5.38	7.1	5.4	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	68.5	83.8	475.9	5.43	6.9	5.7	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	73.7	83.5	489.2	5.46	6.6	5.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	81.1	85.4	525.1	5.52	6.5	6.1	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	33.7	83.0	340.2	5.45	10.1	4.1	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	40.5	86.6	395.8	5.68	9.8	4.6	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	47.1	88.6	438.9	5.83	9.3	5.0	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	53.8	91.4	488.3	5.99	9.1	5.3	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	60.1	93.9	532.7	6.08	8.9	5.7	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	66.5	92.5	545.5	6.04	8.2	5.9	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	74.3	95.2	593.6	6.10	8.0	6.2	9.26	13.79	8.03	.595	.340	4.69

(35T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 15.313 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 6.699 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	44.6	97.6	493.3	6.30	11.1	5.1	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	53.7	102.1	574.1	6.55	10.7	5.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	62.8	105.1	640.7	6.70	10.2	6.1	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	70.9	108.2	704.8	6.86	9.9	6.5	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	79.2	111.5	767.1	6.91	9.7	6.9	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	87.6	114.9	829.8	6.97	9.5	7.2	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	98.7	119.5	911.8	7.01	9.2	7.6	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	120.6	121.5	1014.8	7.18	8.4	8.4	12.99	16.33	10.24	.665	.395	6.45
18 X 6 X 35.0 I-T	26.29	65.5	116.5	760.5	7.26	11.6	6.5	7.73	17.78	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	76.2	121.2	857.9	7.48	11.3	7.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	95.9	127.8	1009.7	7.67	10.5	7.9	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	114.5	135.1	1157.5	7.76	10.1	8.6	12.53	18.24	7.56	.695	.415	7.57
21 X 8 1/4 X 62.0 I-T	44.94	133.4	158.5	1552.1	8.83	11.6	9.8	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	146.2	163.5	1664.4	8.87	11.4	10.2	14.46	21.13	8.27	.685	.430	9.09

(35T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 17.500 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 8.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	4.1	21.5	15.2	1.22	3.7	.7	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.2	23.2	19.0	1.34	3.7	.8	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.2	24.7	22.5	1.43	3.7	.9	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	7.9	26.6	27.8	1.57	3.5	1.0	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	7.0	26.0	25.6	1.51	3.7	1.0	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.6	28.9	25.4	1.56	4.6	.9	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	7.0	31.0	31.5	1.70	4.5	1.0	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.2	32.6	36.6	1.81	4.4	1.1	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.5	34.0	41.9	1.91	4.4	1.2	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.3	34.1	43.7	1.94	4.2	1.3	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.4	36.1	52.1	2.07	4.2	1.4	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.5	37.0	40.1	1.93	5.4	1.1	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.6	38.4	45.7	2.03	5.3	1.2	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.8	36.1	36.7	1.86	5.4	1.0	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.7	40.8	55.8	2.20	5.2	1.4	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.7	42.6	65.1	2.33	5.1	1.5	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.1	39.0	48.1	2.07	5.3	1.2	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.6	41.3	58.8	2.25	5.1	1.4	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.6	43.3	66.1	2.36	5.3	1.5	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	15.8	45.0	78.4	2.51	5.0	1.7	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.2	48.6	81.1	2.61	5.7	1.7	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.3	50.7	96.2	2.77	5.6	1.9	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.6	52.1	109.5	2.89	5.3	2.1	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	23.8	53.5	123.2	3.00	5.2	2.3	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	26.6	54.7	135.1	3.08	5.1	2.5	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 17.500 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 8.750 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT							WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
							LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8		X	21.5	T	20.94	30.1	53.9	141.5	3.08	4.7	2.6	6.16	6.83	8.00	.530	.305	2.08
7	X	8		X	24.0	T	23.53	33.6	55.0	154.3	3.14	4.6	2.8	6.92	6.90	8.03	.595	.340	2.35
8	X	4		X	10.0	I-T	7.19	9.6	50.6	67.8	2.50	7.1	1.3	2.11	7.89	3.94	.205	.170	1.34
8	X	4		X	13.0	I-T	9.52	12.4	53.3	85.3	2.72	6.9	1.6	2.80	7.99	4.00	.255	.230	1.84
8	X	5	1/2	X	13.0	T	12.83	18.7	57.4	117.7	3.07	6.3	2.1	3.77	7.85	5.50	.345	.250	1.96
8	X	4		X	15.0	I-T	10.79	14.6	55.9	99.7	2.89	6.8	1.8	3.17	8.11	4.02	.315	.245	1.99
8	X	5	1/2	X	15.5	T	15.28	22.9	59.8	139.7	3.25	6.1	2.3	4.49	7.94	5.53	.440	.275	2.18
8	X	5	1/4	X	18.0	I-T	12.00	18.0	59.0	119.1	3.12	6.6	2.0	3.53	8.14	5.25	.330	.230	1.87
8	X	7		X	18.0	T	17.73	27.4	61.5	159.6	3.38	5.8	2.6	5.22	7.93	6.99	.430	.295	2.34
8	X	7		X	20.0	T	19.79	31.3	63.1	177.9	3.49	5.7	2.8	5.82	8.01	7.00	.505	.305	2.44
8	X	5	1/4	X	21.0	I-T	13.87	21.4	61.6	139.3	3.30	6.5	2.3	4.08	8.28	5.27	.400	.250	2.07
8	X	7		X	22.5	T	22.32	34.9	64.3	193.8	3.56	5.6	3.0	6.56	8.07	7.04	.565	.345	2.78
8	X	6	1/2	X	24.0	I-T	15.11	23.8	60.5	144.1	3.30	6.0	2.4	4.44	7.93	6.50	.400	.245	1.94
8	X	7	1/8	X	25.0	T	24.83	38.6	65.5	209.5	3.61	5.4	3.2	7.30	8.13	7.07	.630	.380	3.09
8	X	6	1/2	X	28.0	I-T	17.69	27.8	62.5	164.9	3.44	5.9	2.6	5.20	8.06	6.54	.465	.285	2.30
8	X	7	1/8	X	28.5	T	28.28	43.5	67.1	230.1	3.67	5.3	3.4	8.32	8.22	7.12	.715	.430	3.53
8	X	8		X	31.0	I-T	19.16	31.0	63.2	176.7	3.50	5.7	2.8	5.63	8.00	8.00	.435	.285	2.28
9	X	6		X	17.5	T	17.26	27.6	67.9	183.6	3.64	6.6	2.7	5.08	8.85	6.00	.425	.300	2.66
9	X	6		X	20.0	T	19.76	32.7	70.2	210.8	3.80	6.4	3.0	5.81	8.95	6.02	.525	.315	2.82
10	X	4		X	12.0	I-T	9.07	13.8	65.7	117.9	3.21	8.6	1.8	2.67	9.87	3.96	.210	.190	1.88
10	X	4		X	15.0	I-T	11.27	17.3	69.0	145.1	3.47	8.4	2.1	3.32	9.99	4.00	.270	.230	2.30
10	X	4		X	17.0	I-T	12.48	20.0	71.9	165.9	3.65	8.3	2.3	3.67	10.11	4.01	.330	.240	2.43
10	X	4		X	19.0	I-T	13.77	22.9	74.6	188.1	3.83	8.2	2.5	4.05	10.24	4.02	.395	.250	2.56
10	X	5	3/4	X	22.0	I-T	15.04	27.2	77.1	214.4	4.03	7.9	2.8	4.42	10.17	5.75	.360	.240	2.44

(35T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 17.500 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 8.750 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	32.3	80.2	249.6	4.24	7.7	3.1	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	37.5	82.8	283.2	4.39	7.5	3.4	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.0	78.2	266.1	4.22	6.8	3.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	46.7	81.4	309.0	4.40	6.6	3.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	54.2	84.2	349.7	4.53	6.4	4.2	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.0	81.7	191.0	3.99	10.1	2.3	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	21.7	84.2	215.2	4.17	9.9	2.6	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.3	88.6	256.8	4.46	9.8	2.9	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	30.9	92.0	296.5	4.68	9.6	3.2	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	38.3	96.1	348.5	5.00	9.1	3.6	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	44.1	98.7	391.5	5.16	8.9	4.0	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	51.8	102.1	446.9	5.33	8.6	4.4	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	56.6	99.2	448.2	5.25	7.9	4.5	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	63.3	101.7	490.0	5.33	7.7	4.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	70.3	104.3	532.9	5.41	7.6	5.1	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	75.5	104.2	550.0	5.46	7.3	5.3	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	83.2	106.4	592.6	5.54	7.1	5.6	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	34.5	103.7	368.2	5.22	10.7	3.6	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	41.4	108.0	431.2	5.49	10.4	4.0	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	48.1	110.6	480.9	5.67	10.0	4.3	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	55.1	114.0	537.6	5.86	9.8	4.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	61.6	116.8	588.9	5.98	9.6	5.0	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	68.1	115.3	606.4	5.97	8.9	5.3	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	76.2	118.3	662.4	6.06	8.7	5.6	9.26	13.79	8.03	.595	.340	4.69

(35T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 17.500 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 8.750 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	45.6	121.6	536.9	6.09	11.8	4.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	55.0	127.0	628.8	6.38	11.4	5.0	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	64.4	130.6	705.5	6.58	11.0	5.4	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	72.8	134.4	779.3	6.77	10.7	5.8	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	81.4	137.9	851.0	6.86	10.5	6.2	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	90.2	141.6	923.1	6.94	10.2	6.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	101.8	146.5	1017.0	7.03	10.0	6.9	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	124.2	149.5	1141.8	7.25	9.2	7.6	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	141.3	155.4	1259.4	7.28	8.9	8.1	15.00	16.52	10.30	.760	.455	7.52
18 X 6 X 35.0 I-T	26.29	67.3	144.3	835.1	7.12	12.4	5.8	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	78.3	149.9	946.3	7.38	12.1	6.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	98.7	157.3	1121.6	7.64	11.4	7.1	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	118.1	165.2	1290.9	7.79	10.9	7.8	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	137.5	173.5	1455.5	7.84	10.6	8.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	172.7	178.8	1659.6	8.03	9.6	9.3	17.00	18.39	11.09	.770	.480	8.83
21 X 8 1/4 X 62.0 I-T	44.94	137.8	193.0	1727.5	8.87	12.5	9.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	151.1	198.3	1855.1	8.94	12.3	9.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	161.7	202.6	1954.7	8.98	12.1	9.6	15.47	21.24	8.30	.740	.455	9.66
21 X 12 1/4 X 101.0 I-T	68.38	230.3	218.1	2448.8	9.21	10.6	11.2	20.11	21.36	12.29	.800	.500	10.68

(35T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 19.688 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 11.074 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T	4.91	4.1	24.3	16.0	1.13	3.9	.7	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T	6.40	5.3	26.7	20.0	1.24	3.8	.8	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T	7.42	6.3	28.7	23.8	1.34	3.8	.8	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	8.0	31.2	29.6	1.47	3.7	.9	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	7.1	30.3	27.1	1.42	3.8	.9	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.7	33.3	26.6	1.44	4.7	.8	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	7.1	36.1	33.2	1.58	4.6	.9	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.4	38.3	38.7	1.69	4.6	1.0	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	9.7	40.2	44.4	1.79	4.6	1.1	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.5	40.5	46.4	1.82	4.4	1.1	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	12.6	43.1	55.6	1.96	4.4	1.3	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.6	43.2	42.1	1.79	5.5	1.0	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	8.7	45.2	48.0	1.89	5.5	1.1	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	6.9	42.0	38.4	1.73	5.5	.9	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	10.9	48.5	58.9	2.06	5.4	1.2	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	12.9	50.9	69.1	2.20	5.4	1.4	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.2	46.0	50.7	1.94	5.5	1.1	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	11.8	49.2	62.2	2.11	5.3	1.3	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	12.8	51.7	70.1	2.22	5.5	1.4	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.0	54.2	83.7	2.38	5.2	1.5	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.4	58.2	85.9	2.46	6.0	1.5	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	17.6	61.1	102.6	2.63	5.8	1.7	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	20.9	63.0	117.4	2.76	5.6	1.9	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	24.1	65.0	132.9	2.88	5.5	2.0	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	27.1	66.5	146.4	2.97	5.4	2.2	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 19.688 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 11.074 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	30.6	65.8	154.4	2.99	5.0	2.3	6.16	6.83	8.00	.530	.305	2.68
7 X 8 X 24.0 T	23.53	34.2	67.2	169.2	3.07	4.9	2.5	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.7	59.8	70.8	2.32	7.3	1.2	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.6	63.5	89.7	2.54	7.1	1.4	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.0	69.3	125.2	2.90	6.6	1.8	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	14.8	67.0	105.3	2.72	7.1	1.6	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.2	72.5	149.6	3.10	6.4	2.1	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.2	71.2	126.4	2.94	6.9	1.8	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	27.8	74.7	172.1	3.25	6.2	2.3	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	31.8	76.9	192.8	3.38	6.1	2.5	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	21.7	74.5	148.6	3.13	6.8	2.0	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	35.5	78.4	211.0	3.46	5.9	2.7	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.2	73.4	154.5	3.15	6.4	2.1	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	39.3	79.8	229.0	3.53	5.8	2.9	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.3	76.0	177.7	3.30	6.3	2.3	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	44.4	81.8	252.8	3.61	5.7	3.1	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	31.5	77.0	191.3	3.38	6.1	2.5	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.1	82.4	197.2	3.49	7.0	2.4	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.3	85.5	227.7	3.67	6.8	2.7	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	13.9	78.6	123.6	3.00	8.9	1.6	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.6	83.1	153.0	3.26	8.7	1.8	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.3	86.8	175.5	3.45	8.7	2.0	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.2	90.3	199.7	3.63	8.6	2.2	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	27.6	93.7	228.6	3.84	8.3	2.4	4.42	10.17	5.75	.360	.240	2.44

(35T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 19.688 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 11.074 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	32.8	97.6	267.7	4.07	8.2	2.7	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.2	100.8	305.4	4.24	8.0	3.0	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	39.6	95.4	288.1	4.09	7.3	3.0	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	47.5	99.3	336.7	4.29	7.1	3.4	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	55.3	102.8	383.3	4.45	6.9	3.7	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.2	98.6	200.8	3.75	10.4	2.0	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.0	101.8	227.1	3.93	10.3	2.2	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	26.7	107.4	272.3	4.23	10.2	2.5	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.4	111.7	315.9	4.46	10.0	2.8	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	38.9	117.2	373.4	4.79	9.6	3.2	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	44.8	120.4	421.6	4.97	9.4	3.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	52.8	124.4	484.0	5.17	9.2	3.9	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	57.6	121.2	487.9	5.13	8.5	4.0	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	64.5	124.0	535.7	5.23	8.3	4.3	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	71.8	127.0	584.7	5.34	8.1	4.6	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	77.0	127.1	605.4	5.40	7.9	4.8	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	84.9	129.8	654.6	5.50	7.7	5.0	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.0	126.1	391.9	4.98	11.2	3.1	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.1	131.5	461.6	5.27	11.0	3.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.0	134.7	517.3	5.47	10.6	3.8	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	56.1	138.9	580.8	5.68	10.4	4.2	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	62.8	142.2	638.6	5.83	10.2	4.5	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	69.4	140.6	660.9	5.85	9.5	4.7	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	77.8	144.0	724.8	5.97	9.3	5.0	9.26	13.79	8.03	.595	.340	4.69

(35T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 19.688 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 11.074 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	46.4	148.0	574.2	5.84	12.4	3.9	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	56.1	154.5	676.3	6.17	12.1	4.4	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	65.6	158.8	762.7	6.40	11.6	4.8	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	74.2	163.4	845.7	6.61	11.4	5.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	83.1	167.3	927.0	6.74	11.2	5.5	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	92.2	171.3	1008.6	6.85	10.9	5.9	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	104.4	176.6	1114.9	6.97	10.7	6.3	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	127.1	180.9	1261.1	7.24	9.9	7.0	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	144.9	187.1	1395.1	7.32	9.6	7.5	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	166.2	194.7	1552.1	7.38	9.3	8.0	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	68.7	175.0	901.1	6.92	13.1	5.1	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	80.0	181.7	1025.6	7.21	12.8	5.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	101.0	190.3	1224.1	7.54	12.1	6.4	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	121.1	198.9	1415.6	7.74	11.7	7.1	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	141.4	207.7	1600.8	7.85	11.3	7.7	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	177.4	214.3	1839.3	8.09	10.4	8.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	198.9	221.7	2008.0	8.15	10.1	9.1	19.18	18.59	11.15	.870	.535	9.95
21 X 8 1/4 X 62.0 I-T	44.94	141.4	231.7	1892.2	8.63	13.4	8.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	155.2	237.5	2036.0	8.93	13.1	8.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	166.2	242.1	2148.1	9.00	12.9	8.9	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	186.6	250.9	2353.2	9.06	12.6	9.4	17.58	21.43	8.36	.835	.515	11.04
21 X 12 1/4 X 101.0 I-T	68.38	236.8	259.6	2715.0	9.33	11.5	10.5	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	257.9	267.2	2896.6	9.34	11.2	10.8	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 21.875 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 13.672 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5.0 T					4.91	4.2	26.7	16.7	1.05	3.9	.6	1.44	3.95	3.94	.205	.170	.67
4 X 4 X 6.5 T					6.40	5.4	29.7	21.0	1.16	3.9	.7	1.88	4.00	4.00	.255	.230	.92
4 X 4 X 7.5 T					7.42	6.4	32.2	25.0	1.26	3.9	.8	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T					8.82	8.2	35.6	31.2	1.38	3.8	.9	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T					8.39	7.2	34.3	28.6	1.33	4.0	.8	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T					5.88	5.7	37.1	27.7	1.34	4.8	.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T					7.37	7.3	40.8	34.7	1.48	4.8	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T					8.36	8.5	43.6	40.5	1.58	4.8	.9	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T					9.42	9.8	46.2	46.6	1.68	4.7	1.0	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T					9.91	10.7	46.6	48.9	1.72	4.6	1.0	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T					11.69	12.8	50.0	58.8	1.85	4.6	1.2	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T					6.94	7.7	48.9	43.8	1.67	5.7	.9	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T					7.88	8.9	51.5	50.1	1.77	5.7	1.0	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T					6.17	7.0	47.3	39.9	1.60	5.7	.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T					9.34	11.0	55.8	61.6	1.94	5.6	1.1	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T					10.89	13.1	59.1	72.6	2.07	5.6	1.2	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T					8.30	9.4	52.6	52.9	1.81	5.6	1.0	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T					9.78	11.9	56.9	65.2	1.98	5.5	1.1	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T					10.74	13.0	59.9	73.6	2.09	5.7	1.2	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T					12.63	16.3	63.3	88.3	2.25	5.4	1.4	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T					10.81	14.6	67.8	90.1	2.31	6.2	1.3	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T					12.85	17.8	71.6	108.1	2.49	6.1	1.5	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T					14.81	21.2	74.2	124.3	2.63	5.9	1.7	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T					16.77	24.5	76.9	141.3	2.76	5.8	1.8	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T					18.74	27.4	78.8	156.2	2.85	5.7	2.0	5.51	7.05	6.77	.515	.310	2.19

(35T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 21.875 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 13.672 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 21.5 T	20.94	31.1	78.3	165.7	2.89	5.3	2.1	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	34.8	80.1	182.4	2.98	5.2	2.3	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T	7.19	9.9	68.5	73.3	2.16	7.4	1.1	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	12.7	73.6	93.5	2.38	7.3	1.3	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.2	81.3	131.6	2.75	6.9	1.6	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.0	78.0	110.1	2.56	7.3	1.4	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	23.6	85.6	158.2	2.95	6.7	1.8	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.5	83.5	132.6	2.78	7.2	1.6	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.2	88.6	182.9	3.11	6.5	2.1	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	32.2	91.4	205.8	3.25	6.4	2.3	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.0	87.9	156.5	2.97	7.1	1.8	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	36.1	93.3	226.1	3.34	6.3	2.4	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	24.5	86.8	163.3	3.00	6.7	1.9	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	40.0	95.1	246.4	3.43	6.2	2.6	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	28.7	90.2	188.9	3.16	6.6	2.1	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	45.2	97.5	273.1	3.52	6.0	2.8	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	31.9	91.6	204.0	3.25	6.4	2.2	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	28.5	97.6	208.8	3.34	7.3	2.1	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	33.7	101.5	242.4	3.53	7.2	2.4	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.1	91.4	128.3	2.80	9.1	1.4	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	17.8	97.3	159.6	3.07	9.0	1.6	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	20.5	102.0	183.6	3.25	8.9	1.8	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	23.5	106.5	209.5	3.44	8.9	2.0	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	27.9	110.9	240.7	3.65	8.6	2.2	4.42	10.17	5.75	.360	.240	2.44

(35T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 21.875 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 13.672 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 26.0 I-T	17.37	33.3	116.0	283.2	3.88	8.5	2.4	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	38.7	119.9	324.7	4.07	8.4	2.7	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.1	113.7	307.2	3.94	7.7	2.7	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.2	118.6	361.2	4.16	7.5	3.0	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	56.1	122.9	413.3	4.34	7.4	3.4	8.29	10.18	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.5	115.9	209.0	3.52	10.7	1.8	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.3	120.0	237.1	3.70	10.6	2.0	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.1	127.1	285.3	4.00	10.5	2.2	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	31.9	132.5	332.4	4.24	10.4	2.5	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.4	139.6	394.6	4.57	10.0	2.8	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.4	143.6	447.5	4.77	9.8	3.1	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	53.5	148.5	516.5	5.00	9.6	3.5	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	58.4	144.9	522.8	4.97	9.0	3.6	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	65.5	148.2	576.3	5.10	8.8	3.9	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	73.0	151.6	631.2	5.23	8.7	4.2	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	78.2	152.1	655.3	5.29	8.4	4.3	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	86.3	155.3	711.0	5.42	8.2	4.6	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.5	149.9	412.0	4.73	11.6	2.7	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	42.7	156.6	487.5	5.04	11.4	3.1	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	49.7	160.6	548.7	5.25	11.0	3.4	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	56.9	165.7	618.4	5.48	10.9	3.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	63.8	169.7	682.4	5.65	10.7	4.0	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	70.5	168.0	709.1	5.69	10.1	4.2	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	79.0	171.9	780.6	5.83	9.9	4.5	9.26	13.79	8.03	.595	.340	4.69

(35T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 21.875 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 13.672 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 26.0 I-T	19.49	47.1	176.3	606.2	5.59	12.9	3.4	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	56.9	184.1	717.6	5.94	12.6	3.9	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	66.7	189.4	812.8	6.19	12.2	4.3	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	75.4	194.9	904.4	6.42	12.0	4.6	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	84.6	199.3	994.9	6.57	11.8	5.0	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	94.0	203.8	1085.9	6.72	11.6	5.3	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	106.5	209.7	1204.5	6.87	11.3	5.7	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	129.5	215.4	1371.3	7.17	10.6	6.4	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	147.9	222.2	1522.4	7.29	10.3	6.9	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	169.9	230.3	1699.1	7.39	10.0	7.4	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	69.9	208.5	959.0	6.69	13.7	4.6	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	81.4	216.4	1095.8	7.01	13.5	5.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	102.9	226.4	1316.6	7.39	12.8	5.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	123.6	236.0	1530.3	7.64	12.4	6.5	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	144.6	245.3	1737.0	7.79	12.0	7.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	181.3	253.5	2009.8	8.10	11.1	7.9	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	203.6	261.5	2199.2	8.18	10.8	8.4	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	220.2	267.8	2339.8	8.21	10.6	8.7	21.02	18.73	11.20	.940	.590	11.05
21 X 8 1/4 X 62.0 I-T	44.94	144.3	274.3	2044.3	8.72	14.2	7.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	158.6	280.7	2204.7	8.85	13.9	7.9	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	169.9	285.6	2329.6	8.94	13.7	8.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	191.1	294.9	2557.6	9.05	13.4	8.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	212.7	304.5	2785.7	9.12	13.1	9.1	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	242.4	305.5	2971.6	9.38	12.3	9.7	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	264.3	313.4	3173.9	9.41	12.0	10.1	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 24.063 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 16.543 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 7.5 T	7.42	6.5	35.3	26.1	1.18	4.0	.7	2.18	4.06	4.02	.315	.245	.99
4 X 5 1/4 X 9.0 T	8.82	8.3	39.5	32.7	1.31	3.9	.8	2.59	4.07	5.25	.330	.230	.94
4 X 4 X 13.0 I-T	8.39	7.4	37.9	29.9	1.25	4.1	.8	2.47	4.16	4.06	.345	.280	1.16
5 X 4 X 6.0 T	5.88	5.8	40.3	28.7	1.25	4.9	.7	1.73	4.94	3.96	.210	.190	.94
5 X 4 X 7.5 T	7.37	7.4	44.9	36.0	1.39	4.9	.8	2.17	5.00	4.00	.270	.230	1.15
5 X 4 X 8.5 T	8.36	8.6	48.4	42.2	1.49	4.9	.9	2.46	5.06	4.01	.330	.240	1.21
5 X 4 X 9.5 T	9.42	10.0	51.6	48.6	1.59	4.9	.9	2.77	5.12	4.02	.395	.250	1.28
5 X 5 X 16.0 I-T	9.91	10.8	52.3	51.0	1.62	4.7	1.0	2.92	5.01	5.00	.360	.240	1.20
5 X 5 X 19.0 I-T	11.69	13.0	56.6	61.6	1.76	4.7	1.1	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 7.0 T	6.94	7.8	54.0	45.3	1.56	5.8	.8	2.04	5.96	3.97	.225	.200	1.19
6 X 4 X 8.0 T	7.88	9.0	57.3	51.9	1.66	5.8	.9	2.32	6.00	3.99	.265	.220	1.32
6 X 4 X 9.0 I-T	6.17	7.1	51.9	41.2	1.50	5.8	.8	1.81	5.90	3.94	.215	.170	1.00
6 X 4 X 9.5 T	9.34	11.2	62.7	64.1	1.82	5.7	1.0	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.2	66.8	75.7	1.96	5.7	1.1	3.20	6.16	4.03	.425	.260	1.60
6 X 4 X 12.0 I-T	8.30	9.5	58.6	54.9	1.70	5.8	.9	2.44	6.03	4.00	.280	.230	1.39
6 X 6 X 15.0 I-T	9.78	12.1	64.1	67.8	1.87	5.6	1.1	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.1	67.8	76.6	1.97	5.8	1.1	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	16.5	72.3	92.4	2.14	5.6	1.3	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	14.8	77.0	93.7	2.18	6.3	1.2	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.0	82.0	112.9	2.36	6.3	1.4	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	21.4	85.5	130.4	2.50	6.1	1.5	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	24.8	88.9	148.7	2.63	6.0	1.7	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	27.8	91.4	164.9	2.73	5.9	1.8	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	31.5	91.1	175.8	2.78	5.6	1.9	6.16	6.83	8.00	.530	.305	2.08

(35T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 24.063 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 16.543 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7 X 8 X 24.0 T		23.53	35.2	93.4	194.1	2.88	5.5	2.1	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 10.0 I-T		7.19	10.0	76.5	75.6	2.01	7.6	1.0	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T		9.52	12.9	83.1	96.8	2.24	7.5	1.2	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T		12.83	19.4	93.3	137.2	2.60	7.1	1.5	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T		10.79	15.2	88.8	114.2	2.41	7.5	1.3	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T		15.28	23.8	98.8	165.6	2.81	7.0	1.7	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T		12.00	18.7	95.6	137.9	2.62	7.4	1.4	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T		17.73	28.5	102.7	192.4	2.97	6.7	1.9	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T		19.79	32.6	106.3	217.2	3.12	6.7	2.0	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T		13.87	22.2	101.2	163.3	2.81	7.4	1.6	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T		22.32	36.5	108.8	239.4	3.22	6.6	2.2	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T		15.11	24.7	100.4	171.0	2.85	6.9	1.7	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T		24.83	40.5	111.1	261.8	3.31	6.5	2.4	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T		17.69	29.0	104.7	198.6	3.02	6.8	1.9	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T		28.28	45.9	114.0	291.3	3.42	6.4	2.6	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T		19.16	32.3	106.5	215.1	3.11	6.7	2.0	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T		17.26	28.8	113.1	219.0	3.18	7.6	1.9	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T		19.76	34.1	118.2	255.3	3.38	7.5	2.2	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T		9.07	14.3	103.7	132.4	2.62	9.3	1.3	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T		11.27	18.0	111.4	165.3	2.89	9.2	1.5	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T		12.48	20.8	117.3	190.6	3.07	9.2	1.6	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T		13.77	23.8	122.9	218.0	3.25	9.2	1.8	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T		15.04	28.2	128.6	251.1	3.46	8.9	2.0	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T		17.37	33.6	135.0	296.6	3.70	8.8	2.2	5.11	10.33	5.77	.440	.260	2.69

(35T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 24.063 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 16.543 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 5 3/4 X 30.0 I-T	20.23	39.2	139.8	341.4	3.90	8.7	2.4	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	40.6	132.8	323.9	3.78	8.0	2.4	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	48.7	138.9	382.7	4.02	7.9	2.8	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	56.9	144.1	439.9	4.21	7.7	3.1	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.7	133.0	216.0	3.30	11.0	1.6	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.5	138.2	245.6	3.49	10.9	1.8	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.4	147.2	296.5	3.78	10.8	2.0	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.2	153.9	346.5	4.03	10.7	2.3	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	39.8	163.0	412.8	4.36	10.4	2.5	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	45.9	168.0	469.9	4.57	10.2	2.8	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.2	173.9	544.8	4.81	10.1	3.1	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.1	170.0	553.5	4.80	9.4	3.3	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	66.4	173.9	612.3	4.95	9.2	3.5	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	74.0	178.0	672.8	5.09	9.1	3.8	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	79.3	178.8	700.1	5.16	8.8	3.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	87.5	182.7	761.8	5.30	8.7	4.2	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	35.8	174.6	429.1	4.49	12.0	2.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.2	182.8	509.9	4.80	11.8	2.8	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.2	187.9	575.8	5.03	11.5	3.1	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	57.6	194.1	651.2	5.27	11.3	3.4	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	64.6	198.9	720.8	5.45	11.2	3.6	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	71.3	197.3	751.6	5.51	10.5	3.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	80.1	201.9	830.3	5.67	10.4	4.1	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	47.6	206.0	633.6	5.33	13.3	3.1	5.73	15.69	5.50	.345	.250	3.92

(35T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 24.063 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 16.543 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 5 1/2 X 31.0 I-T	22.70	57.6	215.6	753.4	5.70	13.1	3.5	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	67.5	222.0	856.7	5.96	12.7	3.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	76.4	228.7	956.1	6.21	12.5	4.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	85.8	233.7	1055.3	6.38	12.3	4.5	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	95.4	238.9	1155.2	6.55	12.1	4.8	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	108.3	245.4	1285.9	6.73	11.9	5.2	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	131.5	252.8	1472.1	7.06	11.2	5.8	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	150.4	260.2	1640.4	7.21	10.9	6.3	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	173.1	269.1	1837.1	7.36	10.6	6.8	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	70.8	244.2	1009.6	6.45	14.3	4.1	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	82.5	253.7	1157.7	6.78	14.0	4.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	104.4	265.4	1399.6	7.20	13.4	5.3	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	125.7	276.2	1634.8	7.50	13.0	5.9	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	147.3	286.2	1863.2	7.69	12.6	6.5	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	184.6	296.2	2169.5	8.04	11.8	7.3	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	207.5	304.9	2380.3	8.16	11.5	7.8	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	224.8	311.5	2535.3	8.22	11.3	8.1	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	252.1	321.9	2779.1	8.31	11.0	8.6	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	146.8	320.7	2183.4	8.57	14.9	6.8	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	161.5	327.7	2360.1	8.73	14.6	7.2	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	173.1	333.1	2498.0	8.83	14.4	7.5	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	195.0	343.0	2749.6	8.98	14.1	8.0	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	217.3	353.1	3000.8	9.08	13.8	8.5	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	247.1	355.8	3215.3	9.37	13.0	9.0	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	269.8	364.0	3439.5	9.43	12.7	9.4	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 26.250 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 19.688 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X	5 1/4	X	9.0	T	8.82	8.5	42.9	34.1	1.24	4.0	.8	2.59	4.07	5.25	.330	.230	.94
4	X	4	X	13.0	I-T	8.39	7.5	41.0	31.2	1.19	4.1	.8	2.47	4.16	4.06	.345	.280	1.16
5	X	4	X	7.5	T	7.37	7.5	48.5	37.4	1.31	5.0	.8	2.17	5.00	4.00	.270	.230	1.15
5	X	4	X	8.5	T	8.36	8.8	52.7	43.7	1.41	5.0	.8	2.46	5.06	4.01	.330	.240	1.21
5	X	4	X	9.5	T	9.42	10.2	56.5	50.5	1.50	5.0	.9	2.77	5.12	4.02	.395	.250	1.28
5	X	5	X	16.0	I-T	9.91	11.0	57.5	53.1	1.53	4.8	.9	2.92	5.01	5.00	.360	.240	1.20
5	X	5	X	19.0	I-T	11.69	13.2	62.7	64.1	1.67	4.9	1.0	3.44	5.15	5.03	.430	.270	1.39
6	X	4	X	7.0	T	6.94	7.9	58.3	46.8	1.47	5.9	.8	2.04	5.96	3.97	.225	.200	1.19
6	X	4	X	8.0	T	7.88	9.1	62.3	53.7	1.56	5.9	.9	2.32	6.00	3.99	.265	.220	1.32
6	X	4	X	9.0	I-T	6.17	7.2	55.8	42.5	1.41	5.9	.8	1.81	5.90	3.94	.215	.170	1.00
6	X	4	X	9.5	T	9.34	11.3	68.9	66.3	1.72	5.9	1.0	2.75	6.08	4.01	.350	.235	1.43
6	X	4	X	11.0	T	10.89	13.4	74.1	78.5	1.85	5.9	1.1	3.20	6.16	4.03	.425	.260	1.60
6	X	4	X	12.0	I-T	8.30	9.6	64.0	56.7	1.60	5.9	.9	2.44	6.03	4.00	.280	.230	1.39
6	X	6	X	15.0	I-T	9.78	12.2	70.6	70.3	1.76	5.7	1.0	2.88	5.99	5.99	.260	.230	1.38
6	X	4	X	16.0	I-T	10.74	13.3	75.1	79.4	1.86	6.0	1.1	3.16	6.28	4.03	.405	.260	1.63
6	X	6	X	20.0	I-T	12.63	16.7	80.8	96.0	2.03	5.8	1.2	3.71	6.20	6.02	.365	.260	1.61
7	X	5	X	11.0	T	10.81	15.0	85.6	97.0	2.06	6.5	1.1	3.18	6.87	5.00	.335	.230	1.58
7	X	5	X	13.0	T	12.85	18.2	91.9	117.2	2.23	6.4	1.3	3.78	6.96	5.03	.420	.255	1.77
7	X	6 3/4	X	15.0	T	14.81	21.7	96.4	135.7	2.38	6.3	1.4	4.36	6.92	6.73	.385	.270	1.87
7	X	6 3/4	X	17.0	T	16.77	25.0	100.8	155.2	2.51	6.2	1.5	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	28.1	104.0	172.6	2.62	6.1	1.7	5.51	7.05	6.77	.515	.310	2.19
7	X	8	X	21.5	T	20.94	31.8	104.1	184.7	2.67	5.8	1.8	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T	23.53	35.7	107.0	204.6	2.77	5.7	1.9	6.92	6.90	8.03	.595	.340	2.35

(35T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 26.250 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 19.688 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 10.0 I-T	7.19	10.1	83.5	77.7	1.89	7.7	.9	2.11	7.89	3.94	.205	.170	1.34
8 X 4 X 13.0 I-T	9.52	13.0	92.0	99.7	2.11	7.7	1.1	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	19.6	104.8	142.2	2.46	7.2	1.4	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.4	98.9	117.9	2.27	7.7	1.2	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.1	111.8	172.2	2.67	7.1	1.5	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	18.9	107.3	142.6	2.48	7.6	1.3	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	28.8	116.9	200.7	2.84	7.0	1.7	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.0	121.4	227.2	2.98	6.9	1.9	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.4	114.2	169.4	2.67	7.5	1.5	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	36.9	124.5	251.3	3.09	6.8	2.0	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.0	113.7	177.8	2.71	7.1	1.6	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.0	127.4	275.5	3.19	6.7	2.2	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	29.3	119.1	207.2	2.89	7.1	1.7	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	46.5	131.0	307.7	3.31	6.6	2.3	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	32.6	121.6	224.9	2.98	6.9	1.8	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.1	128.7	228.0	3.03	7.8	1.8	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	34.5	135.0	266.6	3.23	7.7	2.0	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.4	115.3	136.0	2.47	9.4	1.2	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.2	124.9	170.3	2.72	9.4	1.4	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.0	132.1	196.7	2.90	9.4	1.5	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.1	139.0	225.4	3.08	9.4	1.6	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	28.5	146.2	260.2	3.29	9.1	1.8	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.0	154.1	308.3	3.53	9.1	2.0	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	39.6	160.2	356.1	3.73	9.0	2.2	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.0	152.3	338.6	3.62	8.3	2.2	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	49.2	159.9	401.8	3.87	8.2	2.5	7.19	9.92	7.99	.530	.315	3.12

(35T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 26.250 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 19.688 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 45.0 I-T	28.19	57.5	166.3	463.6	4.07	8.1	2.8	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	19.9	149.7	222.1	3.11	11.2	1.5	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	22.7	156.2	252.9	3.29	11.1	1.6	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	27.6	167.2	306.2	3.58	11.1	1.8	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	32.6	175.5	358.8	3.83	11.0	2.0	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.1	186.8	428.6	4.15	10.7	2.3	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	46.4	193.0	489.4	4.37	10.6	2.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	54.7	200.3	569.7	4.62	10.4	2.8	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	59.7	196.2	580.5	4.62	9.7	3.0	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	67.1	200.9	644.2	4.78	9.6	3.2	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	74.8	205.8	710.0	4.94	9.5	3.5	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	80.1	207.0	740.1	5.02	9.2	3.6	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	88.5	211.8	807.6	5.17	9.1	3.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.2	199.6	443.9	4.26	12.3	2.2	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	43.6	209.8	529.2	4.58	12.1	2.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	50.7	216.2	599.5	4.81	11.8	2.8	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	58.2	223.9	679.9	5.05	11.7	3.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	65.3	229.6	754.6	5.25	11.6	3.3	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	72.1	228.1	789.2	5.32	11.0	3.5	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	81.0	233.6	874.5	5.50	10.8	3.7	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.1	236.8	657.4	5.09	13.7	2.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	58.2	248.5	784.5	5.45	13.5	3.2	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	68.2	256.3	895.2	5.73	13.1	3.5	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	77.2	264.4	1001.7	5.99	13.0	3.8	8.26	16.01	7.00	.505	.305	4.88

(35T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 26.250 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 19.688 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 X 45.0 I-T	31.77	86.8	270.2	1108.9	6.18	12.8	4.1	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	96.6	276.1	1217.3	6.36	12.6	4.4	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	109.8	283.6	1359.5	6.57	12.4	4.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	133.2	292.9	1563.8	6.92	11.7	5.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	152.6	301.1	1748.8	7.10	11.5	5.8	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	175.9	310.9	1965.7	7.28	11.2	6.3	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	71.7	281.9	1054.0	6.20	14.7	3.7	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	83.5	293.2	1212.4	6.54	14.5	4.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	105.7	307.0	1473.8	6.99	13.9	4.8	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	127.4	319.2	1729.6	7.33	13.6	5.4	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	149.6	330.1	1979.0	7.56	13.2	6.0	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	187.4	342.3	2317.7	7.95	12.4	6.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	210.9	351.7	2550.1	8.10	12.1	7.3	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	228.7	358.7	2720.7	8.17	11.9	7.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	256.8	369.9	2989.3	8.30	11.6	8.1	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	149.0	370.5	2309.8	8.38	15.5	6.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	163.9	378.3	2502.4	8.56	15.3	6.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	175.9	384.3	2652.9	8.69	15.1	6.9	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	198.3	394.8	2928.0	8.86	14.8	7.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	221.3	405.6	3202.8	9.00	14.5	7.9	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	251.2	410.2	3444.4	9.30	13.7	8.4	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	274.5	418.9	3691.2	9.39	13.4	8.8	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 30.625 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 26.797 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 5 X 19.0 I-T	11.69	13.5	73.3	68.9	1.51	5.1	.9	3.44	5.15	5.03	.430	.270	1.39
6 X 4 X 9.5 T	9.34	11.6	79.3	70.5	1.54	6.1	.9	2.75	6.08	4.01	.350	.235	1.43
6 X 4 X 11.0 T	10.89	13.8	86.5	83.5	1.67	6.1	1.0	3.20	6.16	4.03	.425	.260	1.60
6 X 6 X 15.0 I-T	9.78	12.6	81.7	74.7	1.59	6.0	.9	2.88	5.99	5.99	.260	.230	1.38
6 X 4 X 16.0 I-T	10.74	13.6	87.6	84.5	1.68	6.2	1.0	3.16	6.28	4.03	.405	.260	1.63
6 X 6 X 20.0 I-T	12.63	17.1	95.9	102.5	1.83	6.0	1.1	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 11.0 T	10.81	15.3	100.6	102.8	1.85	6.7	1.0	3.18	6.87	5.00	.335	.230	1.58
7 X 5 X 13.0 T	12.85	18.6	109.8	124.7	2.02	6.7	1.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	22.1	116.7	145.0	2.16	6.6	1.2	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	25.6	123.2	166.6	2.29	6.5	1.4	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	28.7	128.2	185.9	2.40	6.5	1.5	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	32.5	129.3	200.1	2.46	6.2	1.5	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	36.4	133.9	222.7	2.57	6.1	1.7	6.92	6.90	8.03	.595	.340	2.35
8 X 4 X 13.0 I-T	9.52	13.3	106.9	105.0	1.88	7.9	1.0	2.80	7.99	4.00	.255	.230	1.84
8 X 5 1/2 X 13.0 T	12.83	20.0	125.8	150.7	2.22	7.5	1.2	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	15.7	116.5	124.4	2.04	7.9	1.1	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	24.6	136.2	183.5	2.42	7.5	1.3	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.2	128.3	150.8	2.23	7.8	1.2	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.8 T	17.73	29.4	144.1	215.0	2.59	7.3	1.5	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	33.6	151.0	244.4	2.74	7.3	1.6	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	22.9	138.4	179.7	2.41	7.9	1.3	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	37.7	155.8	271.5	2.85	7.2	1.7	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.5	138.8	189.5	2.46	7.4	1.4	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	41.9	160.3	299.0	2.96	7.1	1.9	7.30	8.13	7.07	.630	.380	3.09

(35T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 30.625 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 26.797 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 6 1/2 X 28.0 I-T	17.69	29.9	147.0	221.8	2.63	7.4	1.5	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	47.5	165.8	335.8	3.09	7.1	2.0	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.2	151.1	241.6	2.73	7.3	1.6	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	29.7	158.7	243.2	2.76	8.2	1.5	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.2	168.2	285.9	2.96	8.1	1.7	5.81	8.95	6.02	.525	.315	2.82
10 X 4 X 12.0 I-T	9.07	14.7	135.1	142.3	2.20	9.7	1.1	2.67	9.87	3.96	.210	.190	1.88
10 X 4 X 15.0 I-T	11.27	18.5	149.1	179.0	2.44	9.7	1.2	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.4	159.4	207.1	2.61	9.7	1.3	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.5	169.3	237.9	2.78	9.7	1.4	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	29.0	179.9	275.5	2.97	9.5	1.5	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	34.5	191.7	327.9	3.21	9.5	1.7	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	40.3	200.9	380.8	3.41	9.4	1.9	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	41.7	191.3	363.3	3.32	8.7	1.9	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	50.1	202.7	433.9	3.57	8.7	2.1	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	58.6	212.0	503.9	3.79	8.6	2.4	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.2	180.1	232.4	2.78	11.5	1.3	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.1	189.7	265.4	2.95	11.5	1.4	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.1	205.6	322.4	3.23	11.5	1.6	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.1	217.9	379.3	3.46	11.4	1.7	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	40.8	234.4	454.7	3.77	11.2	1.9	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.1	243.8	521.9	3.99	11.1	2.1	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	55.7	254.7	611.3	4.25	11.0	2.4	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	60.7	250.4	625.8	4.27	10.3	2.5	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	68.3	257.3	698.1	4.45	10.2	2.7	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	76.3	264.2	773.2	4.62	10.1	2.9	9.44	12.19	8.08	.640	.370	4.51

(35T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 30.625 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 26.797 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 10 X 53.0 I-T	33.01	81.6	266.7	808.1	4.70	9.9	3.0	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	90.1	273.4	885.8	4.87	9.8	3.2	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	36.8	249.2	468.3	3.85	12.7	1.9	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	44.3	264.3	561.2	4.17	12.7	2.1	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	51.6	274.0	638.7	4.40	12.4	2.3	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	59.1	285.1	727.5	4.64	12.3	2.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	66.4	293.5	811.0	4.85	12.2	2.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	73.3	292.8	852.2	4.93	11.6	2.9	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	82.5	300.5	949.1	5.13	11.5	3.2	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	48.9	299.6	696.5	4.63	14.2	2.3	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	59.2	316.5	836.0	5.00	14.1	2.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	69.4	328.0	959.1	5.28	13.8	2.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	78.6	339.5	1077.7	5.54	13.7	3.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	88.5	347.6	1199.2	5.76	13.6	3.4	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	98.6	355.8	1322.6	5.96	13.4	3.7	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	112.2	365.7	1485.7	6.20	13.2	4.1	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	135.9	379.8	1722.1	6.58	12.7	4.5	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	156.0	390.3	1938.9	6.81	12.4	5.0	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	180.3	402.4	2194.3	7.05	12.2	5.5	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	73.0	361.1	1127.7	5.71	15.5	3.1	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	85.1	376.9	1303.6	6.07	15.3	3.5	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	107.9	396.2	1599.3	6.55	14.8	4.0	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	130.3	412.3	1892.4	6.94	14.5	4.6	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	153.4	426.0	2181.7	7.23	14.2	5.1	14.93	18.47	7.64	.810	.495	9.14

(35T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 30.625 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 26.797 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/8 X 86.0 I-T	57.79	191.9	443.4	2580.1	7.68	13.4	5.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	216.4	454.8	2854.5	7.88	13.2	6.3	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	235.1	463.0	3056.7	7.99	13.0	6.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	264.5	476.2	3375.0	8.18	12.8	7.1	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	152.4	479.0	2527.4	7.95	16.6	5.3	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	167.9	488.9	2749.6	8.16	16.4	5.6	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	180.2	496.3	2924.0	8.32	16.2	5.9	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	203.7	509.0	3244.6	8.55	15.9	6.4	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.8 I-T	67.42	227.8	521.4	3565.8	8.74	15.7	6.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.8 I-T	68.38	257.7	530.4	3856.2	9.07	15.0	7.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	282.1	540.4	4148.7	9.21	14.7	7.7	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 35.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 35.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
6 X 6 X 20.0 I-T	12.63	17.5	108.1	108.5	1.67	6.2	1.0	3.71	6.20	6.02	.365	.260	1.61
7 X 5 X 13.0 T	12.85	19.1	124.5	131.5	1.84	6.9	1.1	3.78	6.96	5.03	.420	.255	1.77
7 X 6 3/4 X 15.0 T	14.81	22.6	133.9	153.3	1.97	6.8	1.1	4.36	6.92	6.73	.385	.270	1.87
7 X 6 3/4 X 17.0 T	16.77	26.1	143.0	176.4	2.10	6.8	1.2	4.93	6.99	6.75	.455	.285	1.99
7 X 6 3/4 X 19.0 T	18.74	29.3	150.0	197.4	2.21	6.7	1.3	5.51	7.05	6.77	.515	.310	2.19
7 X 8 X 21.5 T	20.94	33.2	152.6	213.3	2.28	6.4	1.4	6.16	6.83	8.00	.530	.305	2.08
7 X 8 X 24.0 T	23.53	37.2	159.2	238.2	2.38	6.4	1.5	6.92	6.90	8.03	.595	.340	2.35
8 X 5 1/2 X 13.0 T	12.83	20.4	143.2	156.3	2.02	7.7	1.1	3.77	7.85	5.50	.345	.250	1.96
8 X 4 X 15.0 I-T	10.79	16.1	130.3	130.3	1.85	8.1	1.0	3.17	8.11	4.02	.315	.245	1.99
8 X 5 1/2 X 15.5 T	15.28	25.1	157.3	193.2	2.21	7.7	1.2	4.49	7.94	5.53	.440	.275	2.18
8 X 5 1/4 X 18.0 I-T	12.00	19.6	145.5	158.1	2.03	8.1	1.1	3.53	8.14	5.25	.330	.230	1.87
8 X 7 X 18.0 T	17.73	30.0	168.6	227.1	2.38	7.6	1.3	5.22	7.93	6.99	.430	.295	2.34
8 X 7 X 20.0 T	19.79	34.3	178.2	258.9	2.52	7.6	1.5	5.82	8.01	7.00	.505	.305	2.44
8 X 5 1/4 X 21.0 I-T	13.87	23.3	159.0	186.7	2.20	8.1	1.2	4.08	8.28	5.27	.400	.250	2.07
8 X 7 X 22.5 T	22.32	38.4	185.3	288.5	2.63	7.5	1.6	6.56	8.07	7.04	.565	.345	2.78
8 X 6 1/2 X 24.0 I-T	15.11	25.9	160.7	199.5	2.25	7.7	1.2	4.44	7.93	6.50	.400	.245	1.94
8 X 7 1/8 X 25.0 T	24.83	42.7	191.8	318.8	2.74	7.5	1.7	7.30	8.13	7.07	.630	.380	3.09
8 X 6 1/2 X 28.0 I-T	17.69	30.4	172.2	234.3	2.41	7.7	1.4	5.20	8.06	6.54	.465	.285	2.30
8 X 7 1/8 X 28.5 T	28.28	48.4	199.8	359.5	2.88	7.4	1.8	8.32	8.22	7.12	.715	.430	3.53
8 X 8 X 31.0 I-T	19.16	33.8	178.1	255.8	2.51	7.6	1.4	5.63	8.00	8.00	.435	.285	2.28
9 X 6 X 17.5 T	17.26	30.2	185.7	256.0	2.53	8.5	1.4	5.08	8.85	6.00	.425	.300	2.66
9 X 6 X 20.0 T	19.76	35.8	199.0	301.9	2.72	8.4	1.5	5.81	8.95	6.02	.525	.315	2.82

(35T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 35.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 35.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 X 15.0 I-T	11.27	18.9	168.9	186.5	2.21	9.9	1.1	3.32	9.99	4.00	.270	.230	2.30
10 X 4 X 17.0 I-T	12.48	21.8	182.4	216.1	2.36	9.9	1.2	3.67	10.11	4.01	.330	.240	2.43
10 X 4 X 19.0 I-T	13.77	24.9	195.6	248.5	2.52	10.0	1.3	4.05	10.24	4.02	.395	.250	2.56
10 X 5 3/4 X 22.0 I-T	15.04	29.4	209.9	288.3	2.70	9.8	1.4	4.42	10.17	5.75	.360	.240	2.44
10 X 5 3/4 X 26.0 I-T	17.37	35.1	226.3	344.2	2.93	9.8	1.5	5.11	10.33	5.77	.440	.260	2.69
10 X 5 3/4 X 30.0 I-T	20.23	40.9	239.4	401.1	3.13	9.8	1.7	5.95	10.47	5.81	.510	.300	3.14
10 X 8 X 33.0 I-T	20.94	42.4	228.4	383.7	3.05	9.1	1.7	6.16	9.73	7.96	.435	.290	2.82
10 X 8 X 39.0 I-T	24.45	50.9	244.5	460.3	3.30	9.0	1.9	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	59.6	257.7	537.0	3.52	9.0	2.1	8.29	10.10	8.02	.620	.350	3.54
12 X 4 X 14.0 I-T	10.98	20.6	205.3	241.2	2.51	11.7	1.2	3.23	11.91	3.97	.225	.200	2.38
12 X 4 X 16.0 I-T	12.37	23.5	218.4	275.9	2.67	11.7	1.3	3.64	11.99	3.99	.265	.220	2.64
12 X 4 X 19.0 I-T	14.20	28.6	239.8	335.8	2.93	11.8	1.4	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	33.7	256.9	396.1	3.15	11.8	1.5	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	41.3	279.3	475.9	3.44	11.5	1.7	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	47.8	292.9	548.1	3.66	11.5	1.9	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	56.5	308.6	644.8	3.92	11.4	2.1	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	61.5	304.6	662.4	3.95	10.8	2.2	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	69.3	314.6	741.9	4.13	10.7	2.4	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	77.5	324.4	824.7	4.31	10.6	2.5	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	82.8	328.3	863.6	4.40	10.4	2.6	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	91.5	337.8	949.8	4.57	10.4	2.8	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.3	295.5	488.0	3.50	13.1	1.7	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.0	316.6	586.9	3.80	13.1	1.9	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	52.3	330.7	670.1	4.03	12.8	2.0	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	60.0	346.3	765.7	4.27	12.8	2.2	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	67.4	358.2	856.3	4.48	12.7	2.4	7.70	14.10	6.77	.515	.310	4.37

(35T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 35.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 35.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
14 X 8 X 43.0 I-T	28.02	74.3	358.8	902.8	4.57	12.1	2.5	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	83.7	369.8	1009.5	4.78	12.1	2.7	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	49.6	361.0	727.7	4.23	14.7	2.0	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	60.1	384.6	877.0	4.59	14.6	2.3	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	70.4	401.1	1010.0	4.87	14.3	2.5	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	79.7	417.1	1138.4	5.13	14.3	2.7	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	89.8	428.5	1271.9	5.36	14.2	3.0	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	100.1	439.7	1407.9	5.57	14.1	3.2	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	114.1	453.2	1588.9	5.82	13.9	3.5	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	138.1	473.3	1852.4	6.21	13.4	3.9	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	158.7	487.1	2097.5	6.48	13.2	4.3	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	183.7	502.6	2388.2	6.75	13.0	4.8	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	74.1	442.4	1186.2	5.27	16.0	2.7	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	86.4	464.2	1376.3	5.62	15.9	3.0	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	109.5	490.9	1700.5	6.12	15.5	3.5	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	132.5	512.6	2025.7	6.53	15.3	4.0	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	156.3	530.2	2350.5	6.86	15.0	4.4	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	195.4	554.4	2801.3	7.34	14.3	5.1	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	220.7	568.7	3114.7	7.58	14.1	5.5	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	240.0	578.5	3347.0	7.73	13.9	5.8	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	270.5	594.6	3713.3	7.96	13.7	6.2	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	155.0	596.9	2705.6	7.49	17.5	4.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	170.9	609.7	2953.8	7.73	17.3	4.8	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	183.6	619.2	3149.5	7.90	17.2	5.1	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	207.8	634.8	3511.9	8.17	16.9	5.5	17.58	21.43	8.36	.835	.515	11.04

(35T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 35.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 35.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 3/8 X 93.0 I-T	67.42	232.8	649.8	3876.7	8.41	16.7	6.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	262.7	664.0	4209.8	8.74	16.0	6.3	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	287.9	675.9	4545.3	8.92	15.8	6.7	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 39.375 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 44.297 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	6 3/4	X	17.0	T	16.77	26.7	159.4	185.5	1.94	7.0	1.2	4.93	6.99	6.75	.455	.285	1.99
7	X	6 3/4	X	19.0	T	18.74	29.9	168.6	207.9	2.04	6.9	1.2	5.51	7.05	6.77	.515	.310	2.19
7	X	8	X	21.5	T	20.94	33.9	172.9	225.2	2.11	6.7	1.3	6.16	6.83	8.00	.530	.305	2.08
7	X	8	X	24.0	T	23.53	38.0	181.8	252.0	2.22	6.6	1.4	6.92	6.90	8.03	.595	.340	2.35
8	X	5 1/2	X	15.5	T	15.28	25.6	174.7	202.2	2.04	7.9	1.2	4.49	7.94	5.53	.440	.275	2.18
8	X	7	X	18.0	T	17.73	30.5	189.4	238.1	2.19	7.8	1.3	5.22	7.93	6.99	.430	.295	2.34
8	X	7	X	20.0	T	19.79	34.9	202.0	271.9	2.33	7.8	1.3	5.82	8.01	7.00	.505	.305	2.44
8	X	7	X	22.5	T	22.32	39.1	211.6	303.6	2.44	7.8	1.4	6.56	8.07	7.04	.565	.345	2.78
8	X	6 1/2	X	24.0	I-T	15.11	26.5	178.7	208.7	2.07	7.9	1.2	4.44	7.93	6.50	.400	.245	1.94
8	X	7 1/8	X	25.0	T	24.83	43.5	220.6	336.1	2.55	7.7	1.5	7.30	8.13	7.07	.630	.380	3.09
8	X	6 1/2	X	28.0	I-T	17.69	31.0	193.6	245.5	2.23	7.9	1.3	5.20	8.06	6.54	.465	.285	2.30
8	X	7 1/8	X	28.5	T	28.28	49.3	231.5	380.1	2.69	7.7	1.6	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T	19.16	34.4	201.6	268.4	2.32	7.8	1.3	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	17.5	T	17.26	30.8	208.7	267.6	2.33	8.7	1.3	5.08	8.85	6.00	.425	.300	2.66
9	X	6	X	20.0	T	19.76	36.4	226.1	316.1	2.51	8.7	1.4	5.81	8.95	6.02	.525	.315	2.82
10	X	4	X	19.0	I-T	13.77	25.4	216.9	258.2	2.31	10.2	1.2	4.05	10.24	4.02	.395	.250	2.56
10	X	5 3/4	X	22.0	I-T	15.04	29.9	235.1	299.7	2.48	10.0	1.3	4.42	10.17	5.75	.360	.240	2.44
10	X	5 3/4	X	26.0	I-T	17.37	35.6	256.4	358.4	2.69	10.1	1.4	5.11	10.33	5.77	.440	.260	2.69
10	X	5 3/4	X	30.0	I-T	20.23	41.6	274.0	418.6	2.89	10.1	1.5	5.95	10.47	5.81	.510	.300	3.14
10	X	8	X	33.0	I-T	20.94	43.0	261.9	401.3	2.82	9.3	1.5	6.16	9.73	7.96	.435	.290	2.82
10	X	8	X	39.0	I-T	24.45	51.7	283.4	482.9	3.06	9.3	1.7	7.19	9.92	7.99	.530	.315	3.12
10	X	8	X	45.0	I-T	28.19	60.5	301.3	565.2	3.28	9.3	1.9	8.29	10.10	8.02	.620	.350	3.54
12	X	4	X	16.0	I-T	12.37	23.9	241.4	285.4	2.44	11.9	1.2	3.64	11.99	3.99	.265	.220	2.64

(35T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 39.375 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 44.297 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X 19.0 I-T	14.20	29.0	268.3	347.7	2.68	12.0	1.3	4.18	12.16	4.01	.350	.235	2.86
12 X 4 X 22.0 I-T	16.33	34.2	290.6	410.8	2.89	12.0	1.4	4.80	12.31	4.03	.425	.260	3.20
12 X 6 1/2 X 26.0 I-T	17.64	41.9	319.4	494.1	3.16	11.8	1.5	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	48.4	338.0	570.3	3.37	11.8	1.7	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	57.3	359.4	673.1	3.62	11.8	1.9	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	62.3	356.3	693.2	3.66	11.1	1.9	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	70.3	370.1	778.6	3.84	11.1	2.1	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	78.5	383.6	867.9	4.02	11.1	2.3	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	83.9	389.3	910.1	4.11	10.8	2.3	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	92.7	402.3	1003.5	4.28	10.8	2.5	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 22.0 I-T	16.18	37.8	336.3	505.0	3.21	13.4	1.5	4.76	13.74	5.00	.335	.230	3.16
14 X 5 X 26.0 I-T	18.87	45.5	364.3	608.7	3.49	13.4	1.7	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	53.0	383.6	696.5	3.71	13.1	1.8	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	60.7	404.5	797.5	3.95	13.1	2.0	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	68.2	420.9	894.0	4.15	13.1	2.1	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	75.3	423.4	944.9	4.24	12.6	2.2	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	84.8	438.6	1059.7	4.45	12.5	2.4	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.2	417.8	753.7	3.88	15.0	1.8	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	60.8	449.6	910.9	4.23	15.0	2.0	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	71.3	472.2	1052.1	4.50	14.8	2.2	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	80.7	493.6	1188.4	4.76	14.7	2.4	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	91.0	509.5	1331.8	4.98	14.6	2.6	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	101.5	524.9	1478.4	5.20	14.6	2.8	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	115.7	543.0	1674.7	5.46	14.5	3.1	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	139.9	570.6	1960.8	5.85	14.0	3.4	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	161.0	588.9	2230.6	6.13	13.9	3.8	15.00	16.52	10.30	.760	.455	7.52

(35T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 39.375 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 44.297 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 3/8 X 89.0 I-T	59.17	186.5	609.1	2552.7	6.43	13.7	4.2	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	75.0	522.2	1234.3	4.87	16.5	2.4	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	87.5	551.3	1436.0	5.21	16.4	2.6	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	110.9	587.7	1783.7	5.71	16.1	3.0	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	134.3	616.7	2136.2	6.13	15.9	3.5	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	158.7	639.9	2491.9	6.49	15.7	3.9	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	198.2	672.6	2988.0	6.98	15.1	4.4	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	224.1	690.9	3336.6	7.25	14.9	4.8	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	244.0	703.0	3596.7	7.42	14.7	5.1	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	275.4	722.9	4007.4	7.68	14.6	5.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	157.1	720.8	2852.9	7.04	18.2	4.0	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	173.3	737.6	3123.6	7.29	18.0	4.2	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.68	186.3	749.8	3337.9	7.47	17.9	4.5	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	211.2	769.6	3737.4	7.77	17.7	4.9	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	236.8	788.1	4141.9	8.04	17.5	5.3	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	266.7	808.4	4509.7	8.37	16.9	5.6	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	292.6	823.1	4886.5	8.58	16.7	5.9	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 43.750 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 54.688 SQ. IN.

NOMINAL SIZE						WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
							FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X	IN	X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X	8	X	24.0	T	23.53	38.8	201.3	265.0	2.07	6.8	1.3	6.92	6.90	8.03	.595	.340	2.35
8	X	7	X	20.0	T	19.79	35.6	221.9	284.1	2.17	8.0	1.3	5.82	8.01	7.00	.505	.305	2.44
8	X	7	X	22.5	T	22.32	39.9	234.2	317.6	2.28	8.0	1.4	6.56	8.07	7.04	.565	.345	2.78
8	X	7 1/8	X	25.0	T	24.83	44.3	245.7	352.0	2.38	7.9	1.4	7.30	8.13	7.07	.630	.380	3.09
8	X	7 1/8	X	28.5	T	28.28	50.3	259.9	398.9	2.52	7.9	1.5	8.32	8.22	7.12	.715	.430	3.53
8	X	8	X	31.0	I-T	19.16	35.1	221.2	280.3	2.16	8.0	1.3	5.63	8.00	8.00	.435	.285	2.28
9	X	6	X	20.0	T	19.76	37.1	248.9	329.4	2.33	8.9	1.3	5.81	8.95	6.02	.525	.315	2.82
10	X	5 3/4	X	26.0	I-T	17.37	36.2	281.4	371.6	2.49	10.3	1.3	5.11	10.33	5.77	.440	.260	2.69
10	X	5 3/4	X	30.0	I-T	20.23	42.2	303.7	434.6	2.68	10.3	1.4	5.95	10.47	5.81	.510	.300	3.14
10	X	8	X	33.0	I-T	20.94	43.7	290.8	417.4	2.62	9.5	1.4	6.16	9.73	7.96	.435	.290	2.82
10	X	8	X	39.0	I-T	24.45	52.5	318.0	503.2	2.85	9.6	1.6	7.19	9.92	7.99	.530	.315	3.12
10	X	8	X	45.0	I-T	28.19	61.3	341.1	590.1	3.06	9.6	1.7	8.29	10.10	8.02	.620	.350	3.54
12	X	4	X	22.0	I-T	16.33	34.7	318.4	424.3	2.67	12.2	1.3	4.80	12.31	4.03	.425	.260	3.20
12	X	6 1/2	X	26.0	I-T	17.64	42.4	353.5	510.4	2.92	12.0	1.4	5.19	12.22	6.49	.380	.230	2.81
12	X	6 1/2	X	30.0	I-T	20.27	49.1	377.4	590.0	3.12	12.0	1.6	5.96	12.34	6.52	.440	.260	3.21
12	X	6 1/2	X	35.0	I-T	23.82	58.0	405.3	697.8	3.36	12.0	1.7	7.01	12.50	6.56	.520	.300	3.75
12	X	8	X	40.0	I-T	25.48	63.1	403.5	720.1	3.40	11.4	1.8	7.49	11.94	8.01	.515	.295	3.52
12	X	8	X	45.0	I-T	28.81	71.2	421.9	810.5	3.58	11.4	1.9	8.47	12.06	8.05	.575	.335	4.04
12	X	8 1/8	X	50.0	I-T	32.11	79.5	439.7	905.3	3.76	11.4	2.1	9.44	12.19	8.08	.640	.370	4.51
12	X	10	X	53.0	I-T	33.01	84.9	447.6	950.3	3.84	11.2	2.1	9.71	12.06	10.00	.575	.345	4.16
12	X	10	X	58.0	I-T	35.92	93.9	464.6	1049.7	4.01	11.2	2.3	10.56	12.19	10.01	.640	.360	4.39
14	X	5	X	22.0	I-T	16.18	38.3	370.3	520.3	2.96	13.6	1.4	4.76	13.74	5.00	.335	.230	3.16

(35T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 43.750 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 54.688 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
14 X 5 X 26.0 I-T	18.87	46.1	405.7	628.0	3.23	13.6	1.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	53.6	430.7	719.6	3.44	13.4	1.7	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	61.5	457.6	825.1	3.66	13.4	1.8	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	69.1	479.0	926.5	3.85	13.4	1.9	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	76.1	484.1	981.1	3.95	12.9	2.0	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	85.8	504.4	1102.7	4.15	12.9	2.2	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	50.8	468.1	776.4	3.58	15.3	1.7	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	61.5	508.9	940.2	3.91	15.3	1.8	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	72.1	538.6	1088.0	4.18	15.1	2.0	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	81.6	566.3	1230.9	4.42	15.1	2.2	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	92.0	587.7	1382.6	4.65	15.0	2.4	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	102.7	608.2	1538.2	4.86	15.0	2.5	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	117.2	632.2	1747.5	5.12	14.9	2.8	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	141.5	668.6	2052.6	5.51	14.5	3.1	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	162.9	692.9	2343.8	5.80	14.4	3.4	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	189.0	719.1	2693.5	6.11	14.3	3.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	75.8	597.5	1275.1	4.52	16.8	2.1	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	88.4	635.0	1486.3	4.85	16.8	2.3	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	112.2	683.2	1853.8	5.33	16.5	2.7	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	135.9	721.6	2229.4	5.76	16.4	3.1	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	160.8	752.1	2611.9	6.13	16.2	3.5	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	200.7	795.2	3147.0	6.63	15.7	4.0	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	227.1	818.7	3526.7	6.91	15.5	4.3	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	247.4	834.0	3811.9	7.10	15.4	4.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	279.4	858.7	4263.1	7.38	15.3	5.0	23.67	18.97	11.27	1.060	.655	12.43

(35T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 43.750 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 54.688 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 8 1/4 X 62.0 I-T	44.94	158.9	847.3	2976.5	6.62	18.7	3.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	175.4	869.1	3266.5	6.67	18.6	3.8	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.56	188.6	884.9	3496.9	7.06	18.5	4.0	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	214.0	910.2	3929.1	7.37	18.4	4.3	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	240.2	933.5	4369.0	7.66	18.2	4.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	270.1	960.9	4766.8	7.98	17.6	5.0	20.11	21.36	12.29	.800	.500	10.66
21 X 12 3/8 X 111.0 I-T	75.30	296.5	979.2	5180.3	8.21	17.5	5.3	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 48.125 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 66.172 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 7 1/8 X 25.0 T	24.83	45.2	267.1	367.3	2.24	8.1	1.4	7.30	8.13	7.07	.630	.380	3.09
8 X 7 1/8 X 28.5 T	28.28	51.2	284.5	416.6	2.36	8.1	1.5	8.32	8.22	7.12	.715	.430	3.53
10 X 8 X 39.0 I-T	24.45	53.3	347.8	522.1	2.67	9.8	1.5	7.19	9.92	7.99	.530	.315	3.12
10 X 8 X 45.0 I-T	28.19	62.3	376.3	613.0	2.87	9.8	1.6	8.29	10.10	8.02	.620	.350	3.54
12 X 6 1/2 X 26.0 I-T	17.64	43.1	381.3	525.9	2.71	12.2	1.4	5.19	12.22	6.49	.380	.230	2.81
12 X 6 1/2 X 30.0 I-T	20.27	49.7	410.6	608.4	2.90	12.2	1.5	5.96	12.34	6.52	.440	.260	3.21
12 X 6 1/2 X 35.0 I-T	23.82	58.8	445.2	720.5	3.14	12.3	1.6	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	64.0	445.2	744.6	3.18	11.6	1.7	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	72.1	468.5	839.2	3.35	11.6	1.8	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	80.6	491.1	938.8	3.52	11.7	1.9	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	86.0	501.3	986.0	3.60	11.5	2.0	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	95.0	522.8	1090.7	3.77	11.5	2.1	10.56	12.19	10.01	.640	.360	4.39
14 X 5 X 26.0 I-T	18.87	46.7	440.2	645.9	3.00	13.8	1.5	5.55	13.91	5.03	.420	.255	3.55
14 X 6 3/4 X 30.0 I-T	21.16	54.3	471.0	740.7	3.20	13.6	1.6	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	62.2	504.0	850.1	3.41	13.7	1.7	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	69.9	531.0	955.6	3.60	13.7	1.8	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	77.0	539.1	1013.3	3.69	13.2	1.9	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	86.8	565.0	1140.7	3.89	13.1	2.0	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 26.0 I-T	19.49	51.4	510.7	797.2	3.33	15.5	1.6	5.73	15.69	5.50	.345	.250	3.92
16 X 5 1/2 X 31.0 I-T	22.70	62.2	560.9	966.5	3.64	15.5	1.7	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	72.9	598.4	1120.0	3.90	15.4	1.9	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	82.5	632.9	1268.4	4.13	15.4	2.0	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	93.0	660.7	1427.2	4.35	15.3	2.2	9.34	16.13	7.04	.565	.345	5.56

(35T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 48.125 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 66.172 SQ. IN.

NOMINAL SIZE IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 7 1/8 X 50.0 I-T	35.34	103.8	687.1	1590.3	4.56	15.3	2.3	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	118.5	718.2	1810.8	4.82	15.3	2.5	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	142.9	764.7	2131.9	5.19	14.9	2.8	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	164.7	796.3	2441.7	5.48	14.8	3.1	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	191.1	830.0	2815.6	5.80	14.7	3.4	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	76.6	666.0	1311.1	4.21	17.1	2.0	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	89.3	712.7	1530.3	4.52	17.1	2.1	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	113.3	774.7	1914.4	5.00	16.9	2.5	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	137.4	824.2	2309.7	5.42	16.8	2.8	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	162.6	863.8	2715.4	5.79	16.7	3.1	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	202.8	919.2	3284.0	6.28	16.2	3.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	229.6	949.2	3691.2	6.58	16.1	3.9	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	250.3	968.6	3998.9	6.77	16.0	4.1	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	282.9	999.5	4486.4	7.07	15.9	4.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	160.6	972.9	3082.2	6.23	19.2	3.2	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	177.2	1000.8	3388.7	6.48	19.1	3.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	190.6	1021.1	3633.1	6.67	19.1	3.6	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	216.4	1053.4	4093.8	6.99	18.9	3.9	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	243.1	1082.9	4565.1	7.29	18.8	4.2	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	273.0	1118.3	4988.5	7.60	18.3	4.5	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	299.9	1141.3	5434.7	7.84	18.1	4.8	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 52.500 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 78.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 8 X 45.0 I-T	28.19	63.2	406.6	634.9	2.70	10.0	1.6	8.29	10.10	8.02	.620	.350	3.54
12 X 6 1/2 X 35.0 I-T	23.82	59.6	478.9	741.9	2.94	12.5	1.5	7.01	12.50	6.56	.520	.300	3.75
12 X 8 X 40.0 I-T	25.48	64.8	480.7	767.6	2.98	11.8	1.6	7.49	11.94	8.01	.515	.295	3.52
12 X 8 X 45.0 I-T	28.81	73.0	509.2	866.0	3.15	11.9	1.7	8.47	12.06	8.05	.575	.335	4.04
12 X 8 1/8 X 50.0 I-T	32.11	81.6	536.8	969.6	3.32	11.9	1.8	9.44	12.19	8.08	.640	.370	4.51
12 X 10 X 53.0 I-T	33.01	87.0	549.4	1018.8	3.39	11.7	1.9	9.71	12.06	10.00	.575	.345	4.16
12 X 10 X 58.0 I-T	35.92	96.2	575.8	1128.0	3.55	11.7	2.0	10.56	12.19	10.01	.640	.360	4.39
14 X 6 3/4 X 30.0 I-T	21.16	55.0	504.3	760.9	2.99	13.8	1.5	6.22	13.84	6.73	.385	.270	3.74
14 X 6 3/4 X 34.0 I-T	23.54	63.0	543.5	873.6	3.19	13.9	1.6	6.92	13.98	6.75	.455	.285	3.98
14 X 6 3/4 X 38.0 I-T	26.17	70.7	576.0	982.6	3.37	13.9	1.7	7.70	14.10	6.77	.515	.310	4.37
14 X 8 X 43.0 I-T	28.02	77.9	587.3	1043.0	3.46	13.4	1.8	8.24	13.66	8.00	.530	.305	4.17
14 X 8 X 48.0 I-T	31.50	87.8	619.4	1175.4	3.65	13.4	1.9	9.26	13.79	8.03	.595	.340	4.69
16 X 5 1/2 X 31.0 I-T	22.70	63.0	605.0	991.0	3.41	15.7	1.6	6.68	15.88	5.53	.440	.275	4.37
16 X 7 X 36.0 I-T	25.69	73.7	650.5	1149.4	3.65	15.6	1.8	7.56	15.86	6.99	.430	.295	4.68
16 X 7 X 40.0 I-T	28.09	83.3	692.0	1302.5	3.87	15.6	1.9	8.26	16.01	7.00	.505	.305	4.88
16 X 7 X 45.0 I-T	31.77	94.0	726.8	1467.3	4.08	15.6	2.0	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	104.9	759.7	1637.1	4.29	15.6	2.2	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	119.7	798.8	1867.1	4.54	15.6	2.3	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	144.3	856.4	2202.0	4.90	15.3	2.6	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	166.3	896.5	2528.0	5.19	15.2	2.8	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	193.1	939.2	2923.1	5.51	15.1	3.1	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 35.0 I-T	26.29	77.5	726.3	1343.8	3.94	17.3	1.9	7.73	17.70	6.00	.425	.300	5.31
18 X 6 X 40.0 I-T	29.35	90.3	782.8	1569.9	4.24	17.4	2.0	8.63	17.90	6.02	.525	.315	5.64

(35T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 52.500 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 78.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 7 1/2 X 50.0 I-T	35.55	114.4	859.6	1968.3	4.70	17.2	2.3	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	138.7	922.0	2380.5	5.11	17.2	2.6	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	164.3	972.4	2806.4	5.47	17.1	2.9	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	204.8	1041.9	3404.0	5.96	16.6	3.3	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	231.9	1079.7	3835.4	6.26	16.5	3.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	252.9	1104.1	4163.1	6.46	16.5	3.8	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	286.1	1142.5	4683.0	6.76	16.4	4.1	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	162.0	1094.5	3174.4	5.88	19.6	2.9	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	178.9	1129.7	3495.3	6.12	19.5	3.1	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	192.5	1155.3	3751.8	6.31	19.5	3.2	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	218.6	1196.2	4237.5	6.63	19.4	3.5	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	245.7	1233.2	4736.4	6.93	19.3	3.8	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	275.6	1277.5	5181.6	7.24	18.8	4.1	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	302.8	1306.4	5657.0	7.49	18.7	4.3	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 61.250 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 107.188 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 10 X 58.0 I-T	35.92	98.6	663.7	1196.8	3.19	12.1	1.8	10.56	12.19	10.01	.640	.360	4.39
14 X 8 X 48.0 I-T	31.50	89.9	707.5	1239.7	3.26	13.8	1.8	9.26	13.79	8.03	.595	.340	4.69
16 X 7 X 45.0 I-T	31.77	96.1	835.2	1540.4	3.64	16.0	1.8	9.34	16.13	7.04	.565	.345	5.56
16 X 7 1/8 X 50.0 I-T	35.34	107.2	882.0	1720.9	3.83	16.1	2.0	10.39	16.26	7.07	.630	.380	6.18
16 X 7 1/8 X 57.0 I-T	40.28	122.3	938.6	1966.8	4.06	16.1	2.1	11.85	16.43	7.12	.715	.430	7.06
16 X 10 1/4 X 67.0 I-T	44.18	147.1	1019.8	2324.1	4.40	15.8	2.3	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	169.5	1080.1	2676.9	4.68	15.8	2.5	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	196.9	1144.2	3107.3	4.99	15.8	2.7	17.40	16.75	10.37	.875	.525	8.79
18 X 6 X 40.0 I-T	29.35	92.1	896.4	1641.8	3.77	17.8	1.8	8.63	17.90	6.02	.525	.315	5.64
18 X 7 1/2 X 50.0 I-T	35.55	116.7	1004.8	2063.4	4.19	17.7	2.1	10.46	17.99	7.50	.570	.355	6.39
18 X 7 1/2 X 60.0 I-T	42.61	141.4	1095.7	2503.5	4.57	17.7	2.3	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	167.5	1171.5	2962.7	4.93	17.7	2.5	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	208.5	1273.5	3607.8	5.39	17.3	2.8	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	236.2	1330.7	4079.5	5.68	17.3	3.1	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	257.7	1368.1	4440.8	5.89	17.2	3.2	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	291.6	1425.5	5015.8	6.19	17.2	3.5	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	164.9	1316.3	3331.7	5.26	20.2	2.5	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	182.0	1368.6	3676.0	5.50	20.2	2.7	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	195.8	1406.9	3952.3	5.68	20.2	2.8	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	222.5	1469.3	4479.3	5.99	20.1	3.0	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	250.3	1525.7	5024.3	6.29	20.1	3.3	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	280.2	1589.9	5504.6	6.58	19.6	3.5	20.11	21.36	12.29	.800	.500	10.68

(35T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 61.250 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 107.188 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
21 X 12 3/8 X 111.0 I-T	75.30	308.1	1634.1	6029.1	6.83	19.6	3.7	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 70.000 IN.) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.) EFFECTIVE PLATE AREA = 140.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 1/4 X 67.0 I-T	44.18	150.0	1151.2	2432.9	3.99	16.2	2.1	12.99	16.33	10.24	.665	.395	6.45
16 X 10 1/4 X 77.0 I-T	50.98	172.8	1233.3	2807.1	4.26	16.2	2.3	15.00	16.52	10.30	.760	.455	7.52
16 X 10 3/8 X 89.0 I-T	59.17	200.6	1321.8	3265.8	4.56	16.3	2.5	17.40	16.75	10.37	.875	.525	8.79
18 X 7 1/2 X 60.0 I-T	42.61	144.2	1234.3	2612.9	4.14	18.1	2.1	12.53	18.24	7.56	.695	.415	7.57
18 X 7 5/8 X 71.0 I-T	50.75	170.7	1338.1	3098.6	4.47	18.2	2.3	14.93	18.47	7.64	.810	.495	9.14
18 X 11 1/8 X 86.0 I-T	57.79	212.1	1475.7	3781.0	4.91	17.8	2.6	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	240.2	1555.9	4284.9	5.19	17.8	2.8	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	262.2	1609.2	4673.4	5.39	17.8	2.9	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	296.7	1690.0	5292.8	5.69	17.8	3.1	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 62.0 I-T	44.94	167.7	1499.6	3467.4	4.76	20.7	2.3	13.22	20.99	8.24	.615	.400	8.40
21 X 8 1/4 X 68.0 I-T	49.15	185.1	1571.1	3830.1	4.98	20.7	2.4	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	199.1	1623.8	4122.1	5.15	20.7	2.5	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	226.2	1711.9	4681.8	5.45	20.7	2.7	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	254.5	1792.0	5263.6	5.74	20.7	2.9	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	284.5	1878.1	5770.9	6.00	20.3	3.1	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	312.9	1941.9	6334.6	6.25	20.2	3.3	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 81.600 LBS. (2.0000 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 78.750 IN.) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.) EFFECTIVE PLATE AREA = 177.188 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
16 X 10 3/8 X 89.0 I-T	59.17	204.6	1467.0	3411.9	4.19	16.7	2.3	17.40	16.75	10.37	.875	.525	8.79
18 X 11 1/8 X 86.0 I-T	57.79	215.9	1641.8	3938.6	4.50	18.2	2.4	17.00	18.39	11.09	.770	.480	8.83
18 X 11 1/8 X 97.0 I-T	65.20	244.5	1746.3	4469.0	4.77	18.3	2.6	19.18	18.59	11.15	.870	.535	9.95
18 X 11 1/4 X 106.0 I-T	71.48	266.7	1817.3	4879.8	4.96	18.3	2.7	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	301.8	1924.3	5535.8	5.25	18.3	2.9	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 1/4 X 68.0 I-T	49.15	188.4	1731.9	3971.7	4.55	21.1	2.3	14.46	21.13	8.27	.685	.430	9.09
21 X 8 1/4 X 73.0 I-T	52.58	202.5	1799.4	4276.4	4.71	21.1	2.4	15.47	21.24	8.30	.740	.455	9.66
21 X 8 3/8 X 83.0 I-T	59.78	230.0	1914.6	4862.9	5.00	21.1	2.5	17.58	21.43	8.36	.835	.515	11.04
21 X 8 3/8 X 93.0 I-T	67.42	258.7	2020.9	5474.8	5.27	21.2	2.7	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	288.8	2129.0	6003.4	5.52	20.8	2.8	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	317.6	2215.4	6599.3	5.75	20.8	3.0	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 91.800 LBS. (2.2500 IN.)

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TABLE XI. Properties of combined beam and plate, I-T and T (35t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE I-T AND T

(35T = 87.500 IN.) PLATE WEIGHT = 102.000 LBS. (2.5000 IN.) EFFECTIVE PLATE AREA = 218.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 11 1/4 X 106.0 I-T	71.48	271.6	1989.0	5073.5	4.60	18.7	2.6	21.02	18.73	11.20	.940	.590	11.05
18 X 11 1/4 X 119.0 I-T	80.48	307.1	2122.7	5760.4	4.87	18.8	2.7	23.67	18.97	11.27	1.060	.655	12.43
21 X 8 3/8 X 93.0 I-T	67.42	263.2	2208.2	5672.8	4.88	21.6	2.6	19.83	21.62	8.42	.930	.580	12.54
21 X 12 1/4 X 101.0 I-T	68.38	293.3	2337.2	6218.0	5.10	21.2	2.7	20.11	21.36	12.29	.800	.500	10.68
21 X 12 3/8 X 111.0 I-T	75.30	322.5	2447.1	6840.9	5.33	21.2	2.8	22.15	21.51	12.34	.875	.550	11.83

(35T) PLATE WEIGHT = 102.000 LBS. (2.5000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t).

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 7.500 IN.) PLATE WEIGHT = 5.100 LBS. (.1250 IN.) EFFECTIVE PLATE AREA = .938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	1.6	1.0	.80	1.5	.6	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.8	1.7	1.2	.84	1.4	.7	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.1	2.1	1.9	1.05	1.7	.9	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.4	2.6	2.9	1.25	2.0	1.1	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	1.9	2.8	3.5	1.31	1.9	1.3	1.09	3.00	3.00	.187	.187	.56

(60T) PLATE WEIGHT = 5.100 LBS. (.1250 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 9.375 IN.) PLATE WEIGHT = 6.375 LBS. (.1563 IN.) EFFECTIVE PLATE AREA = 1.465 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	2.3	1.1	.74	1.7	.5	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	2.4	1.4	.79	1.6	.6	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.1	3.1	2.2	.99	1.9	.7	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.5	3.8	3.3	1.18	2.3	.9	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.0	4.0	4.1	1.27	2.1	1.0	1.09	3.00	3.00	.187	.187	.56

(60T) PLATE WEIGHT = 6.375 LBS. (.1563 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 11.250 IN.) PLATE WEIGHT = 7.650 LBS. (.1875 IN.) EFFECTIVE PLATE AREA = 2.109 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	3.0	1.3	.68	1.8	.4	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	3.2	1.5	.73	1.7	.5	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.2	4.1	2.5	.92	2.1	.6	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.5	5.1	3.7	1.11	2.5	.7	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.0	5.4	4.7	1.21	2.3	.9	1.09	3.00	3.00	.187	.187	.56

(60T) PLATE WEIGHT = 7.650 LBS. (.1875 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 13.125 IN.) PLATE WEIGHT = 8.925 LBS. (.2188 IN.) EFFECTIVE PLATE AREA = 2.871 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	3.7	1.4	.62	1.8	.4	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	3.9	1.6	.68	1.8	.4	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.2	5.2	2.7	.85	2.2	.5	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.5	6.4	4.0	1.03	2.6	.6	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.1	6.9	5.1	1.14	2.5	.7	1.09	3.00	3.00	.187	.187	.56

(60T) PLATE WEIGHT = 8.925 LBS. (.2188 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 15.000 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 3.750 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.8	4.3	1.4	.57	1.9	.3	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	4.7	1.7	.62	1.9	.4	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L	2.76	.9	4.6	1.8	.62	1.9	.4	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L	3.19	1.2	4.9	2.1	.67	1.8	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.2	6.2	2.8	.79	2.3	.5	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.6	6.5	3.5	.85	2.2	.5	1.06	2.50	2.00	.250	.250	.63
3 X 2 X 3/16 X 3.07 L	3.06	1.6	7.8	4.2	.95	2.7	.5	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.1	8.5	5.5	1.06	2.6	.6	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.0	8.2	5.2	1.03	2.6	.6	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.3	8.5	6.0	1.08	2.6	.7	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.7	8.8	6.7	1.13	2.5	.8	1.44	3.00	3.00	.250	.250	.75
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	2.9	10.2	8.4	1.27	2.9	.8	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.3	10.5	9.3	1.33	2.9	.9	1.56	3.50	3.00	.250	.250	.88
4 X 3 X 1/4 X 5.8 L	5.74	3.9	12.2	12.5	1.52	3.2	1.0	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.3	12.5	13.7	1.57	3.2	1.1	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	4.8	12.8	14.8	1.61	3.1	1.2	1.94	4.00	4.00	.250	.250	1.00
5 X 3 X 1/4 X 6.6 L	6.59	5.2	15.7	20.5	1.90	3.9	1.3	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	5.8	16.0	22.3	1.96	3.9	1.4	2.06	5.00	3.50	.250	.250	1.25
10 X 2 5/8 X X 15.3 C	11.66	16.2	34.9	113.6	3.98	7.0	3.3	3.43	10.00	2.60	.436	.240	2.40

(60T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 16.875 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 4.746 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.8	4.8	1.5	.53	2.0	.3	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	5.3	1.8	.58	1.9	.3	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L	2.76	1.0	5.2	1.9	.58	1.9	.4	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L	3.19	1.2	5.7	2.3	.63	1.9	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.3	7.2	3.0	.73	2.4	.4	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.6	7.6	3.7	.79	2.3	.5	1.06	2.50	2.00	.250	.250	.63
3 X 2 X 3/16 X 3.07 L	3.06	1.6	9.1	4.4	.89	2.8	.5	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.1	10.8	5.8	.99	2.7	.6	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.0	9.6	5.5	.96	2.7	.6	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.4	10.1	6.3	1.02	2.7	.6	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.7	10.5	7.1	1.07	2.6	.7	1.44	3.00	3.00	.250	.250	.75
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	2.9	12.2	8.9	1.20	3.1	.7	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.3	12.6	9.9	1.25	3.0	.8	1.56	3.50	3.00	.250	.250	.88
4 X 3 X 1/4 X 5.8 L	5.74	3.9	14.7	13.3	1.44	3.4	.9	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.4	15.1	14.6	1.49	3.3	1.0	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	4.9	15.5	15.9	1.54	3.3	1.0	1.94	4.00	4.00	.250	.250	1.00
5 X 3 X 1/4 X 6.6 L	6.59	5.3	19.0	21.8	1.81	4.1	1.2	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	5.9	19.4	23.8	1.87	4.1	1.2	2.06	5.00	3.50	.250	.250	1.25
10 X 2 5/8 X X 15.3 C	11.66	16.6	42.3	122.6	3.87	7.4	2.9	3.43	10.00	2.60	.436	.240	2.40

(60T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 18.750 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 5.859 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 2 X 3/16 X 2.44 L	2.42	1.0	5.8	1.9	.54	2.0	.3	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L	2.76	1.0	5.8	2.0	.54	2.0	.3	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L	3.19	1.2	6.4	2.4	.59	1.9	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.3	8.0	3.1	.68	2.4	.4	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.6	8.7	3.8	.75	2.4	.4	1.06	2.50	2.00	.250	.250	.63
3 X 2 X 3/16 X 3.07 L	3.06	1.6	10.3	4.6	.83	2.9	.4	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.2	11.5	6.0	.93	2.8	.5	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.1	11.0	5.8	.90	2.8	.5	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.4	11.7	6.6	.96	2.7	.6	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.8	12.2	7.5	1.01	2.7	.6	1.44	3.00	3.00	.250	.250	.75
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	2.9	14.2	9.3	1.13	3.2	.7	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.4	14.7	10.4	1.19	3.1	.7	1.56	3.50	3.00	.250	.250	.88
4 X 3 X 1/4 X 5.8 L	5.74	4.0	17.3	14.0	1.36	3.5	.8	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.5	17.8	15.4	1.42	3.4	.9	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	4.9	18.3	16.8	1.47	3.4	.9	1.94	4.00	4.00	.250	.250	1.00
5 X 3 X 1/4 X 6.6 L	6.59	5.4	22.4	23.0	1.72	4.3	1.0	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	5.9	23.0	25.1	1.78	4.2	1.1	2.06	5.00	3.50	.250	.250	1.25
10 X 2 5/8 X X 15.3 C	11.66	16.9	50.4	130.5	3.75	7.7	2.6	3.43	10.00	2.60	.436	.240	2.40
12 X 3 X X 20.7 C	16.03	26.5	63.6	230.4	4.67	8.7	3.6	4.72	12.00	2.94	.501	.282	3.38

(60T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 20.625 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 7.090 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 2 X 1/4 X 3.19 L	3.19	1.3	7.0	2.5	.56	2.0	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.3	8.8	3.2	.64	2.5	.4	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.7	9.6	4.0	.70	2.4	.4	1.06	2.50	2.00	.250	.250	.63
2 1/2 X 2 X 5/16 X 4.5 L	4.46	2.0	10.2	4.7	.75	2.4	.5	1.31	2.50	2.00	.313	.313	.78
3 X 2 X 3/16 X 3.07 L	3.06	1.6	11.4	4.8	.77	2.9	.4	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.2	12.9	6.2	.87	2.9	.5	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.1	12.4	6.0	.85	2.9	.5	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.4	13.2	6.9	.91	2.8	.5	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.8	13.8	7.8	.96	2.8	.6	1.44	3.00	3.00	.250	.250	.75
3 X 2 X 5/16 X 5.0 L	4.99	2.5	13.0	7.0	.91	2.8	.5	1.47	3.00	2.00	.313	.313	.94
3 X 2 1/2 X 5/16 X 5.6 L	5.52	2.9	13.8	8.1	.96	2.8	.6	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.4	14.4	9.2	1.02	2.7	.6	1.78	3.00	3.00	.313	.313	.94
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.0	16.1	9.7	1.07	3.2	.6	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.4	16.8	10.9	1.12	3.2	.6	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.6	16.8	11.4	1.13	3.2	.7	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.1	17.5	12.8	1.19	3.1	.7	1.94	3.50	3.00	.313	.313	1.10
4 X 3 X 1/4 X 5.8 L	5.74	4.0	19.8	14.6	1.29	3.6	.7	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.5	20.5	16.1	1.34	3.6	.8	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.0	21.1	17.5	1.39	3.5	.8	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	4.9	20.5	17.2	1.37	3.5	.8	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.5	21.2	18.9	1.42	3.5	.9	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.1	21.7	20.6	1.47	3.4	.9	2.41	4.00	4.00	.313	.313	1.25

(60T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 20.625 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 7.090 SQ. IN.

NOMINAL SIZE IN X IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 3 X 1/4 X 6.6 L		6.59	5.4	25.9	24.0	1.63	4.4	.9	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L		7.01	6.0	26.7	26.2	1.69	4.4	1.0	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2 L		8.18	6.6	26.7	28.2	1.72	4.3	1.1	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7 L		8.71	7.3	27.5	30.9	1.79	4.2	1.1	2.56	5.00	3.50	.313	.313	1.57
6 X 3 1/2 X 5/16 X 9.8 L		9.78	9.3	33.8	46.4	2.16	5.0	1.4	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L		10.31	10.2	34.5	49.9	2.22	4.9	1.4	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3 C		11.07	10.7	34.6	52.0	2.24	4.8	1.5	3.26	6.00	3.50	.385	.340	2.04
10 X 2 5/8 X X 15.3 C		11.66	17.1	59.1	137.4	3.61	8.0	2.3	3.43	10.00	2.60	.436	.240	2.40
10 X 3 1/2 X X 21.9 C		16.36	24.2	62.4	160.2	3.89	7.5	2.9	4.81	10.00	3.45	.500	.325	3.25
12 X 3 X X 20.7 C		16.03	27.0	74.4	244.4	4.55	9.1	3.3	4.72	12.00	2.94	.501	.282	3.38

(60T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 2 X 3/8 X 4.7 L	4.62	1.7	8.4	3.4	.59	2.0	.4	1.36	2.00	2.00	.375	.375	.75
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.7	10.4	4.2	.66	2.5	.4	1.06	2.50	2.00	.250	.250	.63
2 1/2 X 2 X 5/16 X 4.5 L	4.46	2.0	11.1	4.9	.71	2.4	.4	1.31	2.50	2.00	.313	.313	.78
2 1/2 X 2 X 3/8 X 5.3 L	5.26	2.3	11.6	5.5	.74	2.4	.5	1.55	2.50	2.00	.375	.375	.94
3 X 2 X 3/16 X 3.07 L	3.06	1.7	12.3	4.9	.73	3.0	.4	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.2	14.1	6.5	.82	2.9	.5	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.1	13.5	6.2	.80	2.9	.5	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.5	14.5	7.2	.86	2.9	.5	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.8	15.3	8.1	.91	2.8	.5	1.44	3.00	3.00	.250	.250	.75
3 X 2 X 5/16 X 5.0 L	4.99	2.6	14.4	7.3	.86	2.9	.5	1.47	3.00	2.00	.313	.313	.94
3 X 2 1/2 X 5/16 X 5.6 L	5.52	3.0	15.3	8.4	.92	2.8	.6	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.4	16.1	9.5	.97	2.8	.6	1.78	3.00	3.00	.313	.313	.94
3 X 2 1/2 X 3/8 X 6.6 L	6.53	3.4	15.9	9.6	.96	2.8	.6	1.92	3.00	2.50	.375	.375	1.13
3 X 3 X 3/8 X 7.2 L	7.17	4.0	16.7	10.8	1.01	2.7	.6	2.11	3.00	3.00	.375	.375	1.13
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.0	17.8	10.0	1.01	3.3	.6	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.4	18.8	11.3	1.06	3.3	.6	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.7	18.8	11.9	1.08	3.2	.6	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.2	19.6	13.3	1.13	3.2	.7	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.2	19.4	13.4	1.13	3.2	.7	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	4.8	20.3	15.1	1.19	3.1	.7	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.1	22.3	15.1	1.22	3.7	.7	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.6	23.1	16.7	1.28	3.7	.7	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.1	23.8	18.2	1.33	3.6	.8	1.94	4.00	4.00	.250	.250	1.00

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X 3	X 5/16	X 7.2 L	7.12	5.0	23.2	17.9	1.30	3.6	.8	2.09	4.00	3.00	.313	.313	1.25
4	X 3 1/2	X 5/16	X 7.7 L	7.65	5.5	24.0	19.7	1.36	3.6	.8	2.25	4.00	3.50	.313	.313	1.25
4	X 4	X 5/16	X 8.2 L	8.18	6.1	24.7	21.5	1.41	3.5	.9	2.41	4.00	4.00	.313	.313	1.25
4	X 3	X 3/8	X 8.5 L	8.45	5.7	23.9	20.3	1.36	3.5	.8	2.48	4.00	3.00	.375	.375	1.50
4	X 3 1/2	X 3/8	X 9.1 L	9.08	6.4	24.7	22.3	1.42	3.5	.9	2.67	4.00	3.50	.375	.375	1.50
4	X 4	X 3/8	X 9.8 L	9.72	7.1	25.3	24.3	1.47	3.4	1.0	2.86	4.00	4.00	.375	.375	1.50
5	X 3	X 1/4	X 6.6 L	6.59	5.5	29.3	24.8	1.55	4.5	.8	1.94	5.00	3.00	.250	.250	1.25
5	X 3 1/2	X 1/4	X 7.0 L	7.01	6.1	30.3	27.2	1.61	4.5	.9	2.06	5.00	3.50	.250	.250	1.25
5	X 3	X 5/16	X 8.2 L	8.18	6.7	30.4	29.4	1.65	4.4	1.0	2.41	5.00	3.00	.313	.313	1.57
5	X 3 1/2	X 5/16	X 8.7 L	8.71	7.4	31.4	32.2	1.71	4.3	1.0	2.56	5.00	3.50	.313	.313	1.57
5	X 3	X 3/8	X 9.8 L	9.72	7.8	31.2	33.4	1.72	4.3	1.1	2.86	5.00	3.00	.375	.375	1.88
5	X 3 1/2	X 3/8	X 10.4 L	10.36	8.6	32.1	36.5	1.78	4.2	1.1	3.05	5.00	3.50	.375	.375	1.88
5	X 5	X 3/8	X 12.3 L	12.27	11.2	34.2	45.3	1.94	4.0	1.3	3.61	5.00	5.00	.375	.375	1.88
6	X 3 1/2	X 5/16	X 9.8 L	9.78	9.4	38.7	48.4	2.07	5.1	1.2	2.88	6.00	3.50	.313	.313	1.88
6	X 4	X 5/16	X 10.3 L	10.31	10.3	39.6	52.2	2.13	5.1	1.3	3.03	6.00	4.00	.313	.313	1.88
6	X 3 1/2	X	X 15.3 C	11.07	10.9	39.8	54.4	2.16	5.0	1.4	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7 L	11.63	11.0	39.6	54.9	2.15	5.0	1.4	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3 L	12.27	12.0	40.5	59.2	2.22	4.9	1.5	3.61	6.00	4.00	.375	.375	2.25
7	X 4	X 3/8	X 13.6 L	13.55	14.8	48.2	83.5	2.59	5.6	1.7	3.98	7.00	4.00	.375	.375	2.63
10	X 2 5/8	X	X 15.3 C	11.66	17.3	68.3	143.5	3.48	8.3	2.1	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2	X	X 21.9 C	16.36	24.5	72.1	189.8	3.78	7.7	2.6	4.81	10.00	3.45	.500	.325	3.25

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 22.500 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 8.438 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 3 X X 20.7 C	16.03	27.4	86.0	256.9	4.42	9.4	3.0	4.72	12.00	2.94	.501	.282	3.38
13 X 4 X X 31.8 C	24.09	44.7	100.3	413.4	5.16	9.3	4.1	7.09	13.00	4.00	.610	.375	4.88

(60T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 1/2 X 2 X 3/8 X 5.3 L	5.26	2.4	13.4	6.0	.68	2.5	.4	1.55	2.50	2.00	.375	.375	.94
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.6	16.8	7.6	.77	3.0	.5	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.9	17.9	8.6	.82	3.0	.5	1.44	3.00	3.00	.250	.250	.75
3 X 2 X 5/16 X 5.0 L	4.99	2.6	16.7	7.8	.78	3.0	.5	1.47	3.00	2.00	.313	.313	.94
3 X 2 1/2 X 5/16 X 5.6 L	5.52	3.1	18.0	9.0	.83	2.9	.5	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.5	19.1	10.2	.88	2.9	.5	1.78	3.00	3.00	.313	.313	.94
3 X 2 1/2 X 3/8 X 6.6 L	6.53	3.5	18.9	10.3	.88	2.9	.5	1.92	3.00	2.50	.375	.375	1.13
3 X 3 X 3/8 X 7.2 L	7.17	4.1	20.0	11.6	.92	2.9	.6	2.11	3.00	3.00	.375	.375	1.13
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.1	20.9	10.7	.91	3.4	.5	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.5	22.2	12.0	.96	3.4	.5	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.8	22.4	12.6	.98	3.4	.6	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.3	23.6	14.2	1.03	3.3	.6	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.3	23.4	14.4	1.03	3.3	.6	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	5.0	24.7	16.2	1.09	3.3	.7	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.2	26.7	16.0	1.10	3.8	.6	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.7	27.9	17.7	1.16	3.8	.6	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.2	29.0	19.4	1.20	3.8	.7	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	5.1	28.3	19.1	1.19	3.8	.7	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.7	29.5	21.1	1.24	3.7	.7	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.3	30.5	23.1	1.29	3.7	.8	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	5.9	29.4	21.8	1.25	3.7	.7	2.48	4.00	3.00	.375	.375	1.50
4 X 3 1/2 X 3/8 X 9.1 L	9.08	6.6	30.5	24.0	1.30	3.6	.8	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8 L	9.72	7.3	31.5	26.3	1.35	3.6	.8	2.86	4.00	4.00	.375	.375	1.50

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE IN X IN X IN X LBS/FT							WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
							LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5	X 3	X 1/4	X	6.6	L		6.59	5.6	35.9	26.3	1.40	4.7	.7	1.94	5.00	3.00	.250	.250	1.25
5	X 3 1/2	X 1/4	X	7.0	L		7.01	6.2	37.3	28.9	1.46	4.7	.8	2.06	5.00	3.50	.250	.250	1.25
5	X 3	X 5/16	X	8.2	L		8.18	6.8	37.7	31.4	1.50	4.6	.8	2.41	5.00	3.00	.313	.313	1.57
5	X 3 1/2	X 5/16	X	8.7	L		8.71	7.6	39.1	34.4	1.57	4.6	.9	2.56	5.00	3.50	.313	.313	1.57
5	X 3	X 3/8	X	9.8	L		9.72	7.9	39.0	35.8	1.58	4.5	.9	2.86	5.00	3.00	.375	.375	1.88
5	X 3 1/2	X 3/8	X	10.4	L		10.36	8.8	40.3	39.3	1.64	4.5	1.0	3.05	5.00	3.50	.375	.375	1.88
5	X 5	X 3/8	X	12.3	L		12.27	11.4	43.4	49.2	1.81	4.3	1.1	3.61	5.00	5.00	.375	.375	1.88
6	X 3 1/2	X 5/16	X	9.8	L		9.78	9.6	48.8	51.8	1.90	5.4	1.1	2.88	6.00	3.50	.313	.313	1.88
6	X 4	X 5/16	X	10.3	L		10.31	10.5	50.1	56.0	1.96	5.3	1.1	3.03	6.00	4.00	.313	.313	1.88
6	X 3 1/2	X	X	15.3	C		11.07	11.1	50.4	58.5	1.99	5.3	1.2	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X	11.7	L		11.63	11.2	50.2	59.1	1.99	5.3	1.2	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X	12.3	L		12.27	12.3	51.5	63.9	2.06	5.2	1.2	3.61	6.00	4.00	.375	.375	2.25
6	X 3 1/2	X	X	18.0	C		12.77	12.9	51.9	66.4	2.09	5.2	1.3	3.76	6.00	3.50	.475	.379	2.27
7	X 4	X 3/8	X	13.6	L		13.55	15.1	61.5	90.3	2.42	6.0	1.5	3.98	7.00	4.00	.375	.375	2.63
10	X 2 5/8	X	X	15.3	C		11.66	17.7	87.8	153.6	3.21	8.7	1.7	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2	X	X	21.9	C		16.36	25.1	93.0	206.1	3.56	8.2	2.2	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8	X	X	20.0	C		15.39	21.8	88.4	182.5	3.35	8.4	2.1	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2	X	X	24.9	C		18.73	28.0	94.2	225.5	3.64	8.0	2.4	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2	X	X	25.3	C		19.76	27.9	93.4	224.2	3.60	8.0	2.4	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2	X	X	28.5	C		21.34	32.0	96.3	250.5	3.76	7.8	2.6	6.28	10.00	3.95	.575	.425	4.25
12	X 3	X	X	20.7	C		16.03	28.0	111.1	278.0	4.14	9.9	2.5	4.72	12.00	2.94	.501	.282	3.38
12	X 3	X	X	25.0	C		20.33	32.3	111.8	311.5	4.22	9.7	2.8	5.98	12.00	3.05	.501	.387	4.64

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 26.250 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 11.484 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
13 X 4 X X 31.8 C	24.09	45.9	129.3	455.3	4.95	9.9	3.5	7.09	13.00	4.00	.610	.375	4.88
15 X 3 3/8 X X 33.9 C	27.03	53.8	149.1	610.6	5.61	11.3	4.1	7.95	15.00	3.40	.650	.400	6.00

(60T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
3 X 3 X 5/16 X 6.1 L	6.05	3.6	21.6	10.9	.80	3.0	.5	1.78	3.00	3.00	.313	.313	.94
3 X 2 1/2 X 3/8 X 6.6 L	6.53	3.7	21.4	10.9	.80	3.0	.5	1.92	3.00	2.50	.375	.375	1.13
3 X 3 X 3/8 X 7.2 L	7.17	4.2	22.9	12.4	.85	3.0	.5	2.11	3.00	3.00	.375	.375	1.13
3 X 2 1/2 X 7/16 X 7.6 L	7.54	4.1	22.4	12.2	.84	3.0	.5	2.22	3.00	2.50	.438	.438	1.31
3 X 3 X 7/16 X 8.3 L	8.28	4.7	23.9	13.8	.89	2.9	.6	2.44	3.00	3.00	.438	.438	1.31
3 X 3 X 1/2 X 9.4 L	9.35	5.2	24.7	15.0	.92	2.9	.6	2.75	3.00	3.00	.500	.500	1.50
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.6	25.0	12.7	.87	3.5	.5	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.8	25.4	13.4	.89	3.5	.5	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.4	27.0	15.1	.94	3.4	.6	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.5	26.8	15.3	.95	3.4	.6	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	5.1	28.5	17.2	1.00	3.4	.6	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.3	30.4	16.9	1.01	3.9	.6	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.8	32.0	18.7	1.05	3.9	.6	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.3	33.5	20.5	1.10	3.9	.6	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	5.2	32.7	20.1	1.09	3.9	.6	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.8	34.3	22.3	1.14	3.8	.7	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.4	35.7	24.4	1.18	3.8	.7	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	6.0	34.3	23.1	1.15	3.8	.7	2.48	4.00	3.00	.375	.375	1.50
4 X 3 1/2 X 3/8 X 9.1 L	9.08	6.7	35.9	25.5	1.20	3.8	.7	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8 L	9.72	7.4	37.3	27.9	1.25	3.8	.7	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 7/16 X 9.8 L	9.77	6.8	35.6	25.8	1.20	3.8	.7	2.87	4.00	3.00	.438	.438	1.75
4 X 3 X 1/2 X 11.1 L	11.05	7.6	36.6	28.2	1.24	3.7	.8	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3 L	11.26	8.4	38.6	31.2	1.30	3.7	.8	3.31	4.00	4.00	.438	.438	1.75
4 X 3 1/2 X 1/2 X 11.9 L	11.90	8.5	38.1	31.2	1.30	3.7	.8	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8 L	12.75	9.4	39.5	34.1	1.35	3.6	.9	3.75	4.00	4.00	.500	.500	2.00

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5	X 3	X 1/4	X 6.6 L	6.59	5.7	41.7	27.6	1.28	4.8	.7	1.94	5.00	3.00	.250	.250	1.25
5	X 3 1/2	X 1/4	X 7.0 L	7.01	6.3	43.6	30.3	1.33	4.8	.7	2.06	5.00	3.50	.250	.250	1.25
5	X 3	X 5/16	X 8.2 L	8.18	6.9	44.4	33.0	1.38	4.8	.7	2.41	5.00	3.00	.313	.313	1.57
5	X 3 1/2	X 5/16	X 8.7 L	8.71	7.7	46.3	36.3	1.44	4.7	.8	2.56	5.00	3.50	.313	.313	1.57
5	X 3	X 3/8	X 9.8 L	9.72	8.1	46.3	37.9	1.46	4.7	.8	2.86	5.00	3.00	.375	.375	1.88
5	X 3 1/2	X 3/8	X 10.4 L	10.36	9.0	48.2	41.6	1.52	4.6	.9	3.05	5.00	3.50	.375	.375	1.88
5	X 3	X 7/16	X 11.3 L	11.26	9.2	47.8	42.4	1.52	4.6	.9	3.31	5.00	3.00	.438	.438	2.19
5	X 3 1/2	X 7/16	X 12.0 L	12.01	10.2	49.6	46.6	1.59	4.6	.9	3.53	5.00	3.50	.438	.438	2.19
5	X 5	X 3/8	X 12.3 L	12.27	11.6	52.5	52.4	1.68	4.5	1.0	3.61	5.00	5.00	.375	.375	1.88
5	X 3	X 1/2	X 12.8 L	12.75	10.2	49.0	46.5	1.58	4.6	1.0	3.75	5.00	3.00	.500	.500	2.50
5	X 3 1/2	X 1/2	X 13.6 L	13.60	11.4	50.7	51.1	1.64	4.5	1.0	4.00	5.00	3.50	.500	.500	2.50
6	X 3 1/2	X 5/16	X 9.8 L	9.78	9.8	58.5	54.5	1.75	5.6	.9	2.88	6.00	3.50	.313	.313	1.88
6	X 4	X 5/16	X 10.3 L	10.31	10.7	60.4	59.1	1.81	5.5	1.0	3.03	6.00	4.00	.313	.313	1.88
6	X 3 1/2	X	X 15.3 C	11.07	11.3	60.9	61.9	1.84	5.5	1.0	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7 L	11.63	11.5	60.7	62.6	1.84	5.5	1.0	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3 L	12.27	12.5	62.4	67.8	1.91	5.4	1.1	3.61	6.00	4.00	.375	.375	2.25
6	X 3 1/2	X	X 18.0 C	12.77	13.1	63.1	70.6	1.94	5.4	1.1	3.76	6.00	3.50	.475	.379	2.27
6	X 4	X 7/16	X 14.3 L	14.24	14.3	64.1	75.9	1.99	5.3	1.2	4.19	6.00	4.00	.438	.438	2.63
6	X 4	X 1/2	X 16.2 L	16.15	15.9	65.3	83.3	2.05	5.2	1.3	4.75	6.00	4.00	.500	.500	3.00
7	X 4	X 3/8	X 13.6 L	13.55	15.4	75.2	95.8	2.25	6.2	1.3	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9 L	17.85	19.6	78.4	117.8	2.41	6.0	1.5	5.25	7.00	4.00	.500	.500	3.50
8	X 4	X 1/2	X 19.6 L	19.55	23.6	91.5	159.6	2.77	6.8	1.7	5.75	8.00	4.00	.500	.500	4.00
8	X 6	X 1/2	X 23.0 L	22.95	31.0	97.9	200.0	3.03	6.5	2.0	6.75	8.00	6.00	.500	.500	4.00

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 30.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 15.000 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
9 X 4 X 1/2 X 21.3	L	21.25	27.9	104.7	209.1	3.14	7.5	2.0	6.25	9.00	4.00	.500	.500	4.50
10 X 2 5/8 X X 15.3	C	11.66	17.9	108.1	161.5	2.96	9.0	1.5	3.43	10.00	2.60	.436	.240	2.40
10 X 3 1/2 X X 21.9	C	16.36	25.5	115.4	219.3	3.33	8.6	1.9	4.81	10.00	3.45	.500	.325	3.25
10 X 2 5/8 X X 20.0	C	16.39	22.2	109.2	193.9	3.13	8.7	1.8	4.82	10.00	2.74	.436	.379	3.79
10 X 3 1/2 X X 24.9	C	18.73	28.6	117.0	241.1	3.43	8.4	2.1	5.51	10.00	3.40	.575	.377	3.77
10 X 3 1/2 X X 25.3	C	19.76	28.5	115.8	240.0	3.40	8.4	2.1	5.81	10.00	3.55	.500	.425	4.25
10 X 4 1/2 X X 28.5	C	21.34	32.6	119.8	269.2	3.56	8.3	2.2	6.28	10.00	3.95	.575	.425	4.25
10 X 3 1/2 X X 28.3	C	22.13	31.5	117.5	260.7	3.48	8.3	2.2	6.51	10.00	3.50	.575	.477	4.77
12 X 3 X X 20.7	C	16.03	28.5	138.3	295.0	3.87	10.4	2.1	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0	C	20.33	32.9	138.9	332.9	3.98	10.1	2.4	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X X 30.9	C	24.48	40.5	144.4	395.6	4.22	9.8	2.7	7.20	12.00	3.45	.610	.450	5.40
12 X 3 1/2 X X 32.9	C	26.52	42.5	144.8	411.1	4.25	9.7	2.8	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0	C	26.91	46.6	148.8	443.3	4.40	9.5	3.0	7.92	12.00	3.77	.700	.467	5.60
13 X 4 X X 31.8	C	24.09	46.9	161.5	490.3	4.71	10.5	3.0	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0	C	27.27	50.2	161.9	517.4	4.74	10.3	3.2	8.02	13.00	4.07	.610	.447	5.81
15 X 3 3/8 X X 33.9	C	27.03	55.1	185.8	658.6	5.36	12.0	3.5	7.95	15.00	3.40	.650	.400	6.00
18 X 4 X X 42.7	C	34.98	80.1	231.0	1100.6	6.60	13.7	4.8	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	84.3	232.2	1144.4	6.61	13.6	4.9	11.19	18.00	4.00	.625	.500	9.00

(60T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	O	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
3 X 3 X 1/2 X 9.4 L	9.35	5.4	27.6	16.0	.86	3.0	.6	2.75	3.00	3.00	.500	.500	1.50
3 1/2 X 3 X 3/8 X 7.9 L	7.81	5.2	31.7	18.2	.92	3.5	.6	2.30	3.50	3.00	.375	.375	1.31
4 X 4 X 1/4 X 6.6 L	6.59	5.4	37.1	21.4	1.01	4.0	.6	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	5.3	36.3	21.1	1.00	4.0	.6	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.9	38.4	23.4	1.05	4.0	.6	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.5	40.2	25.6	1.09	3.9	.6	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	6.2	38.6	24.2	1.06	3.9	.6	2.48	4.00	3.00	.375	.375	1.50
4 X 3 1/2 X 3/8 X 9.1 L	9.08	6.9	40.6	26.8	1.11	3.9	.7	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8 L	9.72	7.6	42.5	29.4	1.16	3.9	.7	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 7/16 X 9.8 L	9.77	7.0	40.4	27.1	1.11	3.9	.7	2.87	4.00	3.00	.438	.438	1.75
4 X 3 X 1/2 X 11.1 L	11.05	7.7	41.7	29.8	1.16	3.8	.7	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3 L	11.26	8.6	44.2	32.9	1.22	3.8	.7	3.31	4.00	4.00	.438	.438	1.75
4 X 3 1/2 X 1/2 X 11.9 L	11.90	8.7	43.8	33.0	1.21	3.8	.8	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8 L	12.75	9.6	45.6	36.1	1.26	3.8	.8	3.75	4.00	4.00	.500	.500	2.00
5 X 3 X 1/4 X 6.6 L	6.59	5.8	46.5	28.7	1.17	4.9	.6	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	6.4	48.9	31.6	1.23	4.9	.6	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2 L	8.18	7.1	50.2	34.5	1.27	4.9	.7	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7 L	8.71	7.8	52.6	37.9	1.33	4.8	.7	2.56	5.00	3.50	.313	.313	1.57
5 X 3 X 3/8 X 9.8 L	9.72	8.2	52.9	39.7	1.35	4.8	.7	2.86	5.00	3.00	.375	.375	1.88
5 X 3 1/2 X 3/8 X 10.4 L	10.36	9.1	55.3	43.7	1.41	4.8	.8	3.05	5.00	3.50	.375	.375	1.88
5 X 3 X 7/16 X 11.3 L	11.26	9.4	55.1	44.6	1.41	4.8	.8	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0 L	12.01	10.4	57.5	49.0	1.48	4.7	.9	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3 L	12.27	11.8	61.1	55.2	1.56	4.7	.9	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8 L	12.75	10.4	56.7	49.0	1.47	4.7	.9	3.75	5.00	3.00	.500	.500	2.50

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5	X 3 1/2	X 1/2	X 13.6	L	13.60	11.6	59.1	53.9	1.53	4.7	.9	4.00	5.00	3.50	.500	.500	2.50
6	X 3 1/2	X 5/16	X 9.8	L	9.78	9.9	67.5	56.9	1.61	5.7	.8	2.88	6.00	3.50	.313	.313	1.88
6	X 4	X 5/16	X 10.3	L	10.31	10.9	69.9	61.7	1.67	5.7	.9	3.03	6.00	4.00	.313	.313	1.88
6	X 3 1/2 X		X 15.3	C	11.07	11.5	70.8	64.8	1.71	5.6	.9	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7	L	11.63	11.6	70.6	65.6	1.71	5.6	.9	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3	L	12.27	12.7	73.0	71.1	1.77	5.6	1.0	3.61	6.00	4.00	.375	.375	2.25
6	X 3 1/2 X		X 18.0	C	12.77	13.3	73.9	74.1	1.81	5.6	1.0	3.76	6.00	3.50	.475	.379	2.27
6	X 4	X 7/16	X 14.3	L	14.24	14.5	75.3	79.9	1.86	5.5	1.1	4.19	6.00	4.00	.438	.438	2.63
6	X 4	X 1/2	X 16.2	L	16.15	16.2	77.1	88.0	1.93	5.4	1.1	4.75	6.00	4.00	.500	.500	3.00
7	X 4	X 3/8	X 13.6	L	13.55	15.6	88.7	100.5	2.09	6.4	1.1	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9	L	17.85	20.0	93.2	124.5	2.27	6.2	1.3	5.25	7.00	4.00	.500	.500	3.50
8	X 4	X 1/2	X 19.6	L	19.55	24.0	109.5	168.8	2.61	7.0	1.5	5.75	8.00	4.00	.500	.500	4.00
8	X 6	X 1/2	X 23.0	L	22.95	31.5	117.9	212.8	2.88	6.8	1.8	6.75	8.00	6.00	.500	.500	4.00
9	X 4	X 1/2	X 21.3	L	21.25	28.4	125.7	221.3	2.96	7.8	1.8	6.25	9.00	4.00	.500	.500	4.50
10	X 2 5/8 X		X 15.3	C	11.66	18.2	128.4	168.0	2.74	9.3	1.3	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9	C	16.36	25.9	138.6	230.2	3.11	8.9	1.7	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0	C	16.39	22.6	130.5	203.3	2.92	9.0	1.6	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9	C	18.73	29.0	140.8	254.0	3.22	8.8	1.8	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3	C	19.76	28.9	139.2	253.1	3.19	8.7	1.8	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2 X		X 28.5	C	21.34	33.1	144.4	284.8	3.36	8.6	2.0	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2 X		X 28.3	C	22.13	32.0	141.4	275.9	3.29	8.6	2.0	6.51	10.00	3.50	.575	.477	4.77

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 33.750 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 18.984 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 3 X	X 20.7 C	16.03	28.8	166.5	308.8	3.61	10.7	1.9	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X	X 25.0 C	20.33	33.5	167.6	350.7	3.75	10.5	2.1	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X	X 30.9 C	24.48	41.3	174.7	419.2	4.00	10.2	2.4	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X	X 30.0 C	25.34	38.8	169.1	396.1	3.87	10.2	2.3	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X	X 32.9 C	26.52	43.4	175.1	436.6	4.04	10.1	2.5	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X	X 35.0 C	26.91	47.4	180.3	471.5	4.19	9.9	2.6	7.92	12.00	3.77	.700	.467	5.60
13 X 4 X	X 31.8 C	24.09	47.6	196.0	519.5	4.46	10.9	2.6	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X	X 35.0 C	27.27	51.1	196.3	550.1	4.51	10.8	2.8	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X	X 40.0 C	32.28	56.6	197.3	596.5	4.58	10.5	3.0	9.49	13.00	4.19	.610	.560	7.28
15 X 3 3/8 X	X 33.9 C	27.03	56.1	225.7	698.9	5.09	12.5	3.1	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X	X 40.0 C	33.15	63.7	227.2	774.3	5.19	12.2	3.4	9.75	15.00	3.52	.650	.520	7.80
18 X 4 X	X 42.7 C	34.98	81.8	280.4	1175.6	6.34	14.4	4.2	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X	X 45.8 C	38.04	86.2	281.5	1225.0	6.37	14.2	4.4	11.19	18.00	4.00	.625	.500	9.00

(60T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 3 1/2 X 3/8 X 9.1	L	9.08	7.0	44.6	28.1	1.04	4.0	.6	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8	L	9.72	7.8	46.8	30.8	1.08	4.0	.7	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 7/16 X 9.8	L	9.77	7.1	44.4	28.4	1.04	4.0	.6	2.87	4.00	3.00	.438	.438	1.75
4 X 3 X 1/2 X 11.1	L	11.05	7.9	46.2	31.3	1.08	3.9	.7	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3	L	11.26	8.8	49.2	34.6	1.14	3.9	.7	3.31	4.00	4.00	.438	.438	1.75
4 X 3 1/2 X 1/2 X 11.9	L	11.90	8.9	48.8	34.6	1.13	3.9	.7	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8	L	12.75	9.8	51.0	38.0	1.18	3.9	.7	3.75	4.00	4.00	.500	.500	2.00
4 X 4 X 5/8 X 15.7	L	15.67	11.6	53.8	44.1	1.25	3.8	.8	4.61	4.00	4.00	.625	.625	2.50
5 X 3 X 5/16 X 8.2	L	8.18	7.2	55.0	35.8	1.18	5.0	.7	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7	L	8.71	8.0	58.0	39.4	1.23	4.9	.7	2.56	5.00	3.50	.313	.313	1.57
5 X 3 X 3/8 X 9.8	L	9.72	8.4	58.6	41.3	1.25	4.9	.7	2.86	5.00	3.00	.375	.375	1.88
5 X 3 1/2 X 3/8 X 10.4	L	10.36	9.3	61.6	45.5	1.31	4.9	.7	3.05	5.00	3.50	.375	.375	1.88
5 X 3 X 7/16 X 11.3	L	11.26	9.6	61.5	46.5	1.32	4.9	.8	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0	L	12.01	10.6	64.5	51.2	1.38	4.8	.8	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3	L	12.27	12.0	69.0	57.7	1.46	4.8	.8	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8	L	12.75	10.6	63.7	51.2	1.37	4.8	.8	3.75	5.00	3.00	.500	.500	2.50
5 X 3 1/2 X 1/2 X 13.6	L	13.60	11.8	66.7	56.4	1.43	4.8	.8	4.00	5.00	3.50	.500	.500	2.50
5 X 3 1/2 X 5/8 X 16.8	L	16.73	14.0	70.1	65.8	1.52	4.7	.9	4.92	5.00	3.50	.625	.625	3.13
6 X 3 1/2 X 5/16 X 9.8	L	9.76	10.1	75.4	59.8	1.50	5.8	.8	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3	L	10.31	11.0	78.5	64.0	1.56	5.8	.8	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3	C	11.07	11.6	79.7	67.3	1.59	5.8	.8	3.26	6.00	3.50	.385	.340	2.04
6 X 3 1/2 X 3/8 X 11.7	L	11.63	11.8	79.6	68.2	1.59	5.8	.9	3.42	6.00	3.50	.375	.375	2.25
6 X 4 X 3/8 X 12.3	L	12.27	12.9	82.6	74.0	1.65	5.7	.9	3.61	6.00	4.00	.375	.375	2.25
6 X 3 1/2 X X 18.0	C	12.77	13.5	83.9	77.2	1.69	5.7	.9	3.76	6.00	3.50	.475	.379	2.27
6 X 4 X 7/16 X 14.3	L	14.24	14.8	85.9	83.4	1.74	5.7	1.0	4.19	6.00	4.00	.438	.438	2.63
6 X 4 X 1/2 X 16.2	L	16.15	16.5	88.4	92.1	1.81	5.6	1.0	4.75	6.00	4.00	.500	.500	3.00

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
6	X 4	X 9/16	X 18.1	L	18.06	18.2	90.5	100.2	1.87	5.5	1.1	5.31	6.00	4.00	.563	.563	3.38
6	X 4	X 5/8	X 20.0	L	19.92	19.7	92.2	107.7	1.92	5.5	1.2	5.86	6.00	4.00	.625	.625	3.75
7	X 4	X 3/8	X 13.6	L	13.55	15.8	101.4	104.5	1.95	6.6	1.0	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9	L	17.85	20.3	107.8	130.2	2.13	6.4	1.2	5.25	7.00	4.00	.500	.500	3.50
7	X 4	X 5/8	X 22.1	L	22.05	24.4	111.9	152.6	2.26	6.3	1.4	6.48	7.00	4.00	.625	.625	4.38
8	X 4	X 1/2	X 19.6	L	19.55	24.4	127.3	176.6	2.46	7.2	1.4	5.75	8.00	4.00	.500	.500	4.00
8	X 4	X 9/16	X 21.9	L	21.89	27.0	129.8	192.5	2.54	7.1	1.5	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	31.9	138.3	223.8	2.72	7.0	1.6	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	29.4	131.9	207.1	2.60	7.1	1.6	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	35.3	140.6	243.6	2.80	6.9	1.7	7.57	8.00	6.00	.563	.563	4.50
9	X 4	X 1/2	X 21.3	L	21.25	28.8	147.0	231.7	2.79	8.0	1.6	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	31.8	149.7	252.6	2.88	7.9	1.7	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	34.7	151.9	271.9	2.95	7.8	1.8	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8 X		X 15.3	C	11.66	18.4	147.8	173.4	2.54	9.5	1.2	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9	C	16.36	26.2	161.7	239.3	2.91	9.1	1.5	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0	C	16.39	22.9	151.7	211.2	2.73	9.2	1.4	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9	C	18.73	29.4	164.9	264.9	3.02	9.0	1.6	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3	C	19.76	29.3	162.9	264.2	3.01	9.0	1.6	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2 X		X 28.5	C	21.34	33.6	169.7	298.0	3.17	8.9	1.8	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2 X		X 28.3	C	22.13	32.5	165.9	288.8	3.11	8.9	1.7	6.51	10.00	3.50	.575	.477	4.77
10	X 4	X	X 33.6	C	26.44	38.2	170.9	331.8	3.26	8.7	1.9	7.78	10.00	4.10	.575	.575	5.75
12	X 3	X	X 20.7	C	16.03	29.2	195.1	320.2	3.37	11.0	1.6	4.72	12.00	2.94	.501	.282	3.38

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 37.500 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 23.438 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12	X 3	X	X 25.0	C	20.33	33.9	197.0	365.6	3.53	10.8	1.9	5.98	12.00	3.05	.501	.387	4.64
12	X 3 1/2	X	X 30.9	C	24.48	41.9	206.1	439.2	3.79	10.5	2.1	7.20	12.00	3.45	.600	.450	5.40
12	X 3	X	X 30.0	C	25.34	39.4	199.2	415.1	3.67	10.5	2.1	7.45	12.00	3.17	.501	.510	6.12
12	X 3 1/2	X	X 32.9	C	26.52	44.0	206.7	458.3	3.83	10.4	2.2	7.80	12.00	3.50	.600	.500	6.00
12	X 4	X	X 35.0	C	26.91	48.1	213.3	495.6	3.98	10.3	2.3	7.92	12.00	3.77	.700	.467	5.60
12	X 3 1/2	X	X 37.0	C	30.60	48.3	208.0	495.2	3.91	10.2	2.4	9.00	12.00	3.60	.600	.600	7.20
12	X 4	X	X 40.0	C	31.93	53.3	214.3	539.2	4.05	10.1	2.5	9.39	12.00	3.89	.700	.590	7.08
13	X 4	X	X 31.8	C	24.09	48.2	232.2	544.0	4.22	11.3	2.3	7.09	13.00	4.00	.610	.375	4.88
13	X 4	X	X 35.0	C	27.27	51.9	232.6	577.8	4.29	11.1	2.5	8.02	13.00	4.07	.610	.447	5.81
13	X 4	X	X 40.0	C	32.28	57.6	233.6	629.2	4.37	10.9	2.7	9.49	13.00	4.19	.610	.560	7.28
15	X 3 3/8	X	X 33.9	C	27.03	56.9	267.8	733.1	4.83	12.9	2.7	7.95	15.00	3.40	.650	.400	6.00
15	X 3 3/8	X	X 40.0	C	33.15	64.8	269.4	816.2	4.96	12.6	3.0	9.75	15.00	3.52	.650	.520	7.80
15	X 4	X	X 50.0	C	41.02	81.3	281.1	985.2	5.27	12.1	3.5	12.06	15.00	4.00	.797	.625	9.38
18	X 4	X	X 42.7	C	34.98	83.2	333.6	1239.9	6.06	14.9	3.7	10.29	18.00	3.95	.625	.450	8.10
18	X 4	X	X 45.8	C	38.04	87.8	334.6	1294.8	6.12	14.8	3.9	11.19	18.00	4.00	.625	.500	9.00
18	X 4	X	X 51.9	C	44.16	96.8	337.0	1400.1	6.20	14.5	4.2	12.99	18.00	4.10	.625	.600	10.80

(60T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 1/2 X 12.8 L	12.75	10.0	55.8	39.8	1.11	4.0	.7	3.75	4.00	4.00	.500	.500	2.00
4 X 4 X 5/8 X 15.7 L	15.67	11.8	59.4	46.3	1.18	3.9	.8	4.61	4.00	4.00	.625	.625	2.50
5 X 3 1/2 X 3/8 X 10.4 L	10.36	9.5	66.9	47.2	1.23	5.0	.7	3.05	5.00	3.50	.375	.375	1.88
5 X 3 X 7/16 X 11.3 L	11.26	9.7	67.0	48.3	1.24	5.0	.7	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0 L	12.01	10.8	70.6	53.3	1.29	4.9	.8	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3 L	12.27	12.2	75.9	59.9	1.37	4.9	.8	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8 L	12.75	10.8	69.9	53.3	1.29	4.9	.8	3.75	5.00	3.00	.500	.500	2.50
5 X 3 1/2 X 1/2 X 13.6 L	13.60	12.0	73.5	58.8	1.35	4.9	.8	4.00	5.00	3.50	.500	.500	2.50
5 X 3 1/2 X 5/8 X 16.8 L	16.73	14.3	78.0	68.8	1.44	4.8	.9	4.92	5.00	3.50	.625	.625	3.13
6 X 3 1/2 X 5/16 X 9.8 L	9.78	10.3	82.1	61.0	1.40	5.9	.7	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L	10.31	11.2	85.8	66.2	1.45	5.9	.8	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X C X 15.3 C	11.07	11.8	87.4	69.6	1.48	5.9	.8	3.26	6.00	3.50	.385	.340	2.04
6 X 3 1/2 X 3/8 X 11.7 L	11.63	12.0	87.5	70.6	1.49	5.9	.8	3.42	6.00	3.50	.375	.375	2.25
6 X 4 X 3/8 X 12.3 L	12.27	13.1	91.2	76.7	1.55	5.8	.8	3.61	6.00	4.00	.375	.375	2.25
6 X 3 1/2 X X 18.0 C	12.77	13.7	92.8	80.0	1.58	5.8	.9	3.76	6.00	3.50	.475	.379	2.27
6 X 4 X 7/16 X 14.3 L	14.24	15.0	95.4	86.6	1.63	5.8	.9	4.19	6.00	4.00	.438	.438	2.63
6 X 4 X 1/2 X 16.2 L	16.15	16.7	98.8	95.7	1.70	5.7	1.0	4.75	6.00	4.00	.500	.500	3.00
6 X 4 X 9/16 X 18.1 L	18.06	18.5	101.6	104.4	1.76	5.7	1.0	5.31	6.00	4.00	.563	.563	3.38
6 X 4 X 5/8 X 20.0 L	19.92	20.1	103.9	112.4	1.81	5.6	1.1	5.86	6.00	4.00	.625	.625	3.75
7 X 4 X 3/8 X 13.6 L	13.55	16.1	113.0	108.1	1.83	6.7	1.0	3.98	7.00	4.00	.375	.375	2.63
7 X 4 X 1/2 X 17.9 L	17.85	20.6	121.5	135.3	2.01	6.6	1.1	5.25	7.00	4.00	.500	.500	3.50
7 X 4 X 5/8 X 22.1 L	22.05	24.7	127.2	159.2	2.14	6.4	1.3	6.48	7.00	4.00	.625	.625	4.38
8 X 4 X 1/2 X 19.6 L	19.55	24.7	144.6	183.4	2.32	7.4	1.3	5.75	8.00	4.00	.500	.500	4.00

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8	X 4	X 9/16	X 21.9	L	21.89	27.3	148.0	200.4	2.40	7.3	1.4	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	32.3	158.5	233.4	2.58	7.2	1.5	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	29.8	150.7	216.1	2.47	7.3	1.4	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	35.8	161.7	254.7	2.66	7.1	1.6	7.57	8.00	6.00	.563	.563	4.50
9	X 4	X 1/2	X 21.3	L	21.25	29.2	168.0	240.7	2.64	8.3	1.4	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	32.3	171.5	263.1	2.73	8.2	1.5	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	35.2	174.5	283.8	2.80	8.1	1.6	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8 X		X 15.3	C	11.66	18.5	165.7	178.2	2.37	9.6	1.1	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9	C	16.36	26.4	184.0	247.1	2.73	9.3	1.3	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0	C	16.39	23.2	172.0	218.1	2.56	9.4	1.3	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9	C	18.73	29.7	188.5	274.2	2.85	9.2	1.5	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3	C	19.76	29.7	186.3	273.8	2.83	9.2	1.5	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2 X		X 28.5	C	21.34	34.0	194.8	309.3	2.99	9.1	1.6	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2 X		X 28.3	C	22.13	32.9	190.3	300.0	2.93	9.1	1.6	6.51	10.00	3.50	.575	.477	4.77
10	X 4	X	X 33.6	C	26.44	38.7	196.8	346.0	3.09	8.9	1.8	7.78	10.00	4.10	.575	.575	5.75
12	X 3	X	X 20.7	C	16.03	29.4	223.0	330.0	3.16	11.2	1.5	4.72	12.00	2.94	.501	.282	3.38
12	X 3	X	X 25.0	C	20.33	34.3	226.3	378.3	3.32	11.0	1.7	5.98	12.00	3.05	.501	.387	4.64
12	X 3 1/2	X	X 30.9	C	24.48	42.4	238.1	456.4	3.58	10.8	1.9	7.20	12.00	3.45	.600	.450	5.40
12	X 3	X	X 30.0	C	25.34	39.9	229.8	431.5	3.47	10.8	1.9	7.45	12.00	3.17	.501	.510	6.12
12	X 3 1/2	X	X 32.9	C	26.52	44.6	239.0	477.0	3.63	10.7	2.0	7.80	12.00	3.50	.600	.500	6.00
12	X 4	X	X 35.0	C	26.91	48.7	247.2	516.2	3.77	10.6	2.1	7.92	12.00	3.77	.700	.467	5.60
12	X 3 1/2	X	X 37.0	C	30.60	49.0	240.8	517.0	3.72	10.5	2.1	9.00	12.00	3.60	.600	.600	7.20
12	X 4	X	X 40.0	C	31.93	54.1	248.6	563.7	3.86	10.4	2.3	9.39	12.00	3.89	.700	.590	7.08

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 41.250 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 28.359 SQ. IN.

NOMINAL SIZE IN X IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
13 X 4 X X 31.8	C	24.09	48.8	269.2	565.0	3.99	11.6	2.1	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0	C	27.27	52.5	269.9	601.6	4.07	11.5	2.2	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0	C	32.28	58.4	271.4	657.5	4.17	11.3	2.4	9.49	13.00	4.19	.610	.560	7.28
15 X 3 3/8 X X 33.9	C	27.03	57.6	311.5	762.1	4.58	13.2	2.4	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0	C	33.15	65.7	313.7	852.4	4.73	13.0	2.7	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	82.6	327.7	1034.6	5.06	12.5	3.2	12.06	15.00	4.00	.797	.625	9.38
18 X 4 X X 42.7	C	34.98	84.3	389.8	1295.3	5.79	15.4	3.3	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	89.0	390.8	1355.2	5.85	15.2	3.5	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	98.4	393.3	1470.5	5.96	14.9	3.7	12.99	18.00	4.10	.625	.600	10.80

(60T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X 4	X 5/8	X 15.7	L	15.67	12.1	64.3	48.4	1.12	4.0	.8	4.61	4.00	4.00	.625	.625	2.50
4	X 4	X 3/4	X 18.5	L	18.49	13.8	67.5	54.4	1.18	3.9	.8	5.44	4.00	4.00	.750	.750	3.00
5	X 5	X 3/8	X 12.3	L	12.27	12.5	81.7	62.1	1.29	5.0	.8	3.61	5.00	5.00	.375	.375	1.88
5	X 3 1/2	X 1/2	X 13.6	L	13.60	12.3	79.3	61.0	1.27	5.0	.8	4.00	5.00	3.50	.500	.500	2.50
5	X 3 1/2	X 5/8	X 16.8	L	16.73	14.6	85.0	71.6	1.36	4.9	.8	4.92	5.00	3.50	.625	.625	3.13
5	X 3 1/2	X 3/4	X 19.8	L	19.76	16.7	89.0	80.8	1.43	4.8	.9	5.81	5.00	3.50	.750	.750	3.75
6	X 3 1/2	X	X 15.3	C	11.07	12.0	94.0	71.8	1.39	6.0	.8	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7	L	11.63	12.2	94.2	72.9	1.40	6.0	.8	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3	L	12.27	13.3	98.5	79.2	1.46	5.9	.8	3.61	6.00	4.00	.375	.375	2.25
6	X 3 1/2	X	X 18.0	C	12.77	13.9	100.6	82.7	1.48	5.9	.8	3.76	6.00	3.50	.475	.379	2.27
6	X 4	X 7/16	X 14.3	L	14.24	15.2	103.9	89.6	1.54	5.9	.9	4.19	6.00	4.00	.438	.438	2.63
6	X 4	X 1/2	X 16.2	L	16.15	17.0	108.2	99.2	1.60	5.8	.9	4.75	6.00	4.00	.500	.500	3.00
6	X 4	X 9/16	X 18.1	L	18.06	18.7	111.8	108.3	1.67	5.8	1.0	5.31	6.00	4.00	.563	.563	3.38
6	X 4	X 5/8	X 20.0	L	19.92	20.4	114.8	116.8	1.72	5.7	1.0	5.86	6.00	4.00	.625	.625	3.75
6	X 4	X 3/4	X 23.6	L	23.59	23.5	119.5	132.4	1.80	5.6	1.1	6.94	6.00	4.00	.750	.750	4.50
7	X 4	X 3/8	X 13.6	L	13.55	16.3	123.3	111.4	1.72	6.8	.9	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9	L	17.85	20.9	134.2	139.9	1.89	6.7	1.0	5.25	7.00	4.00	.500	.500	3.50
7	X 4	X 5/8	X 22.1	L	22.05	25.1	141.6	165.2	2.03	6.6	1.2	6.48	7.00	4.00	.625	.625	4.38
7	X 4	X 3/4	X 26.2	L	26.14	29.0	146.9	187.6	2.13	6.5	1.3	7.69	7.00	4.00	.750	.750	5.25
8	X 4	X 1/2	X 19.6	L	19.55	25.0	160.9	189.6	2.19	7.6	1.2	5.75	8.00	4.00	.500	.500	4.00
8	X 4	X 9/16	X 21.9	L	21.89	27.7	165.3	207.5	2.27	7.5	1.3	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	32.7	178.0	241.9	2.44	7.4	1.4	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	30.2	169.0	224.1	2.34	7.4	1.3	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	36.2	182.2	264.5	2.53	7.3	1.5	7.57	8.00	6.00	.563	.563	4.50

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8	X 4	X 3/4	X 28.7	L	28.69	35.0	174.7	254.9	2.46	7.3	1.5	8.44	8.00	4.00	.750	.750	6.00
8	X 6	X 3/4	X 33.8	L	33.79	46.0	191.1	324.3	2.72	7.1	1.7	9.94	8.00	6.00	.750	.750	6.00
9	X 4	X 1/2	X 21.3	L	21.25	29.5	188.1	248.7	2.49	8.4	1.3	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	32.7	192.8	272.4	2.59	8.3	1.4	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	35.7	196.6	294.3	2.66	8.3	1.5	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8 X		X 15.3	C	11.66	18.7	181.9	182.4	2.22	9.7	1.0	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9	C	16.36	26.7	205.2	254.0	2.57	9.5	1.2	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0	C	16.39	23.4	191.0	224.2	2.41	9.6	1.2	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9	C	18.73	30.0	211.1	282.4	2.68	9.4	1.3	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3	C	19.76	30.0	208.7	282.2	2.67	9.4	1.4	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2 X		X 28.5	C	21.34	34.4	219.2	319.3	2.82	9.3	1.5	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2 X		X 28.3	C	22.13	33.3	214.0	309.8	2.77	9.3	1.4	6.51	10.00	3.50	.575	.477	4.77
10	X 4 X		X 33.6	C	26.44	39.2	222.4	358.4	2.94	9.1	1.6	7.78	10.00	4.10	.575	.575	5.75
12	X 3 X		X 20.7	C	16.03	29.7	249.7	338.5	2.97	11.4	1.4	4.72	12.00	2.94	.501	.282	3.38
12	X 3 X		X 25.0	C	20.33	34.7	255.0	389.4	3.13	11.2	1.5	5.98	12.00	3.05	.501	.387	4.64
12	X 3 1/2 X		X 30.9	C	24.48	42.8	269.9	471.3	3.39	11.0	1.7	7.20	12.00	3.45	.600	.450	5.40
12	X 3 X		X 30.0	C	25.34	40.4	260.1	445.8	3.29	11.0	1.7	7.45	12.00	3.17	.501	.510	6.12
12	X 3 1/2 X		X 32.9	C	26.52	45.1	271.3	493.3	3.45	10.9	1.8	7.80	12.00	3.50	.600	.500	6.00
12	X 4 X		X 35.0	C	26.91	49.2	281.2	534.1	3.58	10.9	1.9	7.92	12.00	3.77	.700	.467	5.60
12	X 3 1/2 X		X 37.0	C	30.60	49.7	273.9	536.1	3.54	10.8	2.0	9.00	12.00	3.60	.600	.600	7.20
12	X 4 X		X 40.0	C	31.93	54.8	283.4	585.2	3.68	10.7	2.1	9.39	12.00	3.89	.700	.590	7.08
12	X 4 X		X 45.0	C	36.90	60.2	285.6	633.7	3.77	10.5	2.2	10.85	12.00	4.01	.700	.712	8.54
13	X 4 X		X 31.8	C	24.09	49.2	306.3	583.1	3.78	11.8	1.9	7.09	13.00	4.00	.618	.375	4.88

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 45.000 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 33.750 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
13 X 4 X X 35.0	C	27.27	53.1	307.6	622.2	3.86	11.7	2.0	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0	C	32.28	59.1	309.9	682.2	3.97	11.5	2.2	9.49	13.00	4.19	.610	.560	7.28
15 X 3 3/8 X X 33.9	C	27.03	58.1	355.9	787.2	4.34	13.5	2.2	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0	C	33.15	66.5	359.1	883.9	4.51	13.3	2.5	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	83.7	376.2	1078.0	4.85	12.9	2.9	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	79.8	364.7	1031.7	4.71	12.9	2.8	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	87.8	377.3	1122.2	4.90	12.8	3.0	13.01	15.00	4.06	.797	.688	10.32
18 X 4 X X 42.7	C	34.98	85.3	448.3	1343.3	5.52	15.8	3.0	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	90.1	449.5	1407.8	5.60	15.6	3.1	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	99.7	452.4	1532.2	5.73	15.4	3.4	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0	C	50.28	109.1	455.7	1651.0	5.83	15.1	3.6	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 52.500 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 45.938 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
6	X 4	X 9/16	X 18.1	L	18.06	19.3	128.9	115.6	1.50	6.0	.9	5.31	6.00	4.00	.563	.563	3.38
6	X 4	X 5/8	X 20.0	L	19.92	21.0	133.4	124.8	1.55	5.9	.9	5.86	6.00	4.00	.625	.625	3.75
6	X 4	X 3/4	X 23.6	L	23.59	24.2	140.7	142.0	1.64	5.9	1.0	6.94	6.00	4.00	.750	.750	4.50
7	X 4	X 1/2	X 17.9	L	17.85	21.4	155.8	148.4	1.70	6.9	1.0	5.25	7.00	4.00	.500	.500	3.50
7	X 4	X 5/8	X 22.1	L	22.05	25.8	167.1	175.9	1.83	6.8	1.1	6.48	7.00	4.00	.625	.625	4.38
7	X 4	X 3/4	X 26.2	L	26.14	29.8	175.4	200.7	1.93	6.7	1.1	7.69	7.00	4.00	.750	.750	5.25
8	X 4	X 1/2	X 19.6	L	19.55	25.6	189.5	200.5	1.97	7.8	1.1	5.75	8.00	4.00	.500	.500	4.00
8	X 4	X 9/16	X 21.9	L	21.89	28.4	196.3	220.0	2.05	7.8	1.1	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	33.4	213.6	256.7	2.21	7.7	1.2	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	31.0	202.1	238.2	2.12	7.7	1.2	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	37.1	220.4	281.6	2.29	7.6	1.3	7.57	8.00	6.00	.563	.563	4.50
8	X 4	X 3/4	X 28.7	L	28.69	35.9	211.3	272.3	2.24	7.6	1.3	8.44	8.00	4.00	.750	.750	6.00
8	X 4	X 7/8	X 33.1	L	33.10	40.5	218.2	302.9	2.33	7.5	1.4	9.73	8.00	4.00	.875	.875	7.00
8	X 6	X 3/4	X 33.8	L	33.79	47.1	235.0	348.2	2.50	7.4	1.5	9.94	8.00	6.00	.750	.750	6.00
8	X 6	X 7/8	X 39.1	L	39.05	53.3	241.6	387.4	2.60	7.3	1.6	11.48	8.00	6.00	.875	.875	7.00
9	X 4	X 1/2	X 21.3	L	21.25	30.2	224.2	262.7	2.24	8.7	1.2	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	33.4	231.7	288.4	2.33	8.6	1.2	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	36.5	237.9	312.5	2.41	8.6	1.3	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8 X		X 15.3	C	11.66	19.1	207.8	190.1	1.96	10.0	.9	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9	C	16.36	27.2	242.0	266.1	2.29	9.8	1.1	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0	C	16.39	23.9	223.6	235.0	2.15	9.8	1.1	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9	C	18.73	30.6	251.6	296.7	2.40	9.7	1.2	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3	C	19.76	30.6	249.0	296.8	2.39	9.7	1.2	5.81	10.00	3.55	.500	.425	4.25

(60T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 52.500 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 45.938 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 1/2 X	X 28.5 C	21.34	35.0	264.0	336.3	2.54	9.6	1.3	6.28	10.00	3.95	.575	.425	4.25
10 X 3 1/2 X	X 28.3 C	22.13	34.0	257.5	326.6	2.50	9.6	1.3	6.51	10.00	3.50	.575	.477	4.77
10 X 4 X	X 33.6 C	26.44	40.1	270.7	379.7	2.66	9.5	1.4	7.78	10.00	4.10	.575	.575	5.75
10 X 4 X	X 41.1 C	33.95	47.4	279.0	440.5	2.81	9.3	1.6	9.99	10.00	4.32	.575	.796	7.96
12 X 3 X	X 20.7 C	16.03	30.2	297.2	353.1	2.64	11.7	1.2	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X	X 25.0 C	20.33	35.4	307.8	408.3	2.80	11.5	1.3	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X	X 30.9 C	24.48	43.7	330.3	496.4	3.06	11.4	1.5	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X	X 30.0 C	25.34	41.2	317.6	470.0	2.97	11.4	1.5	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X	X 32.9 C	26.52	46.0	333.2	520.8	3.11	11.3	1.6	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X	X 35.0 C	26.91	50.2	346.9	564.2	3.24	11.2	1.6	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X	X 37.0 C	30.60	50.8	338.5	568.3	3.22	11.2	1.7	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X	X 40.0 C	31.93	55.9	351.8	621.3	3.35	11.1	1.8	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X	X 45.0 C	36.90	61.6	356.2	675.9	3.45	11.0	1.9	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X	X 50.0 C	41.93	67.3	360.5	729.8	3.54	10.9	2.0	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X	X 31.8 C	24.09	50.0	377.8	613.1	3.40	12.3	1.6	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X	X 35.0 C	27.27	54.0	381.2	656.6	3.49	12.2	1.7	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X	X 40.0 C	32.28	60.3	386.2	723.5	3.61	12.0	1.9	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X	X 50.0 C	42.30	72.4	395.0	849.3	3.81	11.7	2.2	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X	X 33.9 C	27.03	59.2	443.5	828.6	3.92	14.0	1.9	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X	X 40.0 C	33.15	67.8	450.4	936.1	4.10	13.8	2.1	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X	X 50.0 C	41.02	85.5	475.8	1150.4	4.45	13.5	2.4	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X	X 50.0 C	43.15	81.8	460.3	1102.4	4.34	13.5	2.4	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X	X 53.2 C	44.23	89.9	477.8	1200.7	4.51	13.4	2.5	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X	X 56.4 C	47.46	94.3	480.1	1251.3	4.57	13.3	2.6	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X	X 59.6 C	50.61	98.6	482.1	1299.0	4.62	13.2	2.7	14.89	15.00	4.19	.797	.813	12.20

(60T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 52.500 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 45.938 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 4 X X 42.7 C	34.98	86.9	568.3	1422.1	5.03	16.4	2.5	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C	38.04	91.9	570.8	1494.5	5.11	16.3	2.6	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9 C	44.16	102.0	575.7	1635.0	5.27	16.0	2.8	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0 C	50.28	111.8	580.6	1770.0	5.40	15.8	3.0	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 60.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 60.000 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X 4	X 3/4	X 26.2	L	26.14	30.7	199.2	212.5	1.77	6.9	1.1	7.69	7.00	4.00	.750	.750	5.25
8	X 4	X 9/16	X 21.9	L	21.89	29.1	221.4	231.3	1.87	8.0	1.0	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	34.2	243.1	269.8	2.01	7.9	1.1	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	31.7	229.6	250.8	1.93	7.9	1.1	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	37.9	252.9	296.5	2.09	7.8	1.2	7.57	8.00	6.00	.563	.563	4.50
8	X 4	X 3/4	X 28.7	L	28.69	36.8	242.9	287.5	2.05	7.8	1.2	8.44	8.00	4.00	.750	.750	6.00
8	X 4	X 7/8	X 33.1	L	33.10	41.5	253.1	320.9	2.15	7.7	1.3	9.73	8.00	4.00	.875	.875	7.00
8	X 6	X 3/4	X 33.8	L	33.79	48.2	274.6	368.9	2.30	7.7	1.3	9.94	8.00	6.00	.750	.750	6.00
8	X 4	X 1	X 37.4	L	37.40	45.9	261.1	351.2	2.22	7.7	1.3	11.00	8.00	4.00	1.000	1.000	8.00
8	X 6	X 7/8	X 39.1	L	39.05	54.5	284.7	411.8	2.40	7.6	1.4	11.48	8.00	6.00	.875	.875	7.00
8	X 6	X 1	X 44.2	L	44.20	60.5	292.6	451.0	2.49	7.5	1.5	13.00	8.00	6.00	1.000	1.000	8.00
9	X 4	X 1/2	X 21.3	L	21.25	30.8	253.7	275.1	2.04	8.9	1.1	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	34.2	264.4	302.5	2.12	8.9	1.1	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	37.3	273.4	328.4	2.20	8.8	1.2	7.73	9.00	4.00	.625	.625	5.63
10	X 3 1/2	X	X 21.9	C	16.36	27.8	270.7	276.9	2.07	10.0	1.0	4.81	10.00	3.45	.500	.325	3.25
10	X 3 1/2	X	X 24.9	C	18.73	31.2	284.3	309.2	2.17	9.9	1.1	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2	X	X 25.3	C	19.76	31.3	281.8	309.6	2.17	9.9	1.1	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2	X	X 28.5	C	21.34	35.7	301.4	351.1	2.30	9.8	1.2	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2	X	X 28.3	C	22.13	34.7	293.9	341.3	2.27	9.8	1.2	6.51	10.00	3.50	.575	.477	4.77
10	X 4	X	X 33.6	C	26.44	40.9	312.8	397.9	2.42	9.7	1.3	7.78	10.00	4.10	.575	.575	5.75
10	X 4	X	X 41.1	C	33.95	48.5	326.6	464.1	2.58	9.6	1.4	9.99	10.00	4.32	.575	.796	7.96
12	X 3	X	X 20.7	C	16.03	30.7	335.0	365.7	2.38	11.9	1.1	4.72	12.00	2.94	.501	.282	3.38
12	X 3	X	X 25.0	C	20.33	36.0	352.1	424.3	2.54	11.8	1.2	5.98	12.00	3.05	.501	.387	4.64

(60T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 60.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 60.000 SQ. IN.

NOMINAL SIZE			WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
				FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT			LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 3 1/2 X X 30.9 C			24.48	44.4	383.6	517.5	2.77	11.7	1.3	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X X 30.0 C			25.34	42.0	368.2	498.3	2.70	11.7	1.3	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X X 32.9 C			26.52	46.9	388.6	543.7	2.83	11.6	1.4	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0 C			26.91	51.0	406.3	589.1	2.95	11.6	1.4	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0 C			30.60	51.7	397.6	595.0	2.94	11.5	1.5	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0 C			31.93	57.0	415.3	651.1	3.06	11.4	1.6	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X X 45.0 C			36.90	62.8	423.2	710.8	3.17	11.3	1.7	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0 C			41.93	68.7	430.4	770.0	3.26	11.2	1.8	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 31.8 C			24.09	50.8	441.9	637.9	3.08	12.6	1.4	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0 C			27.27	54.9	448.5	684.7	3.17	12.5	1.5	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0 C			32.28	61.3	458.0	757.3	3.30	12.3	1.7	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X X 50.0 C			42.30	73.9	473.3	895.1	3.52	12.1	1.9	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 33.9 C			27.03	60.0	525.0	862.0	3.56	14.4	1.6	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0 C			33.15	69.0	537.8	978.2	3.74	14.2	1.8	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0 C			41.02	87.0	574.4	1208.9	4.10	13.9	2.1	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0 C			43.15	83.4	555.0	1159.8	3.99	13.9	2.1	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2 C			44.23	91.5	578.2	1264.3	4.16	13.8	2.2	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4 C			47.46	96.1	582.0	1320.2	4.22	13.7	2.3	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6 C			50.61	100.6	585.4	1373.0	4.28	13.7	2.3	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8 C			53.82	105.0	588.7	1425.7	4.34	13.6	2.4	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9 C			56.93	109.3	591.8	1476.1	4.39	13.5	2.5	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1 C			60.13	113.7	594.9	1527.1	4.43	13.4	2.6	17.69	15.00	4.37	.797	1.000	15.00
18 X 4 X X 42.7 C			34.98	88.2	687.0	1484.5	4.60	16.8	2.2	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C			38.04	93.4	691.8	1563.4	4.69	16.7	2.3	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9 C			44.16	103.8	700.7	1717.2	4.85	16.5	2.5	12.99	18.00	4.10	.625	.600	10.80

(60T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 60.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 60.000 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 4 X X 58.0 C	50.28	114.0	708.8	1865.9	4.99	16.4	2.6	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 67.500 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 75.938 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4	X 7/8	X 33.1	L		33.10	42.6	282.5	337.5	1.98	7.9	1.2	9.73	8.00	4.00	.875	.875	7.00
8 X 6	X 3/4	X 33.8	L		33.79	49.3	308.5	387.7	2.12	7.9	1.3	9.94	8.00	6.00	.750	.750	6.00
8 X 4	X 1	X 37.4	L		37.40	47.1	293.6	370.1	2.06	7.9	1.3	11.00	8.00	4.00	1.000	1.000	8.00
8 X 6	X 7/8	X 39.1	L		39.05	55.8	322.7	433.9	2.23	7.8	1.3	11.48	8.00	6.00	.875	.875	7.00
8 X 6	X 1	X 44.2	L		44.20	61.9	334.0	476.3	2.31	7.7	1.4	13.00	8.00	6.00	1.000	1.000	8.00
9 X 4	X 5/8	X 26.3	L		26.30	38.2	302.1	343.2	2.03	9.0	1.1	7.73	9.00	4.00	.625	.625	5.63
10 X 4	X	X 33.6	C		26.44	41.7	347.4	414.6	2.23	9.9	1.2	7.78	10.00	4.10	.575	.575	5.75
10 X 4	X	X 41.1	C		33.95	49.5	367.6	485.3	2.38	9.8	1.3	9.99	10.00	4.32	.575	.796	7.96
12 X 3	X	X 25.0	C		20.33	36.6	386.9	439.1	2.32	12.0	1.1	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2	X	X 30.9	C		24.48	45.2	427.6	536.3	2.54	11.9	1.3	7.20	12.00	3.45	.600	.450	5.40
12 X 3	X	X 30.0	C		25.34	42.8	409.9	508.7	2.47	11.9	1.2	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2	X	X 32.9	C		26.52	47.7	435.1	564.0	2.60	11.8	1.3	7.80	12.00	3.50	.600	.500	6.00
12 X 4	X	X 35.0	C		26.91	51.8	456.8	611.1	2.70	11.8	1.3	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2	X	X 37.0	C		30.60	52.7	448.7	618.6	2.70	11.7	1.4	9.00	12.00	3.60	.600	.600	7.20
12 X 4	X	X 40.0	C		31.93	57.9	471.0	677.1	2.82	11.7	1.4	9.39	12.00	3.89	.700	.590	7.08
12 X 4	X	X 45.0	C		36.90	63.9	483.3	741.1	2.92	11.6	1.5	10.85	12.00	4.01	.700	.712	8.54
12 X 4	X	X 50.0	C		41.93	70.0	494.5	804.8	3.02	11.5	1.6	12.33	12.00	4.14	.700	.835	10.02
13 X 4	X	X 31.8	C		24.09	51.6	495.8	659.6	2.82	12.8	1.3	7.09	13.00	4.00	.610	.375	4.88
13 X 4	X	X 35.0	C		27.27	55.7	506.5	709.3	2.91	12.7	1.4	8.02	13.00	4.07	.610	.447	5.81
13 X 4	X	X 40.0	C		32.28	62.3	521.6	786.5	3.03	12.6	1.5	9.49	13.00	4.19	.610	.560	7.28
13 X 4	X	X 50.0	C		42.30	75.3	545.8	934.2	3.25	12.4	1.7	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8	X	X 33.9	C		27.03	60.9	596.4	890.7	3.26	14.6	1.5	7.95	15.00	3.40	.650	.400	6.00

(60T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 67.500 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 75.938 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15 X 3 3/8 X X 40.0	C	33.15	70.0	617.1	1014.0	3.44	14.5	1.6	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	88.3	667.4	1258.0	3.78	14.2	1.9	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	84.8	644.3	1208.2	3.69	14.2	1.9	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	93.0	673.6	1317.7	3.85	14.2	2.0	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4	C	47.46	97.7	679.8	1378.1	3.92	14.1	2.0	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6	C	50.61	102.3	685.3	1435.3	3.98	14.0	2.1	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8	C	53.82	106.9	690.5	1492.4	4.03	14.0	2.2	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9	C	56.93	111.3	695.4	1547.3	4.09	13.9	2.2	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1	C	60.13	115.9	700.2	1602.9	4.14	13.8	2.3	17.69	15.00	4.37	.797	1.000	15.00
15 X 4 X X 75.0	C	66.06	124.3	708.7	1704.7	4.23	13.7	2.4	19.43	15.00	4.49	.797	1.116	16.74
18 X 4 X X 42.7	C	34.98	89.3	798.5	1536.2	4.22	17.2	1.9	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	36.04	94.7	806.8	1620.5	4.31	17.1	2.0	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	105.3	821.6	1785.2	4.48	17.0	2.2	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0	C	50.28	115.8	834.7	1945.4	4.63	16.8	2.3	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 75.000 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 93.750 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 6 X 7/8 X 39.1	L	39.05	57.1	354.9	454.8	2.08	8.0	1.3	11.48	8.00	6.00	.875	.875	7.00
8 X 6 X 1 X 44.2	L	44.20	63.3	369.8	500.0	2.16	7.9	1.4	13.00	8.00	6.00	1.000	1.000	8.00
10 X 4 X X 41.1	C	33.95	50.6	401.5	505.4	2.21	10.0	1.3	9.99	10.00	4.32	.575	.796	7.96
12 X 4 X X 35.0	C	26.91	52.7	497.6	631.7	2.49	12.0	1.3	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0	C	30.60	53.6	490.8	640.6	2.50	11.9	1.3	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0	C	31.93	58.9	517.6	701.2	2.61	11.9	1.4	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X X 45.0	C	36.90	65.1	534.9	768.7	2.71	11.8	1.4	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0	C	41.93	71.3	550.7	836.2	2.81	11.7	1.5	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 31.8	C	24.09	52.3	538.9	679.9	2.60	13.0	1.3	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0	C	27.27	56.6	554.0	731.9	2.68	12.9	1.3	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0	C	32.28	63.3	575.4	813.1	2.81	12.8	1.4	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X X 50.0	C	42.30	76.6	610.1	969.2	3.02	12.7	1.6	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 33.9	C	27.03	61.7	655.8	916.6	3.00	14.9	1.4	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0	C	33.15	71.0	685.6	1046.0	3.18	14.7	1.5	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	89.6	751.2	1301.0	3.51	14.5	1.7	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	86.1	724.9	1250.8	3.43	14.5	1.7	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	94.4	760.6	1364.4	3.57	14.5	1.8	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4	C	47.46	99.2	769.9	1428.5	3.64	14.4	1.9	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6	C	50.61	103.9	778.1	1489.5	3.70	14.3	1.9	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8	C	53.82	108.6	785.9	1550.5	3.76	14.3	2.0	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9	C	56.93	113.2	793.1	1609.2	3.82	14.2	2.0	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1	C	60.13	117.8	800.1	1668.8	3.87	14.2	2.1	17.69	15.00	4.37	.797	1.000	15.00
15 X 4 X X 75.0	C	66.06	126.5	812.5	1778.1	3.96	14.1	2.2	19.43	15.00	4.49	.797	1.116	16.74

(60T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 75.000 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 93.750 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
18 X 4 X X 42.7	C	34.98	90.4	898.6	1581.1	3.90	17.5	1.8	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	95.9	911.3	1669.7	3.99	17.4	1.8	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	106.7	933.9	1843.7	4.16	17.3	2.0	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0	C	50.28	117.5	953.6	2013.5	4.31	17.1	2.1	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 82.500 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 113.438 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS				
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X X 45.0	C	36.90	66.3	577.6	795.0	2.53	12.0	1.4	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0	C	41.93	72.6	598.3	865.9	2.62	11.9	1.4	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 50.0	C	42.30	77.9	665.0	1001.9	2.82	12.9	1.5	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 40.0	C	33.15	72.1	742.4	1075.8	2.96	14.9	1.4	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	90.9	823.9	1340.3	3.27	14.7	1.6	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	87.4	795.0	1289.8	3.20	14.8	1.6	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	95.7	837.0	1406.8	3.34	14.7	1.7	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4	C	47.46	100.7	849.8	1474.2	3.40	14.6	1.7	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6	C	50.61	105.5	861.2	1538.4	3.46	14.6	1.8	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8	C	53.82	110.3	872.1	1602.8	3.52	14.5	1.8	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9	C	56.93	114.9	882.1	1664.8	3.58	14.5	1.9	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1	C	60.13	119.7	891.9	1727.8	3.63	14.4	1.9	17.69	15.00	4.37	.797	1.000	15.00
15 X 4 X X 75.0	C	66.06	128.5	908.9	1843.7	3.73	14.3	2.0	19.43	15.00	4.49	.797	1.116	16.74
18 X 4 X X 42.7	C	34.98	91.5	984.8	1621.6	3.62	17.7	1.6	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	97.0	1002.5	1714.0	3.71	17.7	1.7	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	108.1	1034.3	1895.8	3.87	17.5	1.8	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0	C	50.28	119.0	1061.9	2073.9	4.02	17.4	2.0	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 90.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 135.000 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15	X 4	X	X 50.0	C	41.02	92.2	884.9	1377.5	3.06	14.9	1.6	12.06	15.00	4.00	.797	.625	9.38
15	X 3 3/8	X	X 50.0	C	43.15	88.8	853.9	1326.9	3.00	14.9	1.6	12.69	15.00	3.72	.650	.716	10.74
15	X 4	X	X 53.2	C	44.23	97.1	902.0	1446.8	3.13	14.9	1.6	13.01	15.00	4.06	.797	.688	10.32
15	X 4	X	X 56.4	C	47.46	102.2	918.6	1517.0	3.19	14.8	1.7	13.96	15.00	4.13	.797	.751	11.27
15	X 4	X	X 59.6	C	50.61	107.0	933.5	1584.0	3.25	14.8	1.7	14.89	15.00	4.19	.797	.813	12.20
15	X 4	X	X 62.8	C	53.82	111.9	947.7	1651.3	3.31	14.8	1.7	15.83	15.00	4.25	.797	.876	13.14
15	X 4	X	X 65.9	C	56.93	116.6	960.9	1716.2	3.36	14.7	1.8	16.74	15.00	4.31	.797	.937	14.06
15	X 4	X	X 69.1	C	60.13	121.5	973.7	1782.4	3.42	14.7	1.8	17.69	15.00	4.37	.797	1.000	15.00
15	X 4	X	X 75.0	C	66.06	130.5	996.0	1904.1	3.51	14.6	1.9	19.43	15.00	4.49	.797	1.116	16.74
18	X 4	X	X 42.7	C	34.98	92.6	1056.4	1659.7	3.38	17.9	1.6	10.29	18.00	3.95	.625	.450	8.10
18	X 4	X	X 45.8	C	38.04	98.2	1079.5	1755.3	3.47	17.9	1.6	11.19	18.00	4.00	.625	.500	9.00
18	X 4	X	X 51.9	C	44.16	109.4	1121.8	1944.0	3.62	17.8	1.7	12.99	18.00	4.10	.625	.600	10.80
18	X 4	X	X 58.0	C	50.28	120.6	1157.5	2129.2	3.77	17.7	1.8	14.79	18.00	4.20	.625	.700	12.60

(60T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE XII. Properties of combined beam and plate, C-L and L (60t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(60T = 105.000 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 183.750 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15	X 4	X	X 75.0 C	66.06	134.6	1139.3	2016.5	3.15	15.0	1.8	19.43	15.00	4.49	.797	1.116	16.74

(60T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t).

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 6.250 IN.) PLATE WEIGHT = 5.100 LBS. (.1250 IN.) EFFECTIVE PLATE AREA = .781 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	1.4	.9	.82	1.4	.7	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.8	1.4	1.1	.86	1.3	.8	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.1	1.8	1.8	1.06	1.6	1.0	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.4	2.3	2.7	1.26	1.9	1.2	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	1.8	2.4	3.2	1.32	1.8	1.4	1.09	3.00	3.00	.187	.187	.56

(50T) PLATE WEIGHT = 5.100 LBS. (.1250 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 7.813 IN.) PLATE WEIGHT = 6.375 LBS. (.1563 IN.) EFFECTIVE PLATE AREA = 1.221 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	2.0	1.1	.77	1.6	.6	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	2.0	1.3	.82	1.5	.6	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.1	2.6	2.1	1.02	1.9	.8	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.4	3.2	3.1	1.22	2.2	1.0	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	1.9	3.4	3.9	1.30	2.0	1.1	1.09	3.00	3.00	.187	.187	.56

(50T) PLATE WEIGHT = 6.375 LBS. (.1563 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 9.375 IN.) PLATE WEIGHT = 7.650 LBS. (.1875 IN.) EFFECTIVE PLATE AREA = 1.758 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	2.6	1.2	.72	1.7	.5	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	2.7	1.5	.77	1.7	.5	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.2	3.5	2.4	.96	2.0	.7	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.5	4.3	3.5	1.15	2.4	.8	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.0	4.6	4.4	1.25	2.2	1.0	1.09	3.00	3.00	.187	.187	.56

(50T) PLATE WEIGHT = 7.650 LBS. (.1875 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 10.938 IN.) PLATE WEIGHT = 8.925 LBS. (.2188 IN.) EFFECTIVE PLATE AREA = 2.393 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.7	3.2	1.3	.66	1.8	.4	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	3.4	1.6	.71	1.8	.5	.71	2.00	2.00	.187	.187	.37
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.2	4.5	2.6	.90	2.1	.6	.81	2.50	2.00	.187	.187	.47
3 X 2 X 3/16 X 3.07 L	3.06	1.5	5.5	3.8	1.08	2.5	.7	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.0	5.9	4.9	1.19	2.4	.8	1.09	3.00	3.00	.187	.187	.56

(50T) PLATE WEIGHT = 8.925 LBS. (.2188 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 12.500 IN.) PLATE WEIGHT = 10.200 LBS. (.2500 IN.) EFFECTIVE PLATE AREA = 3.125 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L		2.11	.7	3.8	1.4	.61	1.9	.4	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L		2.42	.9	4.1	1.7	.66	1.8	.4	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L		2.76	.9	4.0	1.7	.66	1.8	.4	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L		3.19	1.2	4.3	2.1	.71	1.8	.5	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L		2.74	1.2	5.4	2.7	.84	2.2	.5	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L		3.61	1.6	5.7	3.3	.89	2.2	.6	1.06	2.50	2.00	.250	.250	.63
3 X 2 X 3/16 X 3.07 L		3.06	1.6	6.8	4.1	1.01	2.6	.6	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L		3.70	2.1	7.3	5.3	1.12	2.5	.7	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L		4.04	2.0	7.0	5.0	1.08	2.5	.7	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L		4.46	2.3	7.3	5.7	1.14	2.5	.8	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L		4.89	2.7	7.6	6.4	1.18	2.4	.8	1.44	3.00	3.00	.250	.250	.75
3 1/2 X 2 1/2 X 1/4 X 4.9 L		4.89	2.8	8.8	8.0	1.33	2.8	.9	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L		5.31	3.2	9.0	8.9	1.38	2.8	1.0	1.56	3.50	3.00	.250	.250	.88
4 X 3 X 1/4 X 5.8 L		5.74	3.8	10.5	11.9	1.57	3.1	1.1	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L		6.16	4.3	10.7	13.0	1.62	3.0	1.2	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L		6.59	4.7	10.9	14.1	1.67	3.0	1.3	1.94	4.00	4.00	.250	.250	1.00
5 X 3 X 1/4 X 6.6 L		6.59	5.1	13.4	19.5	1.96	3.8	1.5	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L		7.01	5.7	13.7	21.2	2.02	3.7	1.5	2.06	5.00	3.50	.250	.250	1.25
10 X 2 5/8 X X 15.3 C		11.66	16.0	30.0	106.9	4.04	6.7	3.6	3.43	10.00	2.60	.436	.240	2.40

(50T) PLATE WEIGHT = 10.200 LBS. (.2500 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 14.063 IN.) PLATE WEIGHT = 11.475 LBS. (.2813 IN.) EFFECTIVE PLATE AREA = 3.955 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.8	4.3	1.5	.57	1.9	.3	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	.9	4.7	1.8	.62	1.9	.4	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L	2.76	1.0	4.6	1.8	.62	1.9	.4	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L	3.19	1.2	5.0	2.2	.67	1.8	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.2	6.3	2.9	.78	2.3	.5	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.6	6.7	3.6	.84	2.2	.5	1.06	2.50	2.00	.250	.250	.63
3 X 2 X 3/16 X 3.07 L	3.06	1.6	8.0	4.3	.94	2.7	.5	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.1	8.7	5.6	1.05	2.6	.6	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.0	8.4	5.3	1.02	2.6	.6	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.4	8.8	6.1	1.08	2.6	.7	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.7	9.1	6.8	1.13	2.5	.8	1.44	3.00	3.00	.250	.250	.75
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	2.9	10.5	8.6	1.26	3.0	.8	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.3	10.9	9.5	1.32	2.9	.9	1.56	3.50	3.00	.250	.250	.88
4 X 3 X 1/4 X 5.8 L	5.74	3.9	12.7	12.8	1.50	3.3	1.0	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.4	13.0	14.0	1.56	3.2	1.1	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	4.8	13.3	15.1	1.60	3.1	1.1	1.94	4.00	4.00	.250	.250	1.00
5 X 3 X 1/4 X 6.6 L	6.59	5.2	16.3	20.9	1.88	4.0	1.3	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	5.8	16.6	22.7	1.94	3.9	1.4	2.06	5.00	3.50	.250	.250	1.25
10 X 2 5/8 X X 15.3 C	11.66	16.4	36.3	115.9	3.96	7.1	3.2	3.43	10.00	2.60	.436	.240	2.40

(50T) PLATE WEIGHT = 11.475 LBS. (.2813 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 15.625 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 4.883 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 1 1/2 X 3/16 X 2.12 L	2.11	.8	4.7	1.5	.53	2.0	.3	.62	2.00	1.50	.187	.187	.37
2 X 2 X 3/16 X 2.44 L	2.42	1.0	5.2	1.9	.58	2.0	.4	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L	2.76	1.0	5.2	1.9	.58	1.9	.4	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L	3.19	1.2	5.7	2.3	.63	1.9	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.3	7.1	3.0	.73	2.4	.4	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.6	7.6	3.7	.79	2.3	.5	1.06	2.50	2.00	.250	.250	.63
3 X 2 X 3/16 X 3.07 L	3.06	1.6	9.1	4.5	.88	2.8	.5	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.1	10.1	5.9	.99	2.7	.6	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.0	9.7	5.6	.96	2.7	.6	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.4	10.2	6.4	1.02	2.7	.6	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.7	10.6	7.2	1.07	2.6	.7	1.44	3.00	3.00	.250	.250	.75
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	2.9	12.3	9.0	1.19	3.1	.7	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.3	12.7	10.1	1.25	3.0	.8	1.56	3.50	3.00	.250	.250	.88
4 X 3 X 1/4 X 5.8 L	5.74	4.0	14.9	13.5	1.43	3.4	.9	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.4	15.3	14.8	1.49	3.3	1.0	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	4.9	15.7	16.1	1.54	3.3	1.0	1.94	4.00	4.00	.250	.250	1.00
5 X 3 X 1/4 X 6.6 L	6.59	5.3	19.2	22.1	1.80	4.2	1.2	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	5.9	19.7	24.1	1.86	4.1	1.2	2.06	5.00	3.50	.250	.250	1.25
10 X 2 5/8 X X 15.3 C	11.66	16.7	43.1	124.0	3.86	7.4	2.9	3.43	10.00	2.60	.436	.240	2.40

(50T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 15.625 IN.) PLATE WEIGHT = 12.750 LBS. (.3125 IN.) EFFECTIVE PLATE AREA = 4.883 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12	X 3	X	X 20.7 C	16.03	26.1	54.7	217.5	4.76	8.3	4.0	4.72	12.00	2.94	.501	.282	3.38

(50T) PLATE WEIGHT = 12.750 LBS. (.3125 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 17.188 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 5.908 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 2 X 3/16 X 2.44 L	2.42	1.0	5.7	2.0	.54	2.0	.3	.71	2.00	2.00	.187	.187	.37
2 X 1 1/2 X 1/4 X 2.77 L	2.76	1.0	5.6	2.0	.55	2.0	.4	.81	2.00	1.50	.250	.250	.50
2 X 2 X 1/4 X 3.19 L	3.19	1.2	6.2	2.4	.60	2.0	.4	.94	2.00	2.00	.250	.250	.50
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.3	7.9	3.1	.68	2.4	.4	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.6	8.5	3.9	.75	2.4	.5	1.06	2.50	2.00	.250	.250	.63
2 1/2 X 2 X 5/16 X 4.5 L	4.46	2.0	9.0	4.6	.80	2.3	.5	1.31	2.50	2.00	.313	.313	.78
3 X 2 X 3/16 X 3.07 L	3.06	1.6	10.1	4.7	.83	2.9	.5	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.2	11.3	6.1	.93	2.8	.5	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.1	10.9	5.8	.91	2.8	.5	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.4	11.5	6.7	.96	2.8	.6	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.8	12.1	7.6	1.02	2.7	.6	1.44	3.00	3.00	.250	.250	.75
3 X 2 X 5/16 X 5.0 L	4.99	2.5	11.4	6.8	.96	2.7	.6	1.47	3.00	2.00	.313	.313	.94
3 X 2 1/2 X 5/16 X 5.6 L	5.52	2.9	12.0	7.9	1.02	2.7	.7	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.4	12.5	8.8	1.07	2.6	.7	1.78	3.00	3.00	.313	.313	.94
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.0	14.0	9.4	1.13	3.2	.7	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.4	14.6	10.6	1.19	3.1	.7	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.6	14.6	11.0	1.20	3.1	.8	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.1	15.1	12.4	1.26	3.0	.8	1.94	3.50	3.00	.313	.313	1.10
4 X 3 X 1/4 X 5.8 L	5.74	4.0	17.2	14.1	1.36	3.5	.8	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.5	17.7	15.6	1.42	3.5	.9	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.0	18.2	16.9	1.47	3.4	.9	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	4.9	17.7	16.6	1.44	3.4	.9	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.4	18.3	18.2	1.49	3.3	1.0	2.25	4.00	3.50	.313	.313	1.25

(50T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 17.188 IN.) PLATE WEIGHT = 14.025 LBS. (.3438 IN.) EFFECTIVE PLATE AREA = 5.908 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 5/16 X 8.2 L	8.18	6.8	18.7	19.7	1.54	3.3	1.1	2.41	4.00	4.00	.313	.313	1.25
5 X 3 X 1/4 X 6.6 L	6.59	5.4	22.3	23.2	1.72	4.3	1.0	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	6.0	22.9	25.3	1.78	4.2	1.1	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2 L	8.18	6.5	23.0	27.2	1.81	4.2	1.2	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7 L	8.71	7.2	23.6	29.6	1.87	4.1	1.3	2.56	5.00	3.50	.313	.313	1.57
6 X 3 1/2 X 5/16 X 9.8 L	9.78	9.2	29.0	44.4	2.25	4.8	1.5	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L	10.31	10.1	29.6	47.7	2.31	4.7	1.6	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3 C	11.07	10.6	29.7	49.6	2.33	4.7	1.7	3.26	6.00	3.50	.385	.340	2.04
10 X 2 5/8 X X 15.3 C	11.66	16.9	50.5	131.3	3.75	7.7	2.6	3.43	10.00	2.60	.436	.240	2.40
10 X 3 1/2 X X 21.9 C	16.36	23.8	53.5	170.5	3.99	7.2	3.2	4.81	10.00	3.45	.500	.325	3.25
12 X 3 X X 20.7 C	16.03	26.6	63.8	231.7	4.67	8.7	3.6	4.72	12.00	2.94	.501	.282	3.38

(50T) PLATE WEIGHT = 14.025 LBS. (.3438 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 X 2 X 1/4 X 3.19 L	3.19	1.3	6.8	2.5	.56	2.0	.4	.94	2.00	2.00	.250	.250	.50
2 X 2 X 3/8 X 4.7 L	4.62	1.7	7.5	3.3	.63	1.9	.4	1.36	2.00	2.00	.375	.375	.75
2 1/2 X 2 X 3/16 X 2.75 L	2.74	1.3	8.5	3.3	.65	2.5	.4	.81	2.50	2.00	.187	.187	.47
2 1/2 X 2 X 1/4 X 3.62 L	3.61	1.7	9.3	4.1	.71	2.4	.4	1.06	2.50	2.00	.250	.250	.63
2 1/2 X 2 X 5/16 X 4.5 L	4.46	2.0	9.9	4.8	.76	2.4	.5	1.31	2.50	2.00	.313	.313	.78
2 1/2 X 2 X 3/8 X 5.3 L	5.26	2.3	10.3	5.4	.79	2.4	.5	1.55	2.50	2.00	.375	.375	.94
3 X 2 X 3/16 X 3.07 L	3.06	1.7	11.0	4.8	.78	2.9	.4	.90	3.00	2.00	.187	.187	.56
3 X 3 X 3/16 X 3.71 L	3.70	2.2	12.5	6.3	.88	2.9	.5	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.1	12.0	6.1	.86	2.9	.5	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.5	12.8	7.0	.92	2.8	.5	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.8	13.5	7.9	.97	2.8	.6	1.44	3.00	3.00	.250	.250	.75
3 X 2 X 5/16 X 5.0 L	4.99	2.5	12.7	7.1	.92	2.8	.6	1.47	3.00	2.00	.313	.313	.94
3 X 2 1/2 X 5/16 X 5.6 L	5.52	3.0	13.4	8.2	.97	2.8	.6	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.4	14.1	9.3	1.02	2.7	.7	1.78	3.00	3.00	.313	.313	.94
3 X 2 1/2 X 3/8 X 6.6 L	6.53	3.4	13.9	9.3	1.02	2.7	.7	1.92	3.00	2.50	.375	.375	1.13
3 X 3 X 3/8 X 7.2 L	7.17	3.9	14.5	10.4	1.07	2.7	.7	2.11	3.00	3.00	.375	.375	1.13
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.0	15.7	9.8	1.07	3.3	.6	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.4	16.4	11.0	1.13	3.2	.7	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.6	16.4	11.5	1.14	3.2	.7	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.1	17.1	12.9	1.20	3.1	.8	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.2	16.9	13.0	1.19	3.1	.8	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	4.8	17.6	14.6	1.25	3.0	.8	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.1	19.4	14.7	1.30	3.6	.8	1.69	4.00	3.00	.250	.250	1.00

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4	X 3 1/2	X 1/4	X 6.2	L	6.16	4.5	20.1	16.2	1.35	3.6	.8	1.81	4.00	3.50	.250	.250	1.00
4	X 4	X 1/4	X 6.6	L	6.59	5.0	20.6	17.7	1.40	3.5	.9	1.94	4.00	4.00	.250	.250	1.00
4	X 3	X 5/16	X 7.2	L	7.12	4.9	20.1	17.3	1.38	3.5	.9	2.09	4.00	3.00	.313	.313	1.25
4	X 3 1/2	X 5/16	X 7.7	L	7.65	5.5	20.8	19.0	1.43	3.5	.9	2.25	4.00	3.50	.313	.313	1.25
4	X 4	X 5/16	X 8.2	L	8.18	6.1	21.3	20.7	1.48	3.4	1.0	2.41	4.00	4.00	.313	.313	1.25
4	X 3	X 3/8	X 8.5	L	8.45	5.7	20.7	19.5	1.43	3.4	.9	2.48	4.00	3.00	.375	.375	1.50
4	X 3 1/2	X 3/8	X 9.1	L	9.08	6.4	21.3	21.5	1.49	3.4	1.0	2.67	4.00	3.50	.375	.375	1.50
4	X 4	X 3/8	X 9.8	L	9.72	7.1	21.8	23.3	1.54	3.3	1.1	2.86	4.00	4.00	.375	.375	1.50
5	X 3	X 1/4	X 6.6	L	6.59	5.5	25.4	24.1	1.64	4.4	1.0	1.94	5.00	3.00	.250	.250	1.25
5	X 3 1/2	X 1/4	X 7.0	L	7.01	6.0	26.2	26.4	1.70	4.4	1.0	2.06	5.00	3.50	.250	.250	1.25
5	X 3	X 5/16	X 8.2	L	8.18	6.6	26.3	28.4	1.73	4.3	1.1	2.41	5.00	3.00	.313	.313	1.57
5	X 3 1/2	X 5/16	X 8.7	L	8.71	7.3	27.0	31.0	1.80	4.2	1.1	2.56	5.00	3.50	.313	.313	1.57
5	X 3	X 3/8	X 9.8	L	9.72	7.7	26.9	32.1	1.80	4.2	1.2	2.86	5.00	3.00	.375	.375	1.88
5	X 3 1/2	X 3/8	X 10.4	L	10.36	8.5	27.6	35.1	1.87	4.1	1.3	3.05	5.00	3.50	.375	.375	1.88
5	X 5	X 3/8	X 12.3	L	12.27	11.1	29.3	43.2	2.02	3.9	1.5	3.61	5.00	5.00	.375	.375	1.88
6	X 3 1/2	X 5/16	X 9.8	L	9.78	9.4	33.3	46.5	2.17	5.0	1.4	2.88	6.00	3.50	.313	.313	1.88
6	X 4	X 5/16	X 10.3	L	10.31	10.2	34.0	50.1	2.23	4.9	1.5	3.03	6.00	4.00	.313	.313	1.88
6	X 3 1/2	X	X 15.3	C	11.07	10.8	34.1	52.2	2.25	4.8	1.5	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7	L	11.63	10.9	34.0	52.6	2.24	4.8	1.5	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3	L	12.27	11.9	34.7	56.6	2.31	4.7	1.6	3.61	6.00	4.00	.375	.375	2.25
7	X 4	X 3/8	X 13.6	L	13.55	14.6	41.3	79.7	2.69	5.4	1.9	3.98	7.00	4.00	.375	.375	2.63
10	X 2 5/8	X	X 15.3	C	11.66	17.2	58.4	137.7	3.63	8.0	2.4	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2	X	X 21.9	C	16.36	24.2	61.7	180.4	3.90	7.5	2.9	4.81	10.00	3.45	.500	.325	3.25

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 18.750 IN.) PLATE WEIGHT = 15.300 LBS. (.3750 IN.) EFFECTIVE PLATE AREA = 7.031 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12	X 3	X	X 20.7 C	16.03	27.0	73.6	244.5	4.56	9.1	3.3	4.72	12.00	2.94	.501	.282	3.38
13	X 4	X	X 31.8 C	24.09	44.0	86.3	389.5	5.25	8.9	4.5	7.09	13.00	4.00	.610	.375	4.88

(50T) PLATE WEIGHT = 15.300 LBS. (.3750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
2 1/2 X 2 X 5/16 X 4.5 L	4.46	2.1	11.5	5.1	.69	2.5	.4	1.31	2.50	2.00	.313	.313	.78
2 1/2 X 2 X 3/8 X 5.3 L	5.26	2.4	12.1	5.8	.72	2.5	.5	1.55	2.50	2.00	.375	.375	.94
3 X 3 X 3/16 X 3.71 L	3.70	2.3	14.5	6.7	.79	3.0	.5	1.09	3.00	3.00	.187	.187	.56
3 X 2 X 1/4 X 4.1 L	4.04	2.2	13.9	6.5	.78	3.0	.5	1.19	3.00	2.00	.250	.250	.75
3 X 2 1/2 X 1/4 X 4.5 L	4.46	2.5	15.0	7.5	.83	2.9	.5	1.31	3.00	2.50	.250	.250	.75
3 X 3 X 1/4 X 4.9 L	4.89	2.9	16.0	8.5	.88	2.9	.5	1.44	3.00	3.00	.250	.250	.75
3 X 2 X 5/16 X 5.0 L	4.99	2.6	14.9	7.6	.83	2.9	.5	1.47	3.00	2.00	.313	.313	.94
3 X 2 1/2 X 5/16 X 5.6 L	5.52	3.1	16.0	8.8	.89	2.9	.6	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.5	17.0	10.0	.94	2.8	.6	1.78	3.00	3.00	.313	.313	.94
3 X 2 1/2 X 3/8 X 6.6 L	6.53	3.5	16.8	10.0	.93	2.8	.6	1.92	3.00	2.50	.375	.375	1.13
3 X 3 X 3/8 X 7.2 L	7.17	4.0	17.7	11.3	.98	2.8	.6	2.11	3.00	3.00	.375	.375	1.13
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.1	18.7	10.4	.97	3.4	.6	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.5	19.7	11.7	1.03	3.3	.6	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.7	19.8	12.3	1.04	3.3	.6	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.2	20.8	13.9	1.10	3.3	.7	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.3	20.6	14.0	1.10	3.3	.7	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	4.9	21.6	15.8	1.15	3.2	.7	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.2	23.6	15.7	1.18	3.8	.7	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.6	24.5	17.3	1.23	3.7	.7	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.1	25.4	18.9	1.28	3.7	.7	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	5.0	24.8	18.6	1.26	3.7	.7	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.6	25.7	20.5	1.32	3.6	.8	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.2	26.6	22.4	1.37	3.6	.8	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	5.8	25.6	21.1	1.32	3.6	.8	2.48	4.00	3.00	.375	.375	1.50

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 3 1/2 X 3/8 X 9.1	L	9.08	6.5	26.6	23.3	1.38	3.6	.9	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8	L	9.72	7.2	27.4	25.4	1.43	3.5	.9	2.86	4.00	4.00	.375	.375	1.50
5 X 3 X 1/4 X 6.6	L	6.59	5.6	31.4	25.7	1.49	4.6	.8	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0	L	7.01	6.2	32.5	28.2	1.56	4.6	.9	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2	L	8.18	6.8	32.8	30.5	1.60	4.5	.9	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7	L	8.71	7.5	33.9	33.4	1.66	4.5	1.0	2.56	5.00	3.50	.313	.313	1.57
5 X 3 X 3/8 X 9.8	L	9.72	7.9	33.8	34.7	1.67	4.4	1.0	2.86	5.00	3.00	.375	.375	1.88
5 X 3 1/2 X 3/8 X 10.4	L	10.36	8.7	34.9	38.0	1.74	4.3	1.1	3.05	5.00	3.50	.375	.375	1.88
5 X 5 X 3/8 X 12.3	L	12.27	11.4	37.3	47.3	1.90	4.2	1.3	3.61	5.00	5.00	.375	.375	1.88
6 X 3 1/2 X 5/16 X 9.8	L	9.78	9.6	42.1	50.2	2.01	5.2	1.2	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3	L	10.31	10.4	43.2	54.2	2.07	5.2	1.3	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3	C	11.07	11.0	43.4	56.5	2.10	5.1	1.3	3.26	6.00	3.50	.385	.340	2.04
6 X 3 1/2 X 3/8 X 11.7	L	11.63	11.2	43.2	57.1	2.10	5.1	1.3	3.42	6.00	3.50	.375	.375	2.25
6 X 4 X 3/8 X 12.3	L	12.27	12.2	44.3	61.6	2.16	5.0	1.4	3.61	6.00	4.00	.375	.375	2.25
6 X 3 1/2 X X 18.0	C	12.77	12.8	44.6	63.9	2.19	5.0	1.4	3.76	6.00	3.50	.475	.379	2.27
7 X 4 X 3/8 X 13.6	L	13.55	15.0	52.8	86.9	2.53	5.8	1.6	3.98	7.00	4.00	.375	.375	2.63
10 X 2 5/8 X X 15.3	C	11.66	17.5	75.2	148.4	3.38	8.5	2.0	3.43	10.00	2.60	.436	.240	2.40
10 X 3 1/2 X X 21.9	C	16.36	24.8	79.6	197.5	3.71	8.0	2.5	4.81	10.00	3.45	.500	.325	3.25
10 X 2 5/8 X X 20.0	C	16.39	21.5	75.8	175.1	3.49	8.1	2.3	4.82	10.00	2.74	.436	.379	3.79
10 X 3 1/2 X X 24.9	C	18.73	27.7	80.7	215.3	3.78	7.8	2.7	5.51	10.00	3.40	.575	.377	3.77
10 X 3 1/2 X X 25.3	C	19.76	27.6	80.0	214.0	3.73	7.8	2.7	5.81	10.00	3.55	.500	.425	4.25
10 X 4 1/2 X X 28.5	C	21.34	31.6	82.5	238.3	3.88	7.5	2.9	6.28	10.00	3.95	.575	.425	4.25

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 21.875 IN.) PLATE WEIGHT = 17.850 LBS. (.4375 IN.) EFFECTIVE PLATE AREA = 9.570 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X TN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 3 X X 20.7 C	16.03	27.7	95.0	266.7	4.32	9.6	2.8	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0 C	20.33	31.8	95.8	297.3	4.37	9.3	3.1	5.98	12.00	3.05	.501	.387	4.64
13 X 4 X X 31.8 C	24.09	45.3	110.8	432.0	5.09	9.5	3.9	7.09	13.00	4.00	.610	.375	4.88
15 X 3 3/8 X X 33.9 C	27.03	53.0	128.0	578.6	5.75	10.9	4.5	7.95	15.00	3.40	.650	.400	6.00

(50T) PLATE WEIGHT = 17.850 LBS. (.4375 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE IN X IN X IN X LBS/FT	WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
3 X 2 1/2 X 5/16 X 5.6 L	5.52	3.1	18.2	9.4	.82	3.0	.5	1.62	3.00	2.50	.313	.313	.94
3 X 3 X 5/16 X 6.1 L	6.05	3.6	19.4	10.6	.86	3.0	.5	1.78	3.00	3.00	.313	.313	.94
3 X 2 1/2 X 3/8 X 6.6 L	6.53	3.6	19.2	10.7	.86	2.9	.6	1.92	3.00	2.50	.375	.375	1.13
3 X 3 X 3/8 X 7.2 L	7.17	4.2	20.5	12.1	.91	2.9	.6	2.11	3.00	3.00	.375	.375	1.13
3 X 2 1/2 X 7/16 X 7.6 L	7.54	4.1	20.0	11.9	.90	2.9	.6	2.22	3.00	2.50	.438	.438	1.31
3 X 3 X 7/16 X 8.3 L	8.28	4.7	21.2	13.4	.95	2.9	.6	2.44	3.00	3.00	.438	.438	1.31
3 X 3 X 1/2 X 9.4 L	9.35	5.2	21.8	14.6	.98	2.8	.7	2.75	3.00	3.00	.500	.500	1.50
3 1/2 X 2 1/2 X 1/4 X 4.9 L	4.89	3.2	21.1	11.0	.89	3.5	.5	1.44	3.50	2.50	.250	.250	.88
3 1/2 X 3 X 1/4 X 5.4 L	5.31	3.6	22.5	12.4	.94	3.4	.6	1.56	3.50	3.00	.250	.250	.88
3 1/2 X 2 1/2 X 5/16 X 6.1 L	6.05	3.8	22.7	13.1	.96	3.4	.6	1.78	3.50	2.50	.313	.313	1.10
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.4	24.1	14.7	1.01	3.4	.6	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.4	23.9	14.9	1.01	3.4	.6	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	5.0	25.3	16.8	1.07	3.3	.7	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.2	27.2	16.5	1.08	3.9	.6	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.7	28.5	18.3	1.13	3.9	.6	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.2	29.7	20.0	1.18	3.8	.7	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	5.2	29.0	19.7	1.16	3.8	.7	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.8	30.3	21.8	1.21	3.8	.7	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.4	31.5	23.8	1.26	3.7	.8	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	6.0	30.3	22.5	1.22	3.8	.7	2.48	4.00	3.00	.375	.375	1.50
4 X 3 1/2 X 3/8 X 9.1 L	9.08	6.7	31.6	24.8	1.28	3.7	.8	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8 L	9.72	7.4	32.7	27.2	1.33	3.7	.8	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 7/16 X 9.8 L	9.77	6.8	31.3	25.1	1.28	3.7	.8	2.87	4.00	3.00	.438	.438	1.75
4 X 3 X 1/2 X 11.1 L	11.05	7.5	32.1	27.3	1.32	3.6	.9	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3 L	11.26	8.4	33.7	30.2	1.38	3.6	.9	3.31	4.00	4.00	.438	.438	1.75

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 3 1/2 X 1/2 X 11.9 L	11.90	8.4	33.3	30.2	1.37	3.6	.9	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8 L	12.75	9.3	34.4	33.0	1.42	3.5	1.0	3.75	4.00	4.00	.500	.500	2.00
5 X 3 X 1/4 X 6.6 L	6.59	5.7	36.9	27.0	1.37	4.8	.7	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	6.3	38.5	29.7	1.43	4.7	.8	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2 L	8.18	6.9	39.0	32.2	1.47	4.7	.8	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7 L	8.71	7.7	40.5	35.4	1.53	4.6	.9	2.56	5.00	3.50	.313	.313	1.57
5 X 3 X 3/8 X 9.8 L	9.72	8.0	40.5	36.9	1.55	4.6	.9	2.86	5.00	3.00	.375	.375	1.88
5 X 3 1/2 X 3/8 X 10.4 L	10.36	8.9	42.0	40.5	1.61	4.5	1.0	3.05	5.00	3.50	.375	.375	1.88
5 X 3 X 7/16 X 11.3 L	11.26	9.1	41.7	41.2	1.61	4.5	1.0	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0 L	12.01	10.1	43.2	45.2	1.68	4.5	1.0	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3 L	12.27	11.6	45.5	50.8	1.78	4.4	1.1	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8 L	12.75	10.1	42.6	45.1	1.67	4.4	1.1	3.75	5.00	3.00	.500	.500	2.50
5 X 3 1/2 X 1/2 X 13.6 L	13.60	11.3	44.0	49.4	1.73	4.4	1.1	4.00	5.00	3.50	.500	.500	2.50
6 X 3 1/2 X 5/16 X 9.8 L	9.78	9.7	51.0	53.1	1.86	5.5	1.0	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L	10.31	10.6	52.5	57.5	1.92	5.4	1.1	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3 C	11.07	11.2	52.8	60.1	1.95	5.4	1.1	3.26	6.00	3.50	.385	.340	2.04
6 X 3 1/2 X 3/8 X 11.7 L	11.63	11.4	52.6	60.8	1.95	5.3	1.2	3.42	6.00	3.50	.375	.375	2.25
6 X 4 X 3/8 X 12.3 L	12.27	12.4	54.1	65.7	2.02	5.3	1.2	3.61	6.00	4.00	.375	.375	2.25
6 X 3 1/2 X X 18.0 C	12.77	13.0	54.6	68.4	2.05	5.2	1.3	3.76	6.00	3.50	.475	.379	2.27
6 X 4 X 7/16 X 14.3 L	14.24	14.2	55.3	73.4	2.10	5.2	1.3	4.19	6.00	4.00	.438	.438	2.63
6 X 4 X 1/2 X 16.2 L	16.15	15.8	56.3	80.2	2.16	5.1	1.4	4.75	6.00	4.00	.500	.500	3.00
7 X 4 X 3/8 X 13.6 L	13.55	15.3	64.9	92.8	2.37	6.1	1.4	3.98	7.00	4.00	.375	.375	2.63
7 X 4 X 1/2 X 17.9 L	17.85	19.5	67.4	113.3	2.53	5.8	1.7	5.25	7.00	4.00	.500	.500	3.50

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 25.000 IN.) PLATE WEIGHT = 20.400 LBS. (.5000 IN.) EFFECTIVE PLATE AREA = 12.500 SQ. IN.

NOMINAL SIZE IN X IN X IN X LBS/FT		WT/FT	SECTION MODULUS FLANGE	SECTION MODULUS PLATE	I	R	YF	YP	BEAM DIMENSIONS A	D	WF	TF	TW	ASH
		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 4 X 1/2 X 19.6	L	19.55	23.4	78.6	153.2	2.90	6.6	1.9	5.75	8.00	4.00	.500	.500	4.00
8 X 6 X 1/2 X 23.0	L	22.95	30.7	83.9	190.9	3.15	6.2	2.3	6.75	8.00	6.00	.500	.500	4.00
9 X 4 X 1/2 X 21.3	L	21.25	27.6	89.9	200.4	3.27	7.3	2.2	6.25	9.00	4.00	.500	.500	4.50
10 X 2 5/8 X X 15.3	C	11.66	17.8	92.9	157.0	3.14	8.8	1.7	3.43	10.00	2.60	.436	.240	2.40
10 X 3 1/2 X X 21.9	C	16.36	25.3	98.9	211.5	3.50	8.4	2.1	4.81	10.00	3.45	.500	.325	3.25
10 X 2 5/8 X X 20.0	C	16.39	22.0	93.8	187.2	3.29	8.5	2.0	4.82	10.00	2.74	.436	.379	3.79
10 X 3 1/2 X X 24.9	C	18.73	28.3	100.3	231.7	3.59	8.2	2.3	5.51	10.00	3.40	.575	.377	3.77
10 X 3 1/2 X X 25.3	C	19.76	28.2	99.3	230.5	3.55	8.2	2.3	5.81	10.00	3.55	.500	.425	4.25
10 X 4 1/2 X X 28.5	C	21.34	32.3	102.6	257.8	3.71	8.0	2.5	6.28	10.00	3.95	.575	.425	4.25
10 X 3 1/2 X X 28.3	C	22.13	31.1	100.7	249.7	3.62	8.0	2.5	6.51	10.00	3.50	.575	.477	4.77
12 X 3 X X 20.7	C	16.03	28.2	118.3	284.7	4.07	10.1	2.4	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0	C	20.33	32.6	119.0	319.8	4.16	9.8	2.7	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X X 30.9	C	24.48	40.0	123.8	378.1	4.38	9.4	3.1	7.20	12.00	3.45	.600	.450	5.40
12 X 3 1/2 X X 32.9	C	26.52	42.0	124.2	392.2	4.40	9.3	3.2	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0	C	26.91	46.0	127.5	422.4	4.55	9.2	3.3	7.92	12.00	3.77	.700	.467	5.60
13 X 4 X X 31.8	C	24.09	46.3	138.1	468.3	4.69	10.1	3.4	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0	C	27.27	49.6	138.7	493.8	4.90	9.9	3.6	8.02	13.00	4.07	.610	.447	5.81
15 X 3 3/8 X X 33.9	C	27.03	54.4	159.1	628.1	5.54	11.6	3.9	7.95	15.00	3.40	.650	.400	6.00
18 X 4 X X 42.7	C	34.98	78.8	198.5	1044.0	6.77	13.2	5.3	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	82.9	199.9	1083.8	6.76	13.1	5.4	11.19	18.00	4.00	.625	.500	9.00

(50T) PLATE WEIGHT = 20.400 LBS. (.5000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
3 X 3 X 3/8 X 7.2 L	7.17	4.3	22.8	12.8	.85	3.0	.6	2.11	3.00	3.00	.375	.375	1.13
3 X 2 1/2 X 7/16 X 7.6 L	7.54	4.2	22.3	12.6	.84	3.0	.6	2.22	3.00	2.50	.438	.438	1.31
3 X 3 X 7/16 X 8.3 L	8.28	4.8	23.9	14.3	.88	3.0	.6	2.44	3.00	3.00	.438	.438	1.31
3 X 3 X 1/2 X 9.4 L	9.35	5.3	24.7	15.6	.92	2.9	.6	2.75	3.00	3.00	.500	.500	1.50
3 1/2 X 3 X 5/16 X 6.6 L	6.58	4.5	26.9	15.5	.94	3.5	.6	1.94	3.50	3.00	.313	.313	1.10
3 1/2 X 2 1/2 X 3/8 X 7.2 L	7.17	4.5	26.7	15.8	.94	3.5	.6	2.11	3.50	2.50	.375	.375	1.31
3 1/2 X 3 X 3/8 X 7.9 L	7.81	5.2	28.5	17.8	.99	3.4	.6	2.30	3.50	3.00	.375	.375	1.31
4 X 3 X 1/4 X 5.8 L	5.74	4.3	30.1	17.3	.99	4.0	.6	1.69	4.00	3.00	.250	.250	1.00
4 X 3 1/2 X 1/4 X 6.2 L	6.16	4.8	31.9	19.2	1.04	4.0	.6	1.81	4.00	3.50	.250	.250	1.00
4 X 4 X 1/4 X 6.6 L	6.59	5.3	33.4	21.0	1.09	3.9	.6	1.94	4.00	4.00	.250	.250	1.00
4 X 3 X 5/16 X 7.2 L	7.12	5.3	32.6	20.7	1.07	3.9	.6	2.09	4.00	3.00	.313	.313	1.25
4 X 3 1/2 X 5/16 X 7.7 L	7.65	5.9	34.3	22.9	1.13	3.9	.7	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.5	35.8	25.1	1.17	3.9	.7	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	6.1	34.4	23.7	1.14	3.9	.7	2.48	4.00	3.00	.375	.375	1.50
4 X 3 1/2 X 3/8 X 9.1 L	9.08	6.8	36.1	26.2	1.19	3.8	.7	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8 L	9.72	7.6	37.6	28.7	1.24	3.8	.8	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 7/16 X 9.8 L	9.77	6.9	35.9	26.5	1.19	3.8	.7	2.87	4.00	3.00	.438	.438	1.75
4 X 3 X 1/2 X 11.1 L	11.05	7.7	37.0	29.0	1.23	3.8	.8	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3 L	11.26	8.6	39.0	32.1	1.29	3.7	.8	3.31	4.00	4.00	.438	.438	1.75
4 X 3 1/2 X 1/2 X 11.9 L	11.90	8.6	38.6	32.1	1.29	3.7	.8	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8 L	12.75	9.5	40.1	35.1	1.34	3.7	.9	3.75	4.00	4.00	.500	.500	2.00
5 X 3 X 1/4 X 6.6 L	6.59	5.8	41.7	28.2	1.26	4.9	.7	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	6.4	43.7	31.0	1.32	4.9	.7	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2 L	8.18	7.0	44.6	33.8	1.36	4.8	.8	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7 L	8.71	7.8	46.6	37.1	1.42	4.8	.8	2.56	5.00	3.50	.313	.313	1.57

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5	X 3	X 3/8	X 9.8 L	9.72	8.2	46.8	38.8	1.44	4.7	.8	2.86	5.00	3.00	.375	.375	1.88
5	X 3 1/2	X 3/8	X 10.4 L	10.36	9.1	48.8	42.6	1.50	4.7	.9	3.05	5.00	3.50	.375	.375	1.88
5	X 3	X 7/16	X 11.3 L	11.26	9.3	48.5	43.5	1.51	4.7	.9	3.31	5.00	3.00	.438	.438	2.19
5	X 3 1/2	X 7/16	X 12.0 L	12.01	10.3	50.4	47.7	1.57	4.6	.9	3.53	5.00	3.50	.438	.438	2.19
5	X 5	X 3/8	X 12.3 L	12.27	11.8	53.5	53.7	1.66	4.6	1.0	3.61	5.00	5.00	.375	.375	1.88
5	X 3	X 1/2	X 12.8 L	12.75	10.4	49.8	47.7	1.56	4.6	1.0	3.75	5.00	3.00	.500	.500	2.50
5	X 3 1/2	X 1/2	X 13.6 L	13.60	11.5	51.7	52.4	1.63	4.5	1.0	4.00	5.00	3.50	.500	.500	2.50
6	X 3 1/2	X 5/16	X 9.8 L	9.78	9.9	59.3	55.6	1.73	5.6	.9	2.88	6.00	3.50	.313	.313	1.88
6	X 4	X 5/16	X 10.3 L	10.31	10.8	61.3	60.3	1.79	5.6	1.0	3.03	6.00	4.00	.313	.313	1.88
6	X 3 1/2	X	X 15.3 C	11.07	11.4	61.9	63.2	1.82	5.5	1.0	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7 L	11.63	11.6	61.7	64.0	1.82	5.5	1.0	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3 L	12.27	12.6	63.7	69.3	1.89	5.5	1.1	3.61	6.00	4.00	.375	.375	2.25
6	X 3 1/2	X	X 18.0 C	12.77	13.3	64.4	72.1	1.92	5.4	1.1	3.76	6.00	3.50	.475	.379	2.27
6	X 4	X 7/16	X 14.3 L	14.24	14.4	65.5	77.6	1.97	5.4	1.2	4.19	6.00	4.00	.438	.438	2.63
6	X 4	X 1/2	X 16.2 L	16.15	16.1	66.9	85.2	2.03	5.3	1.3	4.75	6.00	4.00	.500	.500	3.00
7	X 4	X 3/8	X 13.6 L	13.55	15.5	77.0	97.8	2.22	6.3	1.3	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9 L	17.85	19.8	80.6	120.3	2.39	6.1	1.5	5.25	7.00	4.00	.500	.500	3.50
8	X 4	X 1/2	X 19.6 L	19.55	23.8	94.3	162.9	2.75	6.8	1.7	5.75	8.00	4.00	.500	.500	4.00
8	X 6	X 1/2	X 23.0 L	22.95	31.2	101.3	204.4	3.01	6.5	2.0	6.75	8.00	6.00	.500	.500	4.00
9	X 4	X 1/2	X 21.3 L	21.25	28.1	108.2	213.3	3.11	7.6	2.0	6.25	9.00	4.00	.500	.500	4.50
10	X 2 5/8	X	X 15.3 C	11.66	18.1	111.0	164.0	2.92	9.1	1.5	3.43	10.00	2.60	.436	.240	2.40

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 28.125 IN.) PLATE WEIGHT = 22.950 LBS. (.5625 IN.) EFFECTIVE PLATE AREA = 15.820 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 3 1/2 X X 21.9 C	16.36	25.7	119.2	223.1	3.29	8.7	1.9	4.81	10.00	3.45	.500	.325	3.25
10 X 2 5/8 X X 20.0 C	16.39	22.4	112.5	197.3	3.09	8.8	1.8	4.82	10.00	2.74	.436	.379	3.79
10 X 3 1/2 X X 24.9 C	18.73	28.8	121.0	245.5	3.39	8.5	2.0	5.51	10.00	3.40	.575	.377	3.77
10 X 3 1/2 X X 25.3 C	19.76	28.7	119.6	244.5	3.36	8.5	2.0	5.81	10.00	3.55	.500	.425	4.25
10 X 4 1/2 X X 28.5 C	21.34	32.9	123.9	274.3	3.52	8.3	2.2	6.28	10.00	3.95	.575	.425	4.25
10 X 3 1/2 X X 28.3 C	22.13	31.7	121.5	265.7	3.45	8.4	2.2	6.51	10.00	3.50	.575	.477	4.77
12 X 3 X X 20.7 C	16.03	28.6	142.9	299.6	3.82	10.5	2.1	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0 C	20.33	33.2	143.8	338.7	3.94	10.2	2.4	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X X 30.9 C	24.48	40.8	149.7	403.0	4.18	9.9	2.7	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X X 30.0 C	25.34	38.3	145.2	380.7	4.04	9.9	2.6	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X X 32.9 C	26.52	42.9	150.2	418.9	4.21	9.8	2.8	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0 C	26.91	46.9	154.4	451.9	4.36	9.6	2.9	7.92	12.00	3.77	.700	.467	5.60
13 X 4 X X 31.8 C	24.09	47.2	167.7	499.2	4.67	10.6	3.0	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0 C	27.27	50.6	168.1	527.2	4.70	10.4	3.1	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0 C	32.28	55.9	169.2	569.7	4.74	10.2	3.4	9.49	13.00	4.19	.610	.560	7.28
15 X 3 3/8 X X 33.9 C	27.03	55.5	193.1	670.4	5.31	12.1	3.5	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0 C	33.15	62.8	194.8	739.4	5.38	11.8	3.8	9.75	15.00	3.52	.650	.520	7.80
18 X 4 X X 42.7 C	34.98	80.7	240.3	1121.2	6.55	13.9	4.7	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C	38.04	84.9	241.5	1166.4	6.57	13.7	4.8	11.19	18.00	4.00	.625	.500	9.00

(50T) PLATE WEIGHT = 22.950 LBS. (.5625 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 3 1/2 X 5/16 X 7.7 L	7.65	6.0	37.7	24.0	1.05	4.0	.6	2.25	4.00	3.50	.313	.313	1.25
4 X 4 X 5/16 X 8.2 L	8.18	6.6	39.6	26.2	1.09	4.0	.7	2.41	4.00	4.00	.313	.313	1.25
4 X 3 X 3/8 X 8.5 L	8.45	6.3	38.0	24.8	1.06	4.0	.7	2.48	4.00	3.00	.375	.375	1.50
4 X 3 1/2 X 3/8 X 9.1 L	9.08	7.0	40.1	27.5	1.11	3.9	.7	2.67	4.00	3.50	.375	.375	1.50
4 X 4 X 3/8 X 9.8 L	9.72	7.7	42.0	30.1	1.16	3.9	.7	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 7/16 X 9.8 L	9.77	7.1	39.8	27.8	1.11	3.9	.7	2.87	4.00	3.00	.438	.438	1.75
4 X 3 X 1/2 X 11.1 L	11.05	7.9	41.3	30.5	1.16	3.9	.7	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3 L	11.26	8.8	43.8	33.7	1.22	3.9	.8	3.31	4.00	4.00	.438	.438	1.75
4 X 3 1/2 X 1/2 X 11.9 L	11.90	8.8	43.4	33.8	1.21	3.8	.8	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8 L	12.75	9.7	45.3	37.0	1.26	3.8	.8	3.75	4.00	4.00	.500	.500	2.00
4 X 4 X 5/8 X 15.7 L	15.67	11.5	47.5	42.8	1.33	3.7	.9	4.61	4.00	4.00	.625	.625	2.50
5 X 3 X 1/4 X 6.6 L	6.59	5.9	45.6	29.3	1.17	5.0	.6	1.94	5.00	3.00	.250	.250	1.25
5 X 3 1/2 X 1/4 X 7.0 L	7.01	6.5	48.1	32.2	1.22	5.0	.7	2.06	5.00	3.50	.250	.250	1.25
5 X 3 X 5/16 X 8.2 L	8.18	7.2	49.5	35.2	1.27	4.9	.7	2.41	5.00	3.00	.313	.313	1.57
5 X 3 1/2 X 5/16 X 8.7 L	8.71	7.9	52.0	38.7	1.32	4.9	.7	2.56	5.00	3.50	.313	.313	1.57
5 X 3 X 3/8 X 9.8 L	9.72	8.3	52.4	40.5	1.34	4.9	.8	2.86	5.00	3.00	.375	.375	1.88
5 X 3 1/2 X 3/8 X 10.4 L	10.36	9.3	54.8	44.5	1.40	4.8	.8	3.05	5.00	3.50	.375	.375	1.88
5 X 3 X 7/16 X 11.3 L	11.26	9.5	54.6	45.5	1.41	4.8	.8	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0 L	12.01	10.5	57.1	50.0	1.47	4.7	.9	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3 L	12.27	12.0	60.9	56.3	1.56	4.7	.9	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8 L	12.75	10.6	56.4	50.0	1.47	4.7	.9	3.75	5.00	3.00	.500	.500	2.50
5 X 3 1/2 X 1/2 X 13.6 L	13.60	11.7	58.9	55.0	1.53	4.7	.9	4.00	5.00	3.50	.500	.500	2.50
5 X 3 1/2 X 5/8 X 16.9 L	16.73	13.9	61.5	63.9	1.62	4.6	1.0	4.92	5.00	3.50	.625	.625	3.13
6 X 3 1/2 X 5/16 X 9.8 L	9.78	10.0	67.0	57.9	1.61	5.8	.9	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L	10.31	11.0	69.5	62.8	1.67	5.7	.9	3.03	6.00	4.00	.313	.313	1.88

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
6	X 3 1/2	X	X 15.3	C	11.07	11.6	70.4	65.9	1.70	5.7	.9	3.26	6.00	3.50	.385	.340	2.04
6	X 3 1/2	X 3/8	X 11.7	L	11.63	11.8	70.2	66.7	1.71	5.7	.9	3.42	6.00	3.50	.375	.375	2.25
6	X 4	X 3/8	X 12.3	L	12.27	12.8	72.7	72.3	1.77	5.6	1.0	3.61	6.00	4.00	.375	.375	2.25
6	X 3 1/2	X	X 18.0	C	12.77	13.5	73.7	75.4	1.80	5.6	1.0	3.76	6.00	3.50	.475	.379	2.27
6	X 4	X 7/16	X 14.3	L	14.24	14.7	75.3	81.3	1.85	5.5	1.1	4.19	6.00	4.00	.438	.438	2.63
6	X 4	X 1/2	X 16.2	L	16.15	16.4	77.3	89.5	1.92	5.5	1.2	4.75	6.00	4.00	.500	.500	3.00
6	X 4	X 9/16	X 18.1	L	18.06	18.0	78.9	97.2	1.98	5.4	1.2	5.31	6.00	4.00	.563	.563	3.38
6	X 4	X 5/8	X 20.0	L	19.92	19.6	80.2	104.3	2.03	5.3	1.3	5.86	6.00	4.00	.625	.625	3.75
7	X 4	X 3/8	X 13.6	L	13.55	15.8	88.7	102.0	2.08	6.5	1.1	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9	L	17.85	20.2	93.7	126.5	2.26	6.3	1.3	5.25	7.00	4.00	.500	.500	3.50
7	X 4	X 5/8	X 22.1	L	22.05	24.2	97.0	147.5	2.38	6.1	1.5	6.48	7.00	4.00	.625	.625	4.38
8	X 4	X 1/2	X 19.6	L	19.55	24.2	110.3	171.3	2.60	7.1	1.6	5.75	8.00	4.00	.500	.500	4.00
8	X 4	X 9/16	X 21.9	L	21.89	26.7	112.3	186.2	2.68	7.0	1.7	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	31.7	119.2	216.0	2.87	6.8	1.8	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	29.1	113.9	199.9	2.74	6.9	1.8	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	35.1	121.1	234.5	2.94	6.7	1.9	7.57	8.00	6.00	.563	.563	4.50
9	X 4	X 1/2	X 21.3	L	21.25	28.6	126.9	224.4	2.95	7.9	1.8	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	31.5	129.1	244.0	3.03	7.7	1.9	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	34.4	130.9	262.0	3.10	7.6	2.0	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8	X	X 15.3	C	11.66	18.3	128.7	169.9	2.72	9.3	1.3	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2	X	X 21.9	C	16.36	26.0	139.7	233.0	3.09	9.0	1.7	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8	X	X 20.0	C	16.39	22.7	131.4	205.8	2.91	9.1	1.6	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2	X	X 24.9	C	18.73	29.2	142.2	257.2	3.20	8.8	1.8	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2	X	X 25.3	C	19.76	29.1	140.6	256.4	3.18	8.8	1.8	5.81	10.00	3.55	.500	.425	4.25

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 31.250 IN.) PLATE WEIGHT = 25.500 LBS. (.6250 IN.) EFFECTIVE PLATE AREA = 19.531 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 4 1/2 X X 28.5 C	21.34	33.3	146.1	288.5	3.34	8.7	2.0	6.28	10.00	3.95	.575	.425	4.25
10 X 3 1/2 X X 28.3 C	22.13	32.2	143.0	279.5	3.28	8.7	2.0	6.51	10.00	3.50	.575	.477	4.77
10 X 4 X X 33.6 C	26.44	37.9	147.0	319.8	3.42	8.4	2.2	7.78	10.00	4.10	.575	.575	5.75
12 X 3 X X 20.7 C	16.03	29.0	168.2	312.1	3.59	10.8	1.9	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0 C	20.33	33.7	169.5	354.7	3.73	10.5	2.1	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X X 30.9 C	24.48	41.5	177.0	424.3	3.98	10.2	2.4	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X X 30.0 C	25.34	39.0	171.4	400.9	3.85	10.3	2.3	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X X 32.9 C	26.52	43.6	177.6	442.0	4.02	10.1	2.5	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0 C	26.91	47.7	183.0	477.4	4.17	10.0	2.6	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0 C	30.50	47.8	178.7	476.1	4.08	10.0	2.7	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0 C	31.93	52.7	184.0	517.5	4.23	9.8	2.8	9.39	12.00	3.89	.700	.590	7.08
13 X 4 X X 31.8 C	24.09	47.8	199.0	525.4	4.44	11.0	2.6	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0 C	27.27	51.4	199.3	556.7	4.49	10.8	2.8	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0 C	32.28	56.9	200.4	604.0	4.56	10.6	3.0	9.49	13.00	4.19	.610	.560	7.28
15 X 3 3/8 X X 33.9 C	27.03	56.3	229.3	706.7	5.07	12.5	3.1	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0 C	33.15	64.0	231.0	783.4	5.17	12.2	3.4	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0 C	41.82	80.2	241.1	940.4	5.46	11.7	3.9	12.06	15.00	4.00	.797	.625	9.38
18 X 4 X X 42.7 C	34.98	82.2	285.5	1188.6	6.31	14.5	4.2	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C	38.04	86.6	286.6	1239.0	6.35	14.3	4.3	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9 C	44.16	95.3	289.3	1335.4	6.41	14.0	4.6	12.99	18.00	4.10	.625	.600	10.80

(50T) PLATE WEIGHT = 25.500 LBS. (.6250 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 3/8 X 9.8 L	9.72	7.9	45.7	31.5	1.09	4.0	.7	2.86	4.00	4.00	.375	.375	1.50
4 X 3 X 1/2 X 11.1 L	11.05	8.0	45.1	32.0	1.09	4.0	.7	3.25	4.00	3.00	.500	.500	2.00
4 X 4 X 7/16 X 11.3 L	11.26	8.9	48.1	35.3	1.15	4.0	.7	3.31	4.00	4.00	.438	.438	1.75
4 X 3 1/2 X 1/2 X 11.9 L	11.90	9.0	47.7	35.4	1.14	3.9	.7	3.50	4.00	3.50	.500	.500	2.00
4 X 4 X 1/2 X 12.8 L	12.75	9.9	50.0	38.8	1.19	3.9	.8	3.75	4.00	4.00	.500	.500	2.00
4 X 4 X 5/8 X 15.7 L	15.67	11.7	52.8	45.0	1.26	3.8	.9	4.61	4.00	4.00	.625	.625	2.50
5 X 3 1/2 X 5/16 X 8.7 L	8.71	8.1	56.5	40.2	1.24	5.0	.7	2.56	5.00	3.50	.313	.313	1.57
5 X 3 X 3/8 X 9.8 L	9.72	8.5	57.1	42.1	1.26	5.0	.7	2.86	5.00	3.00	.375	.375	1.88
5 X 3 1/2 X 3/8 X 10.4 L	10.36	9.4	60.2	46.3	1.32	4.9	.8	3.05	5.00	3.50	.375	.375	1.88
5 X 3 X 7/16 X 11.3 L	11.26	9.7	60.1	47.4	1.33	4.9	.8	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0 L	12.01	10.7	63.1	52.1	1.39	4.9	.8	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3 L	12.27	12.2	67.6	58.7	1.47	4.8	.9	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8 L	12.75	10.8	62.4	52.2	1.38	4.9	.8	3.75	5.00	3.00	.500	.500	2.50
5 X 3 1/2 X 1/2 X 13.6 L	13.60	11.9	65.4	57.4	1.44	4.8	.9	4.00	5.00	3.50	.500	.500	2.50
5 X 3 1/2 X 5/8 X 16.8 L	16.73	14.2	69.8	67.0	1.53	4.7	1.0	4.92	5.00	3.50	.625	.625	3.13
6 X 3 1/2 X 5/16 X 9.8 L	9.78	10.2	73.6	59.9	1.50	5.9	.8	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L	10.31	11.1	76.7	65.0	1.56	5.8	.8	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3 C	11.07	11.8	78.0	68.3	1.59	5.8	.9	3.26	6.00	3.50	.385	.340	2.04
6 X 3 1/2 X 3/8 X 11.7 L	11.63	11.9	77.9	69.2	1.60	5.8	.9	3.42	6.00	3.50	.375	.375	2.25
6 X 4 X 3/8 X 12.3 L	12.27	13.0	81.0	75.1	1.66	5.8	.9	3.61	6.00	4.00	.375	.375	2.25
6 X 3 1/2 X X 18.0 C	12.77	13.7	82.3	78.4	1.69	5.7	1.0	3.76	6.00	3.50	.475	.379	2.27
6 X 4 X 7/16 X 14.3 L	14.24	14.9	84.4	84.7	1.74	5.7	1.0	4.19	6.00	4.00	.438	.438	2.63
6 X 4 X 1/2 X 16.2 L	16.15	16.6	87.1	93.4	1.81	5.6	1.1	4.75	6.00	4.00	.500	.500	3.00
6 X 4 X 9/16 X 18.1 L	18.06	18.3	89.3	101.7	1.87	5.5	1.1	5.31	6.00	4.00	.563	.563	3.38
6 X 4 X 5/8 X 20.0 L	19.92	19.9	91.0	109.3	1.92	5.5	1.2	5.86	6.00	4.00	.625	.625	3.75

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X 4	X 3/8	X 13.6	L	13.55	16.0	99.7	105.8	1.96	6.6	1.1	3.98	7.00	4.00	.375	.375	2.63
7	X 4	X 1/2	X 17.9	L	17.85	20.5	106.4	131.9	2.14	6.4	1.2	5.25	7.00	4.00	.500	.500	3.50
7	X 4	X 5/8	X 22.1	L	22.05	24.6	110.8	154.5	2.27	6.3	1.4	6.48	7.00	4.00	.625	.625	4.38
8	X 4	X 1/2	X 19.6	L	19.55	24.6	126.0	178.6	2.47	7.3	1.4	5.75	8.00	4.00	.500	.500	4.00
8	X 4	X 9/16	X 21.9	L	21.89	27.1	128.6	194.6	2.54	7.2	1.5	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0	L	22.95	32.1	137.3	226.2	2.73	7.0	1.6	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2	L	24.17	29.6	130.8	209.4	2.61	7.1	1.6	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7	L	25.72	35.6	139.8	246.2	2.81	6.9	1.8	7.57	8.00	6.00	.563	.563	4.50
9	X 4	X 1/2	X 21.3	L	21.25	29.0	145.8	234.0	2.80	8.1	1.6	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8	L	23.81	32.0	148.6	255.2	2.89	8.0	1.7	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3	L	26.30	34.9	150.9	274.7	2.96	7.9	1.8	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8 X		X 15.3	C	11.66	18.5	145.5	175.0	2.54	9.5	1.2	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9	C	16.36	26.3	160.0	241.4	2.91	9.2	1.5	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0	C	16.39	23.0	150.0	213.2	2.74	9.3	1.4	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9	C	18.73	29.5	163.5	267.3	3.03	9.1	1.6	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3	C	19.76	29.5	161.5	266.6	3.01	9.0	1.7	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2 X		X 28.5	C	21.34	33.8	168.5	300.6	3.17	8.9	1.8	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2 X		X 28.3	C	22.13	32.7	164.7	291.5	3.11	8.9	1.8	6.51	10.00	3.50	.575	.477	4.77
10	X 4	X	X 33.6	C	26.44	38.4	169.9	334.9	3.27	8.7	2.0	7.78	10.00	4.10	.575	.575	5.75
12	X 3	X	X 20.7	C	16.03	29.3	193.3	322.7	3.37	11.0	1.7	4.72	12.00	2.94	.501	.282	3.38
12	X 3	X	X 25.0	C	20.33	34.1	195.6	368.5	3.53	10.8	1.9	5.98	12.00	3.05	.501	.387	4.64
12	X 3 1/2 X		X 30.9	C	24.48	42.0	205.2	442.7	3.79	10.5	2.2	7.20	12.00	3.45	.600	.450	5.40
12	X 3	X	X 30.0	C	25.34	39.6	198.2	418.5	3.67	10.6	2.1	7.45	12.00	3.17	.501	.510	6.12
12	X 3 1/2 X		X 32.9	C	26.52	44.2	205.8	462.0	3.83	10.4	2.2	7.80	12.00	3.50	.600	.500	6.00

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 34.375 IN.) PLATE WEIGHT = 28.050 LBS. (.6875 IN.) EFFECTIVE PLATE AREA = 23.633 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X X 35.0	C	26.91	48.3	212.6	499.5	3.98	10.3	2.3	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0	C	30.60	48.6	207.3	499.2	3.91	10.3	2.4	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0	C	31.93	53.6	213.7	543.5	4.06	10.1	2.5	9.39	12.00	3.89	.700	.590	7.08
13 X 4 X X 31.8	C	24.09	48.4	231.4	548.0	4.22	11.3	2.4	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0	C	27.27	52.1	231.9	582.1	4.29	11.2	2.5	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0	C	32.28	57.8	233.2	634.0	4.37	11.0	2.7	9.49	13.00	4.19	.610	.560	7.28
15 X 3 3/8 X X 33.9	C	27.03	57.1	267.3	737.9	4.83	12.9	2.8	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0	C	33.15	65.0	269.1	821.8	4.96	12.6	3.1	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	81.6	281.1	992.1	5.27	12.2	3.5	12.06	15.00	4.00	.797	.625	9.38
18 X 4 X X 42.7	C	34.98	83.4	333.7	1247.3	6.06	14.9	3.7	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	88.8	334.7	1302.7	6.12	14.8	3.9	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	97.1	337.3	1408.8	6.20	14.5	4.2	12.99	18.00	4.10	.625	.600	10.80

(50T) PLATE WEIGHT = 28.050 LBS. (.6875 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
4 X 4 X 1/2 X 12.8 L	12.75	10.1	54.1	40.6	1.13	4.0	.8	3.75	4.00	4.00	.500	.500	2.00
4 X 4 X 5/8 X 15.7 L	15.67	12.0	57.7	47.2	1.20	3.9	.8	4.61	4.00	4.00	.625	.625	2.50
4 X 4 X 3/4 X 18.5 L	18.49	13.7	60.2	52.9	1.26	3.9	.9	5.44	4.00	4.00	.750	.750	3.00
5 X 3 X 7/16 X 11.3 L	11.26	9.9	64.8	49.2	1.25	5.0	.8	3.31	5.00	3.00	.438	.438	2.19
5 X 3 1/2 X 7/16 X 12.0 L	12.01	10.9	68.3	54.1	1.31	5.0	.8	3.53	5.00	3.50	.438	.438	2.19
5 X 5 X 3/8 X 12.3 L	12.27	12.4	73.5	60.9	1.39	4.9	.8	3.61	5.00	5.00	.375	.375	1.88
5 X 3 X 1/2 X 12.8 L	12.75	11.0	67.7	54.3	1.30	4.9	.8	3.75	5.00	3.00	.500	.500	2.50
5 X 3 1/2 X 1/2 X 13.6 L	13.60	12.2	71.3	59.7	1.36	4.9	.8	4.00	5.00	3.50	.500	.500	2.50
5 X 3 1/2 X 5/8 X 16.8 L	16.73	14.5	75.8	69.9	1.45	4.8	.9	4.92	5.00	3.50	.625	.625	3.13
5 X 3 1/2 X 3/4 X 19.8 L	19.76	16.6	78.9	78.7	1.52	4.8	1.0	5.81	5.00	3.50	.750	.750	3.75
6 X 3 1/2 X 5/16 X 9.8 L	9.78	10.4	79.3	61.9	1.41	6.0	.8	2.88	6.00	3.50	.313	.313	1.88
6 X 4 X 5/16 X 10.3 L	10.31	11.3	83.0	67.1	1.47	5.9	.8	3.03	6.00	4.00	.313	.313	1.88
6 X 3 1/2 X X 15.3 C	11.07	11.9	84.6	70.6	1.50	5.9	.8	3.26	6.00	3.50	.385	.340	2.04
6 X 3 1/2 X 3/8 X 11.7 L	11.63	12.1	84.7	71.6	1.51	5.9	.8	3.42	6.00	3.50	.375	.375	2.25
6 X 4 X 3/8 X 12.3 L	12.27	13.2	88.3	77.7	1.56	5.9	.9	3.61	6.00	4.00	.375	.375	2.25
6 X 3 1/2 X X 18.0 C	12.77	13.9	90.0	81.1	1.59	5.8	.9	3.76	6.00	3.50	.475	.379	2.27
6 X 4 X 7/16 X 14.3 L	14.24	15.1	92.7	87.7	1.65	5.8	.9	4.19	6.00	4.00	.438	.438	2.63
6 X 4 X 1/2 X 16.2 L	16.15	16.9	96.1	97.0	1.72	5.7	1.0	4.75	6.00	4.00	.500	.500	3.00
6 X 4 X 9/16 X 18.1 L	18.06	18.6	99.0	105.7	1.78	5.7	1.1	5.31	6.00	4.00	.563	.563	3.38
6 X 4 X 5/8 X 20.0 L	19.92	20.2	101.3	113.8	1.83	5.6	1.1	5.86	6.00	4.00	.625	.625	3.75
6 X 4 X 3/4 X 23.6 L	23.59	23.3	105.0	128.6	1.92	5.5	1.2	6.94	6.00	4.00	.750	.750	4.50
7 X 4 X 3/8 X 13.6 L	13.55	16.2	109.7	109.3	1.85	6.8	1.0	3.98	7.00	4.00	.375	.375	2.63
7 X 4 X 1/2 X 17.9 L	17.85	20.7	118.4	136.7	2.02	6.6	1.2	5.25	7.00	4.00	.500	.500	3.50
7 X 4 X 5/8 X 22.1 L	22.05	24.9	124.2	160.8	2.16	6.5	1.3	6.48	7.00	4.00	.625	.625	4.38

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
7	X 4	X 3/4	X 26.2 L	26.14	28.8	128.3	182.1	2.25	6.3	1.4	7.69	7.00	4.00	.750	.750	5.25
8	X 4	X 1/2	X 19.6 L	19.55	24.9	141.1	185.1	2.34	7.4	1.3	5.75	8.00	4.00	.500	.500	4.00
8	X 4	X 9/16	X 21.9 L	21.89	27.5	144.6	202.2	2.42	7.4	1.4	6.44	8.00	4.00	.563	.563	4.50
8	X 6	X 1/2	X 23.0 L	22.95	32.5	155.1	235.3	2.60	7.2	1.5	6.75	8.00	6.00	.500	.500	4.00
8	X 4	X 5/8	X 24.2 L	24.17	30.0	147.4	218.0	2.49	7.3	1.5	7.11	8.00	4.00	.625	.625	5.00
8	X 6	X 9/16	X 25.7 L	25.72	36.0	158.3	256.7	2.68	7.1	1.6	7.57	8.00	6.00	.563	.563	4.50
8	X 4	X 3/4	X 28.7 L	28.69	34.7	151.9	247.0	2.60	7.1	1.6	8.44	8.00	4.00	.750	.750	6.00
8	X 6	X 3/4	X 33.8 L	33.79	45.6	165.3	312.7	2.87	6.9	1.9	9.94	8.00	6.00	.750	.750	6.00
9	X 4	X 1/2	X 21.3 L	21.25	29.3	164.2	242.6	2.66	8.3	1.5	6.25	9.00	4.00	.500	.500	4.50
9	X 4	X 9/16	X 23.8 L	23.81	32.4	167.9	265.1	2.75	8.2	1.6	7.00	9.00	4.00	.563	.563	5.07
9	X 4	X 5/8	X 26.3 L	26.30	35.4	170.9	285.9	2.82	8.1	1.7	7.73	9.00	4.00	.625	.625	5.63
10	X 2 5/8 X		X 15.3 C	11.66	18.6	161.0	179.5	2.39	9.6	1.1	3.43	10.00	2.60	.436	.240	2.40
10	X 3 1/2 X		X 21.9 C	16.36	26.6	179.6	248.9	2.75	9.4	1.4	4.81	10.00	3.45	.500	.325	3.25
10	X 2 5/8 X		X 20.0 C	16.39	23.3	167.7	219.8	2.58	9.4	1.3	4.82	10.00	2.74	.436	.379	3.79
10	X 3 1/2 X		X 24.9 C	18.73	29.8	184.2	276.1	2.87	9.3	1.5	5.51	10.00	3.40	.575	.377	3.77
10	X 3 1/2 X		X 25.3 C	19.76	29.8	182.1	275.7	2.85	9.2	1.5	5.81	10.00	3.55	.500	.425	4.25
10	X 4 1/2 X		X 28.5 C	21.34	34.2	190.6	311.4	3.01	9.1	1.6	6.28	10.00	3.95	.575	.425	4.25
10	X 3 1/2 X		X 28.3 C	22.13	33.1	186.2	302.0	2.95	9.1	1.6	6.51	10.00	3.50	.575	.477	4.77
10	X 4	X	X 33.6 C	26.44	38.9	192.9	348.2	3.11	8.9	1.8	7.78	10.00	4.10	.575	.575	5.75
12	X 3	X	X 20.7 C	16.03	29.6	217.9	331.9	3.18	11.2	1.5	4.72	12.00	2.94	.501	.282	3.38
12	X 3	X	X 25.0 C	20.33	34.5	221.5	380.5	3.34	11.0	1.7	5.98	12.00	3.05	.501	.387	4.64
12	X 3 1/2 X		X 30.9 C	24.48	42.5	233.5	458.8	3.60	10.8	2.0	7.20	12.00	3.45	.600	.450	5.40
12	X 3	X	X 30.0 C	25.34	40.1	225.3	433.9	3.49	10.8	1.9	7.45	12.00	3.17	.501	.510	6.12

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 37.500 IN.) PLATE WEIGHT = 30.600 LBS. (.7500 IN.) EFFECTIVE PLATE AREA = 28.125 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN X IN	X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12	X 3 1/2 X	X 32.9	C	26.52	44.8	234.5	479.5	3.65	10.7	2.0	7.80	12.00	3.50	.600	.500	6.00
12	X 4 X	X 35.0	C	26.91	48.9	242.7	518.8	3.79	10.6	2.1	7.92	12.00	3.77	.700	.467	5.60
12	X 3 1/2 X	X 37.0	C	30.60	49.2	236.5	519.6	3.74	10.6	2.2	9.00	12.00	3.60	.600	.600	7.20
12	X 4 X	X 40.0	C	31.93	54.3	244.3	566.4	3.89	10.4	2.3	9.39	12.00	3.89	.700	.590	7.08
12	X 4 X	X 45.0	C	36.90	59.6	246.1	611.5	3.96	10.3	2.5	10.85	12.00	4.01	.700	.712	8.54
13	X 4 X	X 31.8	C	24.09	48.9	264.3	567.6	4.01	11.6	2.1	7.09	13.00	4.00	.610	.375	4.88
13	X 4 X	X 35.0	C	27.27	52.7	265.1	604.3	4.09	11.5	2.3	8.02	13.00	4.07	.610	.447	5.81
13	X 4 X	X 40.0	C	32.28	58.6	266.9	660.4	4.19	11.3	2.5	9.49	13.00	4.19	.610	.560	7.28
15	X 3 3/8 X	X 33.9	C	27.03	57.7	306.2	765.0	4.61	13.3	2.5	7.95	15.00	3.40	.650	.400	6.00
15	X 3 3/8 X	X 40.0	C	33.15	65.9	308.7	855.5	4.75	13.0	2.8	9.75	15.00	3.52	.650	.520	7.80
15	X 4 X	X 50.0	C	41.02	82.8	322.9	1038.0	5.08	12.5	3.2	12.06	15.00	4.00	.797	.625	9.38
15	X 3 3/8 X	X 50.0	C	43.15	78.9	313.5	992.9	4.93	12.6	3.2	12.69	15.00	3.72	.650	.716	10.74
15	X 4 X	X 53.2	C	44.23	86.9	324.0	1078.7	5.12	12.4	3.3	13.01	15.00	4.06	.797	.688	10.32
18	X 4 X	X 42.7	C	34.98	84.5	384.2	1298.7	5.81	15.4	3.4	10.29	18.00	3.95	.625	.450	8.10
18	X 4 X	X 45.8	C	38.04	89.2	385.3	1358.7	5.88	15.2	3.5	11.19	18.00	4.00	.625	.500	9.00
18	X 4 X	X 51.9	C	44.16	98.6	388.0	1473.9	5.99	15.0	3.8	12.99	18.00	4.10	.625	.600	10.80
18	X 4 X	X 58.0	C	50.28	107.7	391.2	1583.6	6.07	14.7	4.0	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 30.600 LBS. (.7500 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS				
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
5 X 3 1/2 X 3/4 X 19.8 L	19.76	17.2	92.1	85.1	1.39	5.0	.9	5.81	5.00	3.50	.750	.750	3.75
6 X 4 X 7/16 X 14.3 L	14.24	15.6	106.4	93.5	1.48	6.0	.9	4.19	6.00	4.00	.438	.438	2.63
6 X 4 X 1/2 X 16.2 L	16.15	17.4	111.5	103.5	1.55	5.9	.9	4.75	6.00	4.00	.500	.500	3.00
6 X 4 X 9/16 X 18.1 L	18.06	19.2	115.8	113.2	1.61	5.9	1.0	5.31	6.00	4.00	.563	.563	3.38
6 X 4 X 5/8 X 20.0 L	19.92	20.9	119.5	122.1	1.66	5.9	1.0	5.86	6.00	4.00	.625	.625	3.75
6 X 4 X 3/4 X 23.6 L	23.59	24.0	125.3	138.6	1.75	5.8	1.1	6.94	6.00	4.00	.750	.750	4.50
7 X 4 X 3/8 X 13.6 L	13.55	16.6	126.4	115.7	1.65	7.0	.9	3.98	7.00	4.00	.375	.375	2.63
7 X 4 X 1/2 X 17.9 L	17.85	21.3	139.5	145.5	1.83	6.8	1.0	5.25	7.00	4.00	.500	.500	3.50
7 X 4 X 5/8 X 22.1 L	22.05	25.6	148.6	172.0	1.96	6.7	1.2	6.48	7.00	4.00	.625	.625	4.38
7 X 4 X 3/4 X 26.2 L	26.14	29.6	155.1	195.7	2.06	6.6	1.3	7.69	7.00	4.00	.750	.750	5.25
8 X 4 X 1/2 X 19.6 L	19.55	25.5	168.5	196.5	2.11	7.7	1.2	5.75	8.00	4.00	.500	.500	4.00
8 X 4 X 9/16 X 21.9 L	21.89	28.2	174.0	215.3	2.19	7.6	1.2	6.44	8.00	4.00	.563	.563	4.50
8 X 6 X 1/2 X 23.0 L	22.95	33.3	188.4	250.9	2.36	7.5	1.3	6.75	8.00	6.00	.500	.500	4.00
8 X 4 X 5/8 X 24.2 L	24.17	30.8	178.6	232.8	2.26	7.6	1.3	7.11	8.00	4.00	.625	.625	5.00
8 X 6 X 9/16 X 25.7 L	25.72	36.8	193.8	274.8	2.45	7.5	1.4	7.57	8.00	6.00	.563	.563	4.50
8 X 4 X 3/4 X 28.7 L	28.69	35.6	185.8	265.3	2.38	7.4	1.4	8.44	8.00	4.00	.750	.750	6.00
8 X 4 X 7/8 X 33.1 L	33.10	40.1	191.2	294.4	2.48	7.3	1.5	9.73	8.00	4.00	.875	.875	7.00
8 X 6 X 3/4 X 33.8 L	33.79	46.8	205.1	338.1	2.65	7.2	1.6	9.94	8.00	6.00	.750	.750	6.00
8 X 6 X 7/8 X 39.1 L	39.05	52.9	210.3	374.9	2.74	7.1	1.8	11.48	8.00	6.00	.875	.875	7.00
9 X 4 X 1/2 X 21.3 L	21.25	30.0	198.3	257.3	2.40	8.6	1.3	6.25	9.00	4.00	.500	.500	4.50
9 X 4 X 9/16 X 23.8 L	23.81	33.2	204.2	282.1	2.50	8.5	1.4	7.00	9.00	4.00	.563	.563	5.07
9 X 4 X 5/8 X 26.3 L	26.30	36.3	209.1	305.2	2.58	8.4	1.5	7.73	9.00	4.00	.625	.625	5.63

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE	WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
		FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT	LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 2 5/8 X X 15.3 C	11.66	19.0	187.1	187.6	2.12	9.9	1.0	3.43	10.00	2.60	.436	.240	2.40
10 X 3 1/2 X X 21.9 C	16.36	27.1	215.0	261.6	2.46	9.7	1.2	4.81	10.00	3.45	.500	.325	3.25
10 X 2 5/8 X X 20.0 C	16.39	23.8	199.3	231.1	2.32	9.7	1.2	4.82	10.00	2.74	.436	.379	3.79
10 X 3 1/2 X X 24.9 C	18.73	30.4	222.6	291.2	2.58	9.6	1.3	5.51	10.00	3.40	.575	.377	3.77
10 X 3 1/2 X X 25.3 C	19.76	30.5	220.1	291.2	2.57	9.6	1.3	5.81	10.00	3.55	.500	.425	4.25
10 X 4 1/2 X X 28.5 C	21.34	34.8	232.4	329.6	2.72	9.5	1.4	6.28	10.00	3.95	.575	.425	4.25
10 X 3 1/2 X X 28.3 C	22.13	33.8	226.8	319.9	2.67	9.5	1.4	6.51	10.00	3.50	.575	.477	4.77
10 X 4 X X 33.6 C	26.44	39.8	237.3	370.9	2.84	9.3	1.6	7.78	10.00	4.10	.575	.575	5.75
10 X 4 X X 41.1 C	33.95	47.0	243.4	428.5	2.98	9.1	1.8	9.99	10.00	4.32	.575	.796	7.96
12 X 3 X X 20.7 C	16.03	30.1	262.9	347.5	2.84	11.6	1.3	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0 C	20.33	35.2	270.7	400.7	3.01	11.4	1.5	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X X 30.9 C	24.48	43.4	288.8	485.8	3.27	11.2	1.7	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X X 30.0 C	25.34	41.0	278.0	459.8	3.17	11.2	1.7	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X X 32.9 C	26.52	45.7	290.9	509.0	3.32	11.1	1.7	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0 C	26.91	49.9	302.2	551.1	3.45	11.1	1.8	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0 C	30.60	50.4	294.7	554.1	3.42	11.0	1.9	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0 C	31.93	55.5	305.7	605.1	3.56	10.9	2.0	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X X 45.0 C	36.90	61.1	309.0	656.5	3.66	10.8	2.1	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0 C	41.93	66.6	312.3	707.1	3.74	10.6	2.3	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 31.8 C	24.09	49.8	329.1	600.1	3.64	12.1	1.8	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0 C	27.27	53.7	331.4	641.3	3.72	11.9	1.9	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0 C	32.28	59.9	335.1	704.7	3.84	11.8	2.1	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X X 50.0 C	42.30	71.8	341.6	823.0	4.03	11.5	2.4	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 33.9 C	27.03	58.8	384.7	809.8	4.19	13.8	2.1	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0 C	33.15	67.4	389.6	911.7	4.36	13.5	2.3	9.75	15.00	3.52	.650	.520	7.80

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 43.750 IN.) PLATE WEIGHT = 35.700 LBS. (.8750 IN.) EFFECTIVE PLATE AREA = 38.281 SQ. IN.

NOMINAL SIZE			WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
				FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT			LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15 X 4 X X 50.0 C			41.02	84.8	410.2	1115.3	4.71	13.2	2.7	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0 C			43.15	81.0	397.3	1068.1	4.58	13.2	2.7	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2 C			44.23	89.1	411.8	1162.3	4.76	13.1	2.8	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4 C			47.46	93.4	413.6	1209.5	4.81	13.0	2.9	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6 C			50.61	97.5	415.3	1253.9	4.86	12.9	3.0	14.89	15.00	4.19	.797	.813	12.20
18 X 4 X X 42.7 C			34.98	86.2	489.4	1383.8	5.34	16.0	2.8	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C			38.04	91.2	491.2	1452.0	5.42	15.9	3.0	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9 C			44.16	101.0	495.1	1583.7	5.56	15.7	3.2	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0 C			50.28	110.7	499.2	1709.8	5.68	15.4	3.4	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 35.700 LBS. (.8750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 50.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 50.000 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
6 X 4 X 3/4 X 23.6	L	23.59	24.8	142.3	147.8	1.61	6.0	1.0	6.94	6.00	4.00	.750	.750	4.50
7 X 4 X 5/8 X 22.1	L	22.05	26.3	169.0	182.2	1.80	6.9	1.1	6.48	7.00	4.00	.625	.625	4.38
7 X 4 X 3/4 X 26.2	L	26.14	30.4	178.5	207.9	1.90	6.8	1.2	7.69	7.00	4.00	.750	.750	5.25
8 X 4 X 1/2 X 19.6	L	19.55	26.1	191.1	206.8	1.93	7.9	1.1	5.75	8.00	4.00	.500	.500	4.00
8 X 4 X 9/16 X 21.9	L	21.89	28.9	198.9	227.0	2.01	7.9	1.1	6.44	8.00	4.00	.563	.563	4.50
8 X 6 X 1/2 X 23.0	L	22.95	34.0	217.4	264.6	2.16	7.8	1.2	6.75	8.00	6.00	.500	.500	4.00
8 X 4 X 5/8 X 24.2	L	24.17	31.5	205.5	245.9	2.07	7.8	1.2	7.11	8.00	4.00	.625	.625	5.00
8 X 6 X 9/16 X 25.7	L	25.72	37.7	225.3	290.4	2.25	7.7	1.3	7.57	8.00	6.00	.563	.563	4.50
8 X 4 X 3/4 X 28.7	L	28.69	36.5	216.2	281.2	2.19	7.7	1.3	8.44	8.00	4.00	.750	.750	6.00
8 X 4 X 7/8 X 33.1	L	33.10	41.2	224.3	313.2	2.29	7.6	1.4	9.73	8.00	4.00	.875	.875	7.00
8 X 6 X 3/4 X 33.8	L	33.79	47.9	242.4	359.7	2.45	7.5	1.5	9.94	8.00	6.00	.750	.750	6.00
8 X 4 X 1 X 37.4	L	37.40	45.5	230.6	342.1	2.37	7.5	1.5	11.00	8.00	4.00	1.000	1.000	8.00
8 X 6 X 7/8 X 39.1	L	39.05	54.1	250.3	400.6	2.55	7.4	1.6	11.48	8.00	6.00	.875	.875	7.00
8 X 6 X 1 X 44.2	L	44.20	60.0	256.4	437.6	2.64	7.3	1.7	13.00	8.00	6.00	1.000	1.000	8.00
9 X 4 X 1/2 X 21.3	L	21.25	30.7	227.5	270.2	2.19	8.8	1.2	6.25	9.00	4.00	.500	.500	4.50
9 X 4 X 9/16 X 23.8	L	23.81	34.0	236.1	296.8	2.28	8.7	1.3	7.00	9.00	4.00	.563	.563	5.07
9 X 4 X 5/8 X 26.3	L	26.30	37.1	243.3	321.8	2.36	8.7	1.3	7.73	9.00	4.00	.625	.625	5.63
10 X 3 1/2 X X 21.9	C	16.36	27.6	243.9	272.8	2.23	9.9	1.1	4.81	10.00	3.45	.500	.325	3.25
10 X 2 5/8 X X 20.0	C	16.39	24.3	224.7	241.2	2.10	9.9	1.1	4.82	10.00	2.74	.436	.373	3.79
10 X 3 1/2 X X 24.9	C	18.73	31.0	255.0	304.3	2.34	9.8	1.2	5.51	10.00	3.40	.575	.377	3.77
10 X 3 1/2 X X 25.3	C	19.76	31.1	252.6	304.5	2.34	9.8	1.2	5.81	10.00	3.55	.500	.425	4.25
10 X 4 1/2 X X 28.5	C	21.34	35.5	268.9	345.1	2.48	9.7	1.3	6.28	10.00	3.95	.575	.425	4.25

(50T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 50.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 50.000 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
10 X 3 1/2 X X 28.3 C		22.13	34.5	262.3	335.3	2.44	9.7	1.3	6.51	10.00	3.50	.575	.477	4.77
10 X 4 X X 33.6 C		26.44	40.7	277.5	390.1	2.60	9.6	1.4	7.78	10.00	4.10	.575	.575	5.75
10 X 4 X X 41.1 C		33.95	48.1	287.9	453.4	2.75	9.4	1.6	9.99	10.00	4.32	.575	.796	7.96
12 X 3 X X 20.7 C		16.03	30.6	300.6	360.8	2.57	11.8	1.2	4.72	12.00	2.94	.501	.282	3.38
12 X 3 X X 25.0 C		20.33	35.8	313.7	417.6	2.73	11.7	1.3	5.98	12.00	3.05	.501	.387	4.64
12 X 3 1/2 X X 30.9 C		24.48	44.2	339.3	508.2	2.98	11.5	1.5	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X X 30.0 C		25.34	41.8	326.0	481.4	2.89	11.5	1.5	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X X 32.9 C		26.52	46.6	343.1	533.3	3.04	11.4	1.6	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0 C		26.91	50.7	357.9	577.7	3.16	11.4	1.6	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0 C		30.60	51.4	349.8	582.6	3.14	11.3	1.7	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0 C		31.93	56.6	364.5	636.9	3.27	11.3	1.7	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X X 45.0 C		36.90	62.4	370.3	693.8	3.38	11.1	1.9	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0 C		41.93	68.1	375.7	749.9	3.47	11.0	2.0	12.33	12.00	4.14	.730	.835	10.02
13 X 4 X X 31.8 C		24.09	50.6	389.4	626.6	3.31	12.4	1.6	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0 C		27.27	54.6	394.2	671.5	3.40	12.3	1.7	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0 C		32.28	61.0	401.0	741.0	3.53	12.2	1.8	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X X 50.0 C		42.30	73.3	412.4	871.8	3.74	11.9	2.1	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 33.9 C		27.03	59.7	460.0	845.9	3.82	14.2	1.8	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0 C		33.15	68.6	469.3	957.1	4.00	14.0	2.0	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0 C		41.02	86.4	498.6	1178.1	4.36	13.6	2.4	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0 C		43.15	82.7	482.1	1129.6	4.24	13.7	2.3	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2 C		44.23	90.8	501.3	1230.5	4.42	13.5	2.5	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4 C		47.46	95.4	504.2	1283.2	4.48	13.5	2.5	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6 C		50.61	99.7	506.8	1332.9	4.53	13.4	2.6	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8 C		53.82	104.0	509.4	1382.3	4.58	13.3	2.7	15.83	15.00	4.25	.797	.876	13.14

(50T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 50.000 IN.) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.) EFFECTIVE PLATE AREA = 50.000 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15 X 4 X X 65.9 C				56.93	108.2	511.8	1429.5	4.63	13.2	2.8	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1 C				60.13	112.5	514.3	1477.2	4.67	13.1	2.9	17.69	15.00	4.37	.797	1.000	15.00
18 X 4 X X 42.7 C				34.98	87.6	595.3	1451.5	4.91	16.6	2.4	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C				38.04	92.8	599.2	1526.6	4.99	16.5	2.5	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9 C				44.16	103.0	605.8	1672.3	5.15	16.2	2.8	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0 C				50.28	113.0	612.0	1812.6	5.29	16.0	3.0	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 40.800 LBS. (1.0000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 56.250 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 63.281 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X	IN X	IN X	LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X	6 X	1/2 X	23.0	L	22.95	34.8	241.2	277.2	1.99	8.0	1.1	6.75	8.00	6.00	.500	.500	4.00
8 X	4 X	5/8 X	24.2	L	24.17	32.3	227.6	258.1	1.91	8.0	1.1	7.11	8.00	4.00	.625	.625	5.00
8 X	6 X	9/16 X	25.7	L	25.72	38.5	251.8	304.7	2.07	7.9	1.2	7.57	8.00	6.00	.563	.563	4.50
8 X	4 X	3/4 X	28.7	L	28.69	37.4	242.0	295.8	2.03	7.9	1.2	8.44	8.00	4.00	.750	.750	6.00
8 X	4 X	7/8 X	33.1	L	33.10	42.2	253.2	330.2	2.13	7.8	1.3	9.73	8.00	4.00	.875	.875	7.00
8 X	6 X	3/4 X	33.8	L	33.79	48.9	275.4	379.2	2.28	7.7	1.4	9.94	8.00	6.00	.750	.750	6.00
8 X	4 X	1 X	37.4	L	37.40	46.7	262.1	361.6	2.21	7.7	1.4	11.00	8.00	4.00	1.000	1.000	8.00
8 X	6 X	7/8 X	39.1	L	39.05	55.4	286.7	423.5	2.38	7.6	1.5	11.48	8.00	6.00	.875	.875	7.00
8 X	6 X	1 X	44.2	L	44.20	61.4	295.6	464.0	2.47	7.6	1.6	13.00	8.00	6.00	1.000	1.000	8.00
9 X	4 X	9/16 X	23.8	L	23.81	34.7	262.4	310.4	2.10	8.9	1.2	7.00	9.00	4.00	.563	.563	5.07
9 X	4 X	5/8 X	26.3	L	26.30	37.9	272.1	336.9	2.18	8.9	1.2	7.73	9.00	4.00	.625	.625	5.63
10 X	3 1/2 X		24.9	C	18.73	31.6	280.7	316.4	2.14	10.0	1.1	5.51	10.00	3.40	.575	.377	3.77
10 X	3 1/2 X		25.3	C	19.76	31.7	278.5	316.9	2.14	10.0	1.1	5.81	10.00	3.55	.500	.425	4.25
10 X	4 1/2 X		28.5	C	21.34	36.2	298.9	359.2	2.27	9.9	1.2	6.28	10.00	3.95	.575	.425	4.25
10 X	3 1/2 X		28.3	C	22.13	35.2	291.5	349.3	2.24	9.9	1.2	6.51	10.00	3.50	.575	.477	4.77
10 X	4 X		33.6	C	26.44	41.5	311.9	407.3	2.39	9.8	1.3	7.78	10.00	4.10	.575	.575	5.75
10 X	4 X		41.1	C	33.95	49.1	327.7	475.4	2.55	9.7	1.5	9.99	10.00	4.32	.575	.796	7.96
12 X	3 X		20.7	C	16.03	31.1	330.0	373.0	2.34	12.0	1.1	4.72	12.00	2.94	.501	.282	3.38
12 X	3 X		25.0	C	20.33	36.4	349.2	432.9	2.50	11.9	1.2	5.98	12.00	3.05	.501	.387	4.64
12 X	3 1/2 X		30.9	C	24.48	44.9	382.9	527.8	2.74	11.7	1.4	7.20	12.00	3.45	.600	.450	5.40
12 X	3 X		30.0	C	25.34	42.5	367.4	500.5	2.66	11.8	1.4	7.45	12.00	3.17	.501	.510	6.12
12 X	3 1/2 X		32.9	C	26.52	47.4	388.7	554.7	2.79	11.7	1.4	7.80	12.00	3.50	.600	.500	6.00
12 X	4 X		35.0	C	26.91	51.6	407.2	600.8	2.90	11.6	1.5	7.92	12.00	3.77	.700	.467	5.60
12 X	3 1/2 X		37.0	C	30.60	52.3	399.2	607.4	2.90	11.6	1.5	9.00	12.00	3.60	.600	.600	7.20

(50T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 56.250 IN.) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.) EFFECTIVE PLATE AREA = 63.281 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT					LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12	X 4	X	X 40.0	C	31.93	57.6	417.9	684.4	3.02	11.5	1.6	9.39	12.00	3.89	.700	.590	7.08
12	X 4	X	X 45.0	C	36.90	63.5	427.2	725.8	3.13	11.4	1.7	10.85	12.00	4.01	.700	.712	8.54
12	X 4	X	X 50.0	C	41.93	69.5	435.7	786.7	3.23	11.3	1.8	12.33	12.00	4.14	.700	.835	10.02
13	X 4	X	X 31.8	C	24.09	51.3	442.3	649.5	3.04	12.7	1.5	7.09	13.00	4.00	.610	.375	4.88
13	X 4	X	X 35.0	C	27.27	55.5	450.3	697.5	3.13	12.6	1.5	8.02	13.00	4.07	.610	.447	5.81
13	X 4	X	X 40.0	C	32.28	62.0	461.6	771.9	3.26	12.5	1.7	9.49	13.00	4.19	.610	.560	7.28
13	X 4	X	X 50.0	C	42.30	74.7	479.8	913.4	3.47	12.2	1.9	12.44	13.00	4.41	.610	.787	10.23
15	X 3 3/8	X	X 33.9	C	27.03	60.6	528.4	876.4	3.51	14.5	1.7	7.95	15.00	3.40	.650	.400	6.00
15	X 3 3/8	X	X 40.0	C	33.15	69.6	543.8	995.4	3.69	14.3	1.8	9.75	15.00	3.52	.650	.520	7.80
15	X 4	X	X 50.0	C	41.02	87.8	584.1	1230.8	4.04	14.0	2.1	12.06	15.00	4.00	.797	.625	9.38
15	X 3 3/8	X	X 50.0	C	43.15	84.2	564.2	1181.5	3.94	14.0	2.1	12.69	15.00	3.72	.650	.716	10.74
15	X 4	X	X 53.2	C	44.23	92.4	588.7	1287.7	4.11	13.9	2.2	13.01	15.00	4.06	.797	.688	10.32
15	X 4	X	X 56.4	C	47.46	97.1	593.3	1345.1	4.17	13.9	2.3	13.96	15.00	4.13	.797	.751	11.27
15	X 4	X	X 59.6	C	50.61	101.5	597.4	1399.4	4.23	13.8	2.3	14.89	15.00	4.19	.797	.813	12.20
15	X 4	X	X 62.8	C	53.82	106.0	601.4	1453.6	4.29	13.7	2.4	15.83	15.00	4.25	.797	.876	13.14
15	X 4	X	X 65.9	C	56.93	110.4	605.1	1505.5	4.34	13.6	2.5	16.74	15.00	4.31	.797	.937	14.06
15	X 4	X	X 69.1	C	60.13	114.8	608.8	1557.9	4.39	13.6	2.6	17.69	15.00	4.37	.797	1.000	15.00
15	X 4	X	X 75.0	C	66.06	123.1	615.4	1653.8	4.47	13.4	2.7	19.43	15.00	4.49	.797	1.116	16.74
18	X 4	X	X 42.7	C	34.98	88.8	698.5	1507.4	4.53	17.0	2.2	10.29	18.00	3.95	.625	.450	8.10
18	X 4	X	X 45.8	C	38.04	94.1	704.5	1588.2	4.62	16.9	2.3	11.19	18.00	4.00	.625	.500	9.00
18	X 4	X	X 51.9	C	44.16	104.6	715.4	1745.6	4.78	16.7	2.4	12.99	18.00	4.10	.625	.600	10.80
18	X 4	X	X 58.0	C	50.28	115.0	725.2	1898.1	4.93	16.5	2.6	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 45.900 LBS. (1.1250 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 62.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 78.125 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 6 X 3/4 X 33.8	L	33.79	50.1	303.7	397.5	2.12	7.9	1.3	9.94	8.00	6.00	.750	.750	6.00
8 X 4 X 1 X 37.4	L	37.40	47.9	289.3	380.0	2.06	7.9	1.3	11.00	8.00	4.00	1.000	1.000	8.00
8 X 6 X 7/8 X 39.1	L	39.05	56.6	318.7	444.9	2.23	7.9	1.4	11.48	8.00	6.00	.875	.875	7.00
8 X 6 X 1 X 44.2	L	44.20	62.8	330.7	488.3	2.31	7.8	1.5	13.00	8.00	6.00	1.000	1.000	8.00
10 X 4 X X 41.1	C	33.95	50.2	361.8	495.9	2.37	9.9	1.4	9.99	10.00	4.32	.575	.796	7.96
12 X 3 1/2 X X 30.9	C	24.48	45.7	418.7	546.1	2.53	11.9	1.3	7.20	12.00	3.45	.600	.450	5.40
12 X 3 X X 30.0	C	25.34	43.3	401.3	518.3	2.46	12.0	1.3	7.45	12.00	3.17	.501	.510	6.12
12 X 3 1/2 X X 32.9	C	26.52	48.2	426.8	574.4	2.59	11.9	1.3	7.80	12.00	3.50	.600	.500	6.00
12 X 4 X X 35.0	C	26.91	52.4	448.6	622.0	2.69	11.9	1.4	7.92	12.00	3.77	.700	.467	5.60
12 X 3 1/2 X X 37.0	C	30.60	53.3	441.5	630.0	2.69	11.8	1.4	9.00	12.00	3.60	.600	.600	7.20
12 X 4 X X 40.0	C	31.93	58.6	464.2	689.3	2.81	11.8	1.5	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X X 45.0	C	36.90	64.7	477.7	754.6	2.91	11.7	1.6	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0	C	41.93	70.8	489.9	819.6	3.01	11.6	1.7	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 31.8	C	24.09	52.1	486.4	670.4	2.80	12.9	1.4	7.09	13.00	4.00	.610	.375	4.88
13 X 4 X X 35.0	C	27.27	56.3	498.1	720.9	2.89	12.8	1.4	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0	C	32.28	63.0	514.7	799.7	3.02	12.7	1.6	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X X 50.0	C	42.30	76.0	541.4	950.2	3.24	12.5	1.8	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 33.9	C	27.03	61.4	587.5	903.6	3.24	14.7	1.5	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0	C	33.15	70.7	610.4	1029.0	3.42	14.6	1.7	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	89.1	663.6	1276.5	3.76	14.3	1.9	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	85.6	640.6	1226.6	3.68	14.3	1.9	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	93.8	670.6	1337.4	3.83	14.3	2.0	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4	C	47.46	98.6	677.6	1398.8	3.90	14.2	2.1	13.96	15.00	4.13	.797	.751	11.27

(50T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 62.500 IN.) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.) EFFECTIVE PLATE AREA = 78.125 SQ. IN.

NOMINAL SIZE				WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
					FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT				LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15 X 4 X X 59.6 C				50.61	103.2	683.8	1457.1	3.96	14.1	2.1	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8 C				53.82	107.8	689.7	1515.4	4.02	14.1	2.2	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9 C				56.93	112.3	695.2	1571.3	4.07	14.0	2.3	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1 C				60.13	116.9	700.5	1628.0	4.12	13.9	2.3	17.69	15.00	4.37	.797	1.000	15.00
15 X 4 X X 75.0 C				66.06	125.4	710.0	1731.8	4.21	13.8	2.4	19.43	15.00	4.49	.797	1.116	16.74
18 X 4 X X 42.7 C				34.98	90.0	793.6	1555.4	4.19	17.3	2.0	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8 C				38.04	95.4	803.0	1640.9	4.29	17.2	2.0	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9 C				44.16	106.1	819.9	1808.3	4.46	17.0	2.2	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0 C				50.28	116.7	834.7	1971.1	4.61	16.9	2.4	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 51.000 LBS. (1.2500 IN.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 68.750 IN.) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.) EFFECTIVE PLATE AREA = 94.531 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
8 X 6 X 1 X 44.2	L	44.20	64.3	361.1	511.7	2.18	8.0	1.4	13.00	8.00	6.00	1.000	1.000	8.00
12 X 4 X X 40.0	C	31.93	59.6	502.9	712.8	2.62	12.0	1.4	9.39	12.00	3.89	.700	.590	7.08
12 X 4 X X 45.0	C	36.90	65.8	521.0	781.5	2.72	11.9	1.5	10.85	12.00	4.01	.700	.712	8.54
12 X 4 X X 50.0	C	41.93	72.1	537.5	850.1	2.82	11.8	1.6	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 35.0	C	27.27	57.2	537.3	743.2	2.69	13.0	1.4	8.02	13.00	4.07	.610	.447	5.81
13 X 4 X X 40.0	C	32.28	64.0	559.6	825.6	2.82	12.9	1.5	9.49	13.00	4.19	.610	.560	7.28
13 X 4 X X 50.0	C	42.30	77.3	595.9	984.0	3.03	12.7	1.7	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 33.9	C	27.03	62.3	636.6	928.9	3.01	14.9	1.5	7.95	15.00	3.40	.650	.400	6.00
15 X 3 3/8 X X 40.0	C	33.15	71.7	667.8	1059.9	3.19	14.8	1.6	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	90.4	734.7	1317.7	3.52	14.6	1.8	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	86.9	709.1	1267.4	3.44	14.6	1.8	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	95.2	744.8	1381.9	3.58	14.5	1.9	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4	C	47.46	100.1	754.7	1446.9	3.65	14.5	1.9	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6	C	50.61	104.8	763.4	1508.6	3.71	14.4	2.0	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8	C	53.82	109.5	771.8	1570.5	3.77	14.3	2.0	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9	C	56.93	114.1	779.5	1629.9	3.83	14.3	2.1	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1	C	60.13	118.8	787.0	1690.3	3.88	14.2	2.1	17.69	15.00	4.37	.797	1.000	15.00
15 X 4 X X 75.0	C	66.06	127.5	800.1	1801.1	3.98	14.1	2.3	19.43	15.00	4.49	.797	1.116	16.74
18 X 4 X X 42.7	C	34.98	91.0	878.3	1598.1	3.90	17.6	1.8	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	96.5	891.8	1687.7	4.00	17.5	1.9	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	107.5	916.1	1863.6	4.16	17.3	2.0	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0	C	50.28	118.3	937.2	2035.4	4.31	17.2	2.2	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 56.100 LBS. (1.3750 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 75.000 IN.) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.) EFFECTIVE PLATE AREA = 112.500 SQ. IN.

NOMINAL SIZE		WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
			FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN X IN X IN X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
12 X 4 X X 50.0	C	41.93	73.4	577.9	879.4	2.65	12.0	1.5	12.33	12.00	4.14	.700	.835	10.02
13 X 4 X X 50.0	C	42.30	78.7	642.5	1016.1	2.85	12.9	1.6	12.44	13.00	4.41	.610	.787	10.23
15 X 3 3/8 X X 40.0	C	33.15	72.7	715.5	1089.3	2.99	15.0	1.5	9.75	15.00	3.52	.650	.520	7.80
15 X 4 X X 50.0	C	41.02	91.6	796.5	1356.0	3.30	14.8	1.7	12.06	15.00	4.00	.797	.625	9.38
15 X 3 3/8 X X 50.0	C	43.15	88.2	768.7	1305.5	3.23	14.8	1.7	12.69	15.00	3.72	.650	.716	10.74
15 X 4 X X 53.2	C	44.23	96.5	810.0	1423.2	3.37	14.7	1.8	13.01	15.00	4.06	.797	.688	10.32
15 X 4 X X 56.4	C	47.46	101.5	823.1	1491.3	3.43	14.7	1.8	13.96	15.00	4.13	.797	.751	11.27
15 X 4 X X 59.6	C	50.61	106.3	834.7	1556.1	3.50	14.6	1.9	14.89	15.00	4.19	.797	.813	12.20
15 X 4 X X 62.8	C	53.82	111.2	845.9	1621.1	3.55	14.6	1.9	15.83	15.00	4.25	.797	.876	13.14
15 X 4 X X 65.9	C	56.93	115.8	856.2	1683.7	3.61	14.5	2.0	16.74	15.00	4.31	.797	.937	14.06
15 X 4 X X 69.1	C	60.13	120.6	866.2	1747.3	3.66	14.5	2.0	17.69	15.00	4.37	.797	1.000	15.00
15 X 4 X X 75.0	C	66.06	129.6	883.7	1864.3	3.76	14.4	2.1	19.43	15.00	4.49	.797	1.116	16.74
18 X 4 X X 42.7	C	34.98	92.1	951.2	1637.6	3.65	17.8	1.7	10.29	18.00	3.95	.625	.450	8.10
18 X 4 X X 45.8	C	38.04	97.7	969.3	1730.7	3.74	17.7	1.8	11.19	18.00	4.00	.625	.500	9.00
18 X 4 X X 51.9	C	44.16	108.8	1001.8	1914.1	3.91	17.6	1.9	12.99	18.00	4.10	.625	.600	10.80
18 X 4 X X 58.0	C	50.28	119.9	1030.3	2093.6	4.06	17.5	2.0	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 61.200 LBS. (1.5000 IN.)

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TABLE XIII. Properties of combined beam and plate, C-L and L (50t). - Continued

PROPERTIES OF COMBINED BEAM AND PLATE C-L AND L

(50T = 87.500 IN.) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.) EFFECTIVE PLATE AREA = 153.125 SQ. IN.

NOMINAL SIZE					WT/FT	SECTION MODULUS		I	R	YF	YP	BEAM DIMENSIONS					
						FLANGE	PLATE					A	D	WF	TF	TW	ASH
IN	X IN	X IN	X LBS/FT		LBS	IN3	IN3	IN4	IN	IN	IN	IN2	IN	IN	IN	IN	IN2
15	X 4	X	X 62.8	C	53.82	114.5	967.6	1715.2	3.19	15.0	1.8	15.83	15.00	4.25	.797	.876	13.14
15	X 4	X	X 65.9	C	56.93	119.4	983.6	1783.1	3.24	14.9	1.8	16.74	15.00	4.31	.797	.937	14.06
15	X 4	X	X 69.1	C	60.13	124.3	999.2	1852.3	3.29	14.9	1.9	17.69	15.00	4.37	.797	1.000	15.00
15	X 4	X	X 75.0	C	66.06	133.6	1026.5	1979.8	3.39	14.8	1.9	19.43	15.00	4.49	.797	1.116	16.74
18	X 4	X	X 51.9	C	44.16	111.6	1139.4	2007.1	3.48	18.0	1.8	12.99	18.00	4.10	.625	.600	10.80
18	X 4	X	X 58.0	C	50.28	122.9	1184.0	2199.6	3.62	17.9	1.9	14.79	18.00	4.20	.625	.700	12.60

(50T) PLATE WEIGHT = 71.400 LBS. (1.7500 IN.)

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TABLE XIV. Properties of tee-beams cut from wide flange shapes.

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES										
SIZE DESCRIPTION			WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 04 X 13										
4 X 4	X	8.4 I-T	8.4	2.47	3.09	3.93	4.16	.280	4.06	.345
CF W 05 X 16										
5 X 5	X	9.9 I-T	9.9	2.92	3.87	6.35	5.01	.240	5.00	.360
CF W 05 X 19										
5 X 5	X	11.7 I-T	11.7	3.44	3.98	7.72	5.15	.270	5.03	.430
CF W 06 X 9										
6 X 4	X	6.2 I-T	6.2	1.81	4.22	6.53	5.90	.170	3.94	.215
CF W 06 X 12										
6 X 4	X	8.3 I-T	8.3	2.44	4.26	9.16	6.03	.230	4.00	.280
CF W 06 X 16										
6 X 4	X	10.7 I-T	10.7	3.16	4.56	12.20	6.28	.260	4.03	.405
CF W 06 X 15										
6 X 6	X	9.8 I-T	9.8	2.88	4.49	10.02	5.99	.230	5.99	.260

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES										
SIZE DESCRIPTION				WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT				LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 06 X 20											
6 X 6	X	12.6	I-T	12.6	3.71	4.75	12.95	6.20	.260	6.02	.365
CF W 06 X 25											
6 X 6	X	15.9	I-T	15.9	4.66	4.86	17.04	6.38	.320	6.08	.455
CF W 08 X 10											
4 X 4	X	4.9	T	4.9	1.44	2.97	2.13	3.95	.170	3.94	.205
8 X 4	X	7.2	I-T	7.2	2.11	5.35	14.20	7.89	.170	3.94	.205
CF W 08 X 13											
4 X 4	X	6.4	T	6.4	1.88	2.95	2.87	4.00	.230	4.00	.255
8 X 4	X	9.5	I-T	9.5	2.80	5.32	19.22	7.99	.230	4.00	.255
CF W 08 X 15											
4 X 4	X	7.4	T	7.4	2.18	3.05	3.26	4.06	.245	4.02	.315
8 X 4	X	10.8	I-T	10.8	3.17	5.51	22.19	8.11	.245	4.02	.315
CF W 08 X 18											
8 X 5 1/4	X	12.0	I-T	12.0	3.53	5.90	23.76	8.14	.230	5.25	.330

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES								
SIZE DESCRIPTION	WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT	LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 08 X 21								
8 X 5 1/4 X 13.9 T-T	13.9	4.08	6.08	27.68	8.28	.250	5.27	.400
CF W 08 X 24								
8 X 6 1/2 X 15.1 T-T	15.1	4.44	6.08	25.71	7.93	.245	6.50	.400
CF W 08 X 28								
8 X 6 1/2 X 17.7 T-T	17.7	5.20	6.15	30.99	8.06	.285	6.54	.465
CF W 08 X 31								
8 X 8 X 19.2 T-T	19.2	5.63	6.25	31.63	8.00	.285	8.00	.435
CF W 08 X 35								
8 X 8 X 21.5 T-T	21.5	6.33	6.36	35.96	8.12	.310	8.02	.495
CF W 08 X 40								
8 X 8 X 24.8 T-T	24.8	7.29	6.40	42.97	8.25	.360	8.07	.560
CF W 08 X 48								
9 X 8 X 29.5 T-T	29.5	8.68	6.63	52.26	8.50	.400	8.11	.685

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 08 X 58									
9 X 8 1/4 X 36.4	I-T	36.4	10.71	6.69	69.83	8.75	.510	8.22	.810
CF W 08 X 67									
9 X 8 1/4 X 42.0	I-T	42.0	12.34	6.86	83.89	9.00	.570	8.28	.935
CF HP 08 X 36									
8 X 8 1/4 X 23.8	HP-T	23.8	7.00	5.87	44.28	8.02	.445	8.16	.445
CF W 10 X 12									
5 X 4 X 5.9	T	5.9	1.73	3.55	4.30	4.94	.190	3.96	.210
10 X 4 X 9.1	I-T	9.1	2.67	6.37	28.21	9.87	.190	3.96	.210
CF W 10 X 15									
5 X 4 X 7.4	T	7.4	2.17	3.61	5.41	5.00	.230	4.00	.270
10 X 4 X 11.3	I-T	11.3	3.32	6.49	35.78	9.99	.230	4.00	.270
CF W 10 X 17									
5 X 4 X 8.4	T	8.4	2.46	3.72	6.02	5.06	.240	4.01	.330
10 X 4 X 12.5	I-T	12.5	3.67	6.71	40.34	10.11	.240	4.01	.330

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 10 X 19									
5 X 4 X 9.4	T	9.4	2.77	3.83	6.66	5.12	.250	4.02	.395
10 X 4 X 13.8	I-T	13.8	4.05	6.93	45.20	10.24	.250	4.02	.395
CF W 10 X 22									
10 X 5 3/4 X 15.0	I-T	15.0	4.42	7.28	47.39	10.17	.240	5.75	.360
CF W 10 X 26									
10 X 5 3/4 X 17.4	I-T	17.4	5.11	7.51	55.08	10.33	.260	5.77	.440
CF W 10 X 30									
10 X 5 3/4 X 20.2	I-T	20.2	5.95	7.59	65.54	10.47	.300	5.81	.510
CF W 10 X 33									
10 X 8 X 20.9	I-T	20.9	6.16	7.38	55.33	9.73	.290	7.96	.435
CF W 10 X 39									
10 X 8 X 24.4	I-T	24.4	7.19	7.61	64.66	9.92	.315	7.99	.530
CF W 10 X 45									
10 X 8 X 28.2	I-T	28.2	8.29	7.77	75.76	10.10	.350	8.02	.620

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION			WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 10 X 49										
10 X 10	X	29.9 T-T	29.9	8.80	7.88	74.56	9.98	.340	10.00	.560
CF W 10 X 54										
10 X 10	X	32.9 T-T	32.9	9.67	7.95	83.32	10.09	.370	10.03	.615
CF W 10 X 60										
10 X 10	X	36.9 T-T	36.9	10.86	7.99	96.68	10.22	.420	10.08	.680
CF W 10 X 68										
10 X 10 1/4	X	41.9 T-T	41.9	12.33	8.11	112.81	10.40	.470	10.13	.770
CF W 10 X 77										
11 X 10 1/4	X	47.7 T-T	47.7	14.02	8.22	132.83	10.60	.530	10.19	.870
CF W 10 X 88										
11 X 10 1/4	X	54.8 T-T	54.8	16.12	8.34	159.36	10.84	.605	10.27	.990
CF W 10 X 100										
11 X 10 1/4	X	62.4 T-T	62.4	18.37	8.49	189.34	11.10	.680	10.34	1.120

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION			WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN Y LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 10 X 112										
11 X 10 1/2 X 70.2		I-T	70.2	20.65	8.64	221.95	11.36	.755	10.42	1.250
CF HP 10 X 42										
10 X 10	X 27.5	HP-T	27.5	8.08	7.18	75.13	9.70	.415	10.08	.420
CF HP 10 X 57										
10 X 10 1/4 X 37.7		HP-T	37.7	11.10	7.31	108.71	9.99	.565	10.23	.565
CF W 12 X 14										
4 X 4	X 5.6	T	5.6	1.64	2.94	2.46	3.96	.200	3.97	.225
5 X 4	X 6.3	T	6.3	1.84	3.57	4.59	4.96	.200	3.97	.225
6 X 4	X 6.9	T	6.9	2.04	4.17	7.59	5.96	.200	3.97	.225
7 X 4	X 7.6	T	7.6	2.24	4.75	11.58	6.96	.200	3.97	.225
8 X 4	X 8.3	T	8.3	2.44	5.32	16.66	7.96	.200	3.97	.225
9 X 4	X 9.0	T	9.0	2.64	5.88	22.94	8.96	.200	3.97	.225
10 X 4	X 9.7	T	9.7	2.84	6.43	30.52	9.96	.200	3.97	.225
11 X 4	X 10.3	T	10.3	3.04	6.97	39.52	10.96	.200	3.97	.225
12 X 4	X 11.0	I-T	11.0	3.23	7.49	49.51	11.91	.200	3.97	.225
CF W 12 X 16										
4 X 4	X 6.4	T	6.4	1.88	2.99	2.80	4.00	.220	3.99	.265
5 X 4	X 7.1	T	7.1	2.10	3.62	5.22	5.00	.220	3.99	.265
6 X 4	X 7.9	T	7.9	2.32	4.23	8.62	6.00	.220	3.99	.265
7 X 4	X 8.6	T	8.6	2.54	4.82	13.14	7.00	.220	3.99	.265

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
8 X 4 X 9.4	T	9.4	2.76	5.40	18.89	8.00	.220	3.99	.265
9 X 4 X 10.1	T	10.1	2.98	5.96	26.00	9.00	.220	3.99	.265
10 X 4 X 10.9	T	10.9	3.20	6.52	34.57	10.00	.220	3.99	.265
11 X 4 X 11.6	T	11.6	3.42	7.07	44.72	11.00	.220	3.99	.265
12 X 4 X 12.4	I-T	12.4	3.64	7.61	56.51	11.99	.220	3.99	.265
CF W 12 X 19									
4 X 4 X 7.7	T	7.7	2.28	3.12	3.27	4.08	.235	4.01	.350
5 X 4 X 8.5	T	8.5	2.51	3.78	6.09	5.08	.235	4.01	.350
6 X 4 X 9.3	T	9.3	2.75	4.42	10.05	6.08	.235	4.01	.350
7 X 4 X 10.1	T	10.1	2.98	5.03	15.30	7.08	.235	4.01	.350
8 X 4 X 10.9	T	10.9	3.22	5.62	21.97	8.08	.235	4.01	.350
9 X 4 X 11.7	T	11.7	3.45	6.21	30.21	9.08	.235	4.01	.350
10 X 4 X 12.5	T	12.5	3.69	6.78	40.13	10.08	.235	4.01	.350
11 X 4 X 13.3	T	13.3	3.92	7.34	51.86	11.08	.235	4.01	.350
12 X 4 X 14.2	I-T	14.2	4.18	7.95	66.70	12.16	.235	4.01	.350
CF W 12 X 22									
4 X 4 X 9.1	T	9.1	2.68	3.19	3.82	4.16	.260	4.03	.425
5 X 4 X 10.0	T	10.0	2.94	3.87	7.07	5.16	.260	4.03	.425
6 X 4 X 10.9	T	10.9	3.20	4.51	11.65	6.16	.260	4.03	.425
7 X 4 X 11.8	T	11.8	3.46	5.13	17.71	7.16	.260	4.03	.425
8 X 4 X 12.7	T	12.7	3.72	5.74	25.41	8.16	.260	4.03	.425
9 X 4 X 13.5	T	13.5	3.98	6.33	34.90	9.16	.260	4.03	.425
10 X 4 X 14.4	T	14.4	4.24	6.91	46.31	10.16	.260	4.03	.425
11 X 4 X 15.3	T	15.3	4.50	7.49	59.81	11.16	.260	4.03	.425
12 X 4 X 16.3	I-T	16.3	4.80	8.14	78.15	12.31	.260	4.03	.425
CF W 12 X 26									
7 X 6 1/2 X 5.5	T	5.5	1.61	3.50	6.57	7.00	.230	6.49	.380

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
8 X 6 1/2 X 6.3	T	6.3	1.84	4.00	9.81	8.00	.230	6.49	.380
9 X 6 1/2 X 7.0	T	7.0	2.07	4.50	13.97	9.00	.230	6.49	.380
10 X 6 1/2 X 7.8	T	7.8	2.30	5.00	19.17	10.00	.230	6.49	.380
12 X 6 1/2 X 17.6	I-T	17.6	5.19	8.82	80.16	12.22	.230	6.49	.380
CF W 12 X 30									
7 X 6 1/2 X 6.2	T	6.2	1.82	3.50	7.43	7.00	.260	6.52	.440
8 X 6 1/2 X 7.1	T	7.1	2.08	4.00	11.09	8.00	.260	6.52	.440
9 X 6 1/2 X 8.0	T	8.0	2.34	4.50	15.80	9.00	.260	6.52	.440
10 X 6 1/2 X 8.8	T	8.8	2.60	5.00	21.67	10.00	.260	6.52	.440
12 X 6 1/2 X 20.3	I-T	20.3	5.96	8.92	93.23	12.34	.260	6.52	.440
CF W 12 X 35									
7 X 6 1/2 X 7.1	T	7.1	2.10	3.50	8.58	7.00	.300	6.56	.520
8 X 6 1/2 X 8.2	T	8.2	2.40	4.00	12.80	8.00	.300	6.56	.520
9 X 6 1/2 X 9.2	T	9.2	2.70	4.50	18.23	9.00	.300	6.56	.520
10 X 6 1/2 X 10.2	T	10.2	3.00	5.00	25.00	10.00	.300	6.56	.520
13 X 6 1/2 X 23.8	I-T	23.8	7.01	9.03	111.42	12.50	.300	6.56	.520
CF W 12 X 40									
8 X 8 X 8.0	T	8.0	2.36	4.00	12.59	8.00	.295	8.01	.515
9 X 8 X 9.0	T	9.0	2.66	4.50	17.92	9.00	.295	8.01	.515
10 X 8 X 10.0	T	10.0	2.95	5.00	24.58	10.00	.295	8.01	.515
12 X 8 X 25.5	I-T	25.5	7.49	9.00	102.84	11.94	.295	8.01	.515
CF W 12 X 45									
8 X 8 X 9.1	T	9.1	2.68	4.00	14.29	8.00	.335	8.05	.575
9 X 8 X 10.3	T	10.3	3.02	4.50	20.35	9.00	.335	8.05	.575

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION			WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
10 X 8	X 11.4	T	11.4	3.35	5.00	27.92	10.00	.335	8.05	.575
12 X 8	X 28.8	T-T	28.8	8.47	9.03	118.79	12.06	.335	8.05	.575
CF W 12 X 50										
8 X 8	X 10.1	T	10.1	2.96	4.00	15.79	8.00	.370	8.08	.640
9 X 8	X 11.3	T	11.3	3.33	4.50	22.48	9.00	.370	8.08	.640
10 X 8	X 12.6	T	12.6	3.70	5.00	30.83	10.00	.370	8.08	.640
12 X 8	X 32.1	T-T	32.1	9.44	9.11	134.61	12.19	.370	8.08	.640
CF W 12 X 53										
10 X 10	X 11.7	T	11.7	3.45	5.00	28.75	10.00	.345	10.00	.575
12 X 10	X 33.0	I-T	33.0	9.71	9.31	128.99	12.06	.345	10.00	.575
CF W 12 X 58										
10 X 10	X 12.2	T	12.2	3.60	5.00	30.00	10.00	.360	10.01	.640
12 X 10	X 35.9	I-T	35.9	10.56	9.47	140.11	12.19	.360	10.01	.640
CF W 12 X 65										
12 X 12	X 40.0	I-T	40.0	11.75	9.50	151.74	12.12	.390	12.00	.605
CF W 12 X 72										
12 X 12	X 44.4	I-T	44.4	13.05	9.58	171.45	12.25	.430	12.04	.670

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
TN X TN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 12 X 79									
12 X 12 X 48.8	I-T	48.8	14.35	9.65	191.99	12.38	.470	12.08	.735
CF W 12 X 87									
13 X 12 1/4 X 53.9	I-T	53.9	15.86	9.74	216.36	12.53	.515	12.13	.810
CF W 12 X 96									
13 X 12 1/4 X 59.3	I-T	59.3	17.44	9.89	240.86	12.71	.550	12.16	.900
CF W 12 X 106									
13 X 12 1/4 X 65.8	I-T	65.8	19.36	9.98	275.10	12.89	.610	12.22	.990
CF W 12 X 120									
13 X 12 1/4 X 75.3	I-T	75.3	22.14	10.04	329.69	13.12	.710	12.32	1.105
CF W 12 X 136									
13 X 12 1/2 X 85.4	I-T	85.4	25.11	10.22	387.02	13.41	.790	12.40	1.250
CF W 12 X 152									
14 X 12 1/2 X 95.8	I-T	95.8	28.18	10.40	450.11	13.71	.870	12.48	1.400

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	O	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 12 X 170									
14 X 12 1/2 X 107.4	I-T	107.4	31.58	10.59	524.90	14.03	.960	12.57	1.560
CF W 12 X 190									
14 X 12 3/4 X 120.3	I-T	120.3	35.39	10.79	614.57	14.38	1.060	12.67	1.735
CF W 12 X 210									
15 X 12 3/4 X 134.0	I-T	134.0	39.42	10.94	718.14	14.71	1.180	12.79	1.900
CF W 12 X 230									
15 X 13 X 147.5	I-T	147.5	43.37	11.12	824.97	15.05	1.285	12.90	2.070
CF W 12 X 252									
15 X 13 X 161.9	I-T	161.9	47.62	11.31	947.00	15.41	1.395	13.01	2.250
CF W 12 X 279									
16 X 13 1/4 X 180.0	I-T	180.0	52.93	11.55	1110.33	15.85	1.530	13.14	2.470
CF W 12 X 305									
16 X 13 1/4 X 196.9	I-T	196.9	57.93	11.85	1274.08	16.32	1.625	13.24	2.705

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES								
SIZE DESCRIPTION	WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT	LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 12 X 336								
17 X 13 1/2 X 218.2 I-T	218.2	64.16	12.12	1496.04	16.82	1.775	13.39	2.955
CF HP 12 X 53								
12 X 12 X 34.6 HP-T	34.6	10.17	8.71	141.18	11.78	.435	12.05	.435
CF HP 12 X 63								
12 X 12 1/4 X 41.2 HP-T	41.2	12.13	8.79	172.11	11.94	.515	12.13	.515
CF HP 12 X 74								
12 X 12 1/4 X 49.0 HP-T	49.0	14.42	8.89	209.78	12.13	.605	12.22	.610
CF HP 12 X 84								
12 X 12 1/4 X 55.6 HP-T	55.6	16.36	8.96	243.42	12.28	.685	12.30	.685
CF HP 13 X 60								
13 X 13 X 39.1 HP-T	39.1	11.49	9.28	180.49	12.54	.460	12.90	.460
CF HP 13 X 73								
13 X 13 X 48.4 HP-T	48.4	14.23	9.38	229.83	12.75	.565	13.01	.565

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION	WT/FT	A	CTOE	IO	O	TW	WF	TF
IN X IN X LBS/FT	LBS	IN2	IN	IN4	IN	IN	IN	IN
CF HP 13 X 87								
13 X 13 X 57.4 HP-T	57.4	16.88	9.48	279.85	12.95	.665	13.11	.665
CF HP 13 X 100								
13 X 13 1/4 X 66.6 HP-T	66.6	19.58	9.59	332.96	13.15	.765	13.21	.765
CF W 14 X 22								
5 X 5 X 9.2 T	9.2	2.72	3.77	5.61	4.87	.230	5.00	.335
6 X 5 X 10.0 T	10.0	2.95	4.44	9.50	5.87	.230	5.00	.335
7 X 5 X 10.8 T	10.8	3.18	5.08	14.71	6.87	.230	5.00	.335
8 X 5 X 11.6 T	11.6	3.41	5.70	21.40	7.87	.230	5.00	.335
9 X 5 X 12.4 T	12.4	3.64	6.31	29.71	8.87	.230	5.00	.335
10 X 5 X 13.2 T	13.2	3.87	6.90	39.76	9.87	.230	5.00	.335
11 X 5 X 13.9 T	13.9	4.10	7.49	51.68	10.87	.230	5.00	.335
12 X 5 X 14.7 T	14.7	4.33	8.06	65.60	11.87	.230	5.00	.335
13 X 5 X 15.5 T	15.5	4.56	8.63	81.64	12.87	.230	5.00	.335
14 X 5 X 16.2 I-T	16.2	4.76	9.12	97.41	13.74	.230	5.00	.335
CF W 14 X 26								
5 X 5 X 11.1 T	11.1	3.27	3.87	6.60	4.96	.255	5.03	.420
6 X 5 X 12.0 T	12.0	3.52	4.55	11.13	5.96	.255	5.03	.420
7 X 5 X 12.8 T	12.8	3.78	5.21	17.22	6.96	.255	5.03	.420
8 X 5 X 13.7 T	13.7	4.03	5.85	25.03	7.96	.255	5.03	.420
9 X 5 X 14.6 T	14.6	4.29	6.47	34.72	8.96	.255	5.03	.420
10 X 5 X 15.4 T	15.4	4.54	7.08	46.44	9.96	.255	5.03	.420
11 X 5 X 16.3 T	16.3	4.80	7.68	60.34	10.96	.255	5.03	.420
12 X 5 X 17.2 T	17.2	5.05	8.26	76.55	11.96	.255	5.03	.420

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	O	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
13 X 5 X 18.0	T	18.0	5.31	8.84	95.22	12.96	.255	5.03	.420
14 X 5 X 18.9	I-T	18.9	5.55	9.39	115.47	13.91	.255	5.03	.420
CF W 14 X 30									
7 X 6 3/4 X 14.8	T	14.8	4.36	5.33	18.88	6.92	.270	6.73	.385
8 X 6 3/4 X 15.7	T	15.7	4.63	5.99	27.53	7.92	.270	6.73	.385
9 X 6 3/4 X 16.6	T	16.6	4.90	6.63	38.28	8.92	.270	6.73	.385
10 X 6 3/4 X 17.6	T	17.6	5.17	7.26	51.31	9.92	.270	6.73	.385
11 X 6 3/4 X 18.5	T	18.5	5.44	7.87	66.76	10.92	.270	6.73	.385
12 X 6 3/4 X 19.4	T	19.4	5.71	8.47	84.81	11.92	.270	6.73	.385
13 X 6 3/4 X 20.3	T	20.3	5.98	9.07	105.59	12.92	.270	6.73	.385
14 X 6 3/4 X 21.2	I-T	21.2	6.22	9.61	127.26	13.84	.270	6.73	.385
CF W 14 X 34									
7 X 6 3/4 X 16.8	T	16.8	4.93	5.44	20.84	6.99	.285	6.75	.455
8 X 6 3/4 X 17.7	T	17.7	5.22	6.12	30.38	7.99	.285	6.75	.455
9 X 6 3/4 X 18.7	T	18.7	5.50	6.78	42.24	8.99	.285	6.75	.455
10 X 6 3/4 X 19.7	T	19.7	5.79	7.42	56.60	9.99	.285	6.75	.455
11 X 6 3/4 X 20.6	T	20.6	6.07	8.05	73.65	10.99	.285	6.75	.455
12 X 6 3/4 X 21.6	T	21.6	6.36	8.66	93.55	11.99	.285	6.75	.455
13 X 6 3/4 X 22.6	T	22.6	6.64	9.27	116.47	12.99	.285	6.75	.455
14 X 6 3/4 X 23.5	I-T	23.5	6.92	9.86	142.30	13.98	.285	6.75	.455
CF W 14 X 38									
7 X 6 3/4 X 18.7	T	18.7	5.51	5.50	23.21	7.05	.310	6.77	.515
8 X 6 3/4 X 19.8	T	19.8	5.82	6.18	33.79	8.05	.310	6.77	.515
9 X 6 3/4 X 20.9	T	20.9	6.13	6.84	46.94	9.05	.310	6.77	.515
10 X 6 3/4 X 21.9	T	21.9	6.44	7.49	62.86	10.05	.310	6.77	.515
11 X 6 3/4 X 23.0	T	23.0	6.75	8.12	81.76	11.05	.310	6.77	.515
12 X 6 3/4 X 24.0	T	24.0	7.06	8.74	103.81	12.05	.310	6.77	.515

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
13 X 6 3/4 X 25.1	T	25.1	7.37	9.35	129.20	13.05	.310	6.77	.515
14 X 6 3/4 X 26.2	I-T	26.2	7.70	9.99	159.65	14.10	.310	6.77	.515
CF W 14 X 43									
8 X 8 X 8.3	T	8.3	2.44	4.00	13.01	8.00	.305	8.00	.530
9 X 8 X 9.3	T	9.3	2.75	4.50	18.53	9.00	.305	8.00	.530
10 X 8 X 10.4	T	10.4	3.05	5.00	25.42	10.00	.305	8.00	.530
14 X 8 X 28.0	I-T	28.0	8.24	10.08	153.68	13.66	.305	8.00	.530
CF W 14 X 48									
8 X 8 X 9.2	T	9.2	2.72	4.00	14.51	8.00	.340	8.03	.595
9 X 8 X 10.4	T	10.4	3.06	4.50	20.66	9.00	.340	8.03	.595
10 X 8 X 11.6	T	11.6	3.40	5.00	28.33	10.00	.340	8.03	.595
14 X 8 X 31.5	I-T	31.5	9.26	10.15	175.23	13.79	.340	8.03	.595
CF W 14 X 53									
8 X 8 X 10.1	T	10.1	2.96	4.00	15.79	8.00	.370	8.06	.660
9 X 8 X 11.3	T	11.3	3.33	4.50	22.48	9.00	.370	8.06	.660
10 X 8 X 12.6	T	12.6	3.70	5.00	30.83	10.00	.370	8.06	.660
14 X 8 X 34.8	I-T	34.8	10.23	10.25	195.72	13.92	.370	8.06	.660
CF W 14 X 61									
10 X 10 X 12.8	T	12.8	3.75	5.00	31.25	10.00	.375	10.00	.645
14 X 10 X 38.8	I-T	38.8	11.41	10.55	208.15	13.89	.375	10.00	.645

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION IN X TN X LBS/FT			WT/FT LBS	A IN2	CTOE IN	IO IN4	D IN	TW IN	WF IN	TF IN
CF W 14 X 68										
10 X 10	X 14.1	T	14.1	4.15	5.00	34.58	10.00	.415	10.04	.720
14 X 10	X 43.4	I-T	43.4	12.75	10.64	236.38	14.04	.415	10.04	.720
CF W 14 X 74										
10 X 10	X 15.3	T	15.3	4.50	5.00	37.50	10.00	.450	10.07	.785
14 X 10	X 47.4	I-T	47.4	13.93	10.71	261.93	14.17	.450	10.07	.785
CF W 14 X 82										
10 X 10 1/4	X 17.3	T	17.3	5.10	5.00	42.50	10.00	.510	10.13	.855
14 X 10 1/4	X 52.8	I-T	52.8	15.52	10.72	300.06	14.31	.510	10.13	.855
CF W 14 X 90										
14 X 14 1/2	X 55.0	I-T	55.0	16.17	11.13	270.42	14.02	.440	14.52	.710
CF W 14 X 99										
14 X 14 1/2	X 60.7	I-T	60.7	17.85	11.20	304.42	14.16	.485	14.57	.780
CF W 14 X 109										
14 X 14 1/2	X 66.7	I-T	66.7	19.63	11.31	339.30	14.32	.525	14.61	.860

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
TN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 14 X 120									
14 X 14 3/4 X 74.0	T-T	74.0	21.78	11.35	388.21	14.48	.590	14.67	.940
CF W 14 X 132									
15 X 14 3/4 X 81.5	T-T	81.5	23.96	11.46	436.47	14.66	.645	14.73	1.030
CF W 14 X 145									
15 X 15 1/2 X 89.1	T-T	89.1	26.20	11.61	474.85	14.78	.680	15.50	1.090
CF W 14 X 159									
15 X 15 1/2 X 97.9	T-T	97.9	28.80	11.71	535.71	14.98	.745	15.57	1.190
CF W 14 X 176									
15 X 15 3/4 X 109.0	T-T	109.0	32.05	11.82	616.82	15.22	.830	15.65	1.310
CF W 14 X 193									
15 X 15 3/4 X 119.4	T-T	119.4	35.12	12.01	691.39	15.48	.890	15.71	1.440
CF W 14 X 211									
16 X 15 3/4 X 131.0	T-T	131.0	38.52	12.11	785.36	15.72	.980	15.80	1.560

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION			WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 14 X 233										
16 X 16	X 145.0	T-T	145.0	42.65	12.30	900.08	16.04	1.070	15.89	1.720
CF W 14 X 257										
16 X 16	X 160.7	T-T	160.7	47.26	12.48	1037.46	16.38	1.175	16.00	1.890
CF W 14 X 283										
17 X 16	X 177.7	T-T	177.7	52.27	12.67	1197.10	16.74	1.290	16.11	2.070
CF W 14 X 311										
17 X 16 1/4	X 196.0	T-T	196.0	57.63	12.88	1378.29	17.12	1.410	16.23	2.260
CF W 14 X 342										
18 X 16 1/4	X 216.3	T-T	216.3	63.62	13.11	1593.57	17.54	1.540	16.36	2.470
CF W 14 X 370										
18 X 16 1/2	X 234.9	T-T	234.9	69.08	13.31	1802.20	17.92	1.655	16.48	2.660
CF W 14 X 398										
18 X 16 1/2	X 253.4	T-T	253.4	74.54	13.51	2023.02	18.29	1.770	16.59	2.845

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 14 X 426									
19 X 16 3/4 X 271.9	I-T	271.9	79.98	13.73	2254.41	18.67	1.875	16.70	3.035
CF W 14 X 455									
19 X 16 3/4 X 292.1	I-T	292.1	85.90	13.89	2522.59	19.02	2.015	16.84	3.210
CF W 14 X 500									
20 X 17 X 322.3	I-T	322.3	94.79	14.20	2949.14	19.60	2.190	17.01	3.500
CF W 14 X 550									
20 X 17 1/4 X 356.3	I-T	356.3	104.78	14.56	3467.57	20.24	2.380	17.20	3.820
CF W 14 X 605									
21 X 17 1/2 X 394.2	I-T	394.2	115.94	14.92	4096.02	20.92	2.595	17.42	4.160
CF W 14 X 665									
22 X 17 3/4 X 436.0	I-T	436.0	128.23	15.29	4848.14	21.64	2.830	17.65	4.520
CF W 14 X 730									
22 X 18 X 481.4	I-T	481.4	141.60	15.71	5740.55	22.42	3.070	17.89	4.910

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION			WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
CF HP 14 X 73										
14 X 14 1/2 X	47.5	HP-T	47.5	13.98	10.14	256.30	13.61	.505	14.59	.505
CF HP 14 X 89										
14 X 14 3/4 X	58.4	HP-T	58.4	17.16	10.25	323.18	13.83	.615	14.70	.615
CF HP 14 X 102										
14 X 14 3/4 X	67.3	HP-T	67.3	19.80	10.34	381.07	14.01	.705	14.79	.705
CF HP 14 X 117										
14 X 15 X	77.4	HP-T	77.4	22.77	10.44	448.86	14.21	.805	14.89	.805
CF W 16 X 26										
6 X 5 1/2 X	11.1	T	11.1	3.27	4.44	10.29	5.85	.250	5.50	.345
7 X 5 1/2 X	12.0	T	12.0	3.52	5.09	15.99	6.85	.250	5.50	.345
8 X 5 1/2 X	12.8	T	12.8	3.77	5.72	23.32	7.85	.250	5.50	.345
9 X 5 1/2 X	13.7	T	13.7	4.02	6.34	32.42	8.85	.250	5.50	.345
10 X 5 1/2 X	14.5	T	14.5	4.27	6.94	43.44	9.85	.250	5.50	.345
11 X 5 1/2 X	15.4	T	15.4	4.52	7.53	56.52	10.85	.250	5.50	.345
12 X 5 1/2 X	16.2	T	16.2	4.77	8.10	71.80	11.85	.250	5.50	.345
13 X 5 1/2 X	17.1	T	17.1	5.02	8.68	89.41	12.85	.250	5.50	.345
14 X 5 1/2 X	17.9	T	17.9	5.27	9.24	109.48	13.85	.250	5.50	.345
15 X 5 1/2 X	18.8	T	18.8	5.52	9.80	132.15	14.85	.250	5.50	.345
16 X 5 1/2 X	19.5	I-T	19.5	5.73	10.27	153.43	15.69	.250	5.50	.345

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 16 X 31									
6 X 5 1/2 X 13.4	T	13.4	3.94	4.58	12.08	5.94	.275	5.53	.440
7 X 5 1/2 X 14.3	T	14.3	4.22	5.25	18.74	6.94	.275	5.53	.440
8 X 5 1/2 X 15.3	T	15.3	4.49	5.90	27.29	7.94	.275	5.53	.440
9 X 5 1/2 X 16.2	T	16.2	4.77	6.53	37.92	8.94	.275	5.53	.440
10 X 5 1/2 X 17.1	T	17.1	5.04	7.15	50.79	9.94	.275	5.53	.440
11 X 5 1/2 X 18.1	T	18.1	5.32	7.75	66.06	10.94	.275	5.53	.440
12 X 5 1/2 X 19.0	T	19.0	5.59	8.34	83.88	11.94	.275	5.53	.440
13 X 5 1/2 X 20.0	T	20.0	5.87	8.93	104.41	12.94	.275	5.53	.440
14 X 5 1/2 X 20.9	T	20.9	6.14	9.51	127.79	13.94	.275	5.53	.440
15 X 5 1/2 X 21.8	T	21.8	6.42	10.08	154.18	14.94	.275	5.53	.440
16 X 5 1/2 X 22.7	I-T	22.7	6.68	10.61	181.85	15.88	.275	5.53	.440
CF W 16 X 36									
7 X 7 X 16.7	T	16.7	4.92	5.36	20.85	6.93	.295	6.99	.430
8 X 7 X 17.7	T	17.7	5.22	6.03	30.45	7.93	.295	6.99	.430
9 X 7 X 18.7	T	18.7	5.51	6.68	42.39	8.93	.295	6.99	.430
10 X 7 X 19.7	T	19.7	5.81	7.32	56.86	9.93	.295	6.99	.430
11 X 7 X 20.7	T	20.7	6.10	7.94	74.05	10.93	.295	6.99	.430
12 X 7 X 21.7	T	21.7	6.40	8.55	94.12	11.93	.295	6.99	.430
13 X 7 X 22.7	T	22.7	6.69	9.15	117.25	12.93	.295	6.99	.430
14 X 7 X 23.8	T	23.8	6.99	9.74	143.59	13.93	.295	6.99	.430
15 X 7 X 24.8	T	24.8	7.28	10.33	173.32	14.93	.295	6.99	.430
16 X 7 X 25.7	I-T	25.7	7.56	10.87	204.15	15.86	.295	6.99	.430
CF W 16 X 40									
7 X 7 X 18.8	T	18.8	5.51	5.49	22.63	7.01	.305	7.00	.505
8 X 7 X 19.8	T	19.8	5.82	6.18	33.04	8.01	.305	7.00	.505
9 X 7 X 20.8	T	20.8	6.12	6.85	46.00	9.01	.305	7.00	.505

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION	WT/FT	A	CTOE	IO	D	TW	WF	TF
TN X TN X LBS/FT	LBS	IN2	IN	IN4	IN	IN	IN	IN
10 X 7 X 21.9 T	21.9	6.43	7.50	61.70	10.01	.305	7.00	.505
11 X 7 X 22.9 T	22.9	6.73	8.14	80.36	11.01	.305	7.00	.505
12 X 7 X 23.9 T	23.9	7.04	8.76	102.14	12.01	.305	7.00	.505
13 X 7 X 25.0 T	25.0	7.34	9.38	127.25	13.01	.305	7.00	.505
14 X 7 X 26.0 T	26.0	7.65	9.98	155.84	14.01	.305	7.00	.505
15 X 7 X 27.0 T	27.0	7.95	10.58	188.10	15.01	.305	7.00	.505
16 X 7 X 28.1 T-T	28.1	8.26	11.18	224.39	16.01	.305	7.00	.505
OF W 16 X 45								
7 X 7 X 21.1 T	21.1	6.22	5.51	25.89	7.07	.345	7.04	.565
8 X 7 X 22.3 T	22.3	6.56	6.19	37.72	8.07	.345	7.04	.565
9 X 7 X 23.5 T	23.5	6.91	6.86	52.43	9.07	.345	7.04	.565
10 X 7 X 24.7 T	24.7	7.25	7.51	70.25	10.07	.345	7.04	.565
11 X 7 X 25.8 T	25.8	7.60	8.14	91.40	11.07	.345	7.04	.565
12 X 7 X 27.0 T	27.0	7.94	8.77	116.09	12.07	.345	7.04	.565
13 X 7 X 28.2 T	28.2	8.29	9.38	144.52	13.07	.345	7.04	.565
14 X 7 X 29.3 T	29.3	8.63	9.99	176.90	14.07	.345	7.04	.565
15 X 7 X 30.5 T	30.5	8.98	10.59	213.42	15.07	.345	7.04	.565
16 X 7 X 31.8 T-T	31.8	9.34	11.21	257.09	16.13	.345	7.04	.565
OF W 16 X 50								
7 X 7 X 23.5 T	23.5	6.92	5.54	29.04	7.13	.380	7.07	.630
8 X 7 X 24.8 T	24.8	7.30	6.23	42.23	8.13	.380	7.07	.630
9 X 7 X 26.1 T	26.1	7.68	6.90	58.61	9.13	.380	7.07	.630
10 X 7 X 27.4 T	27.4	8.06	7.55	78.45	10.13	.380	7.07	.630
11 X 7 X 28.7 T	28.7	8.44	8.19	101.98	11.13	.380	7.07	.630
12 X 7 X 30.0 T	30.0	8.82	8.81	129.45	12.13	.380	7.07	.630
13 X 7 X 31.3 T	31.3	9.20	9.43	161.07	13.13	.380	7.07	.630
14 X 7 X 32.6 T	32.6	9.58	10.03	197.06	14.13	.380	7.07	.630
15 X 7 X 33.9 T	33.9	9.96	10.63	237.65	15.13	.380	7.07	.630
16 X 7 X 35.2 T-T	35.2	10.39	11.30	289.30	16.26	.380	7.07	.630

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 16 X 57									
7 X 7 X 26.8	T	26.8	7.89	5.58	33.54	7.22	.430	7.12	.715
8 X 7 X 28.3	T	28.3	8.32	6.26	48.64	8.22	.430	7.12	.715
9 X 7 X 29.7	T	29.7	8.75	6.93	67.39	9.22	.430	7.12	.715
10 X 7 X 31.2	T	31.2	9.18	7.58	90.06	10.22	.430	7.12	.715
11 X 7 X 32.7	T	32.7	9.61	8.22	116.94	11.22	.430	7.12	.715
12 X 7 X 34.1	T	34.1	10.04	8.85	148.28	12.22	.430	7.12	.715
13 X 7 X 35.6	T	35.6	10.47	9.46	184.35	13.22	.430	7.12	.715
14 X 7 X 37.0	T	37.0	10.90	10.07	225.39	14.22	.430	7.12	.715
15 X 7 X 38.5	T	38.5	11.33	10.67	271.65	15.22	.430	7.12	.715
16 X 7 X 40.3	I-T	40.3	11.85	11.39	335.23	16.43	.430	7.12	.715
CF W 16 X 67									
10 X 10 1/4 X 13.4	T	13.4	3.95	5.00	32.92	10.00	.395	10.24	.665
16 X 10 1/4 X 44.2	I-T	44.2	12.99	12.11	342.86	16.33	.395	10.24	.665
CF W 16 X 77									
17 X 10 1/4 X 51.0	I-T	51.0	15.00	12.19	404.08	16.52	.455	10.30	.760
CF W 16 X 89									
17 X 10 1/4 X 59.2	I-T	59.2	17.40	12.30	480.24	16.75	.525	10.37	.875
CF W 16 X 100									
17 X 10 1/2 X 66.7	I-T	66.7	19.62	12.43	552.31	16.97	.585	10.43	.985

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 18 X 35									
6 X 6 X 14.2	T	14.2	4.18	4.50	12.53	5.85	.300	6.00	.425
7 X 6 X 15.2	T	15.2	4.48	5.16	19.55	6.85	.300	6.00	.425
8 X 6 X 16.2	T	16.2	4.78	5.81	28.59	7.85	.300	6.00	.425
9 X 6 X 17.3	T	17.3	5.08	6.43	39.84	8.85	.300	6.00	.425
10 X 6 X 18.3	T	18.3	5.38	7.05	53.49	9.85	.300	6.00	.425
11 X 6 X 19.3	T	19.3	5.68	7.65	69.70	10.85	.300	6.00	.425
12 X 6 X 20.3	T	20.3	5.98	8.24	88.65	11.85	.300	6.00	.425
13 X 6 X 21.3	T	21.3	6.28	8.82	110.50	12.85	.300	6.00	.425
14 X 6 X 22.4	T	22.4	6.58	9.40	135.41	13.85	.300	6.00	.425
15 X 6 X 23.4	T	23.4	6.88	9.97	163.54	14.85	.300	6.00	.425
16 X 6 X 24.4	T	24.4	7.18	10.53	195.05	15.85	.300	6.00	.425
17 X 6 X 25.4	T	25.4	7.48	11.09	230.09	16.85	.300	6.00	.425
18 X 6 X 26.3	I-T	26.3	7.73	11.56	262.78	17.70	.300	6.00	.425
CF W 18 X 40									
6 X 6 X 16.5	T	16.5	4.87	4.64	14.08	5.95	.315	6.02	.525
7 X 6 X 17.6	T	17.6	5.18	5.33	21.93	6.95	.315	6.02	.525
8 X 6 X 18.7	T	18.7	5.50	6.00	32.05	7.95	.315	6.02	.525
9 X 6 X 19.8	T	19.8	5.81	6.64	44.65	8.95	.315	6.02	.525
10 X 6 X 20.8	T	20.8	6.13	7.28	59.92	9.95	.315	6.02	.525
11 X 6 X 21.9	T	21.9	6.44	7.90	78.07	10.95	.315	6.02	.525
12 X 6 X 23.0	T	23.0	6.76	8.51	99.27	11.95	.315	6.02	.525
13 X 6 X 24.0	T	24.0	7.07	9.10	123.70	12.95	.315	6.02	.525
14 X 6 X 25.1	T	25.1	7.39	9.69	151.54	13.95	.315	6.02	.525
15 X 6 X 26.2	T	26.2	7.70	10.28	182.96	14.95	.315	6.02	.525
16 X 6 X 27.3	T	27.3	8.02	10.85	218.14	15.95	.315	6.02	.525
17 X 6 X 28.3	T	28.3	8.33	11.42	257.24	16.95	.315	6.02	.525
18 X 6 X 29.3	I-T	29.3	8.63	11.96	298.17	17.90	.315	6.02	.525

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 18 X 46									
6 X 6 X 19.1	T	19.1	5.62	4.68	16.48	6.03	.360	6.06	.605
7 X 6 X 20.3	T	20.3	5.98	5.37	25.59	7.03	.360	6.06	.605
8 X 6 X 21.6	T	21.6	6.34	6.03	37.31	8.03	.360	6.06	.605
9 X 6 X 22.8	T	22.8	6.70	6.68	51.89	9.03	.360	6.06	.605
10 X 6 X 24.0	T	24.0	7.06	7.32	69.55	10.03	.360	6.06	.605
11 X 6 X 25.2	T	25.2	7.42	7.94	90.51	11.03	.360	6.06	.605
12 X 6 X 26.4	T	26.4	7.78	8.55	114.98	12.03	.360	6.06	.605
13 X 6 X 27.7	T	27.7	8.14	9.15	143.18	13.03	.360	6.06	.605
14 X 6 X 28.9	T	28.9	8.50	9.74	175.29	14.03	.360	6.06	.605
15 X 6 X 30.1	T	30.1	8.86	10.32	211.53	15.03	.360	6.06	.605
16 X 6 X 31.3	T	31.3	9.22	10.90	252.08	16.03	.360	6.06	.605
17 X 6 X 32.6	T	32.6	9.58	11.47	297.13	17.03	.360	6.06	.605
18 X 6 X 33.8	I-T	33.8	9.95	12.05	348.45	18.06	.360	6.06	.605
CF W 18 X 50									
8 X 7 1/2 X 23.5	T	23.5	6.91	6.18	38.27	8.00	.355	7.50	.570
9 X 7 1/2 X 24.7	T	24.7	7.26	6.86	53.39	9.00	.355	7.50	.570
10 X 7 1/2 X 25.9	T	25.9	7.62	7.52	71.75	10.00	.355	7.50	.570
11 X 7 1/2 X 27.1	T	27.1	7.97	8.16	93.57	11.00	.355	7.50	.570
12 X 7 1/2 X 28.3	T	28.3	8.33	8.79	119.07	12.00	.355	7.50	.570
13 X 7 1/2 X 29.5	T	29.5	8.68	9.41	148.48	13.00	.355	7.50	.570
14 X 7 1/2 X 30.7	T	30.7	9.04	10.02	182.00	14.00	.355	7.50	.570
15 X 7 1/2 X 31.9	T	31.9	9.39	10.62	219.84	15.00	.355	7.50	.570
16 X 7 1/2 X 33.1	T	33.1	9.75	11.22	262.18	16.00	.355	7.50	.570
17 X 7 1/2 X 34.4	T	34.4	10.10	11.81	309.24	17.00	.355	7.50	.570
18 X 7 1/2 X 35.6	I-T	35.6	10.46	12.39	360.93	17.99	.355	7.50	.570

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
Check the source to verify that this is the current version before use.

TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 18 X 55									
8 X 7 1/2 X 26.0	T	26.0	7.64	6.21	42.63	8.06	.390	7.53	.630
9 X 7 1/2 X 27.3	T	27.3	8.03	6.89	59.38	9.06	.390	7.53	.630
10 X 7 1/2 X 28.6	T	28.6	8.42	7.55	79.71	10.06	.390	7.53	.630
11 X 7 1/2 X 30.0	T	30.0	8.81	8.19	103.87	11.06	.390	7.53	.630
12 X 7 1/2 X 31.3	T	31.3	9.20	8.82	132.10	12.06	.390	7.53	.630
13 X 7 1/2 X 32.6	T	32.6	9.59	9.44	164.64	13.06	.390	7.53	.630
14 X 7 1/2 X 33.9	T	33.9	9.98	10.05	201.71	14.06	.390	7.53	.630
15 X 7 1/2 X 35.3	T	35.3	10.37	10.66	243.54	15.06	.390	7.53	.630
16 X 7 1/2 X 36.6	T	36.6	10.76	11.25	290.35	16.06	.390	7.53	.630
17 X 7 1/2 X 37.9	T	37.9	11.15	11.84	342.36	17.06	.390	7.53	.630
18 X 7 1/2 X 39.3	I-T	39.3	11.56	12.46	403.10	18.11	.390	7.53	.630
CF W 18 X 60									
8 X 7 1/2 X 28.3	T	28.3	8.33	6.27	46.38	8.12	.415	7.56	.695
9 X 7 1/2 X 29.7	T	29.7	8.75	6.95	64.53	9.12	.415	7.56	.695
10 X 7 1/2 X 31.2	T	31.2	9.16	7.61	86.56	10.12	.415	7.56	.695
11 X 7 1/2 X 32.6	T	32.6	9.58	8.26	112.72	11.12	.415	7.56	.695
12 X 7 1/2 X 34.0	T	34.0	9.99	8.90	143.28	12.12	.415	7.56	.695
13 X 7 1/2 X 35.4	T	35.4	10.41	9.52	178.50	13.12	.415	7.56	.695
14 X 7 1/2 X 36.8	T	36.8	10.82	10.14	218.62	14.12	.415	7.56	.695
15 X 7 1/2 X 38.2	T	38.2	11.24	10.75	263.89	15.12	.415	7.56	.695
16 X 7 1/2 X 39.6	T	39.6	11.65	11.34	314.53	16.12	.415	7.56	.695
17 X 7 1/2 X 41.0	T	41.0	12.07	11.94	370.78	17.12	.415	7.56	.695
18 X 7 1/2 X 42.6	I-T	42.6	12.53	12.59	440.73	18.24	.415	7.56	.695
CF W 18 X 65									
9 X 7 1/2 X 30.7	T	30.7	9.03	6.29	50.79	8.18	.450	7.59	.750
9 X 7 1/2 X 32.2	T	32.2	9.48	6.97	70.58	9.18	.450	7.59	.750

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	O	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
10 X 7 1/2 X 33.8	T	33.8	9.93	7.63	94.57	10.18	.450	7.59	.750
11 X 7 1/2 X 35.3	T	35.3	10.38	8.28	123.05	11.18	.450	7.59	.750
12 X 7 1/2 X 36.8	T	36.8	10.83	8.91	156.30	12.18	.450	7.59	.750
13 X 7 1/2 X 38.4	T	38.4	11.28	9.54	194.60	13.18	.450	7.59	.750
14 X 7 1/2 X 39.9	T	39.9	11.73	10.15	238.23	14.18	.450	7.59	.750
15 X 7 1/2 X 41.4	T	41.4	12.18	10.76	287.43	15.18	.450	7.59	.750
16 X 7 1/2 X 43.0	T	43.0	12.63	11.36	342.46	16.18	.450	7.59	.750
17 X 7 1/2 X 44.5	T	44.5	13.08	11.95	403.58	17.18	.450	7.59	.750
18 X 7 1/2 X 46.3	I-T	46.3	13.61	12.64	483.51	18.35	.450	7.59	.750
OF W 18 X 71									
8 X 7 3/4 X 33.5	T	33.5	9.86	6.30	56.31	8.24	.495	7.64	.810
9 X 7 3/4 X 35.2	T	35.2	10.35	6.97	78.11	9.24	.495	7.64	.810
10 X 7 3/4 X 36.9	T	36.9	10.85	7.63	104.52	10.24	.495	7.64	.810
11 X 7 3/4 X 38.6	T	38.6	11.34	8.27	135.84	11.24	.495	7.64	.810
12 X 7 3/4 X 40.3	T	40.3	11.84	8.91	172.41	12.24	.495	7.64	.810
13 X 7 3/4 X 41.9	T	41.9	12.33	9.53	214.50	13.24	.495	7.64	.810
14 X 7 3/4 X 43.6	T	43.6	12.83	10.14	262.42	14.24	.495	7.64	.810
15 X 7 3/4 X 45.3	T	45.3	13.32	10.75	316.45	15.24	.495	7.64	.810
16 X 7 3/4 X 47.0	T	47.0	13.82	11.35	376.88	16.24	.495	7.64	.810
17 X 7 3/4 X 48.7	T	48.7	14.31	11.94	443.97	17.24	.495	7.64	.810
18 X 7 3/4 X 50.7	T-T	50.7	14.93	12.66	536.43	18.47	.495	7.64	.810
OF W 18 X 76									
18 X 11 X 50.8	T-T	50.8	14.95	13.33	501.00	18.21	.425	11.04	.680
OF W 18 X 86									
18 X 11 X 57.8	T-T	57.8	17.00	13.43	578.49	18.39	.480	11.09	.770

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:40Z
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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 18 X 97									
19 X 11 1/4 X 65.2	I-T	65.2	19.18	13.56	662.82	18.59	.535	11.15	.870
CF W 18 X 106									
19 X 11 1/4 X 71.5	I-T	71.5	21.02	13.58	738.57	18.73	.590	11.20	.940
CF W 18 X 119									
19 X 11 1/4 X 80.5	I-T	80.5	23.67	13.74	847.07	18.97	.655	11.27	1.060
CF W 21 X 44									
6 X 6 1/2 X 16.9	T	16.9	4.98	4.80	18.08	6.33	.350	6.50	.450
8 X 6 1/2 X 19.3	T	19.3	5.68	6.08	38.95	8.33	.350	6.50	.450
10 X 6 1/2 X 21.7	T	21.7	6.38	7.31	70.45	10.33	.350	6.50	.450
12 X 6 1/2 X 24.1	T	24.1	7.08	8.49	114.21	12.33	.350	6.50	.450
14 X 6 1/2 X 26.5	T	26.5	7.78	9.63	171.77	14.33	.350	6.50	.450
16 X 6 1/2 X 28.8	T	28.8	8.48	10.76	244.61	16.33	.350	6.50	.450
18 X 6 1/2 X 31.2	T	31.2	9.18	11.86	334.20	18.33	.350	6.50	.450
21 X 6 1/2 X 34.0	I-T	34.0	10.00	13.13	461.62	20.66	.350	6.50	.450
CF W 21 X 50									
6 X 6 1/2 X 19.5	T	19.5	5.73	4.90	20.54	6.42	.380	6.53	.535
8 X 6 1/2 X 22.1	T	22.1	6.49	6.21	44.12	8.42	.380	6.53	.535
10 X 6 1/2 X 24.6	T	24.6	7.25	7.45	79.70	10.42	.380	6.53	.535
12 X 6 1/2 X 27.2	T	27.2	8.01	8.65	129.07	12.42	.380	6.53	.535
14 X 6 1/2 X 29.8	T	29.8	8.77	9.81	193.93	14.42	.380	6.53	.535

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
16 X 6 1/2 X 32.4	T	32.4	9.53	10.95	275.94	16.42	.380	6.53	.535
19 X 6 1/2 X 35.0	T	35.0	10.29	12.07	376.70	18.42	.380	6.53	.535
21 X 6 1/2 X 38.1	T-T	38.1	11.21	13.39	525.60	20.83	.380	6.53	.535
CF W 21 X 57									
7 X 6 1/2 X 22.6	T	22.6	6.64	5.03	23.30	6.53	.405	6.56	.650
9 X 6 1/2 X 25.3	T	25.3	7.45	6.38	49.86	8.53	.405	6.56	.650
11 X 6 1/2 X 28.1	T	28.1	8.26	7.66	89.90	10.53	.405	6.56	.650
13 X 6 1/2 X 30.8	T	30.8	9.07	8.88	145.43	12.53	.405	6.56	.650
15 X 6 1/2 X 33.6	T	33.6	9.88	10.07	218.32	14.53	.405	6.56	.650
17 X 6 1/2 X 36.4	T	36.4	10.69	11.23	310.37	16.53	.405	6.56	.650
19 X 6 1/2 X 39.1	T	39.1	11.50	12.37	423.33	18.53	.405	6.56	.650
21 X 6 1/2 X 42.6	T-T	42.6	12.53	13.79	598.84	21.06	.405	6.56	.650
CF W 21 X 62									
8 X 8 1/4 X 27.9	T	27.9	8.22	6.56	51.53	8.50	.400	8.24	.615
10 X 8 1/4 X 30.7	T	30.7	9.02	7.89	93.45	10.50	.400	8.24	.615
12 X 8 1/4 X 33.4	T	33.4	9.82	9.16	151.77	12.50	.400	8.24	.615
14 X 8 1/4 X 36.1	T	36.1	10.62	10.40	228.46	14.50	.400	8.24	.615
16 X 8 1/4 X 38.8	T	38.8	11.42	11.60	325.38	16.50	.400	8.24	.615
18 X 8 1/4 X 41.5	T	41.5	12.22	12.78	444.34	18.50	.400	8.24	.615
21 X 8 1/4 X 44.9	T-T	44.9	13.22	14.21	626.28	20.99	.400	8.24	.615
CF W 21 X 68									
9 X 8 1/4 X 30.8	T	30.8	9.05	6.62	56.64	8.57	.430	8.27	.685
11 X 8 1/4 X 33.7	T	33.7	9.91	7.96	102.53	10.57	.430	8.27	.685
13 X 8 1/4 X 36.6	T	36.6	10.77	9.24	166.32	12.57	.430	8.27	.685
15 X 8 1/4 X 39.6	T	39.6	11.63	10.49	250.18	14.57	.430	8.27	.685
17 X 8 1/4 X 42.5	T	42.5	12.49	11.70	356.12	16.57	.430	8.27	.685
19 X 8 1/4 X 45.4	T	45.4	13.35	12.88	486.09	18.57	.430	8.27	.685

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOF	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
21 X 8 1/4 X 49.2	I-T	49.2	14.46	14.36	690.98	21.13	.430	8.27	.685
CF W 21 X 73									
9 X 8 1/4 X 37.1	T	33.1	9.72	6.66	60.88	8.62	.455	8.30	.740
11 X 8 1/4 X 36.2	T	36.2	10.63	8.01	110.02	10.62	.455	8.30	.740
13 X 8 1/4 X 39.2	T	39.2	11.54	9.30	178.30	12.62	.455	8.30	.740
15 X 8 1/4 X 42.3	T	42.3	12.45	10.54	268.01	14.62	.455	8.30	.740
17 X 8 1/4 X 45.4	T	45.4	13.36	11.76	381.30	16.62	.455	8.30	.740
19 X 8 1/4 X 48.5	T	48.5	14.27	12.94	520.26	18.62	.455	8.30	.740
21 X 8 1/4 X 52.6	I-T	52.6	15.47	14.47	744.47	21.24	.455	8.30	.740
CF W 21 X 83									
9 X 8 1/4 X 37.5	T	37.5	11.03	6.69	70.12	8.72	.515	8.36	.835
11 X 8 1/4 X 41.0	T	41.0	12.06	8.04	126.25	10.72	.515	8.36	.835
13 X 8 1/4 X 44.5	T	44.5	13.09	9.33	204.11	12.72	.515	8.36	.835
15 X 8 1/4 X 48.0	T	48.0	14.12	10.57	306.29	14.72	.515	8.36	.835
17 X 8 1/4 X 51.5	T	51.5	15.15	11.79	435.23	16.72	.515	8.36	.835
19 X 8 1/4 X 55.0	T	55.0	16.18	12.97	593.28	18.72	.515	8.36	.835
21 X 8 1/4 X 59.8	I-T	59.8	17.58	14.55	858.47	21.43	.515	8.36	.835
CF W 21 X 93									
9 X 8 1/2 X 42.2	T	42.2	12.40	6.72	80.21	8.81	.580	8.42	.930
11 X 8 1/2 X 46.1	T	46.1	13.56	8.06	143.85	10.81	.580	8.42	.930
13 X 8 1/2 X 50.1	T	50.1	14.72	9.35	231.97	12.81	.580	8.42	.930
15 X 8 1/2 X 54.0	T	54.0	15.88	10.59	347.47	14.81	.580	8.42	.930
17 X 8 1/2 X 57.9	T	57.9	17.04	11.80	493.10	16.81	.580	8.42	.930
19 X 8 1/2 X 61.9	T	61.9	18.20	12.99	671.50	18.81	.580	8.42	.930
22 X 8 1/2 X 67.4	I-T	67.4	19.83	14.61	982.37	21.62	.580	8.42	.930

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 21 X 101									
14 X 12 1/4 X 23.8	T	23.8	7.00	7.00	114.33	14.00	.500	12.29	.800
16 X 12 1/4 X 27.2	T	27.2	8.00	8.00	170.67	16.00	.500	12.29	.800
18 X 12 1/4 X 30.6	T	30.6	9.00	9.00	243.00	18.00	.500	12.29	.800
20 X 12 1/4 X 34.0	T	34.0	10.00	10.00	333.33	20.00	.500	12.29	.800
21 X 12 1/4 X 68.4	I-T	68.4	20.11	15.50	935.87	21.36	.500	12.29	.800
CF W 21 X 111									
14 X 12 1/4 X 26.2	T	26.2	7.70	7.00	125.77	14.00	.550	12.34	.875
16 X 12 1/4 X 29.9	T	29.9	8.80	8.00	187.73	16.00	.550	12.34	.875
18 X 12 1/4 X 33.7	T	33.7	9.90	9.00	267.30	18.00	.550	12.34	.875
20 X 12 1/4 X 37.4	T	37.4	11.00	10.00	366.67	20.00	.550	12.34	.875
22 X 12 1/4 X 75.3	I-T	75.3	22.15	15.56	1043.43	21.51	.550	12.34	.875
CF W 21 X 122									
14 X 12 1/2 X 28.6	T	28.6	8.40	7.00	137.20	14.00	.600	12.39	.960
16 X 12 1/2 X 32.6	T	32.6	9.60	8.00	204.80	16.00	.600	12.39	.960
18 X 12 1/2 X 36.7	T	36.7	10.80	9.00	291.60	18.00	.600	12.39	.960
20 X 12 1/2 X 40.8	T	40.8	12.00	10.00	400.00	20.00	.600	12.39	.960
22 X 12 1/2 X 82.7	I-T	82.7	24.33	15.66	1159.96	21.68	.600	12.39	.960
CF W 21 X 132									
14 X 12 1/2 X 30.9	T	30.9	9.10	7.00	148.63	14.00	.650	12.44	1.035
16 X 12 1/2 X 35.4	T	35.4	10.40	8.00	221.87	16.00	.650	12.44	1.035
18 X 12 1/2 X 39.8	T	39.8	11.70	9.00	315.90	18.00	.650	12.44	1.035
20 X 12 1/2 X 44.2	T	44.2	13.00	10.00	433.33	20.00	.650	12.44	1.035
22 X 12 1/2 X 89.7	I-T	89.7	26.39	15.72	1273.85	21.83	.650	12.44	1.035

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 21 X 147									
14 X 12 1/2 X 34.3	T	34.3	10.08	7.00	164.64	14.00	.720	12.51	1.150
16 X 12 1/2 X 39.2	T	39.2	11.52	8.00	245.76	16.00	.720	12.51	1.150
18 X 12 1/2 X 44.1	T	44.1	12.96	9.00	349.92	18.00	.720	12.51	1.150
20 X 12 1/2 X 49.0	T	49.0	14.40	10.00	480.00	20.00	.720	12.51	1.150
22 X 12 1/2 X 100.1	T-T	100.1	29.44	15.84	1445.15	22.06	.720	12.51	1.150
CF W 24 X 55									
8 X 7 X 21.8	T	21.8	6.41	5.79	36.81	7.79	.395	7.01	.505
10 X 7 X 24.5	T	24.5	7.20	7.04	69.47	9.79	.395	7.01	.505
12 X 7 X 27.2	T	27.2	7.99	8.25	115.79	11.79	.395	7.01	.505
14 X 7 X 29.9	T	29.9	8.78	9.42	177.54	13.79	.395	7.01	.505
16 X 7 X 32.5	T	32.5	9.57	10.56	256.44	15.79	.395	7.01	.505
18 X 7 X 35.2	T	35.2	10.36	11.68	354.16	17.79	.395	7.01	.505
20 X 7 X 37.9	T	37.9	11.15	12.78	472.36	19.79	.395	7.01	.505
22 X 7 X 40.6	T	40.6	11.94	13.87	612.67	21.79	.395	7.01	.505
24 X 7 X 43.0	I-T	43.0	12.65	14.83	757.88	23.57	.395	7.01	.505
CF W 24 X 62									
8 X 7 X 24.8	T	24.8	7.28	5.88	41.59	7.87	.430	7.04	.590
10 X 7 X 27.7	T	27.7	8.14	7.16	78.32	9.87	.430	7.04	.590
12 X 7 X 30.6	T	30.6	9.00	8.38	130.37	11.87	.430	7.04	.590
14 X 7 X 33.5	T	33.5	9.86	9.56	199.69	13.87	.430	7.04	.590
16 X 7 X 36.5	T	36.5	10.72	10.71	288.19	15.87	.430	7.04	.590
18 X 7 X 39.4	T	39.4	11.58	11.84	397.71	17.87	.430	7.04	.590
20 X 7 X 42.3	T	42.3	12.44	12.96	530.06	19.87	.430	7.04	.590
22 X 7 X 45.2	T	45.2	13.30	14.05	687.03	21.87	.430	7.04	.590
24 X 7 X 48.0	I-T	48.0	14.11	15.07	857.62	23.74	.430	7.04	.590

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES									
SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 24 X 68									
10 X 9	X 30.9 T	30.9	9.10	7.48	81.81	9.87	.415	8.97	.585
12 X 9	X 33.7 T	33.7	9.93	8.77	136.84	11.87	.415	8.97	.585
14 X 9	X 36.6 T	36.6	10.76	10.02	210.29	13.87	.415	8.97	.585
16 X 9	X 39.4 T	39.4	11.59	11.23	304.15	15.87	.415	8.97	.585
18 X 9	X 42.2 T	42.2	12.42	12.41	420.29	17.87	.415	8.97	.585
20 X 9	X 45.0 T	45.0	13.25	13.57	560.54	19.87	.415	8.97	.585
22 X 9	X 47.9 T	47.9	14.08	14.71	726.68	21.87	.415	8.97	.585
24 X 9	X 50.5 I-T	50.5	14.85	15.76	906.49	23.73	.415	8.97	.585
CF W 24 X 76									
10 X 9	X 34.7 T	34.7	10.20	7.63	90.25	9.96	.440	8.99	.680
12 X 9	X 37.7 T	37.7	11.08	8.94	150.82	11.96	.440	8.99	.680
14 X 9	X 40.7 T	40.7	11.96	10.21	231.67	13.96	.440	8.99	.680
16 X 9	X 43.6 T	43.6	12.84	11.44	334.94	15.96	.440	8.99	.680
18 X 9	X 46.6 T	46.6	13.72	12.64	462.69	17.96	.440	8.99	.680
20 X 9	X 49.6 T	49.6	14.60	13.82	616.89	19.96	.440	8.99	.680
22 X 9	X 52.6 T	52.6	15.48	14.98	799.46	21.96	.440	8.99	.680
24 X 9	X 55.6 I-T	55.6	16.34	16.09	1007.74	23.92	.440	8.99	.680
CF W 24 X 84									
10 X 9	X 38.4 T	38.4	11.31	7.73	99.29	10.05	.470	9.02	.770
12 X 9	X 41.6 T	41.6	12.25	9.06	165.70	12.05	.470	9.02	.770
14 X 9	X 44.8 T	44.8	13.19	10.34	254.31	14.05	.470	9.02	.770
16 X 9	X 48.0 T	48.0	14.13	11.59	367.46	16.05	.470	9.02	.770
18 X 9	X 51.2 T	51.2	15.07	12.80	507.37	18.05	.470	9.02	.770
20 X 9	X 54.4 T	54.4	16.01	13.99	676.19	20.05	.470	9.02	.770
22 X 9	X 57.6 T	57.6	16.95	15.16	876.00	22.05	.470	9.02	.770
24 X 9	X 60.9 I-T	60.9	17.91	16.34	1115.11	24.10	.470	9.02	.770

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION		WT/FT	A	CTOE	IO	D	TW	WF	TF
IN X IN X LBS/FT		LBS	IN2	IN	IN4	IN	IN	IN	IN
CF W 24 X 94									
10 X 9 X 43.2	T	43.2	12.71	7.81	111.69	10.16	.515	9.07	.875
12 X 9 X 46.7	T	46.7	13.74	9.15	185.96	12.16	.515	9.07	.875
14 X 9 X 50.2	T	50.2	14.77	10.44	284.98	14.16	.515	9.07	.875
16 X 9 X 53.7	T	53.7	15.80	11.69	411.35	16.16	.515	9.07	.875
18 X 9 X 57.2	T	57.2	16.83	12.92	567.53	18.16	.515	9.07	.875
20 X 9 X 60.7	T	60.7	17.86	14.12	755.88	20.16	.515	9.07	.875
22 X 9 X 64.2	T	64.2	18.89	15.29	978.72	22.16	.515	9.07	.875
24 X 9 X 68.0	I-T	68.0	20.00	16.54	1260.01	24.31	.515	9.07	.875
CF W 24 X 104									
14 X 12 3/4 X 23.8	T	23.8	7.00	7.00	114.33	14.00	.500	12.75	.750
16 X 12 3/4 X 27.2	T	27.2	8.00	8.00	170.67	16.00	.500	12.75	.750
18 X 12 3/4 X 30.6	T	30.6	9.00	9.00	243.00	18.00	.500	12.75	.750
20 X 12 3/4 X 34.0	T	34.0	10.00	10.00	333.33	20.00	.500	12.75	.750
24 X 12 3/4 X 72.1	I-T	72.1	21.22	17.08	1288.37	24.06	.500	12.75	.750
CF W 24 X 117									
14 X 12 3/4 X 26.2	T	26.2	7.70	7.00	125.77	14.00	.550	12.80	.850
16 X 12 3/4 X 29.9	T	29.9	8.80	8.00	187.73	16.00	.550	12.80	.850
18 X 12 3/4 X 33.7	T	33.7	9.90	9.00	267.30	18.00	.550	12.80	.850
20 X 12 3/4 X 37.4	T	37.4	11.00	10.00	366.67	20.00	.550	12.80	.850
24 X 12 3/4 X 80.8	I-T	80.8	23.76	17.26	1456.33	24.26	.550	12.80	.850
CF W 24 X 131									
14 X 13 X 28.8	T	28.8	8.47	7.00	138.34	14.00	.605	12.89	.960
16 X 13 X 32.9	T	32.9	9.68	8.00	206.51	16.00	.605	12.89	.960

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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TABLE XIV. Properties of tee-beams cut from wide flange shapes. - Continued

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

SIZE DESCRIPTION			WT/FT	A	CTOE	IO	O	TW	WF	TF
IN X IN X LBS/FT			LBS	IN2	IN	IN4	IN	IN	IN	IN
18 X 13	X 37.0	T	37.0	10.89	9.00	294.03	18.00	.605	12.89	.960
20 X 13	X 41.1	T	41.1	12.10	10.00	403.33	20.00	.605	12.89	.960
24 X 13	X 90.4	I-T	90.4	26.60	17.45	1648.31	24.48	.605	12.89	.960
CF W 24 X 146										
14 X 13	X 30.9	T	30.9	9.10	7.00	148.63	14.00	.650	12.90	1.090
16 X 13	X 35.4	T	35.4	10.40	8.00	221.87	16.00	.650	12.90	1.090
18 X 13	X 39.8	T	39.8	11.70	9.00	315.90	18.00	.650	12.90	1.090
20 X 13	X 44.2	T	44.2	13.00	10.00	433.33	20.00	.650	12.90	1.090
25 X 13	X 100.1	I-T	100.1	29.43	17.73	1841.63	24.74	.650	12.90	1.090
CF W 24 X 162										
14 X 13	X 33.6	T	33.6	9.87	7.00	161.21	14.00	.705	12.96	1.220
16 X 13	X 38.4	T	38.4	11.28	8.00	240.64	16.00	.705	12.96	1.220
18 X 13	X 43.1	T	43.1	12.69	9.00	342.63	18.00	.705	12.96	1.220
20 X 13	X 47.9	T	47.9	14.10	10.00	470.00	20.00	.705	12.96	1.220
25 X 13	X 110.7	T-T	110.7	32.57	17.96	2063.15	25.00	.705	12.96	1.220
CF W 27 X 84										
11 X 10	X 38.4	T	38.4	11.30	8.56	136.97	11.36	.460	9.96	.640
13 X 10	X 41.6	T	41.6	12.22	9.84	215.02	13.36	.460	9.96	.640
15 X 10	X 44.7	T	44.7	13.14	11.08	315.86	15.36	.460	9.96	.640
17 X 10	X 47.8	T	47.8	14.06	12.29	441.66	17.36	.460	9.96	.640
19 X 10	X 50.9	T	50.9	14.98	13.47	594.50	19.36	.460	9.96	.640
21 X 10	X 54.1	T	54.1	15.90	14.64	776.41	21.36	.460	9.96	.640
23 X 10	X 57.2	T	57.2	16.82	15.78	989.38	23.36	.460	9.96	.640
25 X 10	X 60.3	T	60.3	17.74	16.91	1235.36	25.36	.460	9.96	.640
27 X 10	X 62.4	I-T	62.4	18.37	17.67	1421.75	26.71	.460	9.96	.640

PROPERTIES OF TEE BEAMS CUT FROM WIDE FLANGE SHAPES

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